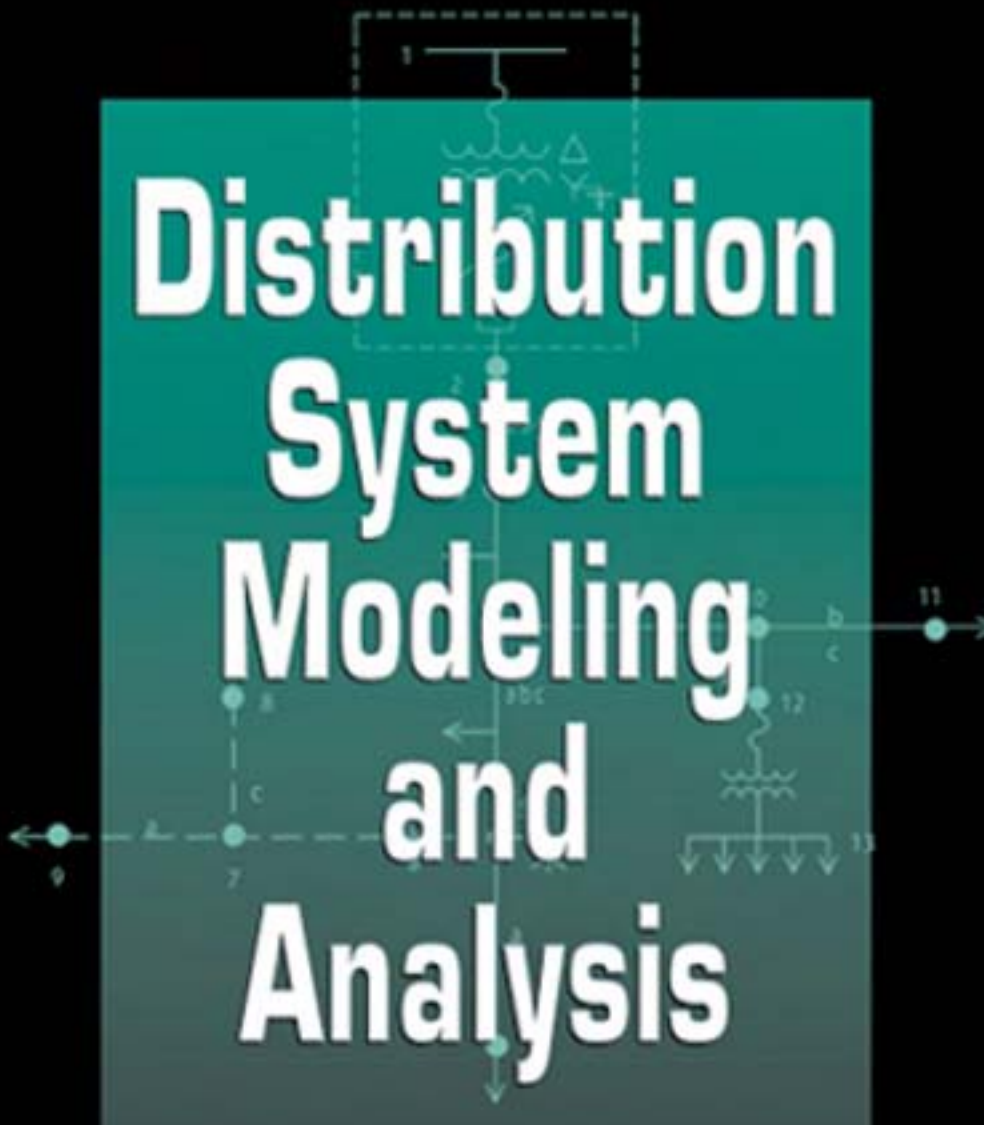


A faint background diagram of a power distribution system. It features a central vertical line with several nodes. At the top, a dashed box encloses a transformer symbol. To the left, a horizontal line with nodes 6, 7, and 8 is shown, with a dashed line connecting node 8 to node 7. To the right, a horizontal line with nodes 10, 11, and 12 is shown, with a dashed line connecting node 12 to node 11. Below node 12, there is a busbar with four downward-pointing arrows, labeled 13. Various other symbols like wavy lines and triangles are scattered throughout the diagram.

Distribution System Modeling and Analysis

William H. Kersting



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Preface

In the last 40 years many papers and textbooks devoted to the computer modeling and analysis of large power system networks have been written. For the most part the models and analysis techniques have been developed for large interconnected transmission systems and synchronous generators. Little, if any, attention was devoted to the distribution system and its major components. As a result, the distribution engineer has not had the same number of tools as the systems engineer to analyze the distribution system under steady-state (power-flow) and fault (short-circuit) conditions. Without these tools the distribution engineer has been left in the dark (no pun intended) as to the operating characteristics of distribution feeders. A lot of “seat of the pants” engineering has had to take place in order to keep the lights on.

In recent years more attention has been devoted to the computer modeling and analysis of distribution systems. Computer programs are now available so that the distribution engineer can develop a real feel for how the distribution system is operating. With the tools, power-flow studies can be run to simulate present loading conditions and to help with the long-range planning of new facilities. The tools also provide an opportunity for the distribution engineer to do such things as optimize capacitor placement in order to minimize losses. Different switching scenarios for normal and emergency conditions can be simulated, and short-circuit studies provide the necessary data for the development of a reliable coordinated protection plan for fuses, reclosers, and relay/circuit breakers. In short, the distribution engineer now has the needed tools.

So what is the problem? “Garbage in, garbage out” is the answer. Armed with a commercially available computer program, it is possible for a user to prepare incorrect data that will lead to results that do not make any sense. Without an understanding of the models and a general “feel” for the operating characteristics of a distribution system, serious design errors and operational procedures may result. The user must fully understand the models and analysis techniques of the program.

Most power systems textbooks and courses are limited to the modeling and analysis of balanced three-phase systems. The models and analyses assume a balance so that only a single-phase equivalent model is required. While this works fine for interconnected systems, it is not sufficient for the modeling and analysis of a distribution system. A distribution system is inherently unbalanced, and therefore three-phase models of all the components must be employed. There is a significant difference between the computer programs developed for interconnected system studies and the programs developed for distribution systems. The data requirements for the

distribution system models are more extensive. In fact, much of the necessary data may not be readily available.

For many years there has been a need for a textbook to assist the student and distribution engineer in developing a basic understanding of the modeling and operating characteristics of the major components of a distribution system. With this knowledge it will be possible to prevent the “garbage in, garbage out” scenario.

This textbook assumes that the student has a basic understanding of transformers, electric machines, transmission lines, and symmetrical components. In many universities all of these topics are crammed into a one-semester course. For that reason a quick review of the theory is presented as needed.

There are many example problems throughout the text. These examples are intended to not only demonstrate the application of the models, but to also teach a “feel” for what the answers should be. The example problems should be studied very carefully. Each chapter will have a series of homework problems that will assist the student in applying the models and developing a better understanding of the operating characteristics of the component being modeled. A word of warning: most of the problems are very number intensive, intensive to the point that most of them cannot be worked easily without using a computing tool such as Mathcad™. Students are urged to learn how to use this very powerful program. They are also encouraged to write their own simple computer programs for many of the problems. A summary of the intent of each chapter follows.

Chapter 1 introduces the basic components of a distribution system. Included is an introduction to the type of data that is necessary to model a distribution system.

Chapter 2 is a discussion of “load.” The attempt here is to make the student understand that the load on a distribution system is constantly changing, and that this must be taken into account in all studies.

Chapter 3 presents some helpful approximate analysis techniques that will help the student know what “ballpark” answers to look for when more precise studies are made.

Chapter 4 is a very important chapter in developing the exact model of line segments. How to take into account the unbalanced loading and unsymmetrical configurations in the calculation of line impedances is presented in great detail. Both overhead and underground lines are included.

Chapter 5 is in many ways a continuation of Chapter 4, except that it is limited to shunt admittance calculations.

Chapter 6 develops the first of the generalized matrices that will be used to model the major components of a distribution system. This chapter is limited to the three-phase, unbalanced line model.

Chapter 7 addresses voltage regulation. Starting with a review of basic transformer theory, the chapter moves to the development of three-phase models of step-voltage regulators and their control. The models developed are in the form of generalized matrices similar to those developed for line segments.

Chapter 8 develops comprehensive models of several of the standard three-phase transformer connections that are common on a distribution system. The models, again, are in the form of generalized matrices.

Chapter 9 develops the models for the various types of loads on a distribution system.

Chapter 10 puts it all together. All of the component models developed in earlier chapters are put together to form a model of a distribution feeder. The ladder iterative technique is developed and demonstrated. Also, the three-phase model for short-circuit studies is developed and demonstrated.

Two student version software packages are available. Students and professors are encouraged to acquire one or both. The packages available are

1. *Radial Distribution Analysis Package (RDAP)*

W. H. Power Consultants
P.O. Box 3903
Las Cruces, NM 88003
(505) 646-2434
E-mail: wkerstin@nmsu.edu
Homepage: www.zianet.com/whpower
or www.nmsu.edu/~wkerstin/

2. *Windmil*

Milsoft Integrated Solutions, Inc.
P.O. Box 7526
Abilene, TX 79608
E-mail: support@milsoft.com
Homepage: www.milsoft.com

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Professor Kersting is a Fellow of the Institute of Electrical and Electronics Engineers; he received the Edison Electric Institute's Power Engineering Educator Award in 1979 and the NMSU Westhafer Award for Excellence in Teaching in 1977. Prior to joining NMSU, he was employed as a distribution engineer by the El Paso Electric Company. Professor Kersting has been an active member of the IEEE Power Engineering Education and Power Engineering Committees.

Acknowledgments

I would be remiss if I didn't acknowledge the patience that my students have displayed over the past many years, as I have taught this material without the aid of a textbook. The students have had to live with taking notes and/or deciphering last-minute notes distributed in class. Their positive attitudes toward the material and what I was trying to accomplish have gone a long way toward making this text possible.

I want to thank Dr. Leonard Bohmann and his students at Michigan Tech for reviewing and correcting the manuscript. Their suggestions were very helpful. My thanks also to Dr. Leo Grigsby for his encouragement and review of the manuscript.

I would like to dedicate this book to my loving wife Joanne for her encouragement and love that has made all of this possible. She spent many lonely evenings practicing the piano as I sat pounding out the text and/or yelling at the computer.

1

Introduction to Distribution Systems

The major components of an electric power system are shown in Figure 1.1. Of these components, the distribution system has traditionally been characterized as the most unglamorous component. In the last half of the twentieth century the design and operation of the generation and transmission components presented many challenges to practicing engineers and researchers. Power plants became larger and larger. Transmission lines crisscrossed the land forming large interconnected networks. The operation of these large interconnected networks required the development of new analysis and operational techniques. Meanwhile, the distribution systems continued to deliver power to the ultimate user's meter with little or no analysis. As a direct result, distribution systems were typically overdesigned.

Times have changed. It has become very important and necessary to operate a distribution system at its maximum capacity. Some of the questions that need to be answered are

1. What is the maximum capacity?
2. How do we determine this capacity?
3. What are the operating limits that must be satisfied?
4. What can be done to operate the distribution system within the operating limits?
5. What can be done to make the distribution system operate more efficiently?

All of these questions can be answered only if the distribution system can be modeled very accurately.

The purpose of this text is to develop accurate models for all of the major components of a distribution system. Once the models have been developed, analysis techniques for steady-state and short-circuit conditions will be developed.

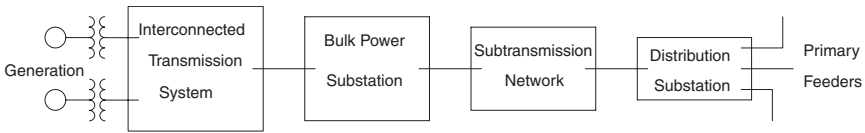


FIGURE 1.1
Major power system components.

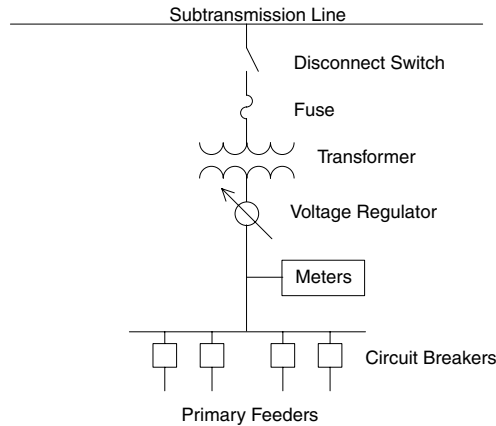


FIGURE 1.2
Simple distribution substation.

1.1 The Distribution System

The distribution system typically starts with the distribution substation that is fed by one or more subtransmission lines. In some cases the distribution substation is fed directly from a high-voltage transmission line, in which case there is likely no subtransmission system. This varies from company to company. Each distribution substation will serve one or more primary feeders. With a rare exception, the feeders are radial, which means that there is only one path for power to flow from the distribution substation to the user.

1.2 Distribution Substations

A diagram of a very simple one-line distribution substation is shown in Figure 1.2. Although Figure 1.2 displays the simplest distribution substation, it illustrates the major components that will be found in all substations.

1. High-side and low-side switching: in Figure 1.2 the high-voltage switching is done with a simple switch. More extensive substations may use high-voltage circuit breakers in a variety of high-voltage bus designs. The low-voltage switching in the figure is accomplished with relay-controlled circuit breakers. In many cases reclosers will be used in place of the relay/circuit breaker combination. Some substation designs will include a low-voltage bus circuit breaker in addition to the circuit breakers for each feeder. As is the case with the high-voltage bus, the low-voltage bus can take on a variety of designs.
2. Voltage transformation: the primary function of a distribution substation is to reduce the voltage to the distribution voltage level. In Figure 1.2 only one transformer is shown. Other substation designs will call for two or more three-phase transformers. The substation transformers can be three-phase units or three single-phase units connected in a standard connection. There are many "standard" distribution voltage levels. Some of the common ones are 34.5 kV, 23.9 kV, 14.4 kV, 13.2 kV, 12.47 kV, and, in older systems, 4.16 kV.
3. Voltage regulation: as the load on the feeders varies, the voltage drop between the substation and the user will vary. In order to maintain the user's voltages within an acceptable range, the voltage at the substation needs to change as the load changes. In Figure 1.2 the voltage is regulated by a "step-type" regulator that will alter the voltage plus or minus 10% on the low-side bus. Sometimes this function is accomplished with a "load tap changing" (LTC) transformer. The LTC changes the taps on the low-voltage windings of the transformer as the load varies. Many substation transformers will have "fixed taps" on the high-voltage winding. These are used when the source voltage is always either above or below the nominal voltage. The fixed tap settings can alter the voltage plus or minus 5%. Many times, instead of a bus regulator, each feeder will have its own regulator. This can be in the form of a three-phase gang-operated regulator or individual phase regulators that operate independently.
4. Protection: the substation must be protected against the occurrence of short circuits. In the simple design of Figure 1.2, the only automatic protection against short circuits inside the substation is by way of the high-side fuses on the transformer. As substation designs become more complex, more extensive protective schemes will be employed to protect the transformer, the high- and low-voltage buses, and any other piece of equipment. Individual feeder circuit breakers or reclosers are used to provide interruption of short circuits that occur outside the substation.

5. Metering: every substation has some form of metering. This may be as simple as an analog ammeter displaying the present value of substation current, as well as the minimum and maximum currents that have occurred over a specific time period. Digital recording meters are becoming very common. These meters record the minimum, average, and maximum values of current, voltage, power, power factor, etc. over a specified time range. Typical time ranges are 15 minutes, 30 minutes, and 1 hour. The digital meters may monitor the output of each substation transformer and/or the output of each feeder.

A more comprehensive substation layout is shown in Figure 1.3. The substation in Figure 1.3 has two load-tap changing transformers, serves four distribution feeders, and is fed from two subtransmission lines. Under normal conditions the circuit breakers (CB) are in the following positions:

Circuit breakers closed: X, Y, 1,3,4,6

Circuit breakers open: Z, 2,5

With the breakers in their normal positions, each transformer is served from a different subtransmission line and serves two feeders. Should one of the subtransmission lines go out of service, then breaker X or Y is opened and breaker Z is closed. Now both transformers are served from the same subtransmission line. The transformers are sized such that each transformer can supply all four feeders under an emergency operating condition. For example, if Transformer T-1 is out of service, then breakers X, 1, and 4 are opened and breakers 2 and 5 are closed. With that breaker arrangement, all four feeders are served by transformer T-2. The low-voltage bus arrangement

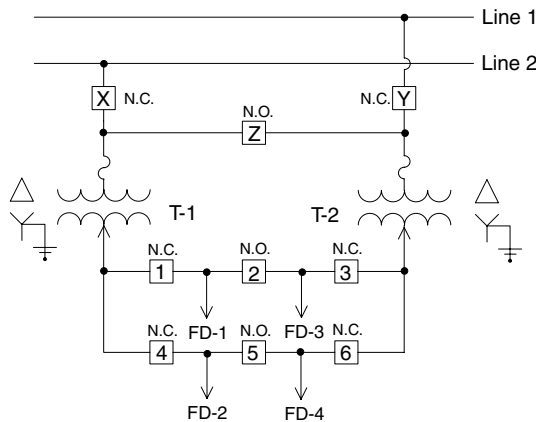


FIGURE 1.3

Two-transformer substation with breaker-and-a-half scheme.

is referred to as a “breaker-and-a-half scheme” since three breakers are required to serve two feeders.

There is an unlimited number of substation configurations possible. It is up to the substation design engineer to create a design that provides the five basic functions and yields the most reliable service economically possible.

1.3 Radial Feeders

Radial distribution feeders are characterized by having only one path for power to flow from the source (distribution substation) to each customer. A typical distribution system will be composed of one or more distribution substations consisting of one or more feeders. Components of the feeder may consist of the following:

1. Three-phase primary “main” feeder
2. Three-phase, two-phase (“V” phase), and single-phase laterals
3. Step-type voltage regulators
4. In-line transformers
5. Shunt capacitor banks
6. Distribution transformers
7. Secondaries
8. Three-phase, two-phase, and single-phase loads

The loading of a distribution feeder is inherently unbalanced because of the large number of unequal single-phase loads that must be served. An additional unbalance is introduced by the nonequilateral conductor spacings of three-phase overhead and underground line segments.

Because of the nature of the distribution system, conventional power-flow and short-circuit programs used for transmission system studies are not adequate. Such programs display poor convergence characteristics for radial systems. The programs also assume a perfectly balanced system so that a single-phase equivalent system is used.

If a distribution engineer is to be able to perform accurate power-flow and short-circuit studies, it is imperative that the distribution feeder be modeled as accurately as possible. This means that three-phase models of the major components must be utilized. Three-phase models for the major components will be developed in the following chapters. They will be developed in the “phase frame” rather than applying the method of symmetrical components.

Figure 1.4 shows a simple “one-line” diagram of a three-phase feeder. Figure 1.4 illustrates the major components of a distribution system. The connecting points of the components will be referred to as “nodes.” Note that

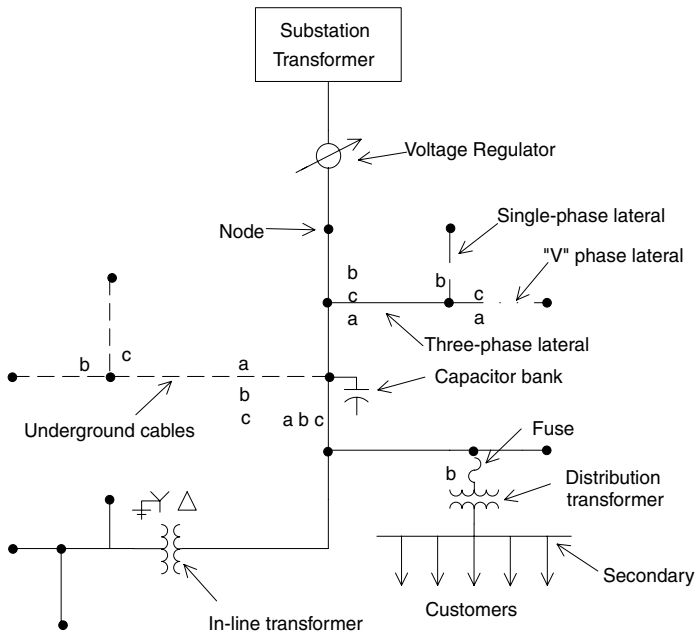


FIGURE 1.4
Simple distribution feeder.

the phasing of the line segments is shown. This is important if the most accurate models are to be developed.

1.4 Distribution Feeder Map

The analysis of a distribution feeder is important to an engineer in order to determine the existing operating conditions of a feeder, and to be able to play the “what if” scenarios of future changes to the feeder. Before the engineer can perform the analysis of a feeder, a detailed map of the feeder must be available. A sample of such a map is shown in Figure 1.5. The map of Figure 1.5 contains most of the following information:

1. Lines (overhead and underground)
 - a. Where
 - b. Distances
 - c. Details
 - i. Conductor sizes (not shown on this map)
 - ii. Phasing



FIGURE 1.5
123-node test feeder.

2. Distribution transformers
 - a. Location
 - b. kVA rating
 - c. Phase connection
 3. In-line transformers
 - a. Location
 - b. kVA rating
 - c. Connection
 4. Shunt capacitors
 - a. Location
 - b. kvar rating
 - c. Phase connection
 5. Voltage regulators
 - a. Location
 - b. Phase connection
 - c. Type (not shown on this map)
 - i. Single-phase
 - ii. Three-phase
 6. Switches
 - a. Location
 - b. Normal open/close status
-

1.5 Distribution Feeder Electrical Characteristics

Information from the map will define the physical location of the various devices. Electrical characteristics for each device will have to be determined before the analysis of the feeder can commence. In order to determine the electrical characteristics, the following data must be available:

1. Overhead and underground spacings
2. Conductor tables
 - a. Geometric mean radius (GMR) (ft.)
 - b. Diameter (inches)
 - c. Resistance (Ω /mile)
3. Voltage regulators
 - a. Potential transformer ratios
 - b. Current transformer ratios

- c. Compensator settings
 - i. Voltage level
 - ii. Bandwidth
 - iii. R and X settings, in volts (V)
 - 4. Transformers
 - a. kVA rating
 - b. Voltage ratings
 - c. Impedance (R and X)
 - d. No-load power loss
-

1.6 Summary

It is becoming increasingly more important to be able to accurately model and analyze distribution systems. There are many different substation designs possible, but, for the most part, the substation serves one or more radial feeders. Each feeder must be modeled as accurately as possible in order for the analysis to have meaning. Sometimes the most difficult task for the engineer is to acquire all of the necessary data. Feeder maps will contain most of the needed data. Additional data such as standard pole configurations, specific conductors used on each line segment, three-phase transformer connections, and voltage regulator settings must come from stored records. Once all of the data has been acquired, the analysis can begin utilizing models of the various devices that will be developed in later chapters.

2

The Nature of Loads

The modeling and analysis of a power system depend upon the load. What is load? The answer to that question depends upon what type of an analysis is desired. For example, the steady-state analysis (power-flow study) of an interconnected transmission system will require a different definition of load than that used in the analysis of a secondary in a distribution feeder. The problem is that the load on a power system is constantly changing. The closer you are to the customer, the more pronounced will be the ever-changing load. There is no such thing as a “steady-state” load. In order to come to grips with load, it is first necessary to look at the load of an individual customer.

2.1 Definitions

The load that an individual customer or a group of customers presents to the distribution system is constantly changing. Every time a light bulb or an electrical appliance is switched on or off, the load seen by the distribution feeder changes. In order to describe the changing load, the following terms are defined:

1. Demand
 - Load averaged over a specific period of time
 - Load can be kW, kvar, kVA, or A
 - Must include the time interval
 - Example: the 15-minute kW demand is 100 kW
2. Maximum Demand
 - Greatest of all demands that occur during a specific time
 - Must include demand interval, period, and units
 - Example: the 15-minute maximum kW demand for the week was 150 kW

3. Average Demand
 - The average of the demands over a specified period (day, week, month, etc.)
 - Must include demand interval, period, and units
 - Example: the 15-minute average kW demand for the month was 350 kW
4. Diversified Demand
 - Sum of demands imposed by a group of loads over a particular period
 - Must include demand interval, period, and units
 - Example: the 15-minute diversified kW demand in the period ending at 9:30 was 200 kW
5. Maximum Diversified Demand
 - Maximum of the sum of the demands imposed by a group of loads over a particular period
 - Must include demand interval, period, and units
 - Example: the 15-minute maximum diversified kW demand for the week was 500 kW
6. Maximum Noncoincident Demand
 - For a group of loads, the sum of the individual maximum demands without any restriction that they occur at the same time
 - Must include demand interval, period, and units
 - Example: the maximum noncoincident 15-minute kW demand for the week was 700 kW
7. Demand Factor
 - Ratio of maximum demand to connected load
8. Utilization Factor
 - Ratio of the maximum demand to rated capacity
9. Load Factor
 - Ratio of the average demand of any individual customer or group of customers over a period to the maximum demand over the same period
10. Diversity Factor
 - Ratio of the maximum noncoincident demand to the maximum diversified demand
11. Load Diversity
 - Difference between maximum noncoincident demand and the maximum diversified demand

2.2 Individual Customer Load

Figure 2.1 illustrates how the instantaneous kW load of a customer changes during two 15-minute intervals.

2.2.1 Demand

In order to define the load, the demand curve is broken into equal time intervals. In Figure 2.1 the selected time interval is 15 minutes. In each interval the average value of the demand is determined. In Figure 2.1 the straight lines represent the average load in a time interval. The shorter the time interval, the more accurate will be the value of the load. This process is very similar to numerical integration. The average value of the load in an interval is defined as the *15-minute kW demand*.

The 24-hour 15-minute kW demand curve for a customer is shown in Figure 2.2. This curve is developed from a spreadsheet that gives the 15-minute kW demand for a period of 24 hours.

2.2.2 Maximum Demand

The demand curve shown in Figure 2.2 represents a typical residential customer. Each bar depicts the 15-minute kW demand. Note that during the 24-hour period there is a great variation in the demand. This particular customer has three periods in which the kW demand exceeds 6.0 kW. The greatest of these is the *15-minute maximum kW demand*. For this customer the 15-minute maximum kW demand occurs at 13:15 and has a value of 6.18 kW.

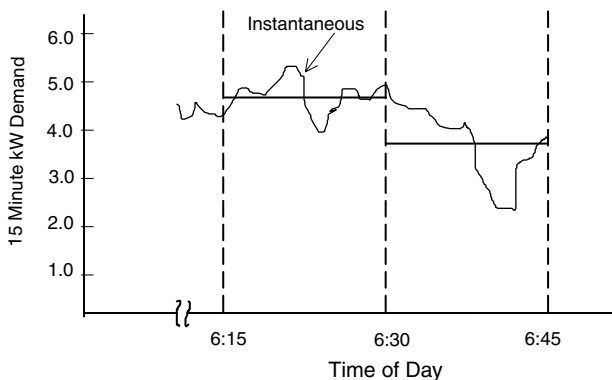
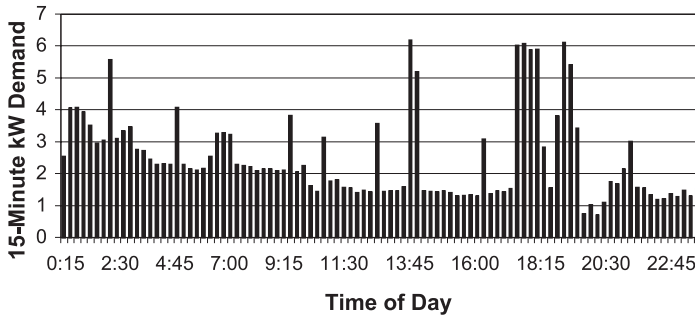


FIGURE 2.1
Customer demand curve.

**FIGURE 2.2**

24-hour demand curve for Customer #1.

2.2.3 Average Demand

During the 24-hour period, energy (kWh) will be consumed. The energy in kWh used during each 15-minute time interval is computed by:

$$\text{kWh} = (\text{15-min kW demand}) \cdot \frac{1}{4} \text{ hour} \quad (2.1)$$

The total energy consumed during the day is the summation of all of the 15-minute interval consumptions. From the spreadsheet, the total energy consumed during the period by Customer #1 is 58.96 kWh. The *15-minute average kW demand* is computed by:

$$\text{Average demand} = \frac{\text{Total energy}}{\text{Hours}} = \frac{58.96}{24} = 2.46 \text{ kW} \quad (2.2)$$

2.2.4 Load Factor

“Load factor” is a term that is often used when describing a load. It is defined as the ratio of the average demand to the maximum demand. In many ways load factor gives an indication of how well the utility’s facilities are being utilized. From the utility’s standpoint, the optimal load factor would be 1.00 since the system has to be designed to handle the maximum demand. Sometimes utility companies will encourage industrial customers to improve their load factors. One method of encouragement is to penalize the customer on the electric bill for having a low load factor.

For Customer #1 in Figure 2.2 the load factor is computed to be

$$\text{Load factor} = \frac{\text{Average 15-min kW demand}}{\text{Max. 15-min kW demand}} = \frac{2.46}{6.18} = 0.40 \quad (2.3)$$

2.3 Distribution Transformer Loading

A distribution transformer will provide service to one or more customers. Each customer will have a demand curve similar to that in Figure 2.2. However, the peaks and valleys and maximum demands will be different for each customer. Figures 2.3, 2.4, and 2.5 give the demand curves for the three additional customers connected to the same distribution transformer. The load curves for the four customers show that each customer has his unique loading characteristic. The customers' individual maximum kW demand occurs at different times of the day. Customer #3 is the only one who will have a high load factor. A summary of individual loads is given in Table 2.1. These four customers demonstrate that there is great diversity among their loads.

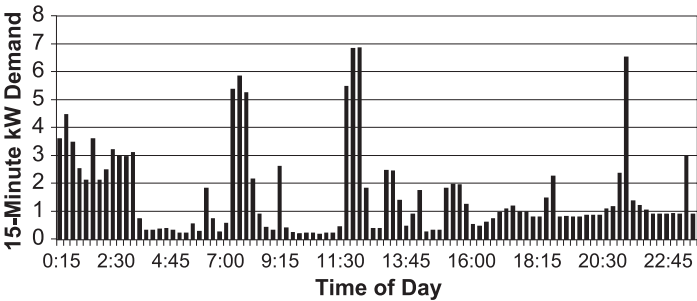


FIGURE 2.3
24-hour demand curve for Customer #2.

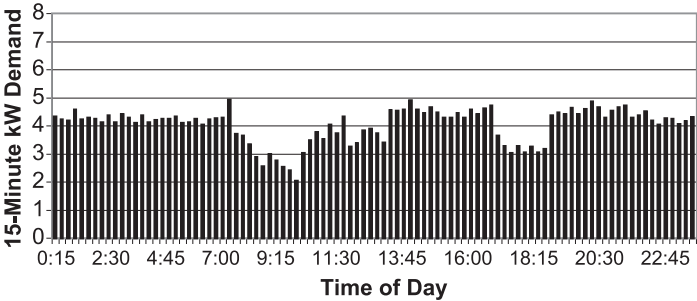


FIGURE 2.4
24-hour demand curve for Customer #3.

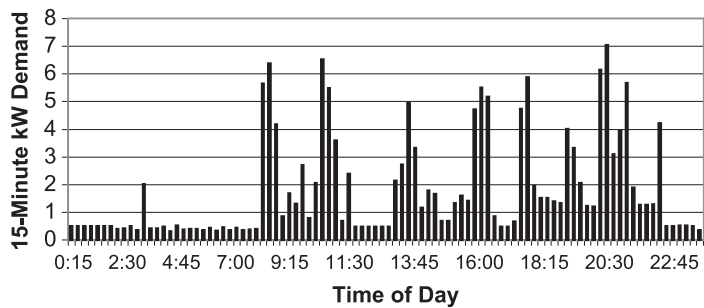


FIGURE 2.5
24-hour demand curve for Customer #4.

TABLE 2.1
Individual Customer Load Characteristics

	Cust. #1	Cust. #2	Cust. #3	Cust. #4
Energy Usage (kWh)	58.57	36.46	95.64	42.75
Maximum kW Demand	6.18	6.82	4.93	7.05
Time of Max. kW Demand	13:15	11:30	6:45	20:30
Average kW Demand	2.44	1.52	3.98	1.78
Load Factor	0.40	0.22	0.81	0.25

2.3.1 Diversified Demand

It is assumed that the same distribution transformer serves the four customers discussed previously. The sum of the four 15 kW demands for each time interval is the *diversified demand* for the group in that time interval, and, in this case, the distribution transformer. The 15-minute diversified kW demand of the transformer for the day is shown in Figure 2.6. Note how the demand curve is beginning to smooth out. There are not as many significant changes as in some of the individual customer curves.

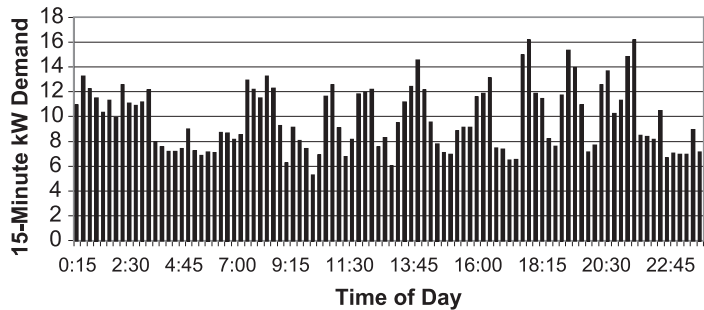


FIGURE 2.6
Transformer diversified demand curve.

2.3.2 Maximum Diversified Demand

The transformer demand curve of Figure 2.6 demonstrates how the combined customer loads begin to smooth out the extreme changes of the individual loads. For the transformer, the 15-minute kW demand exceeds 16 kW twice. The greater of these is the *15-minute maximum diversified kW demand* of the transformer. It occurs at 17:30 and has a value of 16.16 kW. Note that this maximum demand does not occur at the same time as any one of the individual demands, nor is this maximum demand the sum of the individual maximum demands.

2.3.3 Load Duration Curve

A *load duration curve* can be developed for the transformer serving the four customers. Sorting in descending order, the kW demand of the transformer develops the load duration curve shown in Figure 2.7. The load duration curve plots the 15-minute kW demand versus the percent of time the transformer operates at or above the specific kW demand. For example, the load duration curve shows the transformer operates with a 15-minute kW demand of 12 kW or greater 22% of the time. This curve can be used to determine whether a transformer needs to be replaced due to an overloading condition.

2.3.4 Maximum Noncoincident Demand

The *15-minute maximum noncoincident kW demand* for the day is the sum of the individual customer 15-minute maximum kW demands. For the transformer in question, the sum of the individual maximums is

$$\text{Max. noncoincident demand} = 6.18 + 6.82 + 4.93 + 7.05 = 24.98 \text{ kW} \quad (2.4)$$

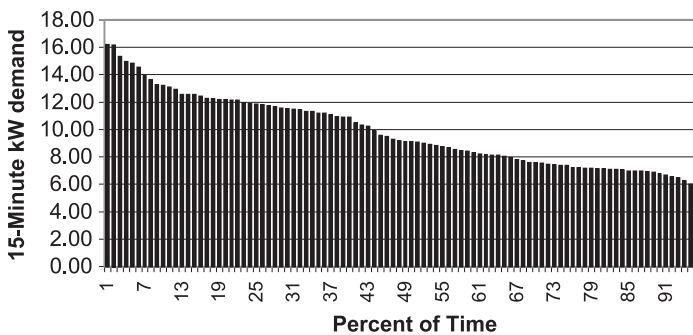


FIGURE 2.7
Transformer load duration curve.

2.3.5 Diversity Factor

By definition, *diversity factor* is the ratio of the maximum noncoincident demand of a group of customers to the maximum diversified demand of the group. With reference to the transformer serving four customers, the diversity factor for the four customers would be

$$\text{Diversity factor} = \frac{\text{Maximum noncoincident demand}}{\text{Maximum diversified demand}} = \frac{24.98}{16.16} = 1.5458 \quad (2.5)$$

The idea behind the diversity factor is that when the maximum demands of the customers are known, then the maximum diversified demand of a group of customers can be computed. There will be a different value of the diversity factor for different numbers of customers. The value computed above would apply for four customers. If there are five customers, then a load survey would have to be set up to determine the diversity factor for five customers. This process would have to be repeated for all practical numbers of customers. Table 2.2 is an example of the diversity factors for the number of customers ranging from one to 70. The table was developed from a different database than the four customers discussed previously. A graph of the diversity factors is shown in Figure 2.8. Note in Table 2.2 and Figure 2.8 that the value of the diversity factor basically leveled out when the number of customers reached 70. This is an important observation because it means, at least for the system from which these diversity factors were determined, that the diversity factor will remain constant at 3.20 from 70 customers up. In other words, as viewed from the substation, the maximum diversified demand of a feeder can be predicted by computing the total noncoincident maximum demand of all of the customers served by the feeder and dividing by 3.2.

TABLE 2.2

Diversity Factors

N	DF	N	DF	N	DF	N	DF	N	DF	N	DF	N	DF
1	1.0	11	2.67	21	2.90	31	3.05	41	3.13	51	3.15	61	3.18
2	1.60	12	2.70	22	2.92	32	3.06	42	3.13	52	3.15	62	3.18
3	1.80	13	2.74	23	2.94	33	3.08	43	3.14	53	3.16	63	3.18
4	2.10	14	2.78	24	2.96	34	3.09	44	3.14	54	3.16	64	3.19
5	2.20	15	2.80	25	2.98	35	3.10	45	3.14	55	3.16	65	3.19
6	2.30	16	2.82	26	3.00	36	3.10	46	3.14	56	3.17	66	3.19
7	2.40	17	2.84	27	3.01	37	3.11	47	3.15	57	3.17	67	3.19
8	2.55	18	2.86	28	3.02	38	3.12	48	3.15	58	3.17	68	3.19
9	2.60	19	2.88	29	3.04	39	3.12	49	3.15	59	3.18	69	3.20
10	2.65	20	2.90	30	3.05	40	3.13	50	3.15	60	3.18	70	3.20

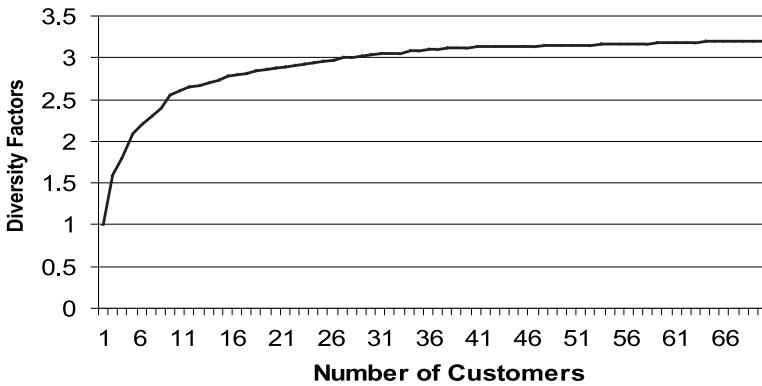


FIGURE 2.8
Diversity factors.

2.3.6 Demand Factor

The demand factor can be defined for an individual customer. For example, the 15-minute maximum kW demand of Customer #1 was found to be 6.18 kW. In order to determine the demand factor, the total connected load of the customer needs to be known. The total connected load will be the sum of the ratings of all of the electrical devices at the customer's location. Assume that this total comes to 35 kW; then, the demand factor is computed to be

$$\text{Demand factor} = \frac{\text{Maximum demand}}{\text{Total connected load}} = \frac{6.18}{35} = 0.1766 \quad (2.6)$$

The demand factor gives an indication of the percentage of electrical devices that are on when the maximum demand occurs. The demand factor can be computed for an individual customer but not for a distribution transformer or the total feeder.

2.3.7 Utilization Factor

The utilization factor gives an indication of how well the capacity of an electrical device is being utilized. For example, the transformer serving the four loads is rated 15 kVA. Using the 16.16-kW maximum diversified demand and assuming a power factor of 0.9, the 15-minute maximum kVA demand on the transformer is computed by dividing the 16.16-kW maximum kW demand by the power factor, and would be 17.96 kVA. The utilization factor is computed to be

$$\text{Utilization factor} = \frac{\text{Maximum kVA demand}}{\text{Transformer kVA rating}} = \frac{17.96}{15} = 1.197 \quad (2.7)$$

2.3.8 Load Diversity

Load diversity is defined as the difference between the noncoincident maximum demand and the maximum diversified demand. For the transformer in question, the load diversity is computed to be

$$\text{Load diversity} = 24.97 - 16.16 = 8.81 \text{ kW} \tag{2.8}$$

2.4 Feeder Load

The load that a feeder serves will display a smoothed-out demand curve as shown in Figure 2.9. The feeder demand curve does not display any of the abrupt changes in demand of an individual customer demand curve or the semi-abrupt changes in the demand curve of a transformer. The simple explanation for this is that with several hundred customers served by the feeder, the odds are good that as one customer is turning off a light bulb another customer will be turning a light bulb on. The feeder load therefore does not experience a jump as would be seen in the individual customer’s demand curve.

2.4.1 Load Allocation

In the analysis of a distribution feeder load, data will have to be specified. The data provided will depend upon how detailed the feeder is to be modeled, and the availability of customer load data. The most comprehensive model of a feeder will represent every distribution transformer. When this is the case, the load allocated to each transformer needs to be determined.

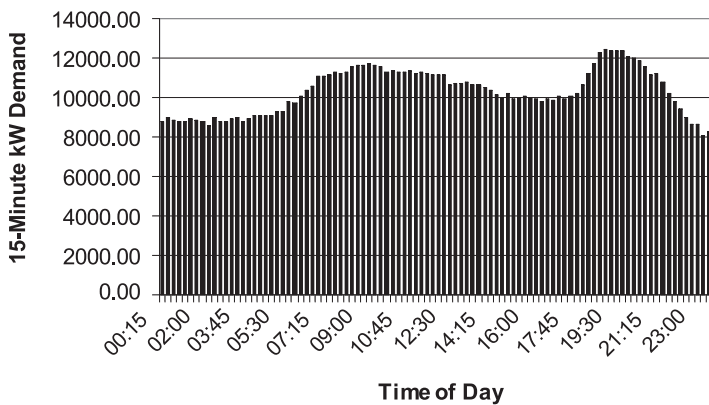


FIGURE 2.9
Feeder demand curve.

2.4.1.1 Application of Diversity Factors

The definition of the diversity factor (DF) is the ratio of the maximum noncoincident demand to the maximum diversified demand. Diversity factors are shown in Table 2.2. When such a table is available, then it is possible to determine the maximum diversified demand of a group of customers such as those served by a distribution transformer; that is, the maximum diversified demand can be computed by:

$$\text{Max. diversified demand} = \frac{\text{Max noncoincident demand}}{DF_n} \quad (2.9)$$

This maximum diversified demand becomes the allocated load for the transformer.

2.4.1.2 Load Survey

Many times the maximum demand of individual customers will be known, either from metering or from a knowledge of the energy (kWh) consumed by the customer. Some utility companies will perform a load survey of similar customers in order to determine the relationship between the energy consumption in kWh and the maximum kW demand. Such a load survey requires the installation of a demand meter at each customer's location. The meter can be the same type used to develop the demand curves previously discussed, or it can be a simple meter that only records the maximum demand during the period. At the end of the survey period the maximum demand vs. kWh for each customer can be plotted on a common graph. Linear regression is used to determine the equation of a straight line that gives the kW demand as a function of kWh. The plot of points for 15 customers, along with the resulting equation derived from a linear regression algorithm, is shown in Figure 2.10. The straight-line equation derived is

$$\text{Max. kW demand} = 0.1058 + 0.005014 \cdot \text{kWh} \quad (2.10)$$

Knowing the maximum demand for each customer is the first step in developing a table of diversity factors as shown in Table 2.2. The next step is to perform a load survey where the maximum diversified demand of groups of customers is metered. This will involve selecting a series of locations where demand meters can be placed that will record the maximum demand for groups of customers ranging from at least 2 to 70. At each meter location the maximum demand of all downstream customers must also be known. With that data, the diversity factor can be computed for the given number of downstream customers.

Example 2.1

A single-phase lateral provides service to three distribution transformers as shown in Figure 2.11. The energy in kWh consumed by each customer during

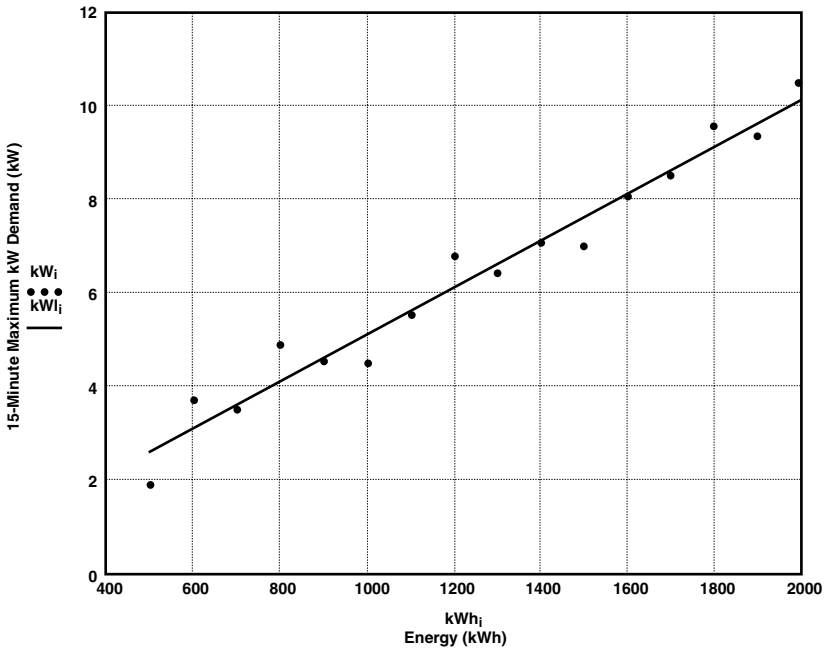


FIGURE 2.10
kW demand vs. kWh for residential customers.

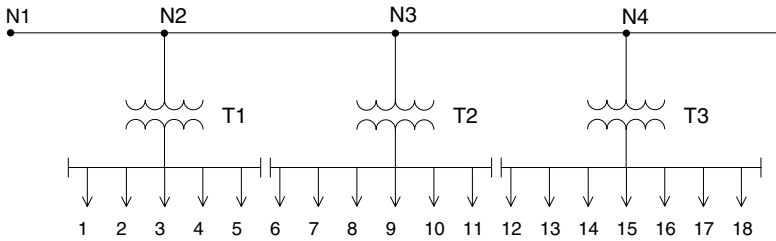


FIGURE 2.11
Single-phase lateral.

a month is known. A load survey has been conducted for customers in this class, and it has been found that the customer 15-minute maximum kW demand is given by the equation:

$$kW_{\text{demand}} = 0.2 + 0.008 \cdot kWh$$

The kWh consumed by Customer #1 is 1523 kWh. The 15-minute maximum kW demand for Customer #1 is then computed as:

$$kW_1 = 0.2 + 0.008 \cdot 1523 = 12.4$$

The results of this calculation for the remainder of the customers is summarized below by transformer.

TRANSFORMER T1

Customer	#1	#2	#3	#4	#5
kWh	1523	1645	1984	1590	1456
kW	12.4	13.4	16.1	12.9	11.9

TRANSFORMER T2

Customer	#6	#7	#8	#9	#10	#11
kWh	1235	1587	1698	1745	2015	1765
kW	10.1	12.9	13.8	14.2	16.3	14.3

TRANSFORMER T3

Customer	#12	#13	#14	#15	#16	#17	#18
kWh	2098	1856	2058	2265	2135	1985	2103
kW	17.0	15.1	16.7	18.3	17.3	16.1	17.0

1. Determine for each transformer the 15-minute noncoincident maximum kW demand and, using the Table of Diversity Factors in Table 2.2, determine the 15-minute maximum diversified kW demand.

$$\mathbf{T1:} \text{ Noncoin. max.} = 12.4 + 13.4 + 16.1 + 12.9 + 11.9 = 66.7 \text{ kW}$$

$$\text{Max. div. demand} = \frac{\text{Noncoincident max.}}{\text{Diversity factor for 5}} = \frac{66.7}{2.20} = 30.3 \text{ kW}$$

$$\mathbf{T2:} \text{ Noncoin. max.} = 12.9 + 13.8 + 14.2 + 16.3 + 14.3 + 17.0 = 81.6 \text{ kW}$$

$$\text{Max. div. demand} = \frac{\text{Noncoincident max.}}{\text{Diversity factor for 6}} = \frac{81.6}{2.30} = 35.5 \text{ kW}$$

$$\mathbf{T3:} \text{ Noncoin. max.} = 17.0 + 15.1 + 16.7 + 18.3 + 17.3 + 16.1 + 17.0 = 117.5 \text{ kW}$$

$$\text{Max. div. demand} = \frac{\text{Noncoincident max.}}{\text{Diversity factor for 7}} = \frac{117.5}{2.40} = 48.9 \text{ kW}$$

Based upon the 15-minute maximum kW diversified demand on each transformer and an assumed power factor of 0.9,

the 15-minute maximum kVA diversified demand on each transformer would be

$$\text{Max. kVA}_{T1} \text{ demand} = \frac{30.3}{.9} = 33.6$$

$$\text{Max. kVA}_{T2} \text{ demand} = \frac{35.5}{.9} = 39.4$$

$$\text{Max. kVA}_{T3} \text{ demand} = \frac{49.0}{.9} = 54.4$$

The kVA ratings selected for the three transformers would be 25 kVA, 37.5 kV, and 50 kVA, respectively. With those selections, only transformer T1 would experience a significant maximum kVA demand greater than its rating (135%).

2. Determine the 15-minute noncoincident maximum kW demand and 15-minute maximum diversified kW demand for each of the line segments.

Segment N1 to N2:

The maximum noncoincident kW demand is the sum of the maximum demands of all 18 customers.

$$\text{Noncoin. max. demand} = 66.7 + 81.6 + 117.5 = 265.5 \text{ kW}$$

The maximum diversified kW demand is the computed by using the diversity factor for 18 customers.

$$\text{Max. div. demand} = \frac{265.8}{2.86} = 92.8 \text{ kW}$$

Segment N2 to N3:

This line segment “sees” 13 customers. The noncoincident maximum demand is the sum of customers number 6 through 18. The diversity factor for 13 (2.74) is used to compute the maximum diversified kW demand.

$$\text{Noncoin. demand} = 81.6 + 117.5 = 199.0 \text{ kW}$$

$$\text{Max. div. demand} = \frac{199.1}{2.74} = 72.6 \text{ kW}$$

Segment N3 to N4:

This line segment sees the same noncoincident demand and diversified demand as that of transformer T3.

$$\text{Noncoin. demand} = 117.4 \text{ kW}$$

$$\text{Max. div. demand} = 48.9 \text{ kW}$$

Example 2.1 demonstrates that Kirchhoff's current law (KCL) is not obeyed when the maximum diversified demands are used as the load flowing through the line segments and through the transformers. For example, at node N1 the maximum diversified demand flowing down the line segment N1-N2 is 92.8 kW, and the maximum diversified demand flowing through transformer T1 is 30.3 kW. KCL would then predict that the maximum diversified demand flowing down line segment N2-N3 would be the difference of these, or 62.5 kW. However, the calculations for the maximum diversified demand in that segment was computed to be 72.6 kW. The explanation is that the maximum diversified demands for the line segments and transformers don't necessarily occur at the same time. At the time that line segment N2-N3 is experiencing its maximum diversified demand, line segment N1-N2 and transformer T1 are not at their maximum values. All that can be said is that, at the time segment N2-N3 is experiencing its maximum diversified demand, the difference between the actual demand on line segment N1-N2 and the demand of transformer T1 will be 72.6 kW. There will be an infinite amount of combinations of line flow down N1-N2 and through transformer T1 that will produce the maximum diversified demand of 72.6 kW on line N2-N3.

2.4.1.3 Transformer Load Management

A transformer load management program is used by utilities to determine the loading on distribution transformers based upon a knowledge of the kWh supplied by the transformer during a peak loading month. The program is primarily used to determine when a distribution transformer needs to be changed out due to a projected overloading condition. The results of the program can also be used to allocate loads to transformers for feeder analysis purposes.

The transformer load management program relates the maximum diversified demand of a distribution transformer to the total kWh supplied by the transformer during a specific month. The usual relationship is the equation of a straight line. Such an equation is determined from a load survey. This type of load survey meters the maximum demand on the transformer in addition to the total energy in kWh of all of the customers connected to the transformer. With the information available from several sample transformers, a curve similar to that shown in Figure 2.10 can be developed, and the constants of the straight-line equation can be computed. This method has an advantage because the utility will have in the billing database the kWh consumed by each customer every month. As long as the utility knows which customers are connected to each transformer by using the developed equation, the maximum diversified demand (allocated load) on each transformer on a feeder can be determined for each billing period.

2.4.1.4 Metered Feeder Maximum Demand

The major disadvantage of allocating load using the diversity factors is that most utilities will not have a table of diversity factors. The process of developing such a table is generally not cost effective. The major disadvantage of

the transformer load management method is that a database is required that specifies which transformers serve which customers. Again, this database is not always available.

Allocating load based upon the metered readings in the substation requires the least amount of data. Most feeders will have metering in the substation that will, at minimum, give either the total three-phase maximum diversified kW or kVA demand and/or the maximum current per phase during a month. The kVA ratings of all distribution transformers is always known for a feeder. The metered readings can be allocated to each transformer based upon the transformer rating. An "allocation factor" (AF) can be determined based upon the metered three-phase kW or kVA demand and the total connected distribution transformer kVA.

$$AF = \frac{\text{Metered demand}}{kVA_{\text{total}}} \quad (2.12)$$

where Metered demand can be either kW or kVA, and

kVA_{total} = sum of the kVA ratings of all distribution transformers

The allocated load per transformer is then determined by:

$$\text{Transformer demand} = AF \cdot kVA_{\text{transformer}} \quad (2.13)$$

The transformer demand will be either kW or kVA depending upon the metered quantity.

When the kW or kVA is metered by phase, then the load can be allocated by phase where it will be necessary to know the phasing of each distribution transformer. When the maximum current per phase is metered, the load allocated to each distribution transformer can be done by assuming nominal voltage at the substation and then computing the resulting kVA. The load allocation will now follow the same procedure as outlined above. If there is no metered information on the reactive power or power factor of the feeder, a power factor will have to be assumed for each transformer load.

Modern substations will have microprocessor-based metering that will provide kW, kvar, kVA, power factor, and current per phase. With this data, the reactive power can also be allocated. Since the metered data at the substation will include losses, an iterative process will have to be followed so that the allocated load plus losses will equal the metered readings.

Example 2.2

Assume that the metered maximum diversified kW demand for the system of Example 2.1 is 92.9 kW. Allocate this load according to the kVA ratings of the three transformers.

$$kVA_{\text{total}} = 25 + 37.5 + 50 = 112.5$$

$$AF = \frac{92.8}{112.5} = 0.8249 \text{ kW/kVA}$$

The allocated kW for each transformer becomes:

$$T1: \quad kW_1 = 0.8249 \cdot 25 = 20.62 \text{ kW}$$

$$T2: \quad kW_1 = 0.8249 \cdot 37.5 = 30.93 \text{ kW}$$

$$T3: \quad kW_1 = 0.8249 \cdot 50 = 41.24 \text{ kW}$$

2.4.1.5 What Method to Use?

Four different methods have been presented for allocating load to distribution transformers:

- Application of diversity factors
- Load survey
- Transformer load management
- Metered feeder maximum demand

Which method to use depends upon the purpose of the analysis. If the purpose is to determine as closely as possible the maximum demand on a distribution transformer, then either the diversity factor or the transformer load management method can be used. Neither of these methods should be employed when the analysis of the total feeder is to be performed. The problem is that using those methods will result in a much larger maximum diversified demand at the substation than actually exists. When the total feeder is to be analyzed, the only method that gives good results is that of allocating load based upon the kVA ratings of the transformers.

2.4.2 Voltage-Drop Calculations Using Allocated Loads

The voltage drops down line segments and through distribution transformers are of interest to the distribution engineer. Four different methods of allocating loads have been presented. The various voltage drops will be computed using the loads allocated by two of the methods in the following examples. For these studies it is assumed that the allocated loads will be modeled as constant real power and reactive power.

2.4.2.1 Application of Diversity Factors

The loads allocated to a line segment or a distribution transformer using diversity factors are a function of the total number of customers downstream from the line segment or distribution transformer. The application of the diversity factors was demonstrated in Example 2.1. With a knowledge of the allocated loads flowing in the line segments and through the transformers and the impedances, the voltage drops can be computed. The assumption is that the allocated loads will be constant real power and reactive power.

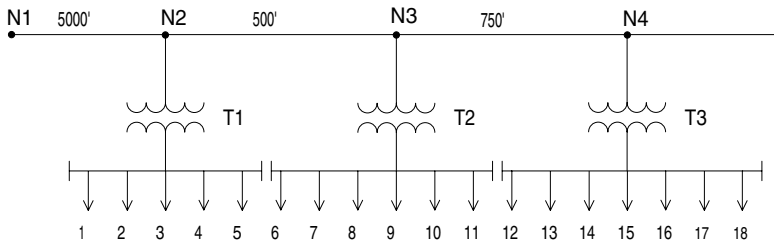


FIGURE 2.12

Single-phase lateral with distances.

In order to avoid an iterative solution, the voltage at the source is assumed and the voltage drops calculated from that point to the last transformer. Example 2.3 demonstrates how the method of load allocation using diversity factors is applied. The same system and allocated loads from Example 2.1 are used.

Example 2.3

For the system of Example 2.1, assume the voltage at N1 is 2400 volts and compute the secondary voltages on the three transformers using the diversity factors. The system of Example 2.1, including segment distances, is shown in Figure 2.12.

Assume that the power factor of the loads is 0.9 lagging.

The impedance of the lines are: $z = 0.3 + j0.6 \, \Omega/\text{mile}$

The ratings of the transformers are

T1: 25 kVA, 2400-240 volts, $Z = 1.8 / \underline{40\%}$

T2: 37.5 kVA, 2400-240 volts, $Z = 1.9 / \underline{45\%}$

T3: 50 kVA, 2400-240 volts, $Z = 2.0 / \underline{50\%}$

From Example 2.1 the maximum diversified kW demands were computed. Using the 0.9 lagging power factor, the maximum diversified kW and kVA demands for the line segments and transformers are

Segment N1-N2: $P_{12} = 92.8 \, \text{kW}$ $S_{12} = 92.8 + j45.0 \, \text{kVA}$

Segment N2-N3: $P_{23} = 72.6 \, \text{kW}$ $S_{23} = 72.6 + j35.2 \, \text{kVA}$

Segment N3-N4: $P_{34} = 48.9 \, \text{kW}$ $S_{34} = 48.9 + j23.7 \, \text{kVA}$

Transformer T1: $P_{T1} = 30.3 \, \text{kW}$ $S_{T1} = 30.3 + j14.7 \, \text{kVA}$

Transformer T2: $P_{T2} = 35.5 \, \text{kW}$ $S_{T2} = 35.5 + j17.2 \, \text{kVA}$

Transformer T3: $P_{T3} = 48.9 \, \text{kW}$ $S_{T3} = 48.9 + j23.7 \, \text{kVA}$

Convert transformer impedances to ohms referred to the high-voltage side

$$T1: Z_{base} = \frac{kV^2 \cdot 1000}{kVA} = \frac{2.4^2 \cdot 1000}{25} = 230.4 \, \Omega$$

$$Z_{T1} = (0.018/40) \cdot 230.4 = 3.18 + j2.67 \, \Omega$$

$$T2: Z_{base} = \frac{kV^2 \cdot 1000}{kVA} = \frac{2.4^2 \cdot 1000}{37.5} = 153.6 \, \Omega$$

$$Z_{T2} = (0.019/45) \cdot 153.6 = 2.06 + j2.06 \, \Omega$$

$$T3: Z_{base} = \frac{kV^2 \cdot 1000}{kVA} = \frac{2.4^2 \cdot 1000}{50} = 115.2 \, \Omega$$

$$Z_{T3} = (0.02/50) \cdot 115.2 = 1.48 + j1.77 \, \Omega$$

Compute the line impedances:

$$N1-N2: Z_{12} = (0.3 + j0.6) \cdot \frac{5000}{5280} = 0.2841 + j0.5682 \, \Omega$$

$$N2-N3: Z_{23} = (0.3 + j0.6) \cdot \frac{500}{5280} = 0.0284 + j0.0568 \, \Omega$$

$$N3-N4: Z_{34} = (0.3 + j0.6) \cdot \frac{750}{5280} = 0.0426 + j0.0852 \, \Omega$$

Calculate the current flowing in segment N1-N2:

$$I_{12} = \left(\frac{kW + jk \, var}{kV} \right)^* = \left(\frac{92.9 + j45.0}{2.4/0} \right)^* = 43.0/\underline{-25.84} \, A$$

Calculate the voltage at N2:

$$V_2 = V_1 - Z_{12} \cdot I_{12}$$

$$V_2 = 2400/\underline{0} - (0.2841 + j0.5682) \cdot 43.0/\underline{-25.84} = 2378.4/\underline{-0.4} \, V$$

Calculate the current flowing into T1:

$$I_{T1} = \left(\frac{kW + jk \, var}{kV} \right)^* = \left(\frac{30.3 + j14.7}{2.378/\underline{-0.4}} \right)^* = 14.16/\underline{-26.24} \, A$$

Calculate the secondary voltage referred to the high side:

$$V_{T1} = V_2 - Z_{T2} \cdot I_{T1}$$

$$V_{T1} = 2378.4/\underline{-0.4} - (3.18 + j2.67) \cdot 14.16/\underline{-26.24} = 2321.5/\underline{-0.8} \, V$$

Compute the secondary voltage by dividing by the turns ratio of 10:

$$V_{low_{T1}} = \frac{2321.5 / -0.8}{10} = 232.15 / -0.8 \text{ V}$$

Calculate the current flowing in line section N2-N3:

$$I_{23} = \left(\frac{\text{kW} + jk \text{ var}}{\text{kV}} \right)^* = \left(\frac{72.6 + j35.2}{2.378 / -0.4} \right)^* = 33.9 / -26.24 \text{ A}$$

Calculate the voltage at N3:

$$V_3 = V_2 - Z_{23} \cdot I_{23}$$

$$V_3 = 2378.4 / -0.4 - (0.0284 + j0.0568) \cdot 33.9 / -26.24 = 2376.7 / -0.4 \text{ V}$$

Calculate the current flowing into T2:

$$I_{T2} = \left(\frac{\text{kW} + jk \text{ var}}{\text{kV}} \right)^* = \left(\frac{35.5 + j17.2}{2.3767 / -0.4} \right)^* = 16.58 / -26.27$$

Calculate the secondary voltage referred to the high side:

$$V_{T2} = V_3 - Z_{T2} \cdot I_{T2}$$

$$V_{T2} = 2376.7 / -0.4 - (2.06 + j2.06) \cdot 16.58 / -26.27 = 2331.1 / -0.8 \text{ V}$$

Compute the secondary voltage by dividing by the turns ratio of 10:

$$V_{low_{T2}} = \frac{2331.1 / -0.8}{10} = 233.1 / -0.8 \text{ V}$$

Calculate the current flowing in line section N3-N4:

$$I_{34} = \left(\frac{\text{kW} + jk \text{ var}}{\text{kV}} \right)^* = \left(\frac{49.0 + j23.7}{2.3767 / -0.4} \right)^* = 22.9 / -26.27 \text{ A}$$

Calculate the voltage at N4:

$$V_4 = V_3 - Z_{34} \cdot I_{34}$$

$$V_4 = 2376.7 / -0.4 - (0.0426 + j0.0852) \cdot 22.9 / -26.27 = 2375.0 / -0.5 \text{ V}$$

The current flowing into T3 is the same as the current from N3 to N4:

$$I_{T3} = 22.91 / \underline{-26.30} \text{ A}$$

Calculate the secondary voltage referred to the high side:

$$V_{T3} = V_4 - Z_{T3} \cdot I_{T3}$$

$$V_{T3} = 2375.0 / \underline{-0.5} - (1.48 + j1.77) \cdot 22.9 / \underline{-26.27} = 2326.9 / \underline{-1.0} \text{ V}$$

Compute the secondary voltage by dividing by the turns ratio of 10:

$$V_{low_{T3}} = \frac{2326.9 / \underline{-1.0}}{10} = 232.7 / \underline{-1.0} \text{ V}$$

Calculate the percent voltage drop to the secondary of transformer T3. Use the secondary voltage referred to the high side:

$$V_{\text{drop}} = \frac{|V_1| - |V_{T3}|}{|V_1|} \cdot 100 = \frac{2400 - 2326.11}{2400} \cdot 100 = 3.0789\%$$

2.4.2.2 Load Allocation Based upon Transformer Ratings

When only the ratings of the distribution transformers are known, the feeder can be allocated based upon the metered demand and the transformer kVA ratings. This method was discussed in Section 2.3.3. Example 2.4 demonstrates this method.

Example 2.4

For the system of Example 2.1, assume the voltage at N1 is 2400 volts and compute the secondary voltages on the three transformers, allocating the loads based upon the transformer ratings. Assume that the metered kW demand at N1 is 92.9 kW.

The impedances of the line segments and transformers are the same as in Example 2.3. Assume the load power factor is 0.9 lagging; compute the kVA demand at N1 from the metered demand:

$$S_{12} = \frac{92.9}{0.9} / \cos^{-1}(0.9) = 92.8 + j45.0 = 103.2 / \underline{25.84} \text{ kVA}$$

Calculate the allocation factor:

$$AF = \frac{103.2 / \underline{25.84}}{25 + 37.5 + 50} = 0.9175 / \underline{25.84}$$

Allocate the loads to each transformer:

$$S_{T1} = AF \cdot kVA_{T1} = (0.9175 / 25.84) \cdot 25 = 20.6 + j10.0 \text{ kVA}$$

$$S_{T2} = AF \cdot kVA_{T2} = (0.9175 / 25.84) \cdot 37.5 = 31.0 + j15.0 \text{ kVA}$$

$$S_{T3} = AF \cdot kVA_{T3} = (0.9175 / 25.84) \cdot 50 = 41.3 + j20.0 \text{ kVA}$$

Calculate the line flows:

$$S_{12} = S_{T1} + S_{T2} + S_{T3} = 92.9 + j45.0 \text{ kVA}$$

$$S_{23} = S_{T2} + S_{T3} = 72.3 + j35 \text{ kVA}$$

$$S_{34} = S_{T3} = 41.3 + j20.0 \text{ kVA}$$

Using these values of line flows and flows into transformers, the procedure for computing the transformer secondary voltages is exactly the same as in Example 2.3. When this procedure is followed, the node and secondary transformer voltages are

$$V_2 = 2378.4 / \underline{-0.4} \text{ V} \quad V_{low_{T1}} = 234.0 / \underline{-0.4} \text{ V}$$

$$V_3 = 2376.7 / \underline{0.4} \text{ V} \quad V_{low_{T2}} = 233.7 / \underline{-0.8} \text{ V}$$

$$V_4 = 2375.3 / \underline{-0.4} \text{ V} \quad V_{low_{T3}} = 233.5 / \underline{-0.9} \text{ V}$$

The percent voltage drop for this case is

$$V_{drop} = \frac{|V_1| - |V_{T3}|}{|V_1|} \cdot 100 = \frac{2400 - 2334.8}{2400} \cdot 100 = 2.7179\%$$

2.5 Summary

This chapter has demonstrated the nature of the loads on a distribution feeder. There is great diversity between individual customer demands, but as the demand is monitored on line segments working back toward the substation, the effect of the diversity between demands becomes very slight. It was shown that the effect of diversity between customer demands must be taken into account when the demand on a distribution transformer is computed. The effect of diversity for short laterals can be taken into account in determining the maximum flow on the lateral. For the diversity factors of Table 2.2, it was shown that when the number of customers exceeds 70 the effect of diversity has pretty much disappeared. This is evidenced by the fact that the diversity factor became almost constant as the number of customers

approached 70. It must be understood that the number 70 will apply only to the diversity factors of Table 2.2. If a utility is going to use diversity factors, that utility must perform a comprehensive load survey in order to develop the table of diversity factors that apply to that particular system.

Examples 2.3 and 2.4 show that the final node and transformer voltages are approximately the same. There was very little difference between the voltages when the loads were allocated using the diversity factors and when the loads were allocated based upon the transformer kVA ratings.

Problems

2.1 Shown below are the 15-minute kW demands for four customers between the hours of 17:00 and 21:00. A 25-kVA single-phase transformer serves the four customers.

Time	Cust #1	Cust #2	Cust #3	Cust #4
	kW	kW	kW	kW
17:00	8.81	4.96	11.04	1.44
17:15	2.12	3.16	7.04	1.62
17:30	9.48	7.08	7.68	2.46
17:45	7.16	5.08	6.08	0.84
18:00	6.04	3.12	4.32	1.12
18:15	9.88	6.56	5.12	2.24
18:30	4.68	6.88	6.56	1.12
18:45	5.12	3.84	8.48	2.24
19:00	10.44	4.44	4.12	1.12
19:15	3.72	8.52	3.68	0.96
19:30	8.72	4.52	0.32	2.56
19:45	10.84	2.92	3.04	1.28
20:00	6.96	2.08	2.72	1.92
20:15	6.62	1.48	3.24	1.12
20:30	7.04	2.33	4.16	1.76
20:45	6.69	1.89	4.96	2.72
21:00	1.88	1.64	4.32	2.41

1. For each of the customers determine:
 - (a) Maximum 15-minute kW demand
 - (b) Average 15-minute kW demand
 - (c) Total kWh usage in the time period
 - (d) Load factor

- 2. For the 25-kVA transformer determine:
 - (a) Maximum 15-minute diversified demand
 - (b) Maximum 15-minute noncoincident demand
 - (c) Utilization factor (assume unity power factor)
 - (d) Diversity factor
 - (e) Load diversity
- 3. Plot the load duration curve for the transformer

2.2 Two transformers each serving four customers are shown in Figure 2.13.

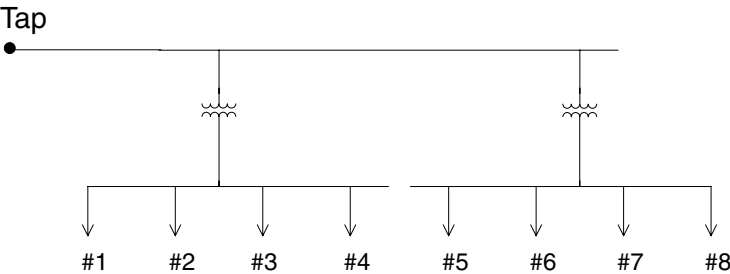


FIGURE 2.13
System for Problem 2.2.

The following table gives the time interval and kVA demand of the four customer demands during the peak load period of the year. Assume a power factor of 0.9 lagging.

Time	#1	#2	#3	#4	#5	#6	#7	#8
3:00–3:30	10	0	10	5	15	10	50	30
3:30–4:00	20	25	15	20	25	20	30	40
4:00–4:30	5	30	30	15	10	30	10	10
4:30–5:00	0	10	20	10	13	40	25	50
5:00–5:30	15	5	5	25	30	30	15	5
5:30–6:00	15	15	10	10	5	20	30	25
6:00–6:30	5	25	25	15	10	10	30	25
6:30–7:00	10	50	15	30	15	5	10	30

- 1. For each transformer determine the following:
 - (a) 30-minute maximum kVA demand
 - (b) Noncoincident maximum kVA demand
 - (c) Load factor
 - (d) Diversity factor
 - (e) Suggested transformer rating (50, 75, 100, 167)

- (f) Utilization factor
- (g) Energy (kWh) during the 4-hour period
- 2. Determine the maximum diversified 30-minute kVA demand at the Tap

2.3 Two single-phase transformers serving 12 customers are shown in Figure 2.14.

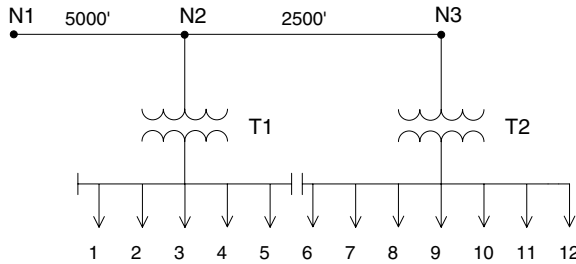


FIGURE 2.14
Circuit for Problem 2.3.

The 15-minute kW demands for the 12 customers between the hours of 5:00 p.m. and 9:00 p.m. are given in the tables that follow. Assume a load power factor of 0.95 lagging. The impedance of the lines are $z = 0.306 + j0.6272 \Omega/\text{mile}$. The voltage at node N1 is $2500\angle 0^\circ$ V. Transformer ratings:

$$\begin{aligned} \text{T1: } & 25 \text{ kVA} \quad 2400\text{-}240 \text{ V} \quad Z_{pu} = 0.018/\underline{40} \\ \text{T2: } & 37.5 \text{ kVA} \quad 2400\text{-}240 \text{ V} \quad Z_{pu} = 0.020/\underline{50} \end{aligned}$$

1. Determine the maximum kW demand for each customer.
2. Determine the average kW demand for each customer.
3. Determine the kWH consumed by each customer in this time period.
4. Determine the load factor for each customer.
5. Determine the maximum diversified demand for each transformer.
6. Determine the maximum noncoincident demand for each transformer.
7. Determine the utilization factor (assume 1.0 power factor) for each transformer.
8. Determine the diversity factor of the load for each transformer.
9. Determine the maximum diversified demand at Node N1.
10. Compute the secondary voltage for each transformer, taking diversity into account.

TRANSFORMER #1-25 kVA

Time	#1	#2	#3	#4	#5
	kW	kW	kW	kW	kW
05:00	2.13	0.19	4.11	8.68	0.39
05:15	2.09	0.52	4.11	9.26	0.36
05:30	2.15	0.24	4.24	8.55	0.43
05:45	2.52	1.80	4.04	9.09	0.33
06:00	3.25	0.69	4.22	9.34	0.46
06:15	3.26	0.24	4.27	8.22	0.34
06:30	3.22	0.54	4.29	9.57	0.44
06:45	2.27	5.34	4.93	8.45	0.36
07:00	2.24	5.81	3.72	10.29	0.38
07:15	2.20	5.22	3.64	11.26	0.39
07:30	2.08	2.12	3.35	9.25	5.66
07:45	2.13	0.86	2.89	10.21	6.37
08:00	2.12	0.39	2.55	10.41	4.17
08:15	2.08	0.29	3.00	8.31	0.85
08:30	2.10	2.57	2.76	9.09	1.67
08:45	3.81	0.37	2.53	9.58	1.30
09:00	2.04	0.21	2.40	7.88	2.70

TRANSFORMER #2-37.5 kVA

Time	#6	#7	#8	#9	#10	#11	#12
	kW	kW	kW	kW	kW	kW	kW
05:00	0.87	2.75	0.63	8.73	0.48	9.62	2.55
05:15	0.91	5.35	1.62	0.19	0.40	7.98	1.72
05:30	1.56	13.39	0.19	5.72	0.70	8.72	2.25
05:45	0.97	13.38	0.05	3.28	0.42	8.82	2.38
06:00	0.76	13.23	1.51	1.26	3.01	7.47	1.73
06:15	1.10	13.48	0.05	7.99	4.92	11.60	2.42
06:30	0.79	2.94	0.66	0.22	3.58	11.78	2.24
06:45	0.60	2.78	0.52	8.97	6.58	8.83	1.74
07:00	0.60	2.89	1.80	0.11	7.96	9.21	2.18
07:15	0.87	2.75	0.07	7.93	6.80	7.65	1.98
07:30	0.47	2.60	0.16	1.07	7.42	7.78	2.19
07:45	0.72	2.71	0.12	1.35	8.99	6.27	2.63
08:00	1.00	3.04	1.39	6.51	8.98	10.92	1.59
08:15	0.47	1.65	0.46	0.18	7.99	5.60	1.81
08:30	0.44	2.16	0.53	2.24	8.01	7.74	2.13
08:45	0.95	0.88	0.56	0.11	7.75	11.72	1.63
09:00	0.79	1.58	1.36	0.95	8.19	12.23	1.68

2.4 On a different day, the metered 15-minute kW demand at node N1 for the system of Problem 2.3 is 72.43 kW. Assume a power factor of 0.95 lagging. Allocate the metered demand to each transformer based upon the transformer kVA rating. Assume the loads are constant current and compute the secondary voltage for each transformer.

2.5 A single-phase lateral serves four transformers as shown in Figure 2.15.

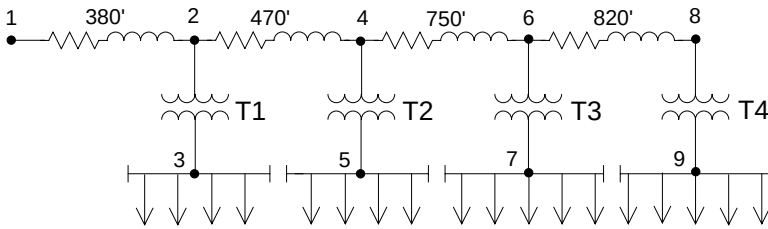


FIGURE 2.15

System for Problem 2.5.

Assume that each customer's maximum demand is $15.5 \text{ kW} + j7.5 \text{ kvar}$. The impedance of the single-phase lateral is $z = 0.4421 + j0.3213 \, \Omega/1000 \text{ ft}$. The four transformers are rated as:

T1 and T2: 37.5 kVA, 2400-240 V, $Z = 0.01 + j0.03$ per-unit

T3 and T4: 50 kVA, 2400-240 V, $Z = 0.015 + j0.035$ per-unit

Use the diversity factors found in Table 2.2 and determine:

- (1) The 15-minute maximum diversified kW and kvar demands on each transformer.
- (2) The 15-minute maximum diversified kW and kvar demands for each line section.
- (3) If the voltage at node 1 is $2600/\underline{0} \text{ V}$, determine the voltage at nodes 2, 3, 4, 5, 6, 7, 8, and 9. In calculating the voltages, take into account diversity using the answers from (1) and (2) above.
- (4) Use the 15-minute maximum diversified demands at the lateral tap (Section 1-2) from Part (2) above. Divide these maximum demands by 18 (number of customers) and assign that as the "instantaneous load" for each customer. Now calculate the voltages at all of the nodes listed in Part (3) using the instantaneous loads.
- (5) Repeat Part (4) above, but assume the loads are "constant current." To do this, take the current flowing from node 1 to node 2 from Part (4) above, divide by 18 (number of customers), and assign that as the instantaneous constant current load for each customer. Again, calculate all of the voltages.

- (6) Take the maximum diversified demand from node 1 to node 2 and allocate that to each of the four transformers based upon their kVA ratings. To do this, take the maximum diversified demand and divide by 175 (total kVA of the four transformers). Now multiply each transformer kVA rating by that number to give how much of the total diversified demand is being served by each transformer. Again, calculate all of the voltages.
- (7) Compute the percent differences in the voltages for Parts (4), (5), and (6) at each of the nodes using the Part (3) answer as the base.

3

Approximate Methods of Analysis

A distribution feeder provides service to unbalanced three-phase, two-phase, and single-phase loads over untransposed three-phase, two-phase, and single-phase line segments. This combination leads to three-phase line currents and line voltages being unbalanced. In order to analyze these conditions as precisely as possible, it will be necessary to model all three phases of the feeder accurately, however, many times only a “ballpark” answer is needed. When this is the case, some approximate methods of modeling and analysis can be employed. It is the purpose of this chapter to develop some of the approximate methods and leave for later chapters the exact models and analysis.

All of the approximate methods of modeling and analysis will assume perfectly balanced three-phase systems. It will be assumed that all loads are balanced three-phase, and all line segments will be three-phase and perfectly transposed. With these assumptions, a single line-to-neutral equivalent circuit for the feeder will be used.

3.1 Voltage Drop

A line-to-neutral equivalent circuit of a three-phase line segment serving a balanced three-phase load is shown in Figure 3.1. Kirchhoff’s voltage law applied to the circuit of Figure 3.1 gives:

$$V_S = V_L + (R + jX) \cdot I = V_L + R \cdot I + jX \cdot I \quad (3.1)$$

The phasor diagram for Equation 3.1 is shown in Figure 3.2. In Figure 3.2 the phasor for the voltage drop through the line resistance (RI) is shown in phase with the current phasor, and the phasor for the voltage drop through the reactance is shown leading the current phasor by 90 degrees. The dashed lines represent the real and imaginary parts of the impedance (ZI) drop. The voltage drop down the line is defined as the difference between the magnitudes of the source and the load voltages.

$$V_{\text{drop}} = |V_S| - |V_L| \quad (3.2)$$

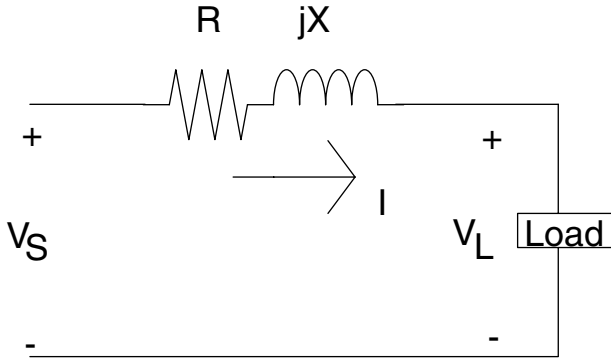


FIGURE 3.1
Line-to-neutral equivalent circuit.

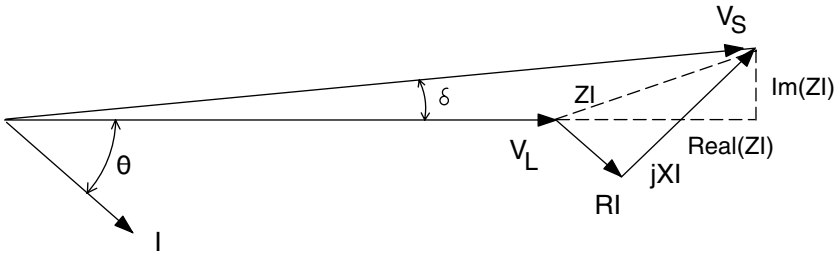


FIGURE 3.2
Phasor diagram.

The angle between the source voltage and the load voltage (δ) is very small. Because of that, the voltage drop between the source and load voltage is approximately equal to the real part of the impedance drop. That is

$$V_{\text{drop}} \cong \text{Re}(Z \cdot I) \quad (3.3)$$

For the purposes of this chapter, Equation 3.3 will be used as the definition of voltage drop.

Example 3.1

In Example 2.3, the impedance of the first line segment is

$$Z_{12} = 0.2841 + j0.5682 \, \Omega$$

The current flowing through the line segment is

$$I_{12} = 43.0093 / \underline{-25.8419} \, \text{A}$$

The voltage at node N1 is

$$V_1 = 2400 / 0.0 \text{ V}$$

The exact voltage at node N2 is computed to be

$$\begin{aligned} V_2 &= 2400 / 0.0 - (0.2841 + j0.5682) \cdot 43.0093 / -25.8419 \\ &= 2378.4098 / -0.4015 \text{ V} \end{aligned}$$

The voltage drop between the nodes is then:

$$V_{\text{drop}} = 2400.0000 - 2378.4098 = 21.5902 \text{ V}$$

Computing the voltage drop according to Equation 3.3 gives:

$$\begin{aligned} V_{\text{drop}} &= \text{Re}[(0.2841 + j0.5682) \cdot 43.0093 / -25.8419] = 21.6486 \text{ V} \\ \text{Error} &= \frac{21.5902 - 21.6486}{21.5902} \cdot 100 = -0.27\% \end{aligned}$$

This example demonstrates the very small error in computing voltage drop when using the approximate equation given by Equation 3.3.

3.2 Line Impedance

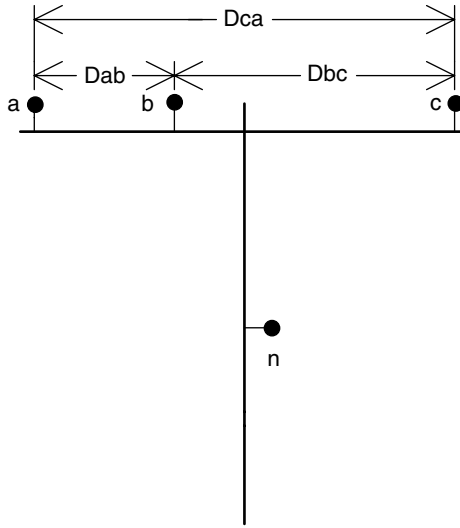
For the approximate modeling of a line segment, it will be assumed that the line segment is transposed. With this assumption, only the positive sequence impedance of the line segment needs to be determined. A typical three-phase line configuration is shown in Figure 3.3. The equation for the positive sequence impedance for the configuration shown in Figure 3.3 is given by:

$$z_{\text{positive}} = r + j0.12134 \cdot \ln\left(\frac{D_{eq}}{\text{GMR}}\right) \Omega/\text{mile} \quad (3.4)$$

where r = conductor resistance (from tables) Ω/mile

$$D_{eq} = \sqrt[3]{D_{ab} \cdot D_{bc} \cdot D_{ca}} \text{ (ft.)} \quad (3.5)$$

GMR = conductor geometric mean radius (from tables) (ft.)

**FIGURE 3.3**

Three-phase line configuration.

Example 3.2

A three-phase line segment has the configuration as shown in Figure 3.3. The spacings between conductors are

$$D_{ab} = 2.5 \text{ ft.}, \quad D_{bc} = 4.5 \text{ ft.}, \quad D_{ca} = 7.0 \text{ ft.}$$

The conductors of the line are 336,400 26/7 ACSR. Determine the positive sequence impedance of the line in ohms/mile:

SOLUTION

From the table of conductor data in Appendix A:

$$r = 0.306 \, \Omega/\text{mile}$$

$$\text{GMR} = 0.0244 \text{ ft}$$

Compute the equivalent spacing:

$$D_{eq} = \sqrt[3]{2.5 \cdot 4.5 \cdot 7.0} = 4.2863 \text{ ft}$$

Using Equation 3.4:

$$z_{\text{positive}} = 0.306 + j0.12134 \cdot \ln\left(\frac{4.2863}{0.0244}\right) = 0.306 + j0.6272 \, \Omega/\text{mile}$$

3.3 “K” Factors

A first approximation for calculating the voltage drop along a line segment is given by Equation 3.3. Another approximation is made by employing a “K” factor. There will be two types of K factors: one for voltage drop and the other for voltage rise calculations.

3.3.1 The K_{drop} Factor

The K_{drop} factor is defined as:

$$K_{\text{drop}} = \frac{\text{Percent voltage drop}}{\text{kVA} \cdot \text{mile}} \quad (3.5)$$

The K_{drop} factor is determined by computing the percent voltage drop down a line that is one mile long and serving a balanced three-phase load of 1 kVA. The percent voltage drop is referenced to the nominal voltage of the line. In order to calculate this factor, the power factor of the load must be assumed.

Example 3.3

For the line of Example 3.2, compute the K_{drop} factor assuming a load power factor of 0.9 lagging and a nominal voltage of 12.47 kV (line-to-line).

SOLUTION

The impedance of one mile of line was computed to be

$$Z = 0.306 + j0.6272 \, \Omega$$

The current taken by 1 kVA at 0.9 lagging power factor is given by:

$$I = \frac{1 \text{ kVA}}{\sqrt{3} \cdot kV_{LL}} \angle -\cos^{-1}(PF) = \frac{1}{\sqrt{3} \cdot 12.47} \angle -\cos^{-1}(0.9) = 0.046299 \angle -25.84^\circ \text{ A}$$

The voltage drop is computed to be

$$V_{\text{drop}} = \text{Re}[Z \cdot I] = \text{Re}[(0.306 + j0.6272) \cdot 0.046299 \angle -25.84^\circ] = 0.025408 \text{ V}$$

The nominal line-to-neutral voltage is

$$V_{LN} = \frac{12470}{\sqrt{3}} = 7199.6 \text{ V}$$

The K_{drop} factor is then:

$$K_{\text{drop}} = \frac{0.025408}{7199.6} \cdot 100 = 0.00035291\% \text{ drop/kVA-mile}$$

The K_{drop} factor computed in Example 3.3 is for the 336,400 26/7 ACSR conductor with the conductor spacings defined in Example 3.2, a nominal voltage of 12.47 kV, and a load power factor of 0.9 lagging. Unique K_{drop} factors can be determined for all standard conductors, spacings, and voltages. Fortunately, most utilities will have a set of standard conductors, standard conductor spacings, and one or two standard distribution voltages. Because of this, a simple spreadsheet program can be written that will compute the K_{drop} factors for the standard configurations. The assumed power factor of 0.9 lagging is a good approximation for a feeder serving a predominately residential load.

The K_{drop} factor can be used to quickly compute the approximate voltage drop down a line section. For example, assume a load of 7500 kVA is to be served at a point 1.5 miles from the substation. Using the K_{drop} factor computed in Example 3.3, the percent voltage drop down the line segment is computed to be

$$V_{\text{drop}} = K_{\text{drop}} \cdot \text{kVA} \cdot \text{mile} = 0.00035291 \cdot 7500 \cdot 1.5 = 3.9702\%$$

This example demonstrates that a load of 7500 kVA can be served 1.5 miles from the substation with a resulting voltage drop of 3.97%. Suppose now that the utility has a maximum allowable voltage drop of 3.0%. How much load can be served 1.5 miles from the substation?

$$\text{kVA}_{\text{load}} = \frac{3.0\%}{0.00035291 \cdot 1.5} = 5667.2 \text{ kVA}$$

The application of the K_{drop} factor is not limited to computing the percent voltage drop down just one line segment. When line segments are in cascade, the total percent voltage drop from the source to the end of the last line segment is the sum of the percent drops in each line segment. This seems logical, but it must be understood that in all cases the percent drop is in reference to the nominal line-to-neutral voltage. That is, the percent voltage drop in a line segment is not referenced to the source end voltage, but rather the nominal line-to-neutral voltage, as would be the usual case. Example 3.4 will demonstrate this application.

Example 3.4

A three-segment feeder is shown in Figure 3.4. The K_{drop} factor for the line segments is

$$K_{\text{drop}} = 0.00035291$$

Determine the percent voltage drop from N0 to N3.

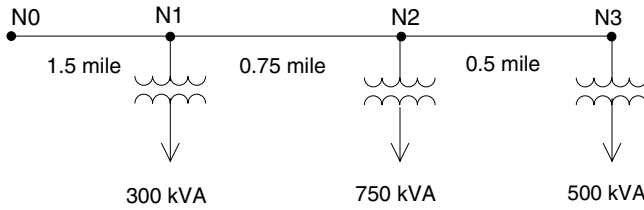


FIGURE 3.4
Three-segment feeder.

SOLUTION

The total kVA flowing in segment N0 to N1 is

$$kVA_{01} = 300 + 750 + 500 = 1550 \text{ kVA}$$

The percent voltage drop from N0 to N1 is

$$Vdrop_{01} = 0.00035291 \cdot 1550 \cdot 1.5 = 0.8205\%$$

The total kVA flowing in segment N1 to N2 is

$$kVA_{12} = 750 + 500 = 1250 \text{ kVA}$$

The percent voltage drop from N1 to N2 is

$$Vdrop_{12} = 0.00035291 \cdot 1250 \cdot 0.75 = 0.3308\%$$

The kVA flowing in segment N2 to N3 is

$$kVA_{23} = 500$$

The percent voltage drop in the last line segment is

$$Vdrop_{23} = 0.00035291 \cdot 500 \cdot 0.5 = 0.0882\%$$

The total percent voltage drop from N0 to N3 is

$$Vdrop_{total} = 0.8205 + 0.3308 + 0.0882 = 1.2396\%$$

The application of the K_{drop} factor provides an easy way of computing the approximate percent voltage drop from a source to a load. It should be kept in mind that the assumption has been a perfectly balanced three-phase load,

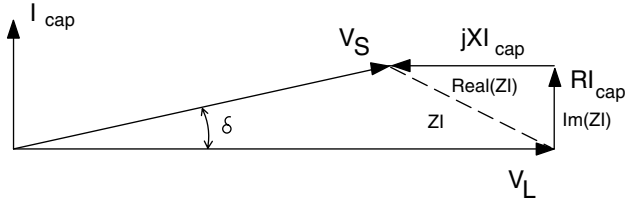


FIGURE 3.5
Voltage rise phasor diagram.

an assumed load power factor, and transposed line segments. Even with these assumptions the results will always provide a “ballpark” result that can be used to verify the results of more sophisticated methods of computing voltage drop.

3.3.2 The K_{rise} Factor

The K_{rise} factor is similar to the K_{drop} factor except that now the load is a shunt capacitor. When a leading current flows through an inductive reactance there will be a voltage rise across the reactance rather than a voltage drop. This is illustrated by the phasor diagram of Figure 3.5. Referring to Figure 3.5, the voltage rise is defined as

$$V_{\text{rise}} = |\text{Re}(ZI_{\text{cap}})| = X \cdot |I_{\text{cap}}| \quad (3.6)$$

In Equation 3.6 it is necessary to take the magnitude of the real part of ZI so that the voltage rise is a positive number. The K_{rise} factor is defined exactly the same as for the K_{drop} factor:

$$K_{\text{rise}} = \frac{\text{Percent voltage rise}}{\text{kvar mile.}} \quad (3.7)$$

Example 3.5

1. Calculate the K_{rise} factor for the line of Example 3.3.
2. Determine the rating of a three-phase capacitor bank to limit the voltage drop in Example 3.3 to 2.5%.

SOLUTION

1. The impedance of one mile of line was computed to be

$$Z = 0.306 + j0.6272 \, \Omega$$

The current taken by a 1-kvar three-phase capacitor bank is given by:

$$I_{\text{cap}} = \frac{1 \text{ kvar}}{\sqrt{3} \cdot kV_{LL}} \angle 90^\circ = \frac{1}{\sqrt{3} \cdot 12.47} \angle 90^\circ = 0.046299 \angle 90^\circ \text{ A}$$

The voltage rise per kvar mile is computed to be

$$\begin{aligned} V_{\text{rise}} &= |\text{Re}[Z \cdot I_{\text{cap}}]| = |\text{Re}[(0.306 + j0.6272) \cdot 0.046299 \angle 90^\circ]| \\ &= 0.029037 \text{ V} \end{aligned}$$

The nominal line-to-neutral voltage is

$$V_{LN} = \frac{12,470}{\sqrt{3}} = 7199.6 \text{ V}$$

The K_{rise} factor is then

$$K_{\text{rise}} = \frac{0.029037}{7199.6} \cdot 100 = 0.00040331\% \text{ rise/kvar mile}$$

2. The percent voltage drop in Example 3.3 was computed to be 3.9702%. To limit the total voltage drop to 2.5%, the required voltage rise due to a shunt capacitor bank is

$$V_{\text{rise}} = 3.9702 - 2.5 = 1.4702\%$$

The required rating of the shunt capacitor is

$$\text{kvar} = \frac{V_{\text{rise}}}{K_{\text{rise}} \cdot \text{mile}} = \frac{1.4702}{0.00040331 \cdot 1.5} = 2430.18 \text{ kvar}$$

3.4 Uniformly Distributed Loads

Many times it can be assumed that loads are uniformly distributed along a line where the line can be a three-phase, two-phase, or single-phase feeder or lateral. This is certainly the case on single-phase laterals where the same rating transformers are spaced uniformly over the length of the lateral. When the loads are uniformly distributed, it is not necessary to model each load in order to determine the total voltage drop from the source end to the last load. Figure 3.6 shows a generalized line with n uniformly distributed loads.

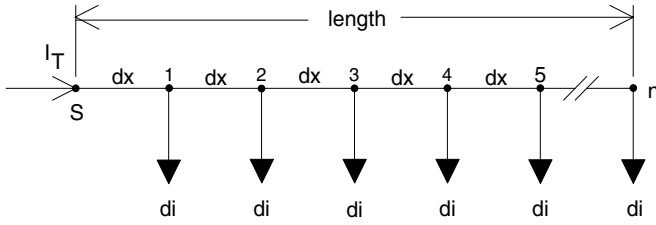


FIGURE 3.6
Uniformly distributed loads.

3.4.1 Voltage Drop

Figure 3.6 shows n uniformly spaced loads dx miles apart. The loads are all equal and will be treated as constant current loads with a value of di . The total current into the feeder is I_T . It is desired to determine the total voltage drop from the source node (S) to the last node n .

Let

- l = length of the feeder
- $z = r + jx$ = impedance of the line in Ω/mile
- dx = length of each line section
- di = load currents at each node
- n = number of nodes and number of line sections
- I_T = total current into the feeder

The load currents are given by:

$$di = \frac{I_T}{n} \quad (3.8)$$

The voltage drop in the first line segment is given by:

$$Vdrop_1 = \text{Re}\{z \cdot dx \cdot (n \cdot di)\} \quad (3.9)$$

The voltage drop in the second line segment is given by:

$$Vdrop_2 = \text{Re}\{z \cdot dx \cdot [(n-1) \cdot di]\} \quad (3.10)$$

The total voltage drop from the source node to the last node is then given by:

$$\begin{aligned} Vdrop_{\text{total}} &= Vdrop_1 + Vdrop_2 + \cdots + Vdrop_n \\ Vdrop_{\text{total}} &= \text{Re}\{z \cdot dx \cdot di \cdot [n + (n-1) + (n-2) + \cdots + (1)]\} \end{aligned} \quad (3.11)$$

Equation 3.11 can be reduced by recognizing the series expansion:

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} \quad (3.12)$$

Using the expansion, Equation 3.11 becomes:

$$V_{drop_{total}} = \text{Re} \left\{ z \cdot dx \cdot di \cdot \left[\frac{n \cdot (n+1)}{2} \right] \right\} \quad (3.13)$$

The incremental distance is

$$dx = \frac{l}{n} \quad (3.14)$$

The incremental current is

$$di = \frac{I_T}{n} \quad (3.15)$$

Substituting Equations 3.14 and 3.15 into Equation 3.13 results in:

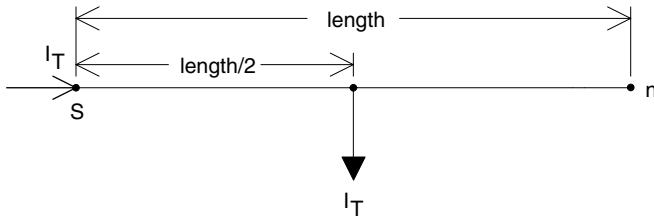
$$\begin{aligned} V_{drop_{total}} &= \text{Re} \left\{ z \cdot \frac{l}{n} \cdot \frac{I_T}{n} \cdot \left[\frac{n \cdot (n+1)}{2} \right] \right\} \\ V_{drop_{total}} &= \text{Re} \left\{ z \cdot l \cdot I_T \cdot \frac{1}{2} \cdot \left(\frac{n+1}{n} \right) \right\} \\ V_{drop_{total}} &= \text{Re} \left\{ \frac{1}{2} \cdot Z \cdot I_T \cdot \left(1 + \frac{1}{n} \right) \right\} \end{aligned} \quad (3.16)$$

where

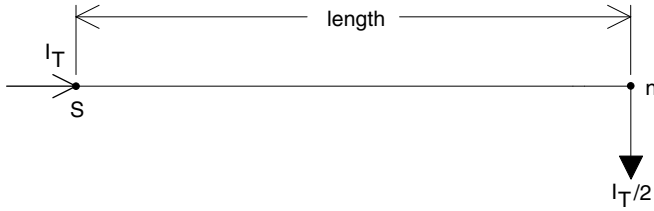
$$Z = z \cdot l$$

Equation 3.16 gives the general equation for computing the total voltage drop from the source to the last node n for a line of length l . In the limiting case where n goes to infinity, the final equation becomes:

$$V_{drop_{total}} = \text{Re} \left\{ \frac{1}{2} \cdot Z \cdot I_T \right\} \quad (3.17)$$

**FIGURE 3.7**

Load lumped at the midpoint.

**FIGURE 3.8**

One-half load lumped at the end.

In Equation 3.17, Z represents the total impedance from the source to the end of the line. The voltage drop is the total from the source to the end of the line. The equation can be interpreted in two ways. The first is to recognize that the total line distributed load can be lumped at the midpoint of the lateral as shown in Figure 3.7. A second interpretation of Equation 3.17 is to lump one-half of the total line load at the end of the line (node n). This model is shown in Figure 3.8. Figures 3.7 and 3.8 give two different models that can be used to calculate the total voltage drop from the source to the end of a line with uniformly distributed loads.

3.4.2 Power Loss

Of equal importance in the analysis of a distribution feeder is the power loss. If the model of Figure 3.7 is used to compute the total three-phase power loss down the line, the result is

$$P_{\text{loss}} = 3 \cdot |I_T|^2 \cdot \frac{R}{2} = \frac{3}{2} \cdot |I_T|^2 \cdot R \quad (3.18)$$

When the model of Figure 3.8 is used to compute the total three-phase power loss, the result is

$$P_{\text{loss}} = 3 \cdot \left| \frac{I_T}{2} \right|^2 \cdot R = \frac{3}{4} \cdot |I_T|^2 \cdot R \quad (3.19)$$

It is obvious that the two models give different results for power loss. The question is, which one is correct? The answer is neither one.

To derive the correct model for power loss, reference is made to Figure 3.6 and the definitions for the parameters in that figure. The total three-phase power loss down the line will be the sum of the power losses in each short segment of the line. For example, the three-phase power loss in the first segment is

$$P_{loss_1} = 3 \cdot (r \cdot dx) \cdot |(n \cdot di)|^2 \quad (3.20)$$

The power loss in the second segment is given by:

$$P_{loss_2} = 3 \cdot (r \cdot dx) \cdot [(n-1) \cdot di]^2 \quad (3.21)$$

The total power loss over the length of the line is then given by:

$$P_{loss_{total}} = 3 \cdot (r \cdot dx) \cdot |di|^2 [n^2 + (n-1)^2 + (n-2)^2 + \dots + 1^2] \quad (3.22)$$

The series inside the brackets of Equation 3.22 is the sum of the squares of n numbers and is equal to:

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n \cdot (n+1) \cdot (2n+1)}{6} \quad (3.23)$$

Substituting Equations 3.14, 3.15, and 3.23 into Equation 3.22 gives:

$$P_{loss_{total}} = 3 \cdot \left(r \cdot \frac{l}{n}\right) \cdot \left(\left|\frac{I_T}{n}\right|\right)^2 \cdot \left[\frac{n \cdot (n+1) \cdot (2n+1)}{6}\right] \quad (3.24)$$

Simplifying Equation 3.24:

$$\begin{aligned} P_{loss_{total}} &= 3 \cdot R \cdot |I_T|^2 \cdot \left[\frac{(n+1) \cdot (2n+1)}{6 \cdot n^2}\right] \\ P_{loss_{total}} &= 3 \cdot R \cdot |I_T|^2 \cdot \left[\frac{2 \cdot n^2 + 3 \cdot n + 1}{6 \cdot n^2}\right] \\ P_{loss_{total}} &= 3 \cdot R \cdot |I_T|^2 \cdot \left[\frac{1}{3} + \frac{1}{2 \cdot n} + \frac{1}{6 \cdot n^2}\right] \end{aligned} \quad (3.25)$$

Where $R = r \cdot l$, the total resistance per phase of the line segment, Equation 3.25 gives the total three-phase power loss for a discrete number of nodes and line segments. For a truly uniformly distributed load, the number of

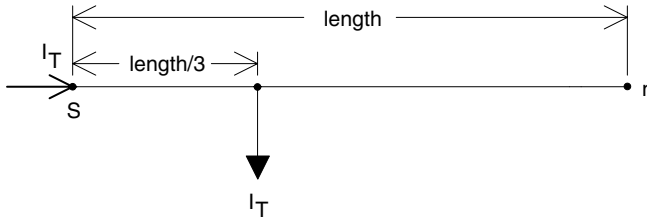


FIGURE 3.9
Power loss model.

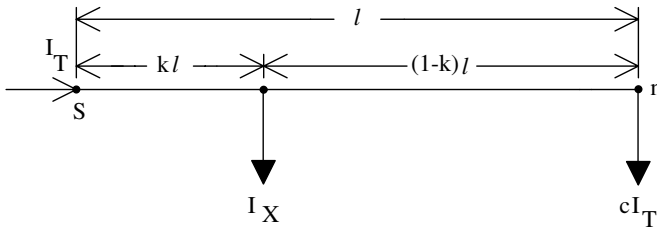


FIGURE 3.10
General exact lumped load model.

nodes goes to infinity. When that limiting case is taken in Equation 3.25, the final equation for computing the total three-phase power loss down the line is given by:

$$P_{loss_{total}} = 3 \cdot \left[\frac{1}{3} \cdot R \cdot |I_T|^2 \right] \quad (3.26)$$

A circuit model for Equation 3.26 is given in Figure 3.9. From a comparison of Figures 3.7 and 3.8, used for voltage drop calculations, to Figure 3.9, used for power loss calculations, it is obvious that the same model cannot be used for both voltage drop and power loss calculations.

3.4.3 The Exact Lumped Load Model

In the previous sections lumped load models were developed. The first models of Section 3.4.1 can be used for the computation of the total voltage drop down the line. It was shown that the same models cannot be used for the computation of the total power loss down the line. Section 3.4.2 developed a model that will give the correct power loss of the line. What is needed is one model that will work for both voltage drop and power loss calculations.

Figure 3.10 shows the general configuration of the exact model that will give correct results for voltage drop and power loss. In Figure 3.10 a portion (I_X) of

the total line current (I_T) will be modeled kl miles from the source end, and the remaining current (cI_T) will be modeled at the end of the line. The values of k and c need to be derived.

In Figure 3.10 the total voltage drop down the line is given by:

$$Vdrop_{total} = \text{Re}[k \cdot Z \cdot I_T + (1 - k) \cdot Z \cdot c \cdot I_T] \quad (3.27)$$

where

Z = total line impedance in ohms

k = factor of the total line length where the first part of the load current is modeled

c = factor of the total current to place at the end of the line such that
 $I_T = I_x + c \cdot I_T$

In Section 3.4.1 it was shown that the total voltage drop down the line is given by:

$$Vdrop_{total} = \text{Re}\left[\frac{1}{2} \cdot Z \cdot I_T\right] \quad (3.28)$$

Set Equation 3.17 equal to Equation 3.27:

$$Vdrop_{total} = \text{Re}\left[\frac{1}{2} \cdot Z \cdot I_T\right] = \text{Re}[k \cdot Z \cdot I_T + (1 - k) \cdot Z \cdot c \cdot I_T] \quad (3.29)$$

Equation 3.29 shows that the terms inside the brackets on both sides of the equal sign need to be set equal, that is

$$\left[\frac{1}{2} \cdot Z \cdot I_T\right] = [k \cdot Z \cdot I_T + (1 - k) \cdot Z \cdot c \cdot I_T] \quad (3.30)$$

Simplify Equation 3.30 by dividing both side of the equation by ZI_T :

$$\left[\frac{1}{2}\right] = [k + (1 - k) \cdot c] \quad (3.31)$$

Solve Equation 3.31 for k :

$$k = \frac{0.5 - c}{1 - c} \quad (3.32)$$

The same procedure can be followed for the power loss model. The total three-phase power loss in Figure 3.10 is given by:

$$Ploss_{total} = 3 \cdot [k \cdot R \cdot |I_T|^2 + (1 - k) \cdot R \cdot (c \cdot |I_T|)^2] \quad (3.33)$$

The model for the power loss of Figure 3.9 gives the total three-phase power loss as:

$$P_{loss_{total}} = 3 \cdot \left[\frac{1}{3} \cdot R \cdot |I_T|^2 \right] \quad (3.34)$$

Equate the terms inside the brackets of Equations 3.33 and 3.34 and simplify:

$$\begin{aligned} \left[\frac{1}{3} \cdot R \cdot |I_T|^2 \right] &= [k \cdot R \cdot |I_T|^2 + (1-k) \cdot R \cdot (c \cdot |I_T|)^2] \\ \left[\frac{1}{3} \right] &= [k + (1-k) \cdot (c)^2] \\ \left[\frac{1}{3} \right] &= [k + c^2 - k \cdot c^2] = [k \cdot (1 - c^2) + c^2] \end{aligned} \quad (3.35)$$

Substitute Equation 3.32 into Equation 3.35:

$$\left[\frac{1}{3} \right] = \left[\frac{0.5 - c}{1 - c} \cdot (1 - c^2) + c^2 \right] \quad (3.36)$$

Solving Equation 3.36 for c results in:

$$c = \frac{1}{3} \quad (3.37)$$

Substitute Equation 3.37 into Equation 3.32 and solve for k :

$$k = \frac{1}{4} \quad (3.38)$$

The interpretation of Equations 3.37 and 3.38 is that one-third of the load should be placed at the end of the line, and two-thirds of the load placed one-fourth of the way from the source end. Figure 3.11 gives the final exact lumped load model.

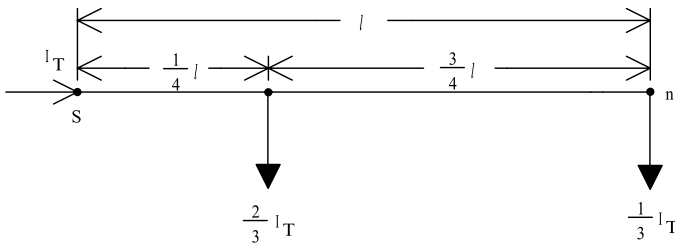


FIGURE 3.11

Exact lumped load model.

3.5 Lumping Loads in Geometric Configurations

Many times feeder areas can be represented by geometric configurations such as rectangles, triangles, and trapezoids. By assuming a constant load density in the configurations, approximate calculations can be made for computing the voltage drop and total power losses. The approximate calculations can aid in the determination of the maximum load that can be served in a specified area at a given voltage level and conductor size. For all of the geographical areas to be evaluated, the following definitions will apply:

- D = load density in $\frac{\text{kVA}}{\text{mile}^2}$
- PF = assumed lagging power factor
- z = line impedance in Ω/mile
- l = length of the area
- w = width of the area
- kV_{LL} = nominal line-to-line voltage in kV

It will also be assumed that the loads are modeled as constant current loads.

3.5.1 The Rectangle

A rectangular area of length l and width w is to be served by a primary main feeder. The feeder area is assumed to have a constant load density with three-phase laterals uniformly tapped off of the primary main. Figure 3.12 is a model for the rectangular area. Figure 3.12 represents a rectangular area of constant load density being served by a three-phase main running from node n to node m . It is desired to determine the total voltage drop and the total three-phase power loss down the primary main from node n to node m .

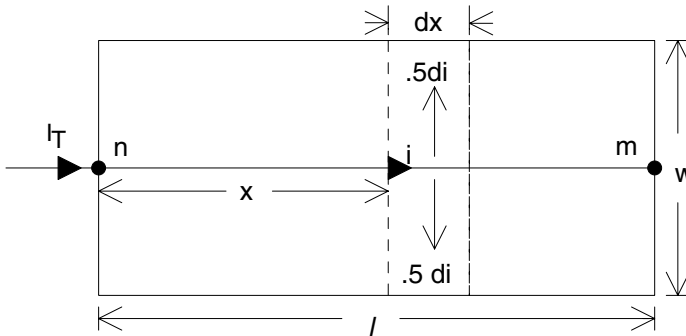


FIGURE 3.12
Constant load density rectangular area.

The total current entering the area is given by:

$$I_T = \frac{D \cdot l \cdot w}{\sqrt{3} \cdot kV_{LL}} / \cos^{-1}(PF) \quad (3.39)$$

An incremental segment is located x miles from node n . The incremental current serving the load in the incremental segment is given by:

$$di = \frac{I_T}{l} \text{ A/mile} \quad (3.40)$$

The current in the incremental segment is given by:

$$i = I_T - x \cdot di = I_T - x \cdot \frac{I_T}{l} = I_T \cdot \left(1 - \frac{x}{l}\right) \quad (3.41)$$

The voltage drop in the incremental segment is

$$dV = \text{Re}(z \cdot i \cdot dx) = \text{Re}\left[z \cdot I_T \cdot \left(1 - \frac{x}{l}\right) \cdot dx\right] \quad (3.42)$$

The total voltage drop down the primary main feeder is

$$V_{\text{drop}} = \int_0^l dV = \text{Re}\left[z \cdot I_T \cdot \int_0^l \left(1 - \frac{x}{l}\right) \cdot dx\right]$$

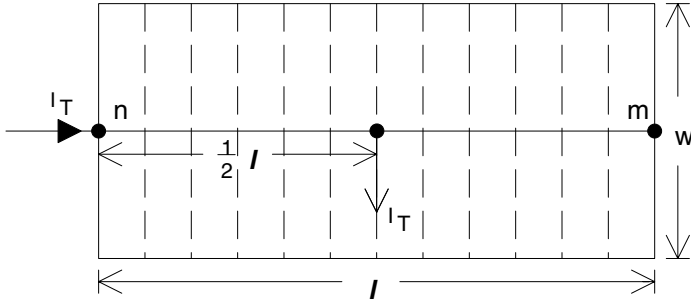
Evaluating the integral and simplifying:

$$V_{\text{drop}} = \text{Re}\left(z \cdot I_T \cdot \frac{1}{2} \cdot l\right) = \text{Re}\left[\frac{1}{2} \cdot Z \cdot I_T\right] \quad (3.43)$$

where

$$Z = z \cdot l$$

Equation 3.43 gives the same result as that of Equation 3.17, which was derived for loads uniformly distributed along a feeder. The only difference is the manner in which the total current (I_T) is determined. The bottom line is that the total load of a rectangular area can be modeled at the centroid of the rectangle as shown in Figure 3.13. It must be understood that in Figure 3.13, with the load modeled at the centroid, the voltage drop computed to the load point will represent the total voltage drop from node n to node m .

**FIGURE 3.13**

Rectangle voltage drop model.

A similar derivation can be done in order to determine the total three-phase power loss down the feeder main. The power loss in the incremental length is

$$\begin{aligned} dp &= 3 \cdot |i|^2 \cdot r \cdot dx = 3 \cdot \left[|I_T|^2 \cdot \left(1 - \frac{x}{l} \right)^2 \cdot r \cdot dx \right] \\ &= 3 \cdot r \cdot |I_T|^2 \cdot \left(1 - 2 \cdot \frac{x}{l} + \frac{x^2}{l^2} \right) \cdot dx \end{aligned}$$

The total three-phase power loss down the primary main is

$$P_{\text{loss}} = \int_0^l dp = 3 \cdot r \cdot |I_T|^2 \cdot \int_0^l \left(1 - 2 \cdot \frac{x}{l} + \frac{x^2}{l^2} \right) \cdot dx$$

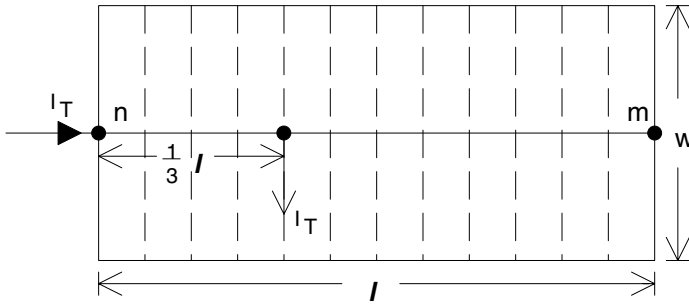
Evaluating the integral and simplifying:

$$P_{\text{loss}} = 3 \cdot \left[\frac{1}{3} \cdot r \cdot l \cdot |I_T|^2 \right] = 3 \cdot \left[\frac{1}{3} \cdot R \cdot |I_T|^2 \right] \quad (3.44)$$

where

$$R = r \cdot l$$

Equation 3.44 gives the same result as that of Equation 3.26. The only difference, again, is the manner in which the total current I_T is determined. The model for computing the total three-phase power loss of the primary main feeder is shown in Figure 3.14. Once again, it must be understood that the power loss computed using the model of Figure 3.14 represents the total power loss from node n to node m .

**FIGURE 3.14**

Rectangle power loss model.

Example 3.6

It is proposed to serve a rectangular area of length 10,000 ft. and width of 6000 ft. The load density of the area is 2500 kVA/mile² with a power factor of 0.9 lagging. The primary main feeder uses 336,400 26/7 ACSR on a pole configured as shown in Example 3.2, Figure 3.3. The question at hand is what minimum standard nominal voltage level can be used to serve this area without exceeding a voltage drop of 3% down the primary main? The choices of nominal voltages are 4.16 kV and 12.47 kV. Compute, also, the total three-phase power loss.

The area to be served is shown in Figure 3.15. From Example 3.2, the impedance of the line was computed to be

$$z = 0.306 + j0.6272 \, \Omega/\text{mile}$$

The length and width of the area in miles are

$$l = \frac{10,000}{5280} = 1.8939 \text{ miles} \quad \text{and} \quad w = \frac{6000}{5280} = 1.1364 \text{ miles}$$

The total area of the rectangular area is

$$A = l \cdot w = 2.1522 \text{ miles}^2$$

The total load of the area is

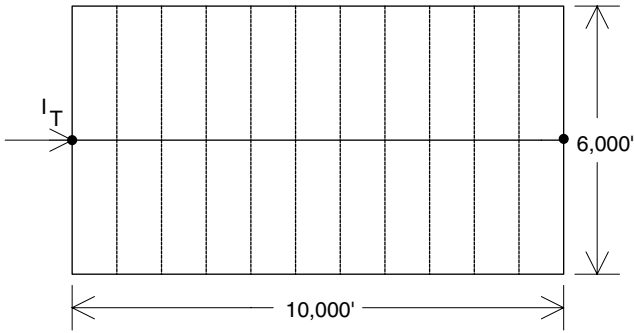
$$\text{kVA} = D \cdot A = 2500 \cdot 2.1522 = 5380.6 \text{ kVA}$$

The total impedance of the line segment is

$$Z = z \cdot l = (0.306 + j0.6272) \cdot 1.8939 = 0.5795 + j1.1879 \, \Omega$$

For a nominal voltage of 4.16 kV, the total area current is

$$I_T = \frac{\text{kVA}}{\sqrt{3} \cdot \text{kV}_{LL}} = \frac{5380.6}{\sqrt{3} \cdot 4.16} / \underline{-\cos^{-1}(0.9)} = 746.7 / \underline{-25.84} \text{ A}$$

**FIGURE 3.15**

Example 3.8 rectangular area.

The total voltage drop down the primary main is

$$\begin{aligned} V_{\text{drop}} &= \text{Re} \left[\frac{1}{2} \cdot Z \cdot I_T \right] = \text{Re} \left[\frac{1}{2} \cdot (0.5795 + j1.1879) \cdot 746.7 / \underline{-25.84} \right] \\ &= 388.1 \text{ V} \end{aligned}$$

The nominal line-to-neutral voltage is

$$V_{LN} = \frac{4160}{\sqrt{3}} = 2401.8 \text{ V}$$

The percent voltage drop is

$$V_{\%} = \frac{V_{\text{drop}}}{V_{LN}} \cdot 100\% = \frac{388.1}{2401.8} \cdot 100\% = 16.16\%$$

It is clear that the nominal voltage of 4.16 kV will not meet the criteria of a voltage drop less than 3.0%. For a nominal voltage of 12.47 kV, the total area current is

$$I_T = \frac{kVA}{\sqrt{3} \cdot kV_{LL}} = \frac{5380.6}{\sqrt{3} \cdot 12.47} \underline{-\cos^{-1}(0.9)} = 249.1 / \underline{-25.84} \text{ A}$$

The total voltage drop down the primary main is

$$\begin{aligned} V_{\text{drop}} &= \text{Re} \left[\frac{1}{2} \cdot Z \cdot I_T \right] = \text{Re} \left[\frac{1}{2} \cdot (0.5795 + j1.1879) \cdot 249.1 / \underline{-25.84} \right] \\ &= 129.5 \text{ V} \end{aligned}$$

The nominal line-to-neutral voltage is

$$V_{LN} = \frac{12,470}{\sqrt{3}} = 7199.6 \text{ V}$$

The percent voltage drop is

$$V_{\%} = \frac{V_{\text{drop}}}{V_{LN}} \cdot 100\% = \frac{129.5}{7199.6} \cdot 100\% = 1.80\%$$

The nominal voltage of 12.47 kV is more than adequate to serve this load. It would be possible at this point to determine how much larger the area could be and still satisfy the 3.0% voltage drop constraint.

For the 12.47 kV selection, the total three-phase power loss down the primary main is

$$P_{\text{loss}} = 3 \cdot \left[\frac{\frac{1}{3} \cdot R \cdot |I_T|^2}{1000} \right] = 3 \cdot \left[\frac{\frac{1}{3} \cdot 0.5795 \cdot 249.1^2}{1000} \right] = 35.965 \text{ kW}$$

3.5.2 The Triangle

Figure 3.16 represents a triangular area of constant load density being served by a three-phase main running from node n to node m . It is desired to determine the total voltage drop and the total three-phase power loss down the primary main from node n to node m .

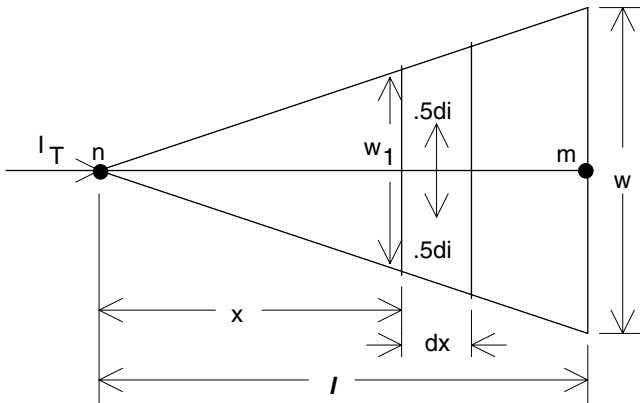


FIGURE 3.16

Constant load density triangular area.

The area of the triangle is

$$\text{Area} = \frac{1}{2} \cdot l \cdot w \quad (3.45)$$

The total current entering the area is given by:

$$I_T = \frac{D \cdot \text{Area}}{\sqrt{3} \cdot k V_{LL}} \frac{-\cos^{-1}(PF)}{A} \quad (3.46)$$

Let

$$di = \frac{I_T}{\text{Area}} = \frac{I_T}{\frac{1}{2} \cdot l \cdot w} = \frac{2 \cdot I_T}{l \cdot w} \text{ A/mile}^2 \quad (3.47)$$

The current entering the incremental line segment is

$$i = I_T - A_1 \cdot di \quad (3.48)$$

where A_1 = area of triangle up to the incremental line segment.

By similar triangles,

$$w_1 = x \cdot \frac{w}{l} \quad (3.49)$$

The area of the small triangle up to the incremental line segment is

$$A_1 = \frac{1}{2} \cdot x \cdot w_1 = \frac{1}{2} \cdot x \cdot \left(x \cdot \frac{w}{l} \right) = \frac{1}{2} \cdot \frac{w}{l} \cdot x^2 \quad (3.50)$$

Substituting Equations 3.47 and 3.50 into Equation 3.48:

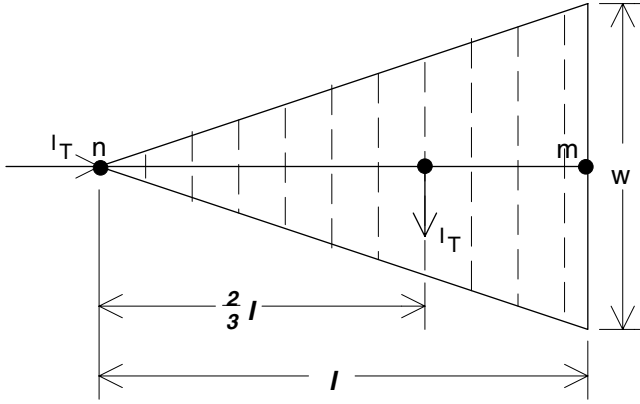
$$i = I_T - \left(\frac{1}{2} \cdot \frac{w}{l} \cdot x^2 \right) \cdot \left(\frac{2}{l \cdot w} \cdot I_T \right) = I_T \cdot \left(1 - \frac{x^2}{l^2} \right) \quad (3.51)$$

The voltage drop in the incremental line segment is given by:

$$dv = \text{Re}[i \cdot z \cdot dx] = \text{Re} \left[z \cdot I_T \cdot \left(1 - \frac{x^2}{l^2} \right) \cdot dx \right] \quad (3.52)$$

The total voltage drop from node n to node m is

$$V_{\text{drop}} = \int_0^l dv = \text{Re} \left[z \cdot I_T \cdot \int_0^l \left(1 - \frac{x^2}{l^2} \right) \cdot dx \right]$$

**FIGURE 3.17**

Triangle voltage drop model.

Evaluating the integral and simplifying:

$$V_{\text{drop}} = \text{Re} \left[z \cdot I_T \cdot \frac{2}{3} \cdot l \right] = \text{Re} \left[\frac{2}{3} \cdot Z_T \cdot I_T \right] \quad (3.53)$$

where

$$Z_T = z \cdot l$$

Equation 3.53 shows that the total voltage drop from the vertex to the base of the triangular area can be computed by modeling the total triangle load two-thirds of the distance between the vertex and the base of the triangle. The model for the voltage drop calculation is shown in Figure 3.17. A similar derivation can be made for the power loss model. The power loss in the incremental line segment is

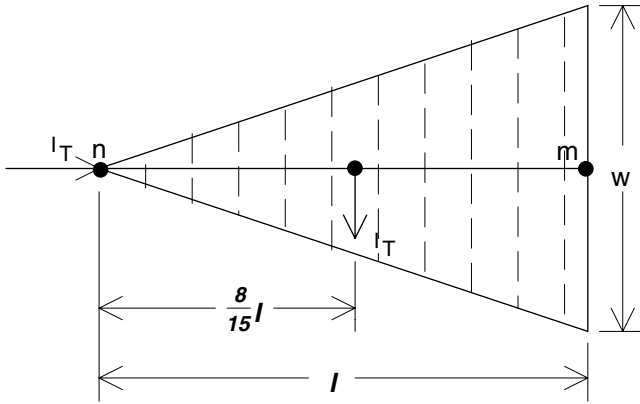
$$dp = 3 \cdot [r \cdot |i|^2 \cdot dx] \quad (3.54)$$

Substituting Equation 3.51 into Equation 3.54:

$$dp = 3 \cdot \left[r \cdot |I_T|^2 \cdot \left(1 - \frac{x^2}{l^2} \right)^2 \cdot dx \right] = 3 \cdot \left[r \cdot |I_T|^2 \cdot \left(1 - 2 \cdot \frac{x^2}{l^2} + \frac{x^4}{l^4} \right) \cdot dx \right]$$

The total three-phase power loss from node n to node m becomes:

$$P_{\text{loss}} = \int_0^l dp = 3 \cdot r \cdot |I_T|^2 \cdot \int_0^l \left(1 - 2 \cdot \frac{x^2}{l^2} + \frac{x^4}{l^4} \right) \cdot dx$$

**FIGURE 3.18**

Triangle power loss model.

Evaluating the integral and simplifying:

$$P_{\text{loss}} = 3 \cdot \left[\frac{8}{15} \cdot R \cdot |I_T|^2 \right] \quad (3.55)$$

Equation 3.55 gives the total three-phase power loss down the primary main from node n to node m . The model for the power loss is given in Figure 3.18.

Example 3.7

The triangular area shown in Figure 3.19 is to be served by a feeder of nominal voltage 12.47 kV. The load density of the area is 3500 kVA/mile² at a power factor of 0.9 lagging. The conductor on the primary main is 336,400 26/7 ACSR, and the configuration of the pole is that of Example 3.2 in Figure 3.3.

Use the K_{drop} factor from the line of Example 3.2 and determine the percent voltage drop from node n to node m .

From Example 3.3 the K_{drop} factor was computed to be

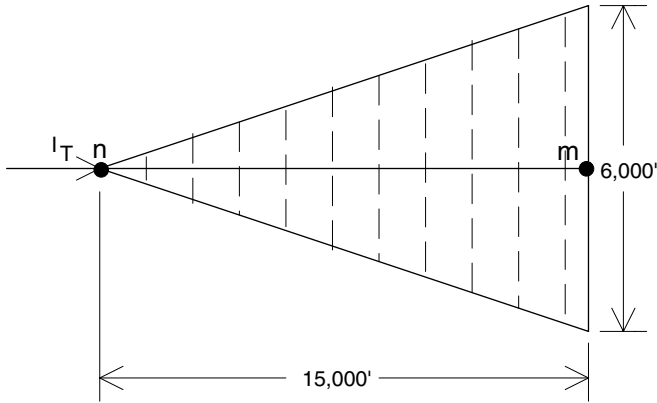
$$K_{\text{drop}} = 0.00035291\% \text{ drop/kVA-mile}$$

The length and width of the triangle in miles is

$$l = \frac{15,000}{5280} = 2.8409 \text{ miles} \quad \text{and} \quad w = \frac{6000}{5280} = 1.1364 \text{ miles}$$

The area of the triangle is

$$\text{Area} = \frac{1}{2} \cdot 2.8509 \cdot 1.1364 = 1.6142 \text{ miles}^2$$

**FIGURE 3.19**

Example 3.9 triangular area.

The total load of the triangular area is

$$kVA = 3500 \cdot 1.6142 = 5649.5 \text{ kVA}$$

The total complex power of the triangular area is

$$\begin{aligned} S &= kVA \angle -\cos^{-1}(PF) = 5649.5 \angle -25.84 \text{ kVA} \\ &= 5084.6 + j2462.6 \end{aligned}$$

Using the K_{drop} factor and lumping the total load at the two-thirds point, the percent drop to node m is

$$V_{\text{drop}} = \frac{2}{3} \cdot K_{\text{drop}} \cdot kVA \cdot \text{miles} = \frac{2}{3} \cdot 0.00035291 \cdot 5649.5 \cdot 2.8409 = 3.7761\%$$

Suppose now that a shunt capacitor bank is to be installed somewhere along the primary main in order to limit the percent voltage drop to node m to 3.0%. Two decisions must be made:

1. Three-phase rating of the capacitor bank
2. Location of the capacitor bank

The total reactive power of the area was computed to be 2462.6 kvar. That means that a capacitor bank rated up to 2462.6 can be used without causing the feeder to go into a leading power factor condition. Since this is assumed

to be the peak load, a capacitor bank rated at 1800 kvar (three-phase) will be used in order to prevent a leading power factor condition for a smaller load. Depending upon the load curve during the day, this bank may or may not have to be switched.

Use the K_{rise} factor from Example 3.5 and determine how far from node n the capacitor bank should be installed in order to limit the voltage drop to 3.0%. From Example 3.5:

$$K_{\text{rise}} = 0.00040331\% \text{ rise/kvar-mile}$$

The needed voltage rise due to the capacitor is

$$V_{\text{rise}} = V_{\text{drop}} - 3.0 = 3.7761 - 3.0 = 0.7761$$

The distance from node n is determined by:

$$\text{dist} = \frac{V_{\text{rise}}}{K_{\text{rise}} \cdot \text{kvar}} = \frac{0.7761}{0.00040331 \cdot 1800} = 1.0691 \text{ miles}$$

The total three-phase power loss down the primary main before the shunt capacitor is added is computed by lumping the total triangular load at:

$$l_{\text{Load}} = \frac{8}{15} \cdot l = 1.5151 \text{ miles from node } n$$

The total load current is

$$I_T = \frac{\text{kVA}}{\sqrt{3} \cdot \text{kV}_{LL}} = \frac{5649.5}{\sqrt{3} \cdot 12.47} / \cos^{-1}(PF) = 261.6 / \underline{-25.84} \text{ A}$$

The total resistance of the primary main is

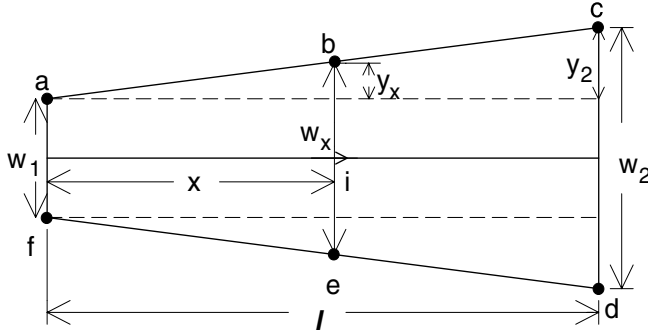
$$R = r \cdot l = 0.306 \cdot 2.8409 = 0.8693 \Omega$$

The total three-phase power loss down the primary main is

$$P_{\text{loss}} = \frac{3}{1000} \cdot \left[\frac{8}{15} \cdot R \cdot |I_T|^2 \right] = \frac{3}{1000} \cdot \left[\frac{8}{15} \cdot 0.8693 \cdot 261.6^2 \right] = 95.16 \text{ kW}$$

3.5.3 The Trapezoid

The final geometric configuration to consider is the trapezoid. As before, it is assumed that the load density is constant throughout the trapezoid. The general model of the trapezoid is shown in Figure 3.20.

**FIGURE 3.21**

Trapezoid dimensions.

Substitute Equation 3.60 into Equation 3.58:

$$I_x = \left(\frac{\sqrt{3} \cdot kV_{LL} \cdot I_T}{\text{Area}_T} \right) \cdot \left(\frac{\text{Area}_x}{\sqrt{3} \cdot kV_{LL}} \right) = \frac{\text{Area}_x}{\text{Area}_T} \cdot I_T \quad (3.61)$$

The current entering the incremental line segment is

$$i = I_T - I_x = I_T \cdot \left(1 - \frac{\text{Area}_x}{\text{Area}_T} \right) \quad (3.62)$$

The only problem at this point is that the area of the small trapezoid cannot be determined since the width w_x is not known. Figure 3.21 will be used to establish the relationship between the unknown width and the known dimensions. Referring to Figure 3.21:

$$w_x = w_1 + 2 \cdot y_x \quad (3.63)$$

From similar triangles:

$$y_x = \frac{x}{l} \cdot y_2 \quad (3.64)$$

But

$$y_2 = \frac{1}{2} \cdot (w_2 - w_1) \quad (3.65)$$

Substitute Equation 3.65 into Equation 3.64:

$$y_x = \frac{x}{l} \cdot \frac{1}{2} \cdot (w_2 - w_1) \quad (3.66)$$

Substitute Equation 3.66 into Equation 3.63:

$$w_x = w_1 + 2 \cdot \frac{x}{l} \cdot \frac{1}{2} \cdot (w_2 - w_1) = w_1 + \frac{x}{l} \cdot (w_2 - w_1) = w_1 \cdot \left(1 - \frac{x}{l}\right) + \frac{x}{l} \cdot w_2 \quad (3.67)$$

Substitute Equation 3.67 into Equation 3.59:

$$\text{Area}_x = \frac{1}{2} \cdot \left[\left(w_1 \cdot \left(1 - \frac{x}{l}\right) + \frac{x}{l} \cdot w_2 \right) + w_1 \right] \cdot x \quad (3.68)$$

Substitute Equations 3.57 and 3.68 into Equation 3.62:

$$\begin{aligned} i &= I_T \cdot \left[1 - \frac{\frac{1}{2} \cdot \left[(w_1 \cdot (2 - \frac{x}{l}) + \frac{x}{l} \cdot w_2) \right] \cdot x}{\frac{1}{2} \cdot (w_2 + w_1) \cdot l} \right] \\ i &= \frac{I_T}{(w_1 + w_2) \cdot l} \cdot \left[(w_1 + w_2) \cdot l - 2 \cdot w_1 \cdot x + w_1 \cdot \frac{x^2}{l} - w_2 \cdot \frac{x^2}{l} \right] \\ i &= \frac{I_T}{(w_1 + w_2) \cdot l} \cdot \left[\left(l - 2 \cdot x + \frac{x^2}{l} \right) \cdot w_1 + \left(l - \frac{x^2}{l} \right) \cdot w_2 \right] \end{aligned} \quad (3.69)$$

The current entering the incremental line segment of Figure 3.20 is given in Equation 3.69 and will be used to compute the voltage drop and power loss in the incremental line segment. The voltage drop in the incremental line segment is given by:

$$dv = \text{Re}[z \cdot i \cdot dx] \quad (3.70)$$

Substitute Equation 3.69 into Equation 3.70:

$$dv = \text{Re} \left\{ z \cdot \frac{I_T}{(w_1 + w_2) \cdot l} \cdot \left[\left(l - 2 \cdot x + \frac{x^2}{l} \right) \cdot w_1 + \left(l - \frac{x^2}{l} \right) \cdot w_2 \right] \cdot dx \right\} \quad (3.71)$$

The total voltage drop down the primary from node n to node m is given by:

$$V_{\text{drop}} = \int_0^l dv = \text{Re} \left\{ \frac{Z \cdot I_T}{(w_1 + w_2) \cdot l} \cdot \int_0^l \left[\left(l - 2 \cdot x + \frac{x^2}{l} \right) \cdot w_1 + \left(l - \frac{x^2}{l} \right) \cdot w_2 \right] \cdot dx \right\}$$

Evaluating the integral and simplifying results in:

$$V_{\text{drop}} = \text{Re} \left[Z \cdot I_T \cdot \left(\frac{w_1 + 2 \cdot w_2}{3 \cdot (w_1 + w_2)} \right) \right] \quad (3.72)$$

Equation 3.72 is very general and can be used to determine the models for the rectangular and triangular areas.

THE RECTANGLE

For a rectangular area the two widths w_1 and w_2 will be equal.

$$\text{Let} \quad w_1 = w_2 = w \quad (3.73)$$

Substitute Equation 3.73 into Equation 3.72:

$$\begin{aligned} V_{\text{drop}} &= \text{Re} \left[Z \cdot I_T \cdot \left(\frac{w + 2 \cdot w}{3 \cdot (w + w)} \right) \right] = \text{Re} \left[Z \cdot I_T \cdot \frac{3 \cdot w}{6 \cdot w} \right] \\ V_{\text{drop}} &= \text{Re} \left[\frac{1}{2} \cdot Z \cdot I_T \right] \end{aligned} \quad (3.74)$$

Equation 3.74 is the same that was initially derived for the rectangular area.

THE TRIANGLE

For a triangular area the width w_1 will be zero.

$$\text{Let} \quad w_1 = 0 \quad (3.75)$$

Substitute Equation 3.75 into Equation 3.72:

$$V_{\text{drop}} = \text{Re} \left[Z \cdot I_T \cdot \left(\frac{0_1 + 2 \cdot w_2}{3 \cdot (0 + w_2)} \right) \right] = \text{Re} \left[\frac{2}{3} \cdot Z \cdot I_T \right] \quad (3.76)$$

Equation 3.76 is the same as was derived for the triangular area.

The total three-phase power loss down the line segment can be developed by starting with the derived current in the incremental segment as given by

Equation 3.69. The three-phase power loss in the incremental segment is

$$dp = 3 \cdot r \cdot i^2 dx \quad (3.77)$$

The total three-phase power loss down the line segment is then:

$$P_{\text{loss}} = 3 \cdot r \cdot \int_0^l i^2 dx \quad (3.78)$$

Substitute Equation 3.69 into Equation 3.78 and simplify:

$$P_{\text{loss}} = 3 \cdot \frac{r \cdot |I_T|^2}{(w_1 + w_2)^2 \cdot l^2} \cdot \int_0^l \left[\left(l - 2 \cdot x + \frac{x^2}{l} \right) \cdot w_1 + \left(l - \frac{x^2}{l} \right) \cdot w_2 \right]^2 dx \quad (3.79)$$

Evaluating the integral and simplifying results in:

$$P_{\text{loss}} = 3 \cdot \left\{ R \cdot |I_T|^2 \cdot \left[\frac{8 \cdot w_2^2 + 9 \cdot w_1 \cdot w_2 + 3 \cdot w_1^2}{15 \cdot (w_1 + w_2)^2} \right] \right\} \quad (3.80)$$

where

$$R = r \cdot l$$

The rectangular and triangular areas are special cases of Equation 3.80.

RECTANGLE

For the rectangle, the two widths w_1 and w_2 are equal.

Let

$$w = w_1 = w_2$$

Substitute into Equation 3.79:

$$P_{\text{loss}} = 3 \cdot \left\{ R \cdot |I_T|^2 \cdot \left[\frac{8 \cdot w^2 + 9 \cdot w \cdot w + 3 \cdot w^2}{15 \cdot (w + w)^2} \right] \right\} = 3 \cdot \left\{ R \cdot |I_T|^2 \cdot \left[\frac{8 + 9 + 3}{15 \cdot (2)^2} \right] \right\}$$

$$P_{\text{loss}} = 3 \cdot \left[\frac{1}{3} \cdot R \cdot |I_T|^2 \right] \quad (3.81)$$

Equation 3.81 is the same as the Equation 3.44 that was previously derived for the rectangular area.

TRIANGLE

For the triangular area, the width w_1 is zero.

Let

$$w_1 = 0$$

Substitute into Equation 3.80:

$$P_{\text{loss}} = 3 \cdot \left\{ R \cdot |I_T|^2 \cdot \left[\frac{8 \cdot w_2^2 + 9 \cdot 0 \cdot w_2 + 3 \cdot 0^2}{15 \cdot (0 + w_2)^2} \right] \right\} = 3 \cdot \left[\frac{8}{15} \cdot R \cdot |I_T|^2 \right] \quad (3.82)$$

Equation 3.82 is the same as Equation 3.55, which was previously derived for the total power loss in a triangular area.

3.6 Summary

This chapter has been devoted to the development of some useful techniques for computing the voltage drop and power loss of line segments with uniformly distributed loads, and for geometric areas with constant load densities. These techniques are very useful for making quick calculations that will be ballpark values. Many times only a ballpark value is needed. More often than not, once inside the ballpark more precise values of voltage drop and power loss are needed. This will be especially true when the unbalanced nature of a distribution feeder is taken into account. The remainder of this text will be devoted to the more precise methods for analyzing a distribution feeder under balanced and unbalanced, steady-state, and short-circuit conditions.

References

1. Glover, J.D. and Sarma, M., *Power System Analysis and Design*, 2nd edition, PWS Publishing Co., Boston, 1994.

Problems

3.1 Shown in Figure 3.22 is the pole configuration of conductors for a three-phase primary feeder. The conductors are 250,000 cm, CON Lay, AA. The nominal line-to-line voltage of the feeder is 14.4 kV.

1. Determine the series impedance per mile of this line.
2. Determine the K_{drop} factor assuming a power factor of 0.88 lag.
3. Determine the K_{rise} factor.

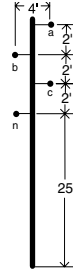


FIGURE 3.22

Problem 3.1 configuration.

3.2 A 4.16 three-phase primary feeder is shown in Figure 3.23.

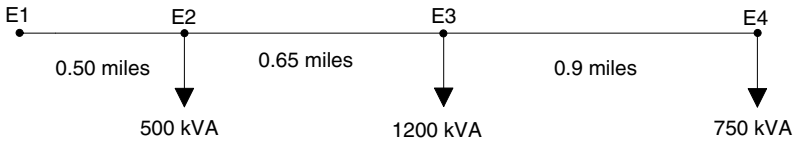


FIGURE 3.23

System for Problem 3.2.

The $K_{\text{drop}} = 0.00298639\%$ drop/kVA-mile

The $K_{\text{rise}} = 0.00334353\%$ rise/kvar-mile

1. Determine the percent voltage drop to node E4.
2. Determine the rating of a three-phase shunt capacitor bank to be placed at E3 to limit the voltage drop to E4 to 5.0%.

3.3 A 4160-V, three-phase feeder is shown in Figure 3.24.

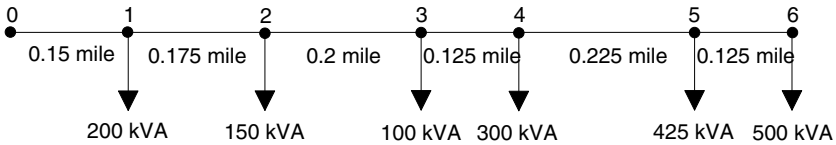


FIGURE 3.24

System for Problem 3.3.

The phase conductors are 4/0 ACSR and are configured on an 8-ft. crossarm with phase spacings of: $D_{ab} = 2.5$ ft., $D_{bc} = 4.5$ ft., and $D_{ca} = 7.0$ ft.

1. Determine the series impedance of the line segment in Ω/mile .
2. Determine the K_{drop} and K_{rise} factors assuming a load power factor of 0.9 lagging.
3. Determine the total percent voltage drop to node 6.

4. Determine the three-phase kvar rating of a shunt capacitor to be placed at node 4 to limit the total percent voltage drop to node 6 to 3.0%.

3.4 Flash Thunderbolt, junior engineer for Tortugas Power and Light, has been given an assignment to design a new 4.16-kV, three-phase feeder that will have the following characteristics:

Total length of feeder = 5000 ft.

Load: 10–500 kVA (three-phase), 0.9 lagging power spaced every 500 ft. with the first load 500 ft. from the substation.

Voltage drop: not to exceed 5% from the sub to the last load.

Figure 3.25 illustrates the new feeder.

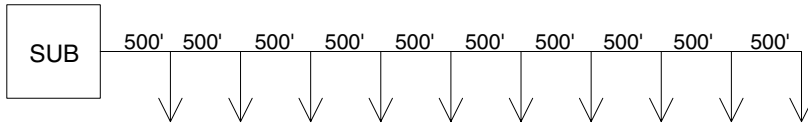


FIGURE 3.25

System for Problem 3.4.

Flash has decided that he will use 336,400 26/7 ACSR (Linnet) conductors constructed on 45-ft. poles with 8-ft. crossarms. The spacings of the conductors on the crossarms are 2.5 ft., 4.5 ft., and 7.0 ft.

1. Determine the percent voltage drop to the last load point and the total three-phase power loss for the feeder shown in Figure 3.25.
2. Lump the total feeder load at the midpoint of the feeder and compute the percent voltage drop to the end of the feeder.
3. Use the exact lumped load model of Figure 3.11 and compute the percent voltage drop to the end of the line, and the total three-phase power loss down the line.

3.5 The rectangular area in Figure 3.26 has a uniform load density of 2000 kVA/mile² at 0.9 lagging power factor. The nominal voltage of the area being served is 4.16 kV. The three-phase primary main conductors are 556,500 26/7 ACSR, while the three-phase lateral conductors are 266,800 26/7 ACSR. The primary main and the laterals are constructed so that the equivalent spacing (D_{eq}) is 3.5 ft. Determine:

1. The % voltage drop to the last customer in the first lateral (point A).
2. The % voltage drop to the last customer in the last lateral (point B).
3. The total three-phase power loss for the total area.

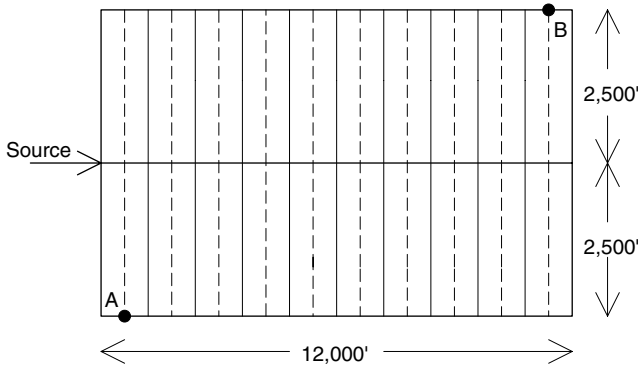


FIGURE 3.26

Rectangular area for Problem 3.5.

3.6 Shown in Figure 3.27 is a rectangle-triangle area that is being fed from a source at point X. Both areas have a load density of 6000 kVA/mile^2 , with loads being uniformly distributed as denoted by the dashed laterals. In addition to the uniformly distributed loads, there is a “spot load” at point Z that is 2000 kVA. The K_{drop} factor for the primary main conductors is $0.00022626\% \text{ drop/kVA-mile}$, and the K_{rise} factor for the primary main conductors is $0.00028436\% \text{ rise/kvar-mile}$.

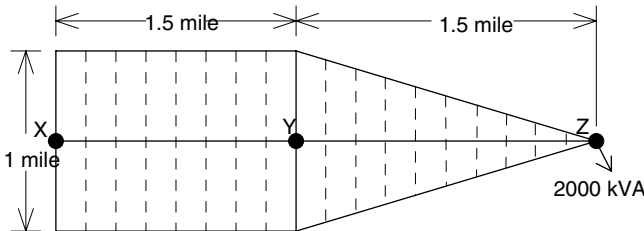


FIGURE 3.27

Rectangular-triangular area of Problem 3.6.

1. Determine the percent drop to point Z.
2. Determine the kVA rating (to the nearest 300 kVA/phase) for a capacitor bank to be placed at point Y in order to limit the voltage drop to Z to 3%.
3. With the capacitor in place, what now is the percent drop to point Z?

3.7 A square area of 20,000 ft. on a side has a load density of 2000 kVA/mile^2 , and 0.9 lagging power factor is to be served from a 12.47-kV substation that is located in the center of the square. Two different plans are being considered for serving the area. The two plans are shown in Figure 3.28.

Plan-A proposes to break the area into four square areas and serve it as shown. The big black line will be the three-phase primary main consisting

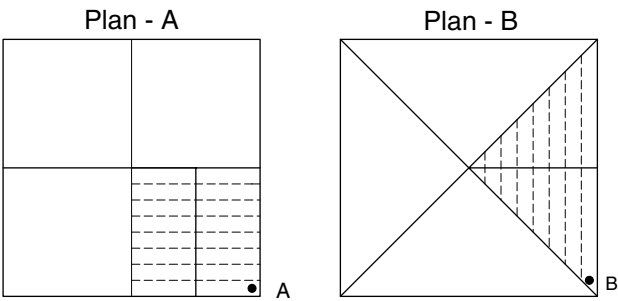


FIGURE 3.28
Two plans for Problem 3.7.

of 336/400 26/7 ACSR conductors, and the dotted lines will be the three-phase laterals consisting of 4/0 ACSR conductors. Both the main and laterals are constructed such that $D_{eq} = 4.3795$ ft. The three-phase laterals will be spaced every 500 ft.

Plan-B proposes to serve the area with four triangularly shaped feeders. Again, the primary main is shown in the dark black line, and the laterals are spaced every 500 ft. and are shown as the dotted lines. The same conductors and D_{eq} will be used in this plan. Determine the percent voltage drop to the “last customer” (points A and B) for the two plans.

3.8 Shown in Figure 3.29 are the areas normally served by two feeders.

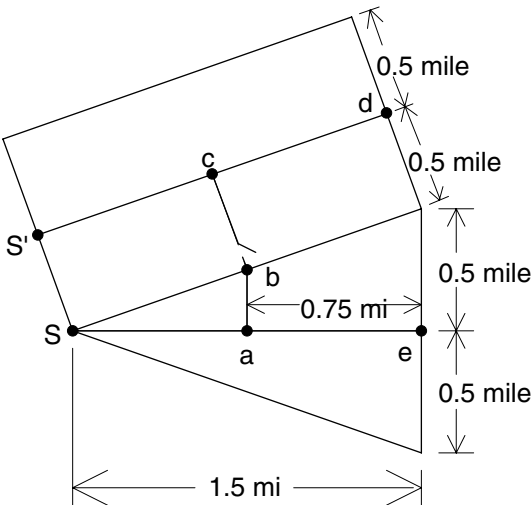


FIGURE 3.29
Areas for Problem 3.8.

Under an emergency condition the switch at **b** is closed so that the feeder normally serving the triangle area must now serve both areas. Assume both

areas have a uniform load density of 2.5 MVA/square mile, and 0.9 lagging power factor. The primary feeder voltage is 13.8 kV. Laterals are uniformly tapped off of the primary main from **S** to **a**. No loads are tapped off the feed from **a** to **b** to **c**, and laterals are tapped off from **c** to **d** and from **c** to **S'**. The primary main conductors are 2/0 ACSR and are placed on a pole such that $D_{eq} = 4.3795$ ft.

1. Determine the K_{drop} and K_{rise} factors.
2. Determine the voltage drop to point **d**.
3. Determine the three-phase kVAr rating of a shunt capacitor bank placed at **c** in order to limit the voltage drop to point **d** to 3.0%.
4. Determine the voltage drop to **e** with the capacitor bank at **c**.
5. Determine the voltage drop to **e** with the source at **S'** and the capacitor at **c**.

4

Series Impedance of Overhead and Underground Lines

The determination of the series impedance for overhead and underground lines is a critical step before the analysis of a distribution feeder can begin. The series impedance of a single-phase, two-phase (V-phase), or three-phase distribution line consists of the resistance of the conductors and the self and mutual inductive reactances resulting from the magnetic fields surrounding the conductors. The resistance component for the conductors will typically come from a table of conductor data such as found in Appendix A.

4.1 Series Impedance of Overhead Lines

The inductive reactance (self and mutual) component of the impedance is a function of the total magnetic fields surrounding a conductor. Figure 4.1 shows conductors 1 through n with the magnetic flux lines created by currents flowing in each of the conductors. The currents in all conductors are assumed to be flowing out of the page. It is further assumed that the sum of the currents will be zero. That is

$$I_1 + I_2 + \dots I_i + \dots I_n = 0 \quad (4.1)$$

The total flux linking conductor i is given by:

$$\lambda_i = 2 \cdot 10^{-7} \cdot \left(I_1 \cdot \ln \frac{1}{D_{i1}} + I_2 \cdot \ln \frac{1}{D_{i2}} + \dots I_i \cdot \ln \frac{1}{\text{GMR}_i} + \dots I_n \cdot \ln \frac{1}{D_{in}} \right) \text{ Wb-t/m} \quad (4.2)$$

where

D_{in} = Distance between conductor i and conductor n (ft.)

GMR_i = Geometric Mean Radius of conductor i (ft.)

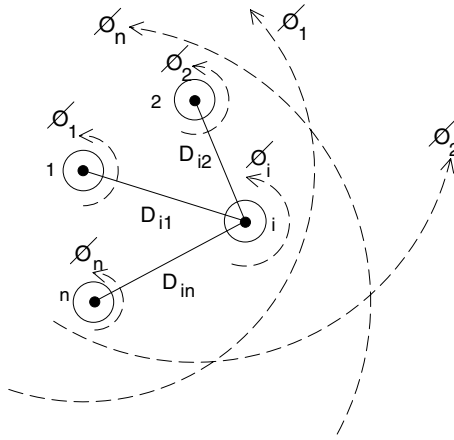


FIGURE 4.1
Magnetic fields.

The inductance of conductor i consists of the self inductance of conductor i and the mutual inductance between conductor i and all of the other $n - 1$ conductors. By definition:

$$\text{Self inductance: } L_{ii} = \frac{\lambda_{ii}}{I_i} = 2 \cdot 10^{-7} \cdot \ln \frac{1}{\text{GMR}_i} \text{ H/m} \quad (4.3)$$

$$\text{Mutual inductance: } L_{in} = \frac{\lambda_{in}}{I_n} = 2 \cdot 10^{-7} \cdot \ln \frac{1}{D_{in}} \text{ H/m} \quad (4.4)$$

4.1.1 Transposed Three-Phase Lines

High-voltage transmission lines are usually assumed to be transposed (each phase occupies the same physical position on the structure for one-third of the length of the line). In addition to the assumption of transposition, it is assumed that the phases are equally loaded (balanced loading). With these two assumptions it is possible to combine the “self” and “mutual” terms into one “phase” inductance.¹

$$\text{Phase inductance: } L_i = 2 \cdot 10^{-7} \cdot \ln \frac{D_{eq}}{\text{GMR}_i} \text{ H/m} \quad (4.5)$$

where

$$D_{eq} = \sqrt[3]{D_{ab} \cdot D_{bc} \cdot D_{ca}} \text{ ft.} \quad (4.6)$$

D_{ab} , D_{bc} , and D_{ca} = distance between phases

Assuming a frequency of 60 Hz, the phase inductive reactance is given by:

$$\text{Phase reactance: } x_i = \omega \cdot L_i = 0.12134 \cdot \ln \frac{D_{eq}}{\text{GMR}_i} \Omega/\text{mile} \quad (4.7)$$

The series impedance per phase of a transposed three-phase line consisting of one conductor per phase is given by:

$$\text{Series impedance: } z_i = r_i + j \cdot 0.12134 \cdot \ln \frac{D_{eq}}{\text{GMR}_i} \Omega/\text{mile} \quad (4.8)$$

4.1.2 Untransposed Distribution Lines

Because distribution systems consist of single-phase, two-phase, and untransposed three-phase lines serving unbalanced loads, it is necessary to retain the identity of the self and mutual impedance terms of the conductors and take into account the ground return path for the unbalanced currents. The ac resistance of the conductors is taken directly from a table of conductor data (Appendix A). Equations 4.3 and 4.4 are used to compute the self and mutual inductive reactances of the conductors. The inductive reactance will be assumed to be at a frequency of 60 Hz, and the length of the conductor will be assumed to be one mile. With those assumptions the self and mutual impedances are given by:

$$\bar{z}_{ii} = r_i + j0.12134 \cdot \ln \frac{1}{\text{GMR}_i} \Omega/\text{mile} \quad (4.9)$$

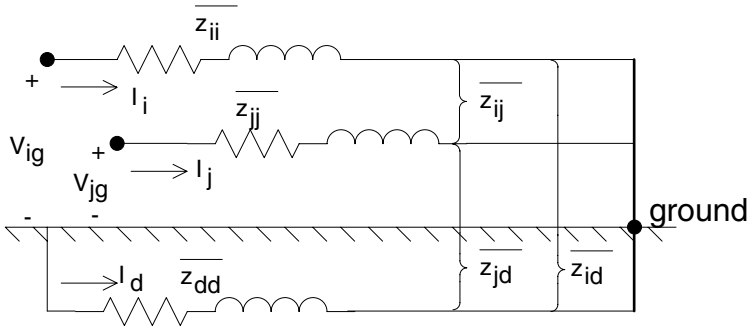
$$\bar{z}_{ij} = j0.12134 \cdot \ln \frac{1}{D_{ij}} \Omega/\text{mile} \quad (4.10)$$

In 1926 John Carson published a paper in which he developed a set of equations for computing the self and mutual impedances of lines, taking into account the return path of current through ground.² Carson's approach was to represent a line with the conductors connected to a source at one end and grounded at the remote end. Figure 4.2 illustrates a line consisting of two conductors (i and j) carrying currents (I_i and I_j) with the remote ends of the conductors tied to ground. A fictitious "dirt" conductor carrying current I_d is used to represent the return path for the currents. In Figure 4.2, Kirchhoff's voltage law (KVL) is used to write the equation for the voltage between conductor i and ground.

$$V_{ig} = \bar{z}_{ii} \cdot I_i + \bar{z}_{ij} \cdot I_j + \bar{z}_{id} \cdot I_d - (\bar{z}_{dd} \cdot I_d + \bar{z}_{di} \cdot I_i + \bar{z}_{dj} \cdot I_j) \quad (4.11)$$

Collect terms in Equation 4.11:

$$V_{ig} = (\bar{z}_{ii} - \bar{z}_{di}) \cdot I_i + (\bar{z}_{ij} - \bar{z}_{dj}) \cdot I_j + (\bar{z}_{id} - \bar{z}_{dd}) \cdot I_d \quad (4.12)$$

**FIGURE 4.2**

Two conductors with dirt return path.

From Kirchhoff's current law:

$$\begin{aligned} I_i + I_j + I_d &= 0 \\ I_d &= -I_i - I_j \end{aligned} \quad (4.13)$$

Substitute Equation 4.13 into Equation 4.12 and collect terms:

$$V_{ig} = (\bar{z}_{ii} + \bar{z}_{dd} - \bar{z}_{di} - \bar{z}_{id}) \cdot I_i + (\bar{z}_{ij} + \bar{z}_{dd} - \bar{z}_{dj} - \bar{z}_{id}) \cdot I_j \quad (4.14)$$

Equation 4.14 is of the general form:

$$V_{ig} = \hat{z}_{ii} \cdot I_i + \hat{z}_{ij} \cdot I_j \quad (4.15)$$

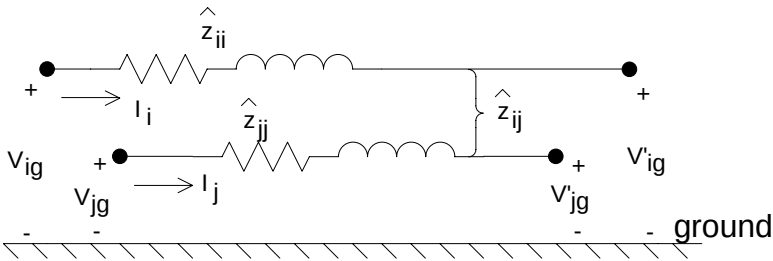
where

$$\hat{z}_{ii} = \bar{z}_{ii} + \bar{z}_{dd} - \bar{z}_{di} - \bar{z}_{id} \quad (4.16)$$

$$\hat{z}_{ij} = \bar{z}_{ij} + \bar{z}_{dd} - \bar{z}_{dj} - \bar{z}_{id} \quad (4.17)$$

In Equations 4.16 and 4.17, the “over bar” impedances are given by Equations 4.9 and 4.10. Note that in these two equations the effect of the ground return path is being folded into what will now be referred to as the “primitive” self and mutual impedances of the line. The equivalent primitive circuit is shown in Figure 4.3. Substituting Equations 4.9 and 4.10 into Equations 4.16 and 4.17, the primitive self impedance is given by:

$$\begin{aligned} \hat{z}_{ii} &= r_i + jx_{ii} + r_d + jx_{dd} - jx_{id} - jx_{di} \\ \hat{z}_{ii} &= r_i + r_d + j0.12134 \cdot \left(\ln \frac{1}{\text{GMR}_i} + \ln \frac{1}{\text{GMR}_d} - \ln \frac{1}{D_{id}} - \ln \frac{1}{D_{di}} \right) \\ \hat{z}_{ii} &= r_i + r_d + j0.12134 \cdot \left(\ln \frac{1}{\text{GMR}_i} + \ln \frac{D_{id} \cdot D_{di}}{\text{GMR}_d} \right) \end{aligned} \quad (4.18)$$

**FIGURE 4.3**

Equivalent primitive circuit.

In similar manner, the primitive mutual impedance can be expanded:

$$\hat{z}_{ij} = jx_{ij} + r_d + jx_{dd} - jx_{dj} - jx_{id}$$

$$\hat{z}_{ij} = r_d + j0.12134 \cdot \left(\ln \frac{1}{D_{ij}} + \ln \frac{1}{\text{GMR}_d} - \ln \frac{1}{D_{dj}} - \ln \frac{1}{D_{id}} \right) \quad (4.19)$$

$$\hat{z}_{ij} = r_d + j0.12134 \left(\ln \frac{1}{D_{ij}} + \ln \frac{D_{dj} \cdot D_{id}}{\text{GMR}_d} \right)$$

The obvious problem in using Equations 4.18 and 4.19 is that we do not know the values of the resistance of dirt (r_d), the Geometric Mean Radius of dirt (GMR_d), and the distances from the conductors to dirt (D_{id} , D_{di} , D_{jd} , D_{dj}). This is where John Carson's work bails us out.

4.1.3 Carson's Equations

Since a distribution feeder is inherently unbalanced, the most accurate analysis should not make any assumptions regarding the spacing between conductors, conductor sizes, and transposition. In Carson's 1926 paper he developed a technique whereby the self and mutual impedances for an arbitrary number of overhead conductors can be determined. The equations can also be applied to underground cables. The technique was not met with a lot of enthusiasm because of the tedious calculations that would have to be done on the slide rule and by hand. With the advent of the digital computer, Carson's equations have become widely used.

In his paper, Carson assumes the earth is an infinite, uniform solid with a flat uniform upper surface and a constant resistivity. Any end effects introduced at the neutral grounding points are not large at power frequencies, and are therefore neglected.

Carson made use of conductor images; that is, every conductor at a given distance above ground has an image conductor the same distance below ground.

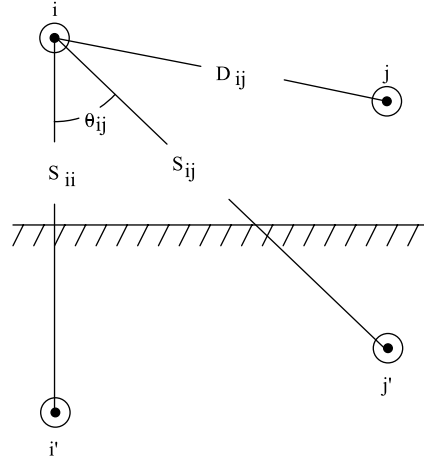


FIGURE 4.4
Conductors and images.

This is illustrated in Figure 4.4. Referring to Figure 4.4, the original Carson equations are given in Equations 4.20 and 4.21.

Self Impedance of Conductor i :

$$\hat{z}_{ii} = r_i + 4\omega P_{ii}G + j\left(X_i + 2\omega G \cdot \ln \frac{S_{ii}}{RD_i} + 4\omega Q_{ii}G\right) \Omega/\text{mile} \quad (4.20)$$

Mutual Impedance between Conductor i and j :

$$\hat{z}_{ij} = 4\omega P_{ij}G + j\left(2\omega G \cdot \ln \frac{S_{ij}}{D_{ij}} + 4\omega Q_{ij}G\right) \Omega/\text{mile} \quad (4.21)$$

where

$\hat{z}_{ii}$ = self impedance of conductor i in Ω/mile

$\hat{z}_{ij}$ = mutual impedance between conductors i and j in Ω/mile

r_i = resistance of conductor i in Ω/mile

$\omega = 2\pi f$ = system angular frequency in radians per second

$G = 0.1609344 \times 10^{-3} \Omega/\text{mile}$

RD_i = radius of conductor i in feet

GMR_i = Geometric Mean Radius of conductor i in feet

f = system frequency in Hertz

ρ = resistivity of earth in $\Omega\text{-meters}$

D_{ij} = distance between conductors i and j in feet (see Figure 4.4)

S_{ij} = distance between conductor i and image j in feet (see Figure 4.4)

θ_{ij} = angle between a pair of lines drawn from conductor i to its own image and to the image of conductor j (see Figure 4.4)

$$X_i = 2\omega G \cdot \ln \frac{RD_i}{GMR_i} \quad \Omega/\text{mile} \quad (4.22)$$

$$P_{ij} = \frac{\pi}{8} - \frac{1}{3\sqrt{2}} k_{ij} \cos(\theta_{ij}) + \frac{k_{ij}^2}{16} \cos(2\theta_{ij}) \cdot \left(0.6728 + \ln \frac{2}{k_{ij}}\right) \quad (4.23)$$

$$Q_{ij} = -0.0386 + \frac{1}{2} \cdot \ln \frac{2}{k_{ij}} + \frac{1}{3\sqrt{2}} k_{ij} \cos(\theta_{ij}) \quad (4.24)$$

$$k_{ij} = 8.565 \times 10^{-4} \cdot S_{ij} \cdot \sqrt{\frac{f}{\rho}} \quad (4.25)$$

4.1.4 Modified Carson's Equations

Only two approximations are made in deriving the modified Carson's equations. These approximations involve the terms associated with P_{ij} and Q_{ij} by using only the first term of the variable P_{ij} and the first two terms of Q_{ij} .

$$P_{ij} = \frac{\pi}{8} \quad (4.26)$$

$$Q_{ij} = -0.03860 + \frac{1}{2} \ln \frac{2}{k_{ij}} \quad (4.27)$$

Substitute X_i (Equation 4.22) into Equation 4.20

$$\hat{z}_{ii} = r_i + 4\omega P_{ii}G + j \left(2\omega G \cdot \ln \frac{RD_i}{GMR_i} + 2\omega G \cdot \ln \frac{S_{ii}}{RD_i} + 4\omega Q_{ii}G \right) \quad (4.28)$$

Combine terms and simplify:

$$\hat{z}_{ii} = r_i + 4\omega P_{ii}G + j2\omega G \left(\ln \frac{S_{ii}}{GMR_i} + 2Q_{ii} \right) \quad (4.29)$$

Simplify Equation 4.21:

$$\hat{z}_{ij} = 4\omega P_{ij}G + j2\omega G \left(\ln \frac{S_{ij}}{D_{ij}} + 2Q_{ij} \right) \quad (4.30)$$

Substitute expressions for P (Equation 4.27) and ω ($2 \cdot \pi \cdot f$):

$$\hat{z}_{ii} = r_i + \pi^2 fG + j4\pi fG \left(\ln \frac{S_{ii}}{\text{GMR}_i} + 2Q_{ii} \right) \quad (4.31)$$

$$\hat{z}_{ij} = \pi^2 fG + j4\pi fG \left(\ln \frac{S_{ij}}{D_{ij}} + 2Q_{ij} \right) \quad (4.32)$$

Substitute the expression for k_{ij} (Equation 4.25) into the approximate expression for Q_{ij} (Equation 4.27):

$$Q_{ij} = -0.03860 + \frac{1}{2} \ln \left(\frac{2}{8.565 \cdot 10^{-4} \cdot S_{ij} \cdot \sqrt{\frac{f}{\rho}}} \right) \quad (4.33)$$

Expand:

$$Q_{ij} = -0.03860 + \frac{1}{2} \ln \left(\frac{2}{8.565 \cdot 10^{-4}} \right) + \frac{1}{2} \ln \frac{1}{S_{ij}} + \frac{1}{2} \ln \sqrt{\frac{\rho}{f}} \quad (4.34)$$

Equation 4.34 can be reduced to

$$Q_{ij} = 3.8393 - \frac{1}{2} \ln S_{ij} + \frac{1}{4} \ln \frac{\rho}{f} \quad (4.35)$$

or

$$2Q_{ij} = 7.6786 - \ln S_{ij} + \frac{1}{2} \ln \frac{\rho}{f} \quad (4.36)$$

Substitute Equation 4.36 into Equation 4.31 and simplify:

$$\begin{aligned} \hat{z}_{ii} &= r_i + \pi^2 fG + j4\pi fG \left(\ln \frac{S_{ii}}{\text{GMR}_i} + 7.6786 - \ln S_{ii} + \frac{1}{2} \ln \frac{\rho}{f} \right) \\ \hat{z}_{ii} &= r_i + \pi^2 fG + 4\pi fG \left(\ln \frac{1}{\text{GMR}_i} + 7.6786 + \frac{1}{2} \ln \frac{\rho}{f} \right) \end{aligned} \quad (4.37)$$

Substitute Equation 4.36 into Equation 4.32 and simplify:

$$\begin{aligned} \hat{z}_{ij} &= \pi^2 fG + j4\pi fG \left(\ln \frac{S_{ij}}{D_{ij}} + 7.6786 - \ln S_{ij} + \frac{1}{2} \ln \frac{\rho}{f} \right) \\ \hat{z}_{ij} &= \pi^2 fG + j4\pi fG \left(\ln \frac{1}{D_{ij}} + 7.6786 + \frac{1}{2} \ln \frac{\rho}{f} \right) \end{aligned} \quad (4.38)$$

Substitute in the values of π and G :

$$\hat{z}_{ii} = r_i + 0.00158836 \cdot f + j0.00202237 \cdot f \left(\ln \frac{1}{\text{GMR}_i} + 7.6786 + \frac{1}{2} \ln \frac{\rho}{f} \right) \quad (4.39)$$

$$\hat{z}_{ij} = 0.00158836 \cdot f + j0.00202237 \cdot f \left(\ln \frac{1}{D_{ij}} + 7.6786 + \frac{1}{2} \ln \frac{\rho}{f} \right) \quad (4.40)$$

It is now assumed:

f = Frequency = 60 Hertz

ρ = Earth resistivity = 100 Ohm-meter

Using these approximations and assumptions, the modified Carson's equations are

$$\hat{z}_{ii} = r_i + 0.09530 + j0.12134 \left(\ln \frac{1}{\text{GMR}_i} + 7.93402 \right) \Omega/\text{mile} \quad (4.41)$$

$$\hat{z}_{ij} = 0.09530 + j0.12134 \left(\ln \frac{1}{D_{ij}} + 7.93402 \right) \Omega/\text{mile}. \quad (4.42)$$

Recall that Equations 4.18 and 4.19 could not be used because the resistance of dirt, the GMR of dirt, and the various distances from conductors to dirt were not known. A comparison of Equations 4.18 and 4.19 to Equations 4.41 and 4.42 shows that the modified Carson's equations have defined the missing parameters. A comparison of the two sets of equations shows that:

$$r_d = 0.09530 \Omega/\text{mile} \quad (4.43)$$

$$\ln \frac{D_{id} \cdot D_{di}}{\text{GMR}_d} = \ln \frac{D_{dj} \cdot D_{jd}}{\text{GMR}_d} = 7.93402 \quad (4.44)$$

The modified Carson's equations will be used to compute the primitive self and mutual impedances of overhead and underground lines.

4.1.5 Primitive Impedance Matrix for Overhead Lines

Equations 4.41 and 4.42 are used to compute the elements of an $n_{\text{cond}} \times n_{\text{cond}}$ primitive impedance matrix. An overhead four-wire grounded wye distribution line segment will result in a 4×4 matrix. For an underground grounded wye line segment consisting of three concentric neutral cables, the

resulting matrix will be 6×6 . The primitive impedance matrix for a three-phase line with m neutrals will be of the form

$$[\hat{z}_{\text{primitive}}] = \begin{bmatrix} \hat{z}_{aa} & \hat{z}_{ab} & \hat{z}_{ac} & | & \hat{z}_{an1} & \hat{z}_{an2} & \hat{z}_{anm} \\ \hat{z}_{ba} & \hat{z}_{bb} & \hat{z}_{bc} & | & \hat{z}_{bn1} & \hat{z}_{bn2} & \hat{z}_{bnm} \\ \hat{z}_{ca} & \hat{z}_{cb} & \hat{z}_{cc} & | & \hat{z}_{cn1} & \hat{z}_{cn2} & \hat{z}_{cnm} \\ --- & --- & --- & --- & --- & --- & --- \\ \hat{z}_{n1a} & \hat{z}_{n1b} & \hat{z}_{n1c} & | & \hat{z}_{n1n1} & \hat{z}_{n1n2} & \hat{z}_{n1nm} \\ \hat{z}_{n2a} & \hat{z}_{n2b} & \hat{z}_{n2c} & | & \hat{z}_{n2n1} & \hat{z}_{n2n2} & \hat{z}_{n2nm} \\ \hat{z}_{nma} & \hat{z}_{nmb} & \hat{z}_{nmc} & | & \hat{z}_{nmn1} & \hat{z}_{nmn2} & \hat{z}_{nmnm} \end{bmatrix} \quad (4.45)$$

In partitioned form, Equation 4.45 becomes

$$[\hat{z}_{\text{primitive}}] = \begin{bmatrix} [\hat{z}_{ij}] & [\hat{z}_{in}] \\ [\hat{z}_{nj}] & [\hat{z}_{nn}] \end{bmatrix} \quad (4.46)$$

4.1.6 Phase Impedance Matrix for Overhead Lines

For most applications the primitive impedance matrix needs to be reduced to a 3×3 phase frame matrix consisting of the self and mutual equivalent impedances for the three phases. Figure 4.5 shows a four-wire grounded neutral line segment. One standard method of reduction is the Kron reduction.³ The assumption is made that the line has a multigrounded neutral as

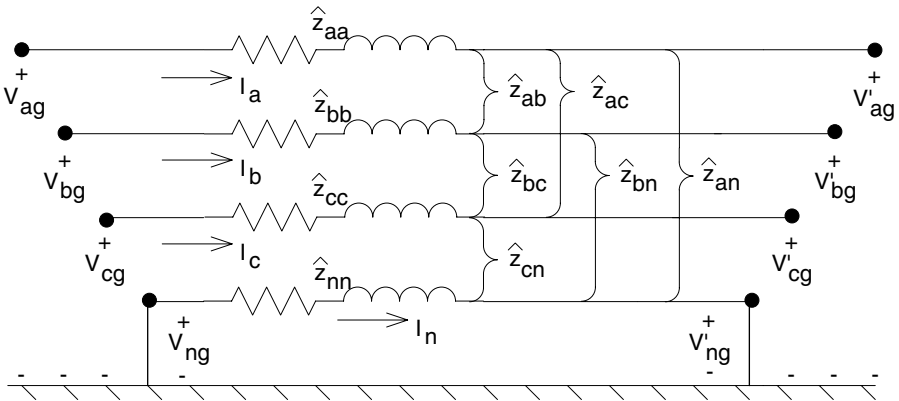


FIGURE 4.5
Four-wire grounded wye line segment.

shown in Figure 4.5. The Kron reduction method applies Kirchhoff's voltage law to the circuit.

$$\begin{bmatrix} V_{ag} \\ V_{bg} \\ V_{cg} \\ V_{ng} \end{bmatrix} = \begin{bmatrix} V'_{ag} \\ V'_{bg} \\ V'_{cg} \\ V'_{ng} \end{bmatrix} + \begin{bmatrix} \hat{z}_{aa} & \hat{z}_{ab} & \hat{z}_{ac} & \hat{z}_{an} \\ \hat{z}_{ba} & \hat{z}_{bb} & \hat{z}_{bc} & \hat{z}_{bn} \\ \hat{z}_{ca} & \hat{z}_{cb} & \hat{z}_{cc} & \hat{z}_{cn} \\ \hat{z}_{na} & \hat{z}_{nb} & \hat{z}_{nc} & \hat{z}_{nn} \end{bmatrix} \cdot \begin{bmatrix} I_a \\ I_b \\ I_c \\ I_n \end{bmatrix} \quad (4.47)$$

In partitioned form, Equation 4.47 becomes

$$\begin{bmatrix} [V_{abc}] \\ [V_{ng}] \end{bmatrix} = \begin{bmatrix} [V'_{abc}] \\ [V'_{ng}] \end{bmatrix} + \begin{bmatrix} [\widehat{z}_{ij}] & [\widehat{z}_{in}] \\ [\widehat{z}_{nj}] & [\widehat{z}_{nn}] \end{bmatrix} \cdot \begin{bmatrix} [I_{abc}] \\ [I_n] \end{bmatrix} \quad (4.48)$$

Because the neutral is grounded, the voltages V_{ng} and V'_{ng} are equal to zero. Substituting those values into Equation 4.48 and expanding results in:

$$[V_{abc}] = [V'_{abc}] + [\widehat{z}_{ij}] \cdot [I_{abc}] + [\widehat{z}_{in}] \cdot [I_n] \quad (4.49)$$

$$[0] = [0] + [\widehat{z}_{nj}] \cdot [I_{abc}] + [\widehat{z}_{nn}] \cdot [I_n] \quad (4.50)$$

Solve Equation 4.50 for $[I_n]$:

$$[I_n] = -[\widehat{z}_{nn}]^{-1} \cdot [\widehat{z}_{nj}] \cdot [I_{abc}] \quad (4.51)$$

Substitute Equation 4.51 into Equation 4.49:

$$\begin{aligned} [V_{abc}] &= [V'_{abc}] + ([\widehat{z}_{ij}] - [\widehat{z}_{in}] \cdot [\widehat{z}_{nn}]^{-1} \cdot [\widehat{z}_{nj}]) \cdot [I_{abc}] \\ [V_{abc}] &= [V'_{abc}] + [z_{abc}] \cdot [I_{abc}] \end{aligned} \quad (4.52)$$

where

$$[z_{abc}] = [\widehat{z}_{ij}] - [\widehat{z}_{in}] \cdot [\widehat{z}_{nn}]^{-1} \cdot [\widehat{z}_{nj}] \quad (4.53)$$

Equation 4.53 is the final form of the Kron reduction technique. The final phase impedance matrix becomes:

$$[z_{abc}] = \begin{bmatrix} z_{aa} & z_{ab} & z_{ac} \\ z_{ba} & z_{bb} & z_{bc} \\ z_{ca} & z_{cb} & z_{cc} \end{bmatrix} \Omega/\text{mile} \quad (4.54)$$

For a distribution line that is not transposed, the diagonal terms of Equation 4.54 will not be equal to each other, and the off-diagonal terms will not be equal to each other. However, the matrix will be symmetrical.

For two-phase (V-phase) and single-phase lines in grounded wye systems, the modified Carson's equations can be applied which will lead to initial 3×3 and 2×2 primitive impedance matrices. Kron reduction will reduce the matrices to 2×2 and a single element. These matrices can be expanded to 3×3 phase frame matrices by the addition of rows and columns consisting of zero elements for the missing phases. For example, a V-phase line consisting of phases a and c , the phase impedance matrix would be

$$[z_{abc}] = \begin{bmatrix} z_{aa} & 0 & z_{ac} \\ 0 & 0 & 0 \\ z_{ca} & 0 & z_{cc} \end{bmatrix} \Omega/\text{mile} \quad (4.55)$$

The phase impedance matrix for a *phase b* single-phase line would be

$$[z_{abc}] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & z_{bb} & 0 \\ 0 & 0 & 0 \end{bmatrix} \Omega/\text{mile} \quad (4.56)$$

The phase impedance matrix for a three-wire delta line is determined by the application of Carson's equations without the Kron reduction step.

The phase impedance matrix can be used to accurately determine the voltage drops on the feeder line segments once the currents have been determined. Since no approximations (transposition, for example) have been made regarding the spacing between conductors, the effect of the mutual coupling between phases is accurately taken into account. The application of the modified Carson's equations and the phase frame matrix leads to the most accurate model of a line segment. Figure 4.6 shows the general three-phase model of

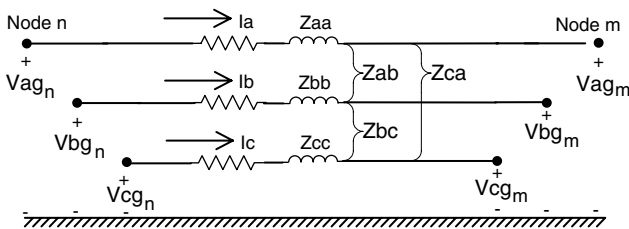


FIGURE 4.6

Three-phase line segment model.

a line segment. Bear in mind that for V-phase and single-phase lines some of the impedance values will be zero. The voltage equation in matrix form for the line segment is

$$\begin{bmatrix} V_{ag} \\ V_{bg} \\ V_{cg} \end{bmatrix}_n = \begin{bmatrix} V_{ag} \\ V_{bg} \\ V_{cg} \end{bmatrix}_m + \begin{bmatrix} Z_{aa} & Z_{ab} & Z_{ac} \\ Z_{ba} & Z_{bb} & Z_{bc} \\ Z_{ca} & Z_{cb} & Z_{cc} \end{bmatrix} \cdot \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \quad (4.57)$$

where $Z_{ij} = z_{ij} \cdot \text{length}$.

Equation 4.57 can be written in condensed form as:

$$[VLG_{abc}]_n = [VLG_{abc}]_m + [Z_{abc}] \cdot [I_{abc}] \quad (4.58)$$

4.1.7 Sequence Impedances

Many times the analysis of a feeder will use only the positive and zero sequence impedances for the line segments. There are two methods for obtaining these impedances. The first incorporates the application of the modified Carson's equations and the Kron reduction to obtain the phase impedance matrix.

The definition of line-to-ground phase voltages as a function of the line-to-ground sequence voltages is given by²:

$$\begin{bmatrix} V_{ag} \\ V_{bg} \\ V_{cg} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a_s^2 & a_s \\ 1 & a_s & a_s^2 \end{bmatrix} \cdot \begin{bmatrix} VLG_0 \\ VLG_1 \\ VLG_2 \end{bmatrix} \quad (4.59)$$

where $a_s = 1.0 / 120$.

In condensed form, Equation 4.59 becomes:

$$[VLG_{abc}] = [A_s] \cdot [VLG_{012}] \quad (4.60)$$

where

$$[A_s] = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a_s^2 & a_s \\ 1 & a_s & a_s^2 \end{bmatrix} \quad (4.61)$$

The phase line currents are defined in the same manner:

$$[I_{abc}] = [A_s] \cdot [I_{012}] \quad (4.62)$$

Equation 4.60 can be used to solve for the sequence line-to-ground voltages as a function of the phase line-to-ground voltages:

$$[VLG_{012}] = [A_s]^{-1} \cdot [VLG_{abc}] \quad (4.63)$$

where

$$[A_s]^{-1} = \frac{1}{3} \cdot \begin{bmatrix} 1 & 1 & 1 \\ 1 & a_s & a_s^2 \\ 1 & a_s^2 & a_s \end{bmatrix} \quad (4.64)$$

Equation 4.58 can be transformed to the sequence domain by multiplying both sides by $[A_s]^{-1}$ and substituting in the definition of the phase currents as given by Equation 4.62.

$$\begin{aligned} [VLG_{012}]_n &= [A_s]^{-1} \cdot [VLG_{abc}]_n \\ [VLG_{012}]_n &= [A_s]^{-1} \cdot [VLG_{abn}]_m + [A_s]^{-1} \cdot [Z_{abc}] \cdot [A_s] \cdot [I_{012}] \\ [VLG_{012}]_n &= [VLG_{012}]_m + [Z_{012}] \cdot [I_{012}] \end{aligned} \quad (4.65)$$

where

$$[Z_{012}] = [A_s]^{-1} \cdot [Z_{abc}] \cdot [A_s] = \begin{bmatrix} Z_{00} & Z_{01} & Z_{02} \\ Z_{10} & Z_{11} & Z_{12} \\ Z_{20} & Z_{21} & Z_{22} \end{bmatrix} \quad (4.66)$$

Equation 4.65 in expanded form is given by:

$$\begin{bmatrix} VLG_0 \\ VLG_1 \\ VLG_2 \end{bmatrix}_n = \begin{bmatrix} VLG_0 \\ VLG_1 \\ VLG_2 \end{bmatrix}_m + \begin{bmatrix} Z_{00} & Z_{01} & Z_{02} \\ Z_{10} & Z_{11} & Z_{12} \\ Z_{20} & Z_{21} & Z_{22} \end{bmatrix} \cdot \begin{bmatrix} I_0 \\ I_1 \\ I_2 \end{bmatrix} \quad (4.67)$$

Equation 4.66 is the defining equation for converting phase impedances to sequence impedances. In Equation 4.66 the diagonal terms of the matrix are the sequence impedances of the line such that

- Z_{00} = zero sequence impedance,
- Z_{11} = positive sequence impedance, and
- Z_{22} = negative sequence impedance.

The off-diagonal terms of Equation 4.66 represent the mutual coupling between sequences. In the idealized state these off-diagonal terms would

be zero. In order for this to happen it must be assumed that the line has been transposed. For high-voltage transmission lines this will sometimes be the case. When the lines are transposed the mutual coupling between phases (off-diagonal terms) are equal and, consequently, the off-diagonal terms of the sequence impedance matrix become zero. Since distribution lines are rarely if ever transposed, the mutual coupling between phases is not equal and, as a result, the off-diagonal terms of the sequence impedance matrix will not be zero.

If a line is assumed to be transposed, the phase impedance matrix is modified so the three diagonal terms are equal and all of the off-diagonal terms are equal. The usual procedure is to set the three diagonal terms of the phase impedance matrix equal to the average of the diagonal terms of Equation 4.54, and the off-diagonal terms equal to the average of the off-diagonal terms of Equation 4.54. When this is done the self and mutual impedances are defined as:

$$z_s = \frac{1}{3} \cdot (z_{aa} + z_{bb} + z_{cc}) \text{ } \Omega/\text{mile} \quad (4.68)$$

$$z_m = \frac{1}{3} \cdot (z_{ab} + z_{bc} + z_{ca}) \text{ } \Omega/\text{mile} \quad (4.69)$$

The phase impedance matrix is now defined as:

$$[z_{abc}] = \begin{bmatrix} z_s & z_m & z_m \\ z_m & z_s & z_m \\ z_m & z_m & z_s \end{bmatrix} \text{ } \Omega/\text{mile} \quad (4.70)$$

When Equation 4.66 is used with this phase impedance matrix the resulting sequence matrix is diagonal (off-diagonal terms are zero). The sequence impedances can be determined directly as:

$$z_{00} = z_s + 2 \cdot z_m \text{ } \Omega/\text{mile} \quad (4.71)$$

$$z_{11} = z_{22} = z_s - z_m \text{ } \Omega/\text{mile} \quad (4.72)$$

A second method commonly used to determine the sequence impedances directly is to employ the concept of Geometric Mean Distances (GMD). The GMD between phases is defined as:

$$D_{ij} = \text{GMD}_{ij} = \sqrt[3]{D_{ab} \cdot D_{bc} \cdot D_{ca}} \text{ ft} \quad (4.73)$$

The GMD between phases and neutral is defined as:

$$D_{in} = \text{GMD}_{in} = \sqrt[3]{D_{an} \cdot D_{bn} \cdot D_{cn}} \text{ ft} \quad (4.74)$$

The GMDs as defined above are used in Equations 4.41 and 4.42 to determine the various self and mutual impedances of the line, resulting in:

$$\hat{z}_{ii} = r_i + 0.0953 + j0.12134 \cdot \left[\ln \left(\frac{1}{\text{GMR}_i} \right) + 7.93402 \right] \Omega/\text{mile} \quad (4.75)$$

$$\hat{z}_{nn} = r_n + 0.0953 + j0.12134 \cdot \left[\ln \left(\frac{1}{\text{GMR}_n} \right) + 7.93402 \right] \Omega/\text{mile} \quad (4.76)$$

$$\hat{z}_{ij} = 0.0953 + j0.12134 \cdot \left[\ln \left(\frac{1}{D_{ij}} \right) + 7.93402 \right] \Omega/\text{mile} \quad (4.77)$$

$$\hat{z}_{in} = 0.0953 + j0.12134 \cdot \left[\ln \left(\frac{1}{D_{in}} \right) + 7.93402 \right] \Omega/\text{mile} \quad (4.78)$$

Equations 4.75 through 4.78 will define a matrix of order $n_{\text{cond}} \times n_{\text{cond}}$, where n_{cond} is the number of conductors (phases plus neutrals) in the line segment. Application of the Kron reduction (Equation 4.53) and the sequence impedance transformation (Equation 4.66) leads to the following expressions for the zero, positive, and negative sequence impedances:

$$z_{00} = \hat{z}_{ii} + 2 \cdot \hat{z}_{ij} - 3 \cdot \left(\frac{\hat{z}_{in}^2}{\hat{z}_{nn}} \right) \Omega/\text{mile} \quad (4.79)$$

$$z_{11} = z_{22} = \hat{z}_{ii} - \hat{z}_{ij} \quad (4.80)$$

$$z_{11} = z_{22} = r_i + j0.12134 \cdot \ln \left(\frac{D_{ij}}{\text{GMR}_i} \right) \Omega/\text{mile}$$

Equation 4.80 is recognized as the standard equation for the calculation of the line impedances when a balanced three-phase system and transposition are assumed.

Example 4.1

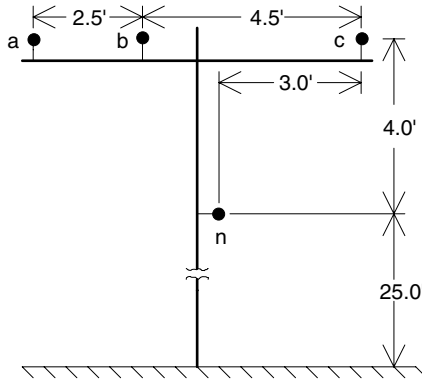
An overhead three-phase distribution line is constructed as shown in Figure 4.7. Determine the phase impedance matrix and the positive and zero sequence impedances of the line. The phase conductors are 336,400 26/7 ACSR (Linnet), and the neutral conductor is 4/0 6/1 ACSR.

SOLUTION

From the table of standard conductor data (Appendix A) it is found that

336,400 26/7 ACSR: $\text{GMR} = 0.0244 \text{ ft}$

Resistance = 0.306 Ω/mile

**FIGURE 4.7**

Three-phase distribution line spacings.

4/0 6/1 ACSR: GMR = 0.00814 ft.

Resistance = 0.5920 Ω /mile

From Figure 4.7, the following distances between conductors can be determined:

$$\begin{aligned} D_{ab} &= 2.5 \text{ ft.} & D_{bc} &= 4.5 \text{ ft.} & D_{ca} &= 7.0 \text{ ft.} \\ D_{an} &= 5.6569 \text{ ft.} & D_{bn} &= 4.272 \text{ ft.} & D_{cn} &= 5.0 \text{ ft.} \end{aligned}$$

Applying the modified Carson's equation for self impedance (Eq. 4.41), the self impedance for *phase a* is

$$\begin{aligned} \hat{z}_{aa} &= 0.0953 + 0.306 + j0.12134 \cdot \left(\ln \frac{1}{0.0244} + 7.93402 \right) \\ &= 0.4013 + j1.4133 \text{ } \Omega/\text{mile} \end{aligned}$$

Applying Equation 4.42 for the mutual impedance between *phases a* and *b*:

$$\begin{aligned} \hat{z}_{ab} &= 0.0953 + j0.12134 \cdot \left(\ln \frac{1}{2.5} + 7.93402 \right) \\ &= 0.0953 + j0.8515 \text{ } \Omega/\text{mile} \end{aligned}$$

Applying the equations for the other self and mutual impedance terms results in the primitive impedance matrix:

$$[\hat{z}] = \begin{bmatrix} 0.4013 + j1.4133 & 0.0953 + j0.8515 & 0.0953 + j0.7266 & 0.0953 + j0.7524 \\ 0.0953 + j0.8515 & 0.4013 + j1.4133 & 0.0953 + j0.7802 & 0.0953 + j0.7865 \\ 0.0953 + j0.7266 & 0.0953 + j0.7802 & 0.4013 + j1.4133 & 0.0953 + j0.7674 \\ 0.0953 + j0.7524 & 0.0953 + j0.7865 & 0.0953 + j0.7674 & 0.6873 + j1.5465 \end{bmatrix} \Omega/\text{mile}$$

The primitive impedance matrix in partitioned form is

$$[\hat{z}_{ij}] = \begin{bmatrix} 0.4013 + j1.4133 & 0.0953 + j0.8515 & 0.0953 + j0.7266 \\ 0.0953 + j0.8515 & 0.4013 + j1.4133 & j0.0943 + j0.7802 \\ 0.0953 + j0.7266 & 0.0953 + j0.7802 & 0.4013 + j1.4133 \end{bmatrix} \Omega/\text{mile}$$

$$[\hat{z}_{in}] = \begin{bmatrix} 0.0953 + j0.7524 \\ 0.0953 + j0.7865 \\ 0.0953 + j0.7674 \end{bmatrix} \Omega/\text{mile}$$

$$[\hat{z}_{nn}] = [0.6873 + j1.5465] \Omega/\text{mile}$$

$$[\hat{z}_{nj}] = \begin{bmatrix} 0.0953 + j0.7524 & 0.0953 + j0.7865 & 0.0953 + j0.7674 \end{bmatrix} \Omega/\text{mile}$$

The Kron reduction of Equation 4.53 results in the phase impedance matrix:

$$[z_{abc}] = [\hat{z}_{ij}] - [\hat{z}_{in}] \cdot [\hat{z}_{nn}]^{-1} \cdot [\hat{z}_{nj}]$$

$$[z_{abc}] = \begin{bmatrix} 0.4576 + j1.0780 & 0.1560 + j.5017 & 0.1535 + j0.3849 \\ 0.1560 + j0.5017 & 0.4666 + j1.0482 & 0.1580 + j0.4236 \\ 0.1535 + j0.3849 & 0.1580 + j0.4236 & 0.4615 + j1.0651 \end{bmatrix} \Omega/\text{mile}$$

The phase impedance matrix can be transformed into the sequence impedance matrix with the application of Equation 4.66:

$$[z_{012}] = [A_s]^{-1} \cdot [z_{abc}] \cdot [A_s]$$

$$[z_{012}] = \begin{bmatrix} 0.7735 + j1.9373 & 0.0256 + j0.0115 & -0.0321 + j0.0159 \\ -0.0321 + j0.0159 & 0.3061 + j0.6270 & -0.0723 - j0.0060 \\ 0.0256 + j0.0115 & 0.0723 - j0.0059 & 0.3061 + j0.6270 \end{bmatrix} \Omega/\text{mile}$$

In the sequence impedance matrix the 1,1 term is the zero sequence impedance; the 2,2 term is the positive sequence impedance; and the 3,3 term is the negative sequence impedance. The 2,2 and 3,3 terms are equal, which demonstrates that for line segments, the positive and negative sequence impedances are equal. Note that the off-diagonal terms are not zero. This implies that there is mutual coupling between sequences. This is a result of the nonsymmetrical spacing between phases. With the off-diagonal terms

nonzero, the three sequence networks representing the line will not be independent. However, it is noted that the off-diagonal terms are small relative to the diagonal terms.

In high-voltage transmission lines, it is usually assumed that the lines are transposed and that the phase currents represent a balanced three-phase set. The transposition can be simulated in Example 4.1 by replacing the diagonal terms of the phase impedance matrix with the average value of the diagonal terms ($0.4619 + j1.0638$), and replacing each off-diagonal term with the average of the off-diagonal terms ($0.1558 + j0.4368$). This modified phase impedance matrix becomes:

$$[z1_{abc}] = \begin{bmatrix} 0.4619 + j1.0638 & 0.1558 + j0.4368 & 0.1558 + j0.4368 \\ 0.1558 + j0.4368 & 0.4619 + j1.0638 & 0.1558 + j0.4368 \\ 0.1558 + j0.4368 & 0.1558 + j0.4368 & 0.4619 + j1.0638 \end{bmatrix} \Omega/\text{mile}$$

Using this modified phase impedance matrix in the symmetrical component transformation equation results in the modified sequence impedance matrix

$$[z1_{012}] = \begin{bmatrix} 0.7735 + j1.9373 & 0 & 0 \\ 0 & 0.3061 + j0.6270 & 0 \\ 0 & 0 & 0.3061 + j0.6270 \end{bmatrix} \Omega/\text{mile}$$

Note that now the off-diagonal terms are all equal to zero, meaning there is no mutual coupling between sequence networks. It should also be noted that the modified zero, positive, and negative sequence impedances are exactly equal to the exact sequence impedances that were first computed.

The results of this example should not be interpreted to mean that a three-phase distribution line can be assumed to have been transposed. The original phase impedance matrix must be used if the correct effect of the mutual coupling between phases is to be modeled.

4.2 Series Impedance of Underground Lines

Figure 4.8 shows the general configuration of three underground cables (concentric neutral or tape shielded) with an additional neutral conductor. The modified Carson's equations can be applied to underground cables in much the same manner as for overhead lines. The circuit of Figure 4.8 will result in a 7×7 primitive impedance matrix. For underground circuits that do not have the additional neutral conductor, the primitive impedance matrix will be 6×6 .

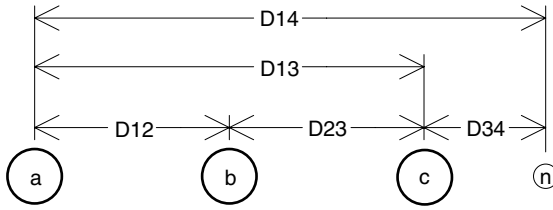


FIGURE 4.8
Three-phase underground with additional neutral.

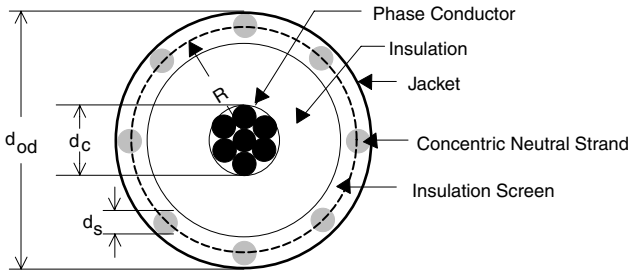


FIGURE 4.9
Concentric neutral cable.

Two popular types of underground cables are the concentric neutral cable and the tape-shielded cable. To apply the modified Carson's equations, the resistance and GMR of the phase conductor and the equivalent neutral must be known.

4.2.1 Concentric Neutral Cable

Figure 4.9 shows a simple detail of a concentric neutral cable. The cable consists of a central phase conductor covered by a thin layer of nonmetallic semiconducting screen, to which is bonded the insulating material. The insulation is then covered by a semiconducting insulation screen. The solid strands of concentric neutral are spiraled around the semiconducting screen with a uniform spacing between strands. Some cables will also have an insulating jacket encircling the neutral strands. In order to apply Carson's equations to this cable, the following data needs to be extracted from a table of underground cables (Appendix B):

- d_c = phase conductor diameter (inches)
- d_{od} = nominal diameter over the concentric neutrals of the cable (inches)
- d_s = diameter of a concentric neutral strand (inches)
- GMR_c = geometric mean radius of the phase conductor (ft.)

GMR_s = geometric mean radius of a neutral strand (ft.)

r_c = resistance of the phase conductor (Ω /mile)

r_s = resistance of a solid neutral strand (Ω /mile)

k = number of concentric neutral strands

The geometric mean radii of the phase conductor and a neutral strand are obtained from a standard table of conductor data (Appendix A). The equivalent geometric mean radius of the concentric neutral is computed using the equation for the geometric mean radius of bundled conductors used in high-voltage transmission lines.¹

$$GMR_{cn} = \sqrt[k]{GMR_s \cdot k \cdot R^{k-1}} \text{ ft} \quad (4.81)$$

where

R = radius of a circle passing through the center of the concentric neutral strands

$$R = \frac{d_{od} - d_s}{24} \text{ ft} \quad (4.82)$$

The equivalent resistance of the concentric neutral is

$$r_{cn} = \frac{r_s}{k} \Omega/\text{mile} \quad (4.83)$$

The various spacings between a concentric neutral and the phase conductors and other concentric neutrals are as follows:

Concentric neutral to its own phase conductor

$$D_{ij} = R \text{ (Equation 4.82 above)}$$

Concentric neutral to an adjacent concentric neutral

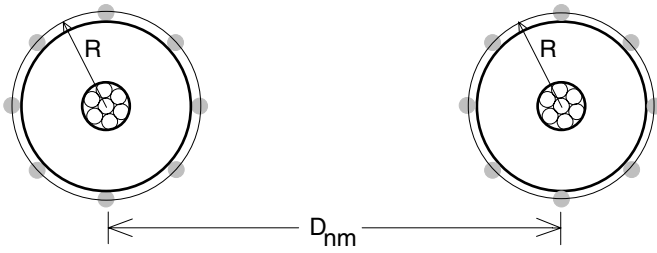
$$D_{ij} = \text{center-to-center distance of the phase conductors}$$

Concentric neutral to an adjacent phase conductor

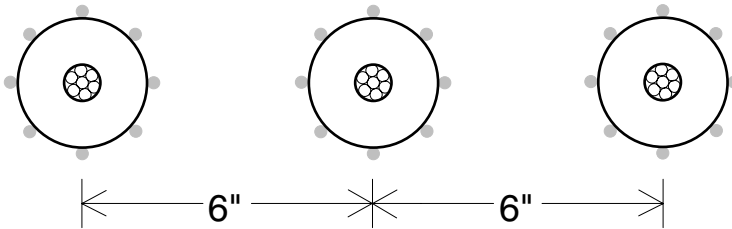
Figure 4.10 shows the relationship between the distance between centers of concentric neutral cables and the radius of a circle passing through the centers of the neutral strands. The geometric mean distance between a concentric neutral and an adjacent phase conductor is given by:

$$D_{ij} = \sqrt[k]{D_{nm}^k - R^k} \text{ ft} \quad (4.84)$$

where D_{nm} = center-to-center distance between phase conductors. For cables buried in a trench the distance between cables will be much greater than the

**FIGURE 4.10**

Distances between concentric neutral cables.

**FIGURE 4.11**

Three-phase concentric neutral cable spacing.

radius R , and therefore it may be assumed that D_{ij} in Equation 4.84 is equal to D_{nm} . For cables in conduit that assumption is not valid.

In applying the modified Carson's equations, the numbering of conductors and neutrals is important. For example, a three-phase underground circuit with an additional neutral conductor must be numbered as:

- 1 = phase conductor #1
- 2 = phase conductor #2
- 3 = phase conductor #3
- 4 = neutral of conductor #1
- 5 = neutral of conductor #2
- 6 = neutral of conductor #3
- 7 = additional neutral conductor (if present)

Example 4.2

Three concentric neutral cables are buried in a trench with spacings as shown in Figure 4.11. The cables are 15 kV, 250,000 CM stranded all-aluminum with 13 strands of #14 annealed, coated copper wires (1/3 neutral). The outside diameter of the cable over the neutral strands is 1.29 inches (Appendix B). Determine the phase impedance matrix and the sequence impedance matrix.

SOLUTION

The data for the phase conductor and neutral strands from a conductor data table (Appendix A) are

250,000 AA phase conductor:

$$\text{GMR} = 0.0171 \text{ ft.}$$

$$\text{Diameter} = 0.567 \text{ inches}$$

$$\text{Resistance} = 0.4100 \text{ } \Omega/\text{mile}$$

#14 copper neutral strands:

$$\text{GMR}_s = 0.00208 \text{ ft.}$$

$$\text{Resistance} = 14.8722 \text{ } \Omega/\text{mile}$$

$$\text{Diameter } (d_s) = 0.0641 \text{ inches}$$

The radius of the circle passing through the center of the strands (Equation 4.82) is

$$R = \frac{d_{od} - d_s}{24} = 0.0511 \text{ ft}$$

The equivalent GMR of the concentric neutral is computed by:

$$\text{GMR}_{cn} = \sqrt[k]{\text{GMR}_s \cdot k \cdot R^{k-1}} = \sqrt[13]{0.00208 \cdot 13 \cdot 0.0511^{13-1}} = 0.0486 \text{ ft}$$

The equivalent resistance of the concentric neutral is

$$r_{cn} = \frac{r_s}{k} = \frac{14.8722}{13} = 1.1438 \text{ } \Omega/\text{mile}$$

The phase conductors are numbered 1, 2, and 3. The concentric neutrals are numbered 4, 5, and 6. The conductor-to-conductor and concentric neutral-to-concentric neutral spacings are

$$D_{12} = D_{21} = D_{45} = D_{54} = 0.5 \text{ ft}$$

$$D_{23} = D_{32} = D_{56} = D_{65} = 0.5 \text{ ft}$$

$$D_{31} = D_{13} = D_{64} = D_{46} = 1.0 \text{ ft}$$

The spacings between conductors and their concentric neutrals are

$$D_{14} = D_{25} = D_{36} = R = 0.0511 \text{ ft}$$

Since the radius R is much smaller than the spacings between cables, the distances between concentric neutrals and adjacent phase conductors are

just the center-to-center distances between conductors:

$$\begin{aligned} D_{15} &= D_{51} = 0.5 \\ D_{26} &= D_{62} = 0.5 \text{ ft} \\ D_{61} &= D_{16} = 1.0. \end{aligned}$$

The self impedance for the cable in position 1 is

$$\begin{aligned} \hat{z}_{11} &= 0.0953 + 0.41 + j0.12134 \cdot \left(\ln \frac{1}{0.0171} + 7.93402 \right) \\ &= 0.5053 + j1.4564 \text{ } \Omega/\text{mile} \end{aligned}$$

The self impedance for the concentric neutral for Cable #1 is

$$\begin{aligned} \hat{z}_{44} &= 0.0953 + 1.144 + j0.12134 \cdot \left(\ln \frac{1}{0.0486} + 7.93402 \right) \\ &= 1.2393 + j1.3296 \text{ } \Omega/\text{mile} \end{aligned}$$

The mutual impedance between Cable #1 and Cable #2 is

$$\hat{z}_{12} = 0.0953 + j0.12134 \cdot \left(\ln \frac{1}{0.5} + 7.93402 \right) = 0.0953 + j1.0468 \text{ } \Omega/\text{mile}$$

The mutual impedance between Cable #1 and its concentric neutral is

$$\hat{z}_{14} = 0.0953 + j0.12134 \cdot \left(\ln \frac{1}{0.0511} + 7.93402 \right) = 0.0953 + j1.3236 \text{ } \Omega/\text{mile}$$

The mutual impedance between the concentric neutral of Cable #1 and the concentric neutral of Cable #2 is

$$\hat{z}_{45} = 0.0953 + j0.12134 \cdot \left(\ln \frac{1}{0.5} + 7.93402 \right) = 0.0953 + j1.0468 \text{ } \Omega/\text{mile}$$

Continuing the application of the modified Carson's equations results in a 6×6 primitive impedance matrix. This matrix is partitioned (Equation 4.33) form is

$$\begin{aligned} [\hat{z}_{ij}] &= \begin{bmatrix} .5053 + j1.4564 & .0953 + j1.0468 & .0953 + j.9627 \\ .0953 + j1.0468 & .5053 + j1.4564 & .0953 + j1.0468 \\ .0953 + j.9627 & .0953 + j1.0468 & .5053 + j1.4564 \end{bmatrix} \text{ } \Omega/\text{mile} \\ [\hat{z}_{in}] &= \begin{bmatrix} .0953 + j1.3236 & .0953 + j1.0468 & .0953 + j.9627 \\ .0953 + j1.0468 & .0953 + j1.3236 & .0953 + j1.0468 \\ .0953 + j.9627 & .0953 + j1.0468 & .0953 + j1.3236 \end{bmatrix} \text{ } \Omega/\text{mile} \end{aligned}$$

$$[\hat{z}_{nj}] = [\hat{z}_{in}]^T$$

$$[\hat{z}_{nn}] = \begin{bmatrix} 1.2391 + j1.3296 & .0953 + j1.0468 & .0953 + j.9627 \\ .0953 + j1.0468 & 1.2391 + j1.3296 & .0953 + j1.0468 \\ .0953 + j.9627 & .0953 + j1.0468 & 1.2391 + j1.3296 \end{bmatrix} \Omega/\text{mile}$$

Using the Kron reduction results in the phase impedance matrix:

$$[z_{abc}] = [\hat{z}_{ij}] - [\hat{z}_{in}] \cdot [\hat{z}_{nn}]^{-1} \cdot [\hat{z}_{nj}]$$

$$[z_{abc}] = \begin{bmatrix} 0.7981 + j0.4463 & 0.3191 + j0.0328 & 0.2849 - j0.0143 \\ 0.3191 + j0.0328 & 0.7891 + j0.4041 & 0.3191 + j0.0328 \\ 0.2849 - j0.0143 & 0.3191 + j0.0328 & 0.7981 + j0.4463 \end{bmatrix} \Omega/\text{mile}$$

The sequence impedance matrix for the concentric neutral three-phase line is determined using Equation 4.66:

$$[z_{012}] = [A_s]^{-1} \cdot [z_{abc}] \cdot [A_s]$$

$$[z_{012}] = \begin{bmatrix} 1.4105 + j0.4665 & -0.0028 - j0.0081 & -0.0056 + j0.0065 \\ -0.0056 + j0.0065 & 0.4874 + j0.4151 & -0.0264 + j0.0451 \\ -0.0028 - j0.0081 & 0.0523 + j0.0003 & 0.4874 + j0.4151 \end{bmatrix} \Omega/\text{mile}$$

4.2.2 Tape-Shielded Cables

Figure 4.12 shows a simple detail of a tape-shielded cable. The cable consists of a central phase conductor covered by a thin layer of nonmetallic semi-conducting screen to which is bonded the insulating material. The insulation is covered by a semiconducting insulation screen. The shield is bare copper

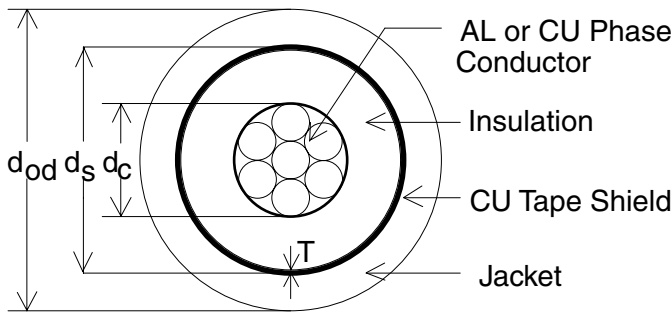


FIGURE 4.12
Tape-shielded cable.

tape helically applied around the insulation screen. An insulating jacket encircles the tape shield. Parameters of the taped shielded cable are

d_c = diameter of phase conductor (inches)

d_s = outside diameter of the tape shield (inches)

d_{od} = outside diameter over jacket (inches)

T = thickness of copper tape shield in mils

Once again, the modified Carson's equations will be applied to calculate the self impedances of the phase conductor and the tape shield, as well as the mutual impedance between the phase conductor and the tape shield. The resistance and GMR of the phase conductor are found in a standard table of conductor data.

The resistance of the tape shield is given by:

$$r_{\text{shield}} = 7.9385 \times 10^8 \frac{\rho}{d_s \cdot T} \Omega/\text{mile} \quad (4.85)$$

The resistivity (ρ) in Equation 4.85 must be expressed in Ω -meters at 50°C. The outside diameter of the tape shield (d_s) is given in inches and the thickness of the tape shield (T) is in mils.

The GMR of the tape shield is the radius of a circle passing through the middle of the shield and is given by:

$$\text{GMR}_{\text{shield}} = \frac{d_s - \frac{T}{1000}}{24} \text{ ft.} \quad (4.86)$$

The various spacings between a tape shield and the conductors and other tape shields are as follows:

Tape shield to its own phase conductor

$$D_{ij} = \text{GMR}_{\text{shield}} = \text{radius to midpoint of the shield (ft.)} \quad (4.87)$$

Tape shield to an adjacent tape shield

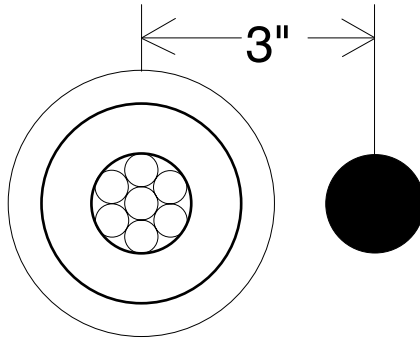
$$D_{ij} = \text{center-to-center distance of the phase conductors (ft.)} \quad (4.88)$$

Tape shield to an adjacent phase or neutral conductor

$$D_{ij} = \text{center-to-center distance between phase conductors (ft.)} \quad (4.89)$$

Example 4.3

A single-phase circuit consists of a 1/0 AA, 220-mil insulation tape shielded cable and a 1/0 Cu neutral conductor, as shown in Figure 4.13. The single-phase line is connected to *phase b*. Determine the phase impedance matrix. Assume $\rho = 2.3715 \times 10^{-8} \Omega\text{-meter}$.

**FIGURE 4.13**

Single-phase tape shield with neutral.

Cable Data: 1/0 AA

Outside diameter of the tape shield = $d_s = 0.88$ inches

Resistance = $0.97 \Omega/\text{mile}$

$\text{GMR}_p = 0.0111$ ft.

Tape shield thickness = $T = 5$ mils

Neutral Data: 1/0 Copper, 7 strand

Resistance = $0.607 \Omega/\text{mile}$

$\text{GMR}_n = 0.01113$ ft.

Distance between cable and neutral = $D_{nm} = 3$ inches

The resistance of the tape shield is computed according to Equation 4.85:

$$r_{\text{shield}} = 7.9385 \times 10^8 \cdot \frac{\rho}{d_s \cdot T} = 7.9385 \times 10^8 \cdot \frac{2.3715 \times 10^{-8}}{0.88 \cdot 5} = 4.2785 \Omega/\text{mile}$$

The GMR of the tape shield is computed according to Equation 4.86:

$$\text{GMR}_{\text{shield}} = \frac{d_s - \frac{T}{1000}}{24} = \frac{0.88 - \frac{5}{1000}}{24} = 0.0365 \text{ ft}$$

The conductors are numbered such that:

#1 = 1/0 AA conductor

#2 = tape shield

#3 = 1/0 copper ground

The spacings used in the modified Carson's equations are

$$D_{12} = \text{GMR}_{\text{shield}} = 0.0365$$

$$D_{13} = \frac{3}{12} = 0.25$$

The self impedance of Conductor #1 is

$$\begin{aligned}\hat{z}_{11} &= 0.0953 + 0.97 + j0.12134 \cdot \left(\ln \frac{1}{0.0111} + 7.93402 \right) \\ &= 1.0653 + j1.5088 \text{ } \Omega/\text{mile}\end{aligned}$$

The mutual impedance between Conductor #1 and the tape shield (Conductor #2) is

$$\begin{aligned}\hat{z}_{12} &= 0.0953 + j0.12134 \cdot \left(\ln \frac{1}{0.0365} + 7.93402 \right) \\ &= 0.0953 + j1.3645 \text{ } \Omega/\text{mile}\end{aligned}$$

The self impedance of the tape shield (Conductor #2) is

$$\begin{aligned}\hat{z}_{22} &= 0.0953 + 4.2786 + j0.12134 \cdot \left(\ln \frac{1}{0.0365} + 7.93402 \right) \\ &= 4.3739 + j1.3645 \text{ } \Omega/\text{mile}\end{aligned}$$

Continuing on, the final primitive impedance matrix is

$$[\hat{z}] = \begin{bmatrix} 1.0653 + j1.5088 & 0.0953 + j1.3645 & 0.0953 + j1.1309 \\ 0.0953 + j1.3645 & 4.3739 + j1.3645 & 0.0953 + j1.1309 \\ 0.0953 + j1.1309 & 0.0953 + j1.1309 & 0.7023 + j1.5085 \end{bmatrix} \Omega/\text{mile}$$

In partitioned form, the primitive impedance matrix is

$$[\hat{z}_{ij}] = [1.0653 + j1.5088]$$

$$[\hat{z}_{in}] = [0.0953 + j1.3645 \quad 0.0953 + j1.1309]$$

$$[\hat{z}_{nj}] = \begin{bmatrix} 0.0953 + j1.3645 \\ 0.0953 + j1.1309 \end{bmatrix}$$

$$[\hat{z}_{nn}] = \begin{bmatrix} 4.3739 + j1.3645 & 0.0953 + j1.1309 \\ 0.0953 + j1.1309 & 0.7023 + j1.5085 \end{bmatrix}$$

Applying Kron's reduction method will result in a single impedance which represents the equivalent single-phase impedance of the tape shield cable and the neutral conductor:

$$z_{1p} = [\hat{z}_{ij}] - [\hat{z}_{in}] \cdot [\hat{z}_{nn}]^{-1} \cdot [\hat{z}_{nj}]$$

$$z_{1p} = 1.3219 + j0.6743 \text{ } \Omega/\text{mile.}$$

Since the single-phase line is on *phase b*, then the phase impedance matrix for the line is

$$[z_{abc}] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1.3219 + j0.6743 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Omega/\text{mile}$$

4.3 Summary

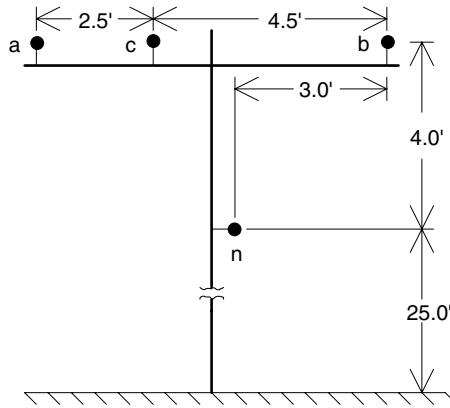
This chapter has been devoted to presenting methods for computing the phase impedances and sequence impedances of overhead lines and underground cables. Carson's equations have been modified in order to simplify the computation of the phase impedances. When using the modified Carson's equations there is no need to make any assumptions, such as transposition of the lines. By assuming an untransposed line and including the actual phasing of the line, the most accurate values of the phase impedances, self and mutual, are determined. Since voltage drop is a primary concern on a distribution line, the impedances used for the line must be as accurate as possible.

References

1. Glover, J. D. and Sarma, M., *Power System Analysis and Design*, 2nd ed., PWS-Kent Publishing, Boston, 1994.
2. Carson, John R., Wave propagation in overhead wires with ground return, *Bell System Technical Journal*, Vol. 5, New York, 1926.
3. Kron, G., Tensorial analysis of integrated transmission systems, Part I, the six basic reference frames, *AIEE Trans.*, Vol. 71, 1952.

Problems

4.1 Determine the phase impedance matrix $[Z_{abc}]$ and the sequence impedance matrix $[Z_{012}]$ in Ω/mile for the three-phase configuration shown in Figure 4.14.

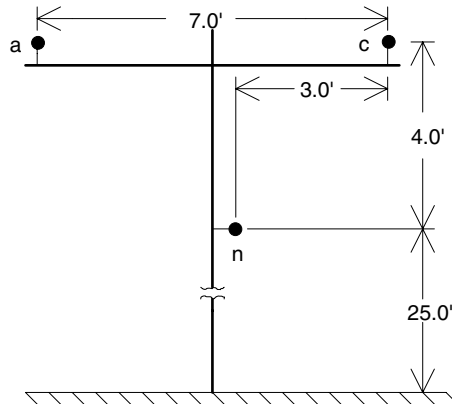
**FIGURE 4.14**

Three-phase configuration for Problem 4.1.

Phase Conductors: 556,500 26/7 ACSR

Neutral Conductor: 4/0 ACSR

4.2 Determine the phase impedance $[Z_{abc}]$ matrix in Ω/mile for the two-phase configuration in Figure 4.15.

**FIGURE 4.15**

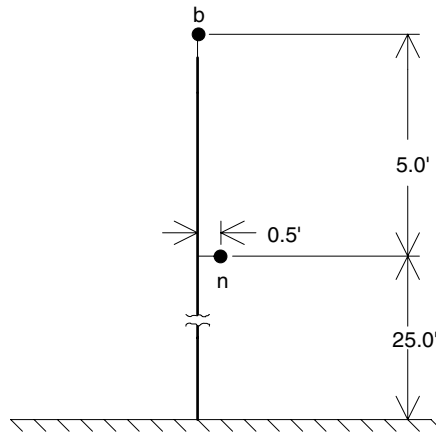
Two-phase configuration for Problem 4.2.

Phase Conductors: 336,400 26/7 ACSR

Neutral Conductor: 4/0 6/1 ACSR

4.3 Determine the phase impedance $[Z_{abc}]$ matrix in Ω/mile for the single-phase configuration shown in Figure 4.16.

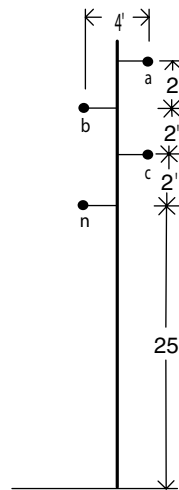
Phase and Neutral Conductors: 1/0 6/1 ACSR

**FIGURE 4.16**

Single-phase pole configuration for Problem 4.3.

4.4 Create the spacings and configurations of Problems 4.1, 4.2, and 4.3 in the Radial Distribution Analysis Package (RDAP). Compare the phase impedance matrices to those computed in the previous problems.

4.5 Determine the phase impedance matrix $[Z_{abc}]$ and sequence impedance matrix $[Z_{012}]$ in Ω/mile for the three-phase pole configuration in Figure 4.17. The phase and neutral conductors are 250,000 all-aluminum.

**FIGURE 4.17**

Three-phase pole configuration for Problem 4.5.

4.6 Compute the positive, negative, and zero sequence impedances in $\Omega/1000 \text{ ft.}$ using the Geometric Mean Distance (GMD) method for the pole configuration shown in Figure 4.17.

4.7 Determine the $[Z_{abc}]$ and $[Z_{012}]$ matrices in Ω/mile for the three-phase configuration shown in Figure 4.18. The phase conductors are 350,000 all-aluminum, and the neutral conductor is 250,000 all-aluminum.

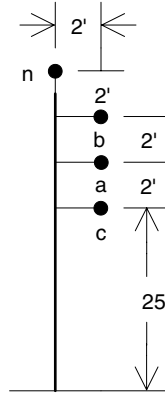


FIGURE 4.18

Three-phase pole configuration for Problem 4.7.

4.8 Compute the positive, negative, and zero sequence impedances in $\Omega/1000$ ft. for the line of Figure 4.18 using the average self and mutual impedances defined in Equations 4.68 and 4.69.

4.9 A 4/0 aluminum concentric neutral cable is to be used for a single-phase lateral. The cable has a full neutral (see Appendix B). Determine the impedance of the cable and the resulting phase impedance matrix in Ω/mile , assuming the cable is connected to *phase b*.

4.10 Three 250,000 CM aluminum concentric cables with one-third neutrals are buried in a trench in a horizontal configuration (see Figure 4.11). Determine the $[Z_{abc}]$ and $[Z_{012}]$ matrices in $\Omega/1000$ ft., assuming phasing of *c-a-b*.

4.11 Create the spacings and configurations of Problems 4.9 and 4.10 in RDAP. Compare the values of the phase impedance matrices to those computed in the previous problems.

4.12 A single-phase underground line is composed of a 350,000 CM aluminum tape-shielded cable. A 4/0 copper conductor is used as the neutral. The cable and neutral are separated by 4 inches. Determine the phase impedance matrix in Ω/mile for this single-phase cable line, assuming *phase c*.

4.13 Three one-third neutral, 2/0 aluminum-jacketed, concentric neutral cables are installed in a 6-inch conduit. Assume the cable jacket has a thickness of 0.2-inch and the cables lie in a triangular configuration inside the conduit. Compute the phase impedance matrix in Ω/mile for this cabled line.

4.14 Create the spacing and configuration of Problem 4.13 in RDAP. Compare results.

5

Shunt Admittance of Overhead and Underground Lines

The shunt admittance of a line consists of the conductance and the capacitive susceptance. The conductance is usually ignored because it is very small compared to the capacitive susceptance. The capacitance of a line is the result of the potential difference between conductors. A charged conductor creates an electric field that emanates outward from the center. Lines of equipotential are created that are concentric to the charged conductor, as illustrated in Figure 5.1.

In Figure 5.1 a difference of potential between two points (P_1 and P_2) is a result of the electric field of the charged conductor. When the potential difference between the two points is known, then the capacitance between the two points can be computed. If there are other charged conductors nearby, the potential difference between the two points will be a function of

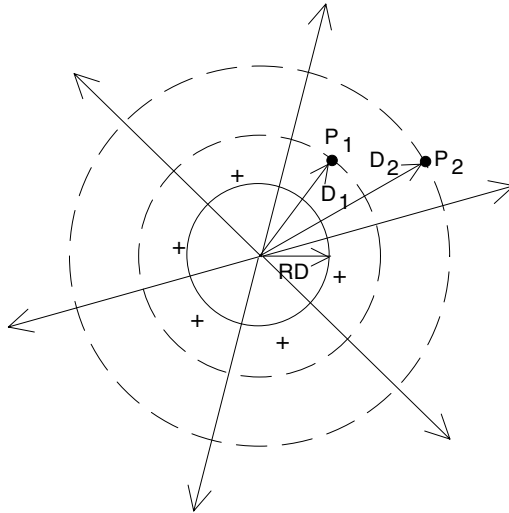


FIGURE 5.1

Electric field of a charged round conductor.

the distance to the other conductors and the charge on each conductor. The principle of superposition is used to compute the total voltage drop between two points, and then the resulting capacitance between the points. The points can be points in space, the surface of two conductors, or the surface of a conductor and ground.

5.1 General Voltage-Drop Equation

Figure 5.2 shows an array of N positively charged solid, round conductors. Each conductor has a unique uniform charge density of q C meter. The voltage drop between Conductor i and Conductor j as a result of all of the charged conductors is given by:

$$V_{ij} = \frac{1}{2\pi\epsilon} \left(q_1 \ln \frac{D_{1j}}{D_{1i}} + \cdots + q_i \ln \frac{D_{ij}}{RD_i} + \cdots + q_j \ln \frac{RD_j}{D_{ij}} + \cdots + q_N \ln \frac{D_{Nj}}{D_{Ni}} \right) \quad (5.1)$$

Equation 5.1 can be written in general form as:

$$V_{ij} = \frac{1}{2\pi\epsilon} \sum_{n=1}^N q_n \ln \frac{D_{nj}}{D_{ni}} \quad (5.2)$$

where

- ϵ = $\epsilon_0\epsilon_r$ = permittivity of the medium
- ϵ_0 = permittivity of free space = 8.85×10^{-12} $\mu\text{F}/\text{meter}$
- ϵ_r = relative permittivity of the medium
- q_n = charge density on Conductor n cb/meter
- D_{ni} = distance between Conductor n and Conductor i (ft.)
- D_{nj} = distance between Conductor n and Conductor j (ft.)
- RD_n = radius of Conductor n

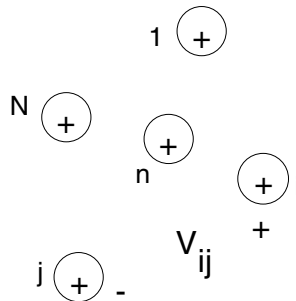


FIGURE 5.2

Array of round conductors.

5.2 Overhead Lines

The method of conductors and their images is employed in the calculation of the shunt capacitance of overhead lines. This is the same concept that was used in Chapter 4 in the general application of Carson's equations. Figure 5.3 illustrates the conductors and their images, and will be used to develop a general voltage-drop equation for overhead lines. In Figure 5.3 it is assumed that:

$$q'_i = -q_i \quad \text{and} \quad q'_j = -q_j \quad (5.3)$$

Applying Equation 5.2 to Figure 5.3:

$$V_{ii} = \frac{1}{2\pi\epsilon} \left(q_i \ln \frac{S_{ii}}{RD_i} + q'_i \ln \frac{RD_i}{S_{ii}} + q_j \ln \frac{S_{ij}}{D_{ij}} + q'_j \ln \frac{D_{ij}}{S_{ij}} \right) \quad (5.4)$$

Because of the assumptions of Equation 5.3, Equation 5.4 can be simplified:

$$\begin{aligned} V_{ii} &= \frac{1}{2\pi\epsilon} \left(q_i \ln \frac{S_{ii}}{RD_i} - q_i \ln \frac{RD_i}{S_{ii}} + q_j \ln \frac{S_{ij}}{D_{ij}} - q_j \ln \frac{D_{ij}}{S_{ij}} \right) \\ V_{ii} &= \frac{1}{2\pi\epsilon} \left(q_i \ln \frac{S_{ii}}{RD_i} + q_i \ln \frac{S_{ii}}{RD_i} + q_j \ln \frac{S_{ij}}{D_{ij}} + q_j \ln \frac{S_{ij}}{D_{ij}} \right) \\ V_{ii} &= \frac{1}{2\pi\epsilon} \left(2 \cdot q_i \ln \frac{S_{ii}}{RD_i} + 2 \cdot q_j \ln \frac{S_{ij}}{D_{ij}} \right) \end{aligned} \quad (5.5)$$

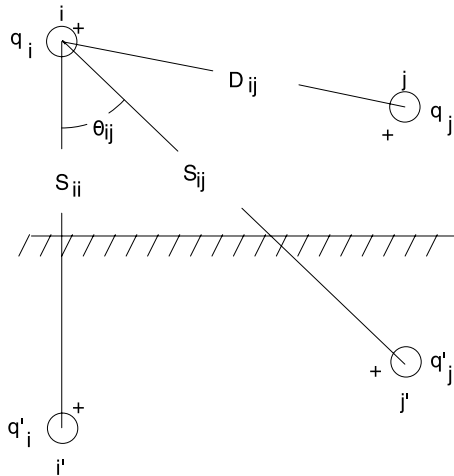


FIGURE 5.3
Conductors and images.

where

S_{ii} = distance from Conductor i to its image i' (ft.)

S_{ij} = distance from Conductor i to the image of Conductor j (ft.)

D_{ij} = distance from Conductor i to Conductor j (ft.)

RD_i = radius of Conductor i in ft.

Equation 5.5 gives the total voltage drop between Conductor i and its image. The voltage drop between Conductor i and ground will be one-half of that given in Equation 5.5:

$$V_{ig} = \frac{1}{2\pi\epsilon} \left(q_i \ln \frac{S_{ii}}{RD_i} + q_j \ln \frac{S_{ij}}{D_{ij}} \right) \quad (5.6)$$

Equation 5.6 can be written in general form as:

$$V_{ig} = \hat{P}_{ii} \cdot q_i + \hat{P}_{ij} \cdot q_j \quad (5.7)$$

where $\hat{P}_{ii}$ and $\hat{P}_{ij}$ are the self and mutual potential coefficients. For overhead lines the relative permittivity of air is assumed to be 1.0 so that:

$$\epsilon_{\text{air}} = 1.0 \times 8.85 \times 10^{-12} \text{ F/meter} \quad (5.8)$$

$$\epsilon_{\text{air}} = 1.4240 \times 10^{-2} \mu\text{F/mile}$$

Using the value of permittivity in $\mu\text{F/mile}$, the self and mutual potential coefficients are defined as:

$$\hat{P}_{ii} = 11.17689 \cdot \ln \frac{S_{ii}}{RD_i} \text{ mile}/\mu\text{F} \quad (5.9)$$

$$\hat{P}_{ij} = 11.17689 \cdot \ln \frac{S_{ij}}{D_{ij}} \text{ mile}/\mu\text{F} \quad (5.10)$$

NOTE In applying Equations 5.9 and 5.10, the values of RD_i , S_{ii} , S_{ij} , and D_{ij} must all be in the same units. For overhead lines the distances between conductors are typically specified in feet, while the value of the conductor diameter from a table will typically be in inches. Care must be taken to assure that the radius in feet is used in applying the two equations.

For an overhead line of n_{cond} conductors, the primitive potential coefficient matrix $[\hat{P}_{\text{primitive}}]$ can be constructed. The primitive potential coefficient

matrix will be an $n_{cond} \times n_{cond}$ matrix. For a four-wire grounded wye line, the primitive coefficient matrix will be of the form:

$$[\hat{P}_{\text{primitive}}] = \begin{bmatrix} \hat{P}_{aa} & \hat{P}_{ab} & \hat{P}_{ac} & \bullet & \hat{P}_{an} \\ \hat{P}_{ba} & \hat{P}_{bb} & \hat{P}_{bc} & \bullet & \hat{P}_{bn} \\ \hat{P}_{ca} & \hat{P}_{cb} & \hat{P}_{cc} & \bullet & \hat{P}_{cn} \\ \bullet & \bullet & \bullet & \bullet & \bullet \\ \hat{P}_{na} & \hat{P}_{nb} & \hat{P}_{nc} & \bullet & \hat{P}_{nn} \end{bmatrix} \quad (5.11)$$

The dots ($\bullet$) in Equation 5.11 are partitioning the matrix between the third and fourth rows and columns. In partitioned form, Equation 5.11 becomes:

$$[\hat{P}_{\text{primitive}}] = \begin{bmatrix} [\hat{P}_{ij}] & [\hat{P}_{in}] \\ [\hat{P}_{nj}] & [\hat{P}_{nn}] \end{bmatrix} \quad (5.12)$$

Because the neutral conductor is grounded, the matrix can be reduced using the Kron reduction method to an $n_{phase} \times n_{phase}$ phase potential coefficient matrix $[P_{abc}]$:

$$[P_{abc}] = [\hat{P}_{ij}] - [\hat{P}_{in}] \cdot [\hat{P}_{nn}]^{-1} \cdot [\hat{P}_{nj}] \quad (5.13)$$

The inverse of the potential coefficient matrix will give the $n_{phase} \times n_{phase}$ capacitance matrix $[C_{abc}]$:

$$[C_{abc}] = [P_{abc}]^{-1} \quad (5.14)$$

For a two-phase line, the capacitance matrix of Equation 5.14 will be 2×2 . A row and column of zeros must be inserted for the missing phase. For a single-phase line, Equation 5.14 will result in a single element. Again, rows and columns of zero must be inserted for the missing phase. In the case of the single-phase line, the only nonzero term will be that of the phase in use.

Neglecting the shunt conductance, the phase shunt admittance matrix is given by:

$$[y_{abc}] = 0 + j \cdot \omega \cdot [C_{abc}] \text{ } \mu\text{S/mile} \quad (5.15)$$

where

$$\omega = 2 \cdot \pi \cdot f = 2 \cdot \pi \cdot 60 = 376.9911$$

Example 5.1

Determine the shunt admittance matrix for the overhead line in Example 4.1. Assume that the neutral conductor is 25 ft. above ground.

The diameters of the phase and neutral conductors from the conductor table (Appendix A) are

$$\begin{aligned} \text{Conductor: } 336,400 \text{ 26/7 ACSR: } d_c &= 0.721 \text{ inches, } RD_c = 0.03004 \text{ ft.,} \\ 4/0 \text{ 6/1 ACSR: } d_s &= 0.563 \text{ inches, } RD_s = 0.02346 \text{ ft.} \end{aligned}$$

For the configuration, the distances between conductors and images in matrix form are:

$$[S] = \begin{bmatrix} 58 & 58.0539 & 58.4209 & 54.1479 \\ 58.0539 & 58 & 58.1743 & 54.0208 \\ 58.4209 & 58.1743 & 58 & 54.0833 \\ 54.1479 & 54.0208 & 54.0833 & 50 \end{bmatrix} \text{ ft}$$

The self-primitive potential coefficient for *phase a* and the mutual primitive potential coefficient between *phases a* and *b* are

$$\hat{P}_{aa} = 11.17689 \ln \frac{58}{0.03004} = 84.5600 \text{ mile}/\mu\text{F}$$

$$\hat{P}_{ab} = 11.17689 \ln \frac{58.0539}{2.5} = 35.1522 \text{ mile}/\mu\text{F}$$

Using Equations 5.9 and 5.10, the total primitive potential coefficient matrix is computed to be

$$[\hat{P}_{\text{primitive}}] = \begin{bmatrix} 84.5600 & 35.1522 & 23.7147 & 25.2469 \\ 35.1522 & 84.5600 & 28.6058 & 28.359 \\ 23.7147 & 28.6058 & 84.5600 & 26.6131 \\ 25.2469 & 28.359 & 26.6131 & 85.6659 \end{bmatrix} \text{ mile}/\mu\text{F}$$

Since the fourth conductor (neutral) is grounded, the Kron reduction method is used to compute the phase potential coefficient matrix. Because only one row and column need to be eliminated, the $[\hat{P}_{44}]$ term is a single element, so the Kron reduction equation for this case can be modified to:

$$P_{ij} = \hat{P}_{ij} - \frac{\hat{P}_{i4} \cdot \hat{P}_{4j}}{\hat{P}_{44}}$$

where $i = 1, 2, 3$ and $j = 1, 2, 3$.

For example, the value of P_{cb} is computed to be

$$P_{cb} = \hat{P}_{3,2} - \frac{\hat{P}_{3,4} \cdot \hat{P}_{4,2}}{\hat{P}_{4,4}} = 28.6058 - \frac{26.6131 \cdot 28.359}{85.6659} = 19.7957$$

Following the Kron reduction, the phase potential coefficient matrix is

$$[P_{abc}] = \begin{bmatrix} 77.1194 & 26.7944 & 15.8714 \\ 26.7944 & 75.172 & 19.7957 \\ 15.8714 & 19.7957 & 76.2923 \end{bmatrix} \text{ mile}/\mu\text{F}$$

Invert $[P_{abc}]$ to determine the shunt capacitance matrix:

$$[C_{abc}] = [P]^{-1} = \begin{bmatrix} 0.015 & -0.0049 & -0.0019 \\ -0.0019 & 0.0159 & -0.0031 \\ -0.0019 & -0.0031 & 0.0143 \end{bmatrix} \mu\text{f}/\text{mile}$$

Multiply $[C_{abc}]$ by the radian frequency to determine the final three-phase shunt admittance matrix:

$$[y_{abc}] = j \cdot 376.9911 \cdot [C_{abc}] = \begin{bmatrix} j5.6712 & -j1.8362 & -j0.7034 \\ -j1.8362 & j5.9774 & -j1.169 \\ -j0.7034 & -j1.169 & j5.3911 \end{bmatrix} \mu\text{S}/\text{mile}$$

5.3 Concentric Neutral Cable Underground Lines

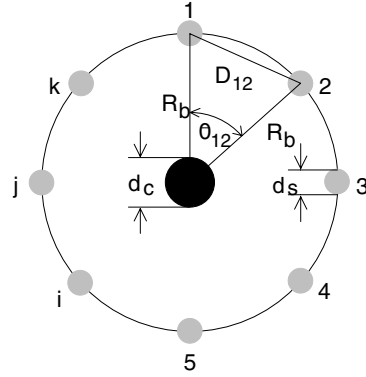
Most underground distribution lines consist of one or more concentric neutral cables. Figure 5.4 illustrates a basic concentric neutral cable with center conductor (black) being the phase conductor and the concentric neutral strands (gray) displaced equally around a circle of radius R_b . Referring to Figure 5.4 the following definitions apply:

R_b = radius of a circle passing through the centers of the neutral strands

d_c = diameter of the phase conductor

d_s = diameter of a neutral strand

k = total number of neutral strands

**FIGURE 5.4**

Basic concentric neutral cable.

The concentric neutral strands are grounded so that they are all at the same potential. Because of the stranding, it is assumed that the electric field created by the charge on the phase conductor will be confined to the boundary of the concentric neutral strands. In order to compute the capacitance between the phase conductor and ground, the general voltage drop of Equation 5.2 will be applied. Since all of the neutral strands are at the same potential, it is only necessary to determine the potential difference between phase conductor p and *Strand 1*.

$$V_{p1} = \frac{1}{2\pi\epsilon} \left(q_p \ln \frac{R_b}{RD_c} + q_1 \ln \frac{RD_s}{R_b} + q_2 \ln \frac{D_{12}}{R_b} + \dots + q_i \ln \frac{D_{1i}}{R_b} + \dots + q_k \ln \frac{D_{k1}}{R_b} \right) \quad (5.16)$$

where

$$RD_c = \frac{d_c}{2}$$

$$RD_s = \frac{d_s}{2}$$

It is assumed that each of the neutral strands carries the same charge such that:

$$q_1 = q_2 = q_i = q_k = -\frac{q_p}{k} \quad (5.17)$$

Equation 5.16 can be simplified:

$$V_{p1} = \frac{1}{2\pi\epsilon} \left[q_p \ln \frac{R_b}{RD_c} - \frac{q_p}{k} \left(\ln \frac{RD_s}{R_b} + \ln \frac{D_{12}}{R_b} + \dots + \ln \frac{D_{1i}}{R_b} + \dots + \ln \frac{D_{1k}}{R_b} \right) \right] \quad (5.18)$$

$$V_{p1} = \frac{q_p}{2\pi\epsilon} \left[\ln \frac{R_b}{RD_c} - \frac{1}{k} \left(\ln \frac{RD_s \cdot D_{12} \cdot D_{1i} \cdot \dots \cdot D_{1k}}{R_b^k} \right) \right]$$

The numerator of the second $\ln$ term in Equation 5.18 needs to be expanded. The numerator represents the product of the radius and the distances between *Strand i* and all of the other strands. Referring to Figure 5.4, the following relations apply:

$$\theta_{12} = \frac{2 \cdot \pi}{k}$$

$$\theta_{13} = 2 \cdot \theta_{12} = \frac{4 \cdot \pi}{k}$$

In general, the angle between *Strand 1* and any other *Strand i* is given by:

$$\theta_{1i} = (i-1) \cdot \theta_{12} = \frac{(i-1) \cdot 2\pi}{k} \quad (5.19)$$

The distances between the various strands are given by:

$$D_{12} = 2 \cdot R_b \cdot \sin\left(\frac{\theta_{12}}{2}\right) = 2 \cdot R_b \cdot \sin\left(\frac{\pi}{k}\right)$$

$$D_{13} = 2 \cdot R_b \cdot \sin\left(\frac{\theta_{13}}{2}\right) = 2 \cdot R_b \cdot \sin\left(\frac{2\pi}{k}\right)$$

The distance between *Strand 1* and any other *Strand i* is given by:

$$D_{1i} = 2 \cdot R_b \cdot \sin\left(\frac{\theta_{1i}}{2}\right) = 2 \cdot R_b \cdot \sin\left(\frac{(i-1) \cdot \pi}{k}\right) \quad (5.20)$$

Equation 5.20 can be used to expand the numerator of the second $\ln$ term of Equation 5.18:

$$RD_s \cdot D_{12} \cdot \dots \cdot D_{1i} \cdot \dots \cdot D_{1k}$$

$$= RD_s \cdot R_b^{k-1} \left[2 \sin\left(\frac{\pi}{k}\right) \cdot 2 \sin\left(\frac{2\pi}{k}\right) \cdot \dots \cdot 2 \sin\left\{\left(\frac{(i-1)\pi}{k}\right)\right\} \cdot \dots \cdot 2 \sin\left\{\left(\frac{(k-1)\pi}{k}\right)\right\} \right] \quad (5.21)$$

The term inside the bracket in Equation 5.21 is a trigonometric identity that is merely equal to the number of strands k .¹ Using that identity, Equation 5.18 becomes:

$$V_{p1} = \frac{q_p}{2\pi\epsilon} \left[\ln \frac{R_b}{RD_c} - \frac{1}{k} \left(\ln \frac{k \cdot RD_s \cdot R_b^{k-1}}{R_b^k} \right) \right] \quad (5.22)$$

$$V_{p1} = \frac{q_p}{2\pi\epsilon} \left[\ln \frac{R_b}{RD_c} - \frac{1}{k} \left(\ln \frac{k \cdot RD_s}{R_b} \right) \right]$$

TABLE 5.1Typical Values of Relative Permittivity (ϵ_r)

Material	Range of Values of Relative Permittivity
Polyvinyl Chloride (PVC)	3.4–8.0
Ethylene-Propylene Rubber (EPR)	2.5–3.5
Polyethylene (PE)	2.5–2.6
Cross-Linked Polyethylene (XLPE)	2.3–6.0

Equation 5.22 gives the voltage drop from the phase conductor to Neutral Strand 1. Care must be taken that the units for the various radii are the same. Typically, underground spacings are given in inches, so the radii of the phase conductor (RD_c) and the strand conductor (RD_s) should be specified in inches.

Since the neutral strands are all grounded, Equation 5.22 gives the voltage drop between the phase conductor and ground. Therefore, the capacitance from phase to ground for a concentric neutral cable is given by:

$$C_{pg} = \frac{q_p}{V_{p1}} = \frac{2\pi\epsilon}{\ln \frac{R_b}{RD_c} - \frac{1}{k} \ln \frac{k \cdot RD_s}{R_b}} \quad (5.23)$$

where

$\epsilon = \epsilon_0\epsilon_r$ = permittivity of the medium

ϵ_0 = permittivity of free space = 0.01420 $\mu\text{F}/\text{mile}$

ϵ_r = relative permittivity of the medium.

The electric field of a cable is confined to the insulation material. Various types of insulation materials are used and each will have a range of values for the relative permittivity. Table 5.1 gives the range of values of relative permittivity for four common insulation materials.² Cross-linked polyethylene is a very popular insulation material. If the minimum value of relative permittivity is assumed (2.3), the equation for the shunt admittance of the concentric neutral cable is given by:

$$y_{ag} = 0 + j \frac{77.3619}{\ln \frac{R_b}{RD_c} - \frac{1}{k} \ln \frac{k \cdot RD_s}{R_b}} \mu\text{S}/\text{mile} \quad (5.24)$$

Example 5.2

Determine the three-phase shunt admittance matrix for the concentric neutral line of Example 4.2 in Chapter 4.

From Example 4.2:

$$R_b = R = 0.0511 \text{ ft.} = 0.6132 \text{ inch}$$

Diameter of the 250,000 AA phase conductor = 0.567 inch,

$$RD_c = \frac{0.567}{2} = 0.2835 \text{ inch}$$

Diameter of the #14 CU concentric neutral strand = 0.0641 inch,

$$RD_s = \frac{0.0641}{2} = 0.03205 \text{ inch}$$

Substitute into Equation 5.24:

$$y_{ag} = j \frac{77.3619}{\ln\left(\frac{R_b}{RD_c}\right) - \frac{1}{k} \cdot \ln\left(\frac{k \cdot RD_s}{R_b}\right)}$$

$$y_{ag} = j \frac{77.3619}{\ln\left(\frac{0.6132}{0.2835}\right) - \frac{1}{13} \cdot \ln\left(\frac{13 \cdot 0.03205}{0.6132}\right)} = j96.5569 \text{ } \mu\text{S/mile}$$

The phase admittance for this three-phase underground line is

$$[y_{abc}] = \begin{bmatrix} j96.5569 & 0 & 0 \\ 0 & j96.5569 & 0 \\ 0 & 0 & j96.5569 \end{bmatrix} \mu\text{S/mile}$$

5.4 Tape-Shielded Cable Underground Lines

A tape-shielded cable is shown in Figure 5.5. Referring to Figure 5.5, R_b is the radius of a circle passing through the center of the tape shield. As with the concentric neutral cable, the electric field is confined to the insulation so that the relative permittivities of Table 5.1 will apply.

The tape-shielded conductor can be visualized as a concentric neutral cable where the number of strands k has become infinite. When k in Equation 5.24

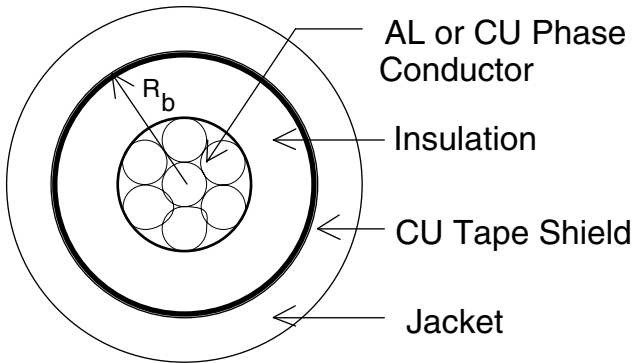


FIGURE 5.5
Tape-shielded conductor.

approaches infinity, the second term in the denominator approaches zero. Therefore, the equation for the shunt admittance of a tape-shielded conductor becomes:

$$y_{ag} = 0 + j \frac{77.3619}{\ln \frac{R_b}{RD_c}} \mu\text{S/mile} \quad (5.25)$$

Example 5.3

Determine the shunt admittance of the single-phase tape-shielded cable of Example 4.3 in Chapter 4. From Example 4.3, the outside diameter of the tape shield is 0.88 inch. The thickness of the tape shield (T) is 5 mils. The radius of a circle passing through the center of the tape shield is

$$T = \frac{5}{1000} = 0.005$$

$$R_b = \frac{d_s - T}{2} = \frac{0.88 - 0.005}{2} = 0.4375 \text{ inch}$$

The diameter of the 1/0 AA phase conductor = 0.368 inch

$$RD_c = \frac{d_p}{2} = \frac{0.368}{2} = 0.1840 \text{ inch}$$

Substitute into Equation 5.25:

$$y_{bg} = j \frac{77.3619}{\ln \left(\frac{R_b}{RD_c} \right)} = j \frac{77.3619}{\ln \left(\frac{0.4375}{0.184} \right)} = j89.3179 \mu\text{S/mile}$$

The line is on *phase b* so that the phase admittance matrix becomes:

$$[y_{abc}] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & j89.3179 & 0 \\ 0 & 0 & 0 \end{bmatrix} \mu\text{S/mile}$$

5.5 Sequence Admittance

The sequence admittances of a three-phase line can be determined in much the same manner as the sequence impedances were determined in Chapter 4. Assume that the 3×3 admittance matrix is given in S/mile. Then the three-phase capacitance currents as a function of the line-to-ground voltages are given by:

$$\begin{bmatrix} Icap_a \\ Icap_b \\ Icap_c \end{bmatrix} = \begin{bmatrix} y_{aa} & y_{ab} & y_{ac} \\ y_{ba} & y_{bb} & y_{bc} \\ y_{ca} & y_{cb} & y_{cc} \end{bmatrix} \cdot \begin{bmatrix} V_{ag} \\ V_{bg} \\ V_{cg} \end{bmatrix} \quad (5.26)$$

$$[Icap_{abc}] = [y_{abc}] \cdot [VLG_{abc}] \quad (5.27)$$

Applying the symmetrical component transformations:

$$[Icap_{012}] = [A_s]^{-1} \cdot [Icap_{abc}] = [A_s]^{-1} \cdot [y_{abc}] \cdot [A_s] \cdot [VLG_{012}] \quad (5.28)$$

From Equation 5.28 the sequence admittance matrix is given by:

$$[y_{012}] = [A_s]^{-1} \cdot [y_{abc}] \cdot [A_s] = \begin{bmatrix} y_{00} & y_{01} & y_{02} \\ y_{10} & y_{11} & y_{12} \\ y_{20} & y_{21} & y_{22} \end{bmatrix} \quad (5.29)$$

For a three-phase overhead line with unsymmetrical spacing, the sequence admittance matrix will be full. That is, the off-diagonal terms will be nonzero. However, a three-phase underground line with three identical cables will only have the diagonal terms since there is no mutual capacitance between phases. In fact, the sequence admittances will be exactly the same as the phase admittances.

5.6 Summary

Methods for computing the shunt capacitive admittance for overhead and underground lines have been presented in this chapter. Distribution lines are typically so short that the shunt admittance can be ignored. However, there are cases of long, lightly loaded overhead lines where the shunt admittance should be included. Underground cables have a much higher shunt admittance per mile than overhead lines. Again, there will be cases where the shunt admittance of an underground cable should be included in the analysis process. When the analysis is being done using a computer, the approach to take is to go ahead and model the shunt admittance for both overhead and underground lines. Why make a simplifying assumption when it is not necessary?

References

1. Glover, J.D. and Sarma, M., *Power System Analysis and Design*, 2nd edition, PWS-Kent Publishing, Boston, 1995.
 2. Arnold, T.P. and Mercier, C.D., *Power Cable Manual*, 2nd ed., Southwire Company, Carrollton, GA, 1997.
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Problems

- 5.1 Determine the phase admittance matrix $[Y_{abc}]$ and sequence admittance matrix $[Y_{012}]$ in $\mu\text{S}/\text{mile}$ for the three-phase overhead line of Problem 4.1.
- 5.2 Determine the phase admittance matrix in $\mu\text{S}/\text{mile}$ for the two-phase line of Problem 4.2.
- 5.3 Determine the phase admittance matrix in $\mu\text{S}/\text{mile}$ for the single-phase line of Problem 4.3.
- 5.4 Verify the results of Problems 5.1, 5.2, and 5.3 using RDAP.
- 5.5 Determine the phase admittance matrix and sequence admittance matrix in $\mu\text{S}/\text{mile}$ for the three-phase line of Problem 4.5.
- 5.6 Determine the phase admittance matrix in $\mu\text{S}/\text{mile}$ for the single-phase concentric neutral cable of Problem 4.9.
- 5.7 Determine the phase admittance matrix and sequence admittance matrix for the three-phase concentric neutral line of Problem 4.10.

- 5.8** Verify the results of Problems 5.6 and 5.7 using RDAP.
- 5.9** Determine the phase admittance matrix in $\mu\text{S}/\text{mile}$ for the single-phase tape-shielded cable line of Problem 4.12.
- 5.10** Determine the phase admittance for the three-phase tape-shielded cable line of Problem 4.13.
- 5.11** Verify the results of Problems 5.9 and 5.10 using RDAP.

6

Distribution System Line Models

The modeling of distribution overhead and underground line segments is a critical step in the analysis of a distribution feeder. It is important to include the actual phasing of the line and the correct spacing between conductors. Chapters 4 and 5 developed the method for the computation of the phase impedance and phase admittance matrices without assuming transposition of the lines. Those matrices will be used in the models for overhead and underground line segments.

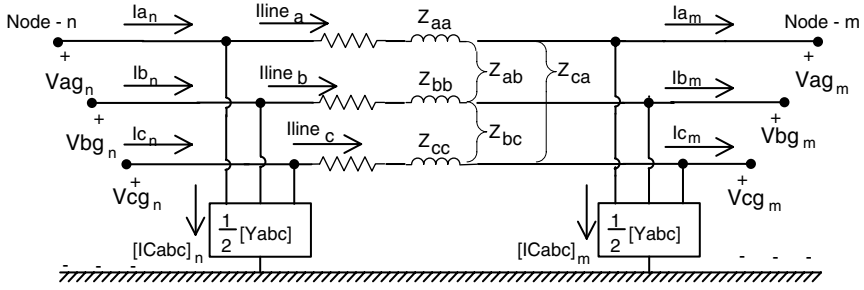
6.1 Exact Line Segment Model

The exact model of a three-phase, two-phase, or single-phase overhead or underground line is shown in Figure 6.1. When a line segment is two-phase (V-phase) or single-phase, some of the impedance and admittance values will be zero. Recall in Chapters 4 and 5 that in all cases the phase impedance and phase admittance matrices were 3×3 . Rows and columns of zeros for the missing phases represent two-phase and single-phase lines. Therefore, one set of equations can be developed to model all overhead and underground line segments. The values of the impedances and admittances in Figure 6.1 represent the total impedances and admittances for the line. That is, the phase impedance matrix derived in Chapter 4 has been multiplied by the length of the line segment. The phase admittance matrix derived in Chapter 5 has also been multiplied by the length of the line segment.

For the line segment of Figure 6.1, the equations relating the input (*Node n*) voltages and currents to the output (*Node m*) voltages and currents are developed as follows:

Kirchhoff's current law applied at *Node m*:

$$\begin{bmatrix} Iline_a \\ Iline_b \\ Iline_c \end{bmatrix}_n = \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}_m + \frac{1}{2} \cdot \begin{bmatrix} Y_{aa} & Y_{ab} & Y_{ac} \\ Y_{ba} & Y_{bb} & Y_{bc} \\ Y_{ca} & Y_{cb} & Y_{cc} \end{bmatrix} \cdot \begin{bmatrix} V_{ag} \\ V_{bg} \\ V_{cg} \end{bmatrix}_m \quad (6.1)$$

**FIGURE 6.1**

Three-phase line segment model.

In condensed form Equation 6.1 becomes:

$$[iline_{abc}]_n = [I_{abc}]_m + \frac{1}{2}[Y_{abc}] \cdot [VLG_{abc}]_m \quad (6.2)$$

Kirchhoff's voltage law applied to the model gives:

$$\begin{bmatrix} Vag \\ Vbg \\ Vcg \end{bmatrix}_n = \begin{bmatrix} Vag \\ Vbg \\ Vcg \end{bmatrix}_m + \begin{bmatrix} Z_{aa} & Z_{ab} & Z_{ac} \\ Z_{ba} & Z_{bb} & Z_{bc} \\ Z_{ca} & Z_{cb} & Z_{cc} \end{bmatrix} \cdot \begin{bmatrix} Iline_a \\ Iline_b \\ Iline_c \end{bmatrix}_m \quad (6.3)$$

In condensed form Equation 6.3 becomes:

$$[VLG_{abc}]_n = [VLG_{abc}]_m + [Z_{abc}] \cdot [iline_{abc}]_m \quad (6.4)$$

Substituting Equation 6.2 into Equation 6.4:

$$[VLG_{abc}]_n = [VLG_{abc}]_m + [Z_{abc}] \cdot \left\{ [I_{abc}]_m + \frac{1}{2}[Y_{abc}] \cdot [VLG_{abc}]_m \right\} \quad (6.5)$$

Collecting terms:

$$[VLG_{abc}]_n = \left\{ [U] + \frac{1}{2} \cdot [Z_{abc}] \cdot [Y_{abc}] \right\} \cdot [VLG_{abc}]_m + [Z_{abc}] \cdot [I_{abc}]_m \quad (6.6)$$

where

$$[U] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (6.7)$$

Equation 6.6 is of the general form:

$$[VLG_{abc}]_n = [a] \cdot [VLG_{abc}]_m + [b] \cdot [I_{abc}]_m \quad (6.8)$$

where
$$[a] = [U] + \frac{1}{2} \cdot [Z_{abc}] \cdot [Y_{abc}] \quad (6.9)$$

$$[b] = [Z_{abc}] \quad (6.10)$$

The input current to the line segment at *Node n* is

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}_n = \begin{bmatrix} Iline_a \\ Iline_b \\ Iline_c \end{bmatrix} + \frac{1}{2} \cdot \begin{bmatrix} Y_{aa} & Y_{ab} & Y_{ac} \\ Y_{ba} & Y_{bb} & Y_{bc} \\ Y_{ca} & Y_{cb} & Y_{cc} \end{bmatrix} \cdot \begin{bmatrix} Vag \\ Vbg \\ Vcg \end{bmatrix}_n \quad (6.11)$$

In condensed form Equation 6.11 becomes:

$$[I_{abc}]_n = [Iline_{abc}]_m + \frac{1}{2} \cdot [Y_{abc}] \cdot [VLG_{abc}]_n \quad (6.12)$$

Substitute Equation 6.2 into Equation 6.12:

$$[I_{abc}]_n = [I_{abc}]_m + \frac{1}{2} [Y_{abc}] \cdot [VLG_{abc}]_m + \frac{1}{2} \cdot [Y_{abc}] \cdot [VLG_{abc}]_n \quad (6.13)$$

Substitute Equation 6.6 into Equation 6.13:

$$\begin{aligned} [I_{abc}]_n &= [I_{abc}]_m + \frac{1}{2} [Y_{abc}] \cdot [VLG_{abc}]_m + \frac{1}{2} \cdot [Y_{abc}] \\ &\times \left(\left\{ [U] + \frac{1}{2} \cdot [Z_{abc}] \cdot [Y_{abc}] \right\} \cdot [VLG_{abc}]_m + [Z_{abc}] \cdot [I_{abc}]_m \right) \end{aligned} \quad (6.14)$$

Collecting terms in Equation 6.14:

$$\begin{aligned} [I_{abc}]_n &= \left\{ [Y_{abc}] + \frac{1}{4} \cdot [Y_{abc}] \cdot [Z_{abc}] \cdot [Y_{abc}] \right\} \cdot [VLG_{abc}]_m \\ &+ \left\{ [U] + \frac{1}{2} \cdot [Y_{abc}] \cdot [Z_{abc}] \right\} [I_{abc}]_m \end{aligned} \quad (6.15)$$

Equation 6.15 is of the form:

$$[I_{abc}]_n = [c] \cdot [VLG_{abc}]_m + [d] \cdot [I_{abc}]_m \quad (6.16)$$

where

$$[c] = [Y_{abc}] + \frac{1}{4} \cdot [Y_{abc}] \cdot [Z_{abc}] \cdot [Y_{abc}] \quad (6.17)$$

$$[d] = [U] + \frac{1}{2} \cdot [Z_{abc}] \cdot [Y_{abc}] \quad (6.18)$$

Equations 6.8 and 6.16 can be put into partitioned matrix form:

$$\begin{bmatrix} [VLG_{abc}]_n \\ [I_{abc}]_n \end{bmatrix} = \begin{bmatrix} [a] & [b] \\ [c] & [d] \end{bmatrix} \cdot \begin{bmatrix} [VLG_{abc}]_m \\ [I_{abc}]_m \end{bmatrix} \quad (6.19)$$

Equation 6.19 is very similar to the equation used in transmission line analysis when the ABCD parameters have been defined.¹ In the case here, the abcd parameters are 3×3 matrices rather than single variables and will be referred to as the “generalized line matrices.”

Equation 6.19 can be turned around to solve for the voltages and currents at *Node m* in terms of the voltages and currents at *Node n*:

$$\begin{bmatrix} [VLG_{abc}]_m \\ [I_{abc}]_m \end{bmatrix} = \begin{bmatrix} [a] & [b] \\ [c] & [d] \end{bmatrix}^{-1} \cdot \begin{bmatrix} [VLG_{abc}]_n \\ [I_{abc}]_n \end{bmatrix} \quad (6.20)$$

The inverse of the abcd matrix is simple because the determinant is

$$[a] \cdot [d] - [b] \cdot [c] = [U] \quad (6.21)$$

Using the relationship of Equation 6.21, Equation 6.20 becomes:

$$\begin{bmatrix} [VLG_{abc}]_m \\ [I_{abc}]_m \end{bmatrix} = \begin{bmatrix} [d] & -[b] \\ -[c] & [a] \end{bmatrix} \cdot \begin{bmatrix} [VLG_{abc}]_n \\ [I_{abc}]_n \end{bmatrix} \quad (6.22)$$

Since the matrix $[a]$ is equal to the matrix $[d]$, Equation 6.22 in expanded form becomes:

$$[VLG_{abc}]_m = [a] \cdot [VLG_{abc}]_n - [b] \cdot [I_{abc}]_n \quad (6.23)$$

$$[I_{abc}]_m = -[c] \cdot [VLG_{abc}]_n + [d] \cdot [I_{abc}]_n \quad (6.24)$$

Sometimes it is necessary to compute the voltages at *Node m* as a function of the voltages at *Node n* and the currents entering *Node m*. This is true in the iterative technique that is developed in Chapter 10.

Solving Equation 6.8 for the *Bus m* voltages gives:

$$[VLG_{abc}]_m = [a]^{-1} \cdot \{[VLG_{abc}]_n - [b] \cdot [I_{abc}]_m\} \quad (6.25)$$

Equation 6.25 is of the form:

$$[VLG_{abc}]_m = [A] \cdot [VLG_{abc}]_n - [B] \cdot [I_{abc}]_m \quad (6.26)$$

where

$$[A] = [a]^{-1} \quad (6.27)$$

$$[B] = [a]^{-1} \cdot [b] \quad (6.28)$$

Because the mutual coupling between phases on the line segments are not equal, there will be different values of voltage drop on each of the three phases. As a result, the voltages on a distribution feeder become unbalanced even when the loads are balanced. A common method of describing the degree of unbalance is to use the National Electrical Manufacturers Association (NEMA) definition of voltage unbalance as given in Equation 6.29²:

$$V_{\text{unbalance}} = \frac{|\text{Maximum deviation from average}|}{|V_{\text{average}}|} \cdot 100\% \quad (6.29)$$

Example 6.1

A balanced three-phase load of 6000 kVA, 12.47 kV, 0.9 lagging power factor is being served at *Node m* of a 10,000-ft. three-phase line segment. The configuration and conductors of the line segment are those of Example 4.1. Determine the generalized line constant matrices $[a]$, $[b]$, $[c]$, and $[d]$. Using the generalized matrices, determine the line-to-ground voltages and line currents at the source end (*Node n*) of the line segment.

SOLUTION

The phase impedance matrix and the shunt admittance matrix for the line segment as computed in Examples 4.1 and 5.1 are

$$[z_{abc}] = \begin{bmatrix} 0.4576 + j1.0780 & 0.1560 + j.5017 & 0.1535 + j0.3849 \\ 0.1560 + j0.5017 & 0.4666 + j1.0482 & 0.1580 + j0.4236 \\ 0.1535 + j0.3849 & 0.1580 + j0.4236 & 0.4615 + j1.0651 \end{bmatrix} \Omega/\text{mile}$$

$$[y_{abc}] = j \cdot 376.9911 \cdot [C_{abc}] = \begin{bmatrix} j5.6712 & -j1.8362 & -j0.7034 \\ -j1.8362 & j5.9774 & -j1.169 \\ -j0.7034 & -j1.169 & j5.3911 \end{bmatrix} \mu\text{S}/\text{mile}$$

For the 10,000-ft. line segment, the total phase impedance matrix and shunt admittance matrix are

$$[Z_{abc}] = \begin{bmatrix} 0.8667 + j2.0417 & 0.2955 + j0.9502 & 0.2907 + j0.7290 \\ 0.2955 + j0.9502 & 0.8837 + j1.9852 & 0.2992 + j0.8023 \\ 0.2907 + j0.7290 & 0.2992 + j0.8023 & 0.8741 + j2.0172 \end{bmatrix} \Omega$$

$$[Y_{abc}] = \begin{bmatrix} j10.7409 & -j3.4777 & -j1.3322 \\ -j3.4777 & j11.3208 & -j2.2140 \\ -j1.3322 & -j2.2140 & j10.2104 \end{bmatrix} \mu\text{S}$$

It should be noted that the elements of the phase admittance matrix are very small. The generalized matrices computed according to Equations 6.9, 6.10, 6.17, and 6.18 are

$$[a] = [U] + \frac{1}{2} \cdot [Z_{abc}] \cdot [Y_{abc}] = \begin{bmatrix} 1.0 & 0 & 0 \\ 0 & 1.0 & 0 \\ 0 & 0 & 1.0 \end{bmatrix}$$

$$[b] = [Z_{abc}] = \begin{bmatrix} 0.8667 + j2.0417 & 0.2955 + j0.9502 & 0.2907 + j0.7290 \\ 0.2955 + j0.9502 & 0.8837 + j1.9852 & 0.2992 + j0.8023 \\ 0.2907 + j0.7290 & 0.2992 + j0.8023 & 0.8741 + j2.0172 \end{bmatrix}$$

$$[c] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[d] = \begin{bmatrix} 1.0 & 0 & 0 \\ 0 & 1.0 & 0 \\ 0 & 0 & 1.0 \end{bmatrix}$$

Because the elements of the phase admittance matrix are so small, the [a] and [d] matrices appear to be the unity matrix. If more significant figures are displayed, the (1,1) element of these matrices is

$$a_{1,1} = 0.99999117 + j0.00000395$$

Also, the elements of the [c] matrix appear to be zero. Again, if more significant figures are displayed, the 1,1 term is

$$c_{1,1} = -0.0000044134 + j0.0000127144$$

The point here is that for all practical purposes, the phase admittance matrix can be neglected.

The magnitude of the line-to-ground voltage at the load is

$$VLG = \frac{12470}{\sqrt{3}} = 7199.56$$

Selecting the phase-a-to-ground voltage as reference, the line-to-ground voltage matrix at the load is

$$\begin{bmatrix} V_{ag} \\ V_{bg} \\ V_{cg} \end{bmatrix}_m = \begin{bmatrix} 7199.56 / 0 \\ 7199.56 / -120 \\ 7199.56 / 120 \end{bmatrix} \text{ V}$$

The magnitude of the load currents is

$$|I|_m = \frac{6000}{\sqrt{3} \cdot 12.47} = 277.79$$

For a 0.9 lagging power factor, the load current matrix is

$$[I_{abc}]_m = \begin{bmatrix} 277.79 / -25.84 \\ 277.79 / -145.84 \\ 277.79 / 94.16 \end{bmatrix} \text{ A}$$

The line-to-ground voltages at *Node n* are computed to be

$$[VLG_{abc}]_n = [a] \cdot [VLG_{abc}]_m + [b] \cdot [I_{abc}]_m = \begin{bmatrix} 7538.70 / 1.57 \\ 7451.25 / -118.30 \\ 7485.11 / 121.93 \end{bmatrix} \text{ V}$$

It is important to note that the voltages at *Node n* are unbalanced even though the voltages and currents at the load (*Node m*) are perfectly balanced. This is a result of the unequal mutual coupling between phases. The degree of voltage unbalance is of concern since, for example, the operating characteristics of a three-phase induction motor are very sensitive to voltage unbalance. Using the NEMA definition for voltage unbalance (Equation 6.29), the

voltage unbalance is

$$|V_{\text{average}}| = \frac{|V_{ag}|_n + |V_{bg}|_n + |V_{cg}|_n}{3} = \frac{7538.70 + 7451.25 + 7485.11}{3} = 7491.69$$

$$V_{\text{deviation}}_{\text{max}} = 7538.70 - 7491.69 = 47.01$$

$$V_{\text{unbalance}} = \frac{47.01}{7491.70} \cdot 100\% = 0.6275\%$$

Although this may not seem like a large unbalance, it does give an indication of how the unequal mutual coupling can generate an unbalance. It is important to know that NEMA standards require that induction motors be de-rated when the voltage unbalance exceeds 1.0%.

Selecting rated line-to-ground voltage as base (7199.56), the per-unit voltages at *Bus n* are

$$\begin{bmatrix} V_{ag} \\ V_{bg} \\ V_{cg} \end{bmatrix}_n = \frac{1}{7199.56} \begin{bmatrix} 7538.70 / 1.577 \\ 7451.25 / -118.30 \\ 7485.11 / 121.93 \end{bmatrix} = \begin{bmatrix} 1.0471 / 1.57 \\ 1.0350 / -118.30 \\ 1.0397 / 121.93 \end{bmatrix} \text{ per-unit}$$

By converting the voltages to per-unit it is easy to see that the voltage drop by phase is 4.71% for *Phase a*, 3.50% for *Phase b*, and 3.97% for *Phase c*.

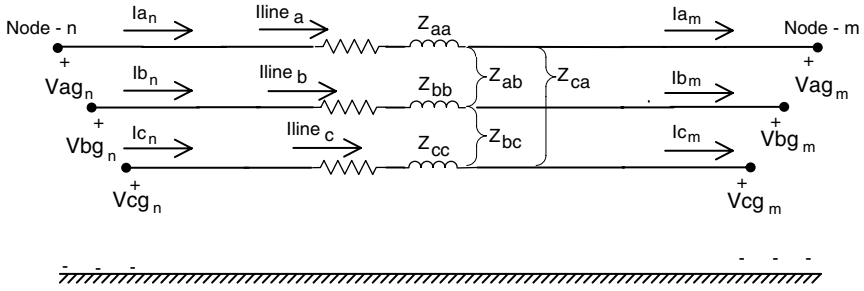
The line currents at *Node n* are computed to be

$$[I_{abc}]_n = [c] \cdot [VLG_{abc}]_m + [d] \cdot [I_{abc}]_m = \begin{bmatrix} 277.71 / -25.83 \\ 277.73 / -148.82 \\ 277.73 / 94.17 \end{bmatrix} \text{ A}$$

Comparing the computed line currents at *Node n* to the balanced load currents at *Node m*, a very slight difference is noted that is another result of the unbalanced voltages at *Node n* and the shunt admittance of the line segment.

6.2 The Modified Line Model

It was demonstrated in Example 6.1 that the shunt admittance of a line is so small that it can be neglected. Figure 6.2 shows the modified line segment model with the shunt admittance neglected.

**FIGURE 6.2**

Modified line segment model.

When the shunt admittance is neglected, the generalized matrices become:

$$[a] = [U] \quad (6.30)$$

$$[b] = [Z_{abc}] \quad (6.31)$$

$$[c] = [0] \quad (6.32)$$

$$[d] = [U] \quad (6.33)$$

$$[A] = [U] \quad (6.34)$$

$$[B] = [Z_{abc}] \quad (6.35)$$

If the line is a three-wire delta, the voltage drops down the line must be in terms of the line-to-line voltages and line currents. However, it is possible to use “equivalent” line-to-neutral voltages so that the equations derived to this point will still apply. Writing the voltage drop equations in terms of line-to-line voltages for the line in Figure 6.2 results in:

$$\begin{bmatrix} V_{ab} \\ V_{bc} \\ V_{ca} \end{bmatrix}_n = \begin{bmatrix} V_{ab} \\ V_{bc} \\ V_{ca} \end{bmatrix}_m + \begin{bmatrix} vdrop_a \\ vdrop_b \\ vdrop_c \end{bmatrix} - \begin{bmatrix} vdrop_b \\ vdrop_c \\ vdrop_a \end{bmatrix} \quad (6.36)$$

where

$$\begin{bmatrix} vdrop_a \\ vdrop_b \\ vdrop_c \end{bmatrix} = \begin{bmatrix} Z_{aa} & Z_{ab} & Z_{ac} \\ Z_{ba} & Z_{bb} & Z_{bc} \\ Z_{ca} & Z_{cb} & Z_{cc} \end{bmatrix} \cdot \begin{bmatrix} Iline_a \\ Iline_b \\ Iline_c \end{bmatrix} \quad (6.37)$$

Expanding Equation 6.36 for *Phase a-b*:

$$Vab_n = Vab_m + vdrop_a - vdrop_b \quad (6.38)$$

but

$$\begin{aligned} Vab_n &= Van_n - Vbn_n \\ Vab_m &= Van_m - Vbn_m \end{aligned} \quad (6.39)$$

Substitute Equations 6.39 into Equation 6.38:

$$Van_n - Vbn_n = Van_m - Vbn_m + vdrop_a - vdrop_b \quad (6.40)$$

Equation 6.40 can be broken into two parts in terms of equivalent line-to-neutral voltages:

$$\begin{aligned} Van_n &= Van_m + vdrop_a \\ Vbn_n &= Vbn_m + vdrop_b \end{aligned} \quad (6.41)$$

The conclusion here is that it is possible to work with equivalent line-to-neutral voltages in a three-wire delta line. This is very important since it makes the development of general analyses techniques the same for four-wire wye and three-wire delta systems.

Example 6.2

The line of Example 6.1 will be used to supply an unbalanced load at *Node m*. Assume that the voltages at the source end (*Node n*) are balanced three-phase at 12.47 kV line-to-line. The balanced line-to-ground voltages are

$$[VLG_{abc}]_n = \begin{bmatrix} 7199.56 / 0 \\ 7199.56 / -120 \\ 7199.56 / 120 \end{bmatrix} \text{ V}$$

The unbalanced currents measured at the source end are given by:

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}_n = \begin{bmatrix} 249.97 / -24.5 \\ 277.56 / -145.8 \\ 305.54 / 95.2 \end{bmatrix} \text{ A}$$

Determine the line-to-ground and line-to-line voltages at the load end (*Node m*) using the modified line model. Determine also the voltage unbalance and the complex powers of the load.

SOLUTION

The $[A]$ and $[B]$ matrices for the modified line model are

$$[A] = [U] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[B] = [Z_{abc}] = \begin{bmatrix} 0.8666 + j2.0417 & 0.2954 + j0.9501 & 0.2907 + j0.7290 \\ 0.2954 + j0.9501 & 0.8838 + j1.9852 & 0.2993 + j0.8024 \\ 0.2907 + j0.7290 & 0.2993 + j0.8024 & 0.8740 + j2.0172 \end{bmatrix} \Omega$$

Since this is the approximate model, $[I_{abc}]_m$ is equal to $[I_{abc}]_n$. Therefore:

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}_m = \begin{bmatrix} 249.97 / -24.5 \\ 277.56 / -145.8 \\ 305.54 / 95.2 \end{bmatrix} \text{ A}$$

The line-to-ground voltages at the load end are

$$[VLG_{abc}]_m = [A] \cdot [VLG_{abc}]_n - [B] \cdot [I_{abc}]_m = \begin{bmatrix} 6942.53 / -1.47 \\ 6918.35 / -121.55 \\ 6887.71 / 117.31 \end{bmatrix} \text{ V}$$

For this condition the average load voltage is

$$|V_{\text{average}}| = \frac{6942.53 + 6918.35 + 6887.71}{3} = 6916.20 \text{ V}$$

The maximum deviation from the average is on *Phase c* so that:

$$V_{\text{deviation}}_{\text{max}} = |6887.71 - 6916.20| = 28.49$$

$$V_{\text{unbalance}} = \frac{28.49}{6916.20} \cdot 100 = 0.4119\%$$

The line-to-line voltages at the load can be computed by:

$$\begin{bmatrix} V_{ab} \\ V_{bc} \\ V_{ca} \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 6942.53 / -1.47 \\ 6918.35 / -121.55 \\ 6887.71 / 117.31 \end{bmatrix} = \begin{bmatrix} 12008.43 / 28.43 \\ 12024.62 / -92.19 \\ 11903.23 / 148.05 \end{bmatrix} \text{ V}$$

The complex powers of the load are

$$\begin{bmatrix} S_a \\ S_b \\ S_c \end{bmatrix} = \frac{1}{1000} \cdot \begin{bmatrix} V_{ag} \cdot I_a^* \\ V_{bg} \cdot I_b^* \\ V_{cg} \cdot I_c^* \end{bmatrix} = \begin{bmatrix} 1735.42 / \underline{23.03} \\ 1920.26 / \underline{24.25} \\ 2104.47 / \underline{22.11} \end{bmatrix} \text{ kVA}$$

6.3 The Approximate Line Segment Model

Many times the only data available for a line segment will be the positive and zero sequence impedances. The approximate line model can be developed by applying the reverse impedance transformation from symmetrical component theory.

Using the known positive and zero sequence impedances, the sequence impedance matrix is given by:

$$[Z_{\text{seq}}] = \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_+ & 0 \\ 0 & 0 & Z_+ \end{bmatrix} \quad (6.42)$$

The reverse impedance transformation results in the following approximate phase impedance matrix:

$$[Z_{\text{approx}}] = [A_s] \cdot [Z_{\text{seq}}] \cdot [A_s]^{-1} \quad (6.43)$$

$$[Z_{\text{approx}}] = \frac{1}{3} \cdot \begin{bmatrix} (2 \cdot Z_+ + Z_0) & (Z_0 - Z_+) & (Z_0 - Z_+) \\ (Z_0 - Z_+) & (2 \cdot Z_+ + Z_0) & (Z_0 - Z_+) \\ (Z_0 - Z_+) & (Z_0 - Z_+) & (2 \cdot Z_+ + Z_0) \end{bmatrix} \quad (6.44)$$

Notice that the approximate impedance matrix is characterized by the three diagonal terms being equal and all mutual terms being equal. This is the same result that is achieved if the line is assumed to be transposed. Applying the approximate impedance matrix, the voltage at *Node n* is computed to be

$$\begin{bmatrix} V_{ag} \\ V_{bg} \\ V_{cg} \end{bmatrix}_n = \begin{bmatrix} V_{ag} \\ V_{bg} \\ V_{cg} \end{bmatrix}_m + \frac{1}{3} \cdot \begin{bmatrix} (2 \cdot Z_+ + Z_0) & (Z_0 - Z_+) & (Z_0 - Z_+) \\ (Z_0 - Z_+) & (2 \cdot Z_+ + Z_0) & (Z_0 - Z_+) \\ (Z_0 - Z_+) & (Z_0 - Z_+) & (2 \cdot Z_+ + Z_0) \end{bmatrix} \cdot \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}_m \quad (6.45)$$

In condensed form, Equation 6.45 becomes:

$$[VLG_{abc}]_n = [VLG_{abc}]_m + [Z_{\text{approx}}] \cdot [I_{abc}]_m \quad (6.46)$$

Note that Equation 6.46 is of the form:

$$[VLG_{abc}]_n = [a][VLG_{abc}]_m + [b] \cdot [I_{abc}]_m \quad (6.47)$$

where

$$[a] = \text{unity matrix}$$

$$[b] = [Z_{\text{approx}}]$$

Equation 6.45 can be expanded and an equivalent circuit for the approximate line segment model can be developed. Solving Equation 6.45 for the *Phase a* voltage at *Node n* results in:

$$Vag_n = Vag_m + \frac{1}{3} \{ (2Z_+ + Z_0)I_a + (Z_0 - Z_+)I_b + (Z_0 + Z_+)I_c \} \quad (6.48)$$

Modify Equation 6.48 by adding and subtracting the term $(Z_0 - Z_+)I_a$ and then combining terms and simplifying:

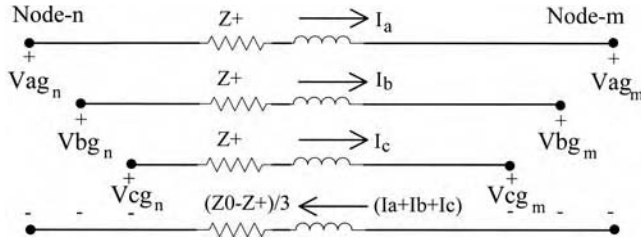
$$\begin{aligned} Vag_n &= Vag_m + \frac{1}{3} \left\{ (2Z_+ + Z_0)I_a + (Z_0 - Z_+)I_b + (Z_0 - Z_+)I_c \right. \\ &\quad \left. + (Z_0 - Z_+)I_a - (Z_0 - Z_+)I_a \right\} \\ Vag_n &= Vag_m + \frac{1}{3} \{ (3Z_+)I_a + (Z_0 - Z_+)(I_a + I_b + I_c) \} \\ Vag_n &= Vag_m + Z_+ \cdot I_a + \frac{(Z_0 - Z_+)}{3} \cdot (I_a + I_b + I_c) \end{aligned} \quad (6.49)$$

The same process can be followed in expanding Equation 6.45 for *Phases b* and *c*. The final results are

$$Vbg_n = Vbg_m + Z_+ \cdot I_b + \frac{(Z_0 - Z_+)}{3} \cdot (I_a + I_b + I_c) \quad (6.50)$$

$$Vcg_n = Vcg_m + Z_+ \cdot I_c + \frac{(Z_0 - Z_+)}{3} \cdot (I_a + I_b + I_c) \quad (6.51)$$

Figure 6.3 illustrates the approximate line segment model. Figure 6.3 is a simple equivalent circuit for the line segment since no mutual coupling has

**FIGURE 6.3**

Approximate line segment model.

to be modeled. It must be understood, however, that the equivalent circuit can only be used when transposition of the line segment has been assumed.

Example 6.3

The line segment of Example 4.1 is to be analyzed assuming that the line has been transposed. In Example 4.1 the positive and zero sequence impedances were computed to be

$$\begin{aligned} z_+ &= 0.3061 + j0.6270 \\ z_0 &= 0.7735 + j1.9373 \end{aligned} \quad \Omega/\text{mile}$$

Assume that the load at *Node m* is the same as in Example 6.1, that is

$$\text{kVA} = 6000, \quad \text{kVLL} = 12.47, \quad \text{Power factor} = 0.8 \text{ lagging}$$

Determine the voltages and currents at the source end (*Node n*) for this loading condition.

SOLUTION

The sequence impedance matrix is

$$[z_{\text{seq}}] = \begin{bmatrix} 0.7735 + j1.9373 & 0 & 0 \\ 0 & 0.3061 + j0.6270 & 0 \\ 0 & 0 & 0.3061 + j0.6270 \end{bmatrix} \quad \Omega/\text{mile}$$

Performing the reverse impedance transformation results in the approximate phase impedance matrix:

$$\begin{aligned} [z_{\text{approx}}] &= [A_s] \cdot [z_{\text{seq}}] \cdot [A_s]^{-1} \\ [Z_{\text{approx}}] &= \begin{bmatrix} 0.4619 + j1.0638 & 0.1558 + j0.4368 & 0.1558 + j0.4368 \\ 0.1558 + j0.4368 & 0.4619 + j1.0638 & 0.1558 + j0.4368 \\ 0.1558 + j0.4368 & 0.1558 + j0.4368 & 0.4619 + j1.0638 \end{bmatrix} \quad \Omega/\text{mile} \end{aligned}$$

For the 10,000-ft. line, the phase impedance matrix and the $[b]$ matrix are

$$[b] = [Z_{\text{approx}}] = [z_{\text{approx}}] \cdot \frac{10,000}{5280}$$

$$[b] = \begin{bmatrix} 0.8748 + j2.0147 & 0.2951 + j0.8272 & 0.2951 + j0.8272 \\ 0.2951 + j0.8272 & 0.8748 + j2.0147 & 0.2951 + j0.8272 \\ 0.2951 + j0.8272 & 0.2951 + j0.8272 & 0.8748 + j2.0147 \end{bmatrix} \Omega$$

Note in the approximate phase impedance matrix that the three diagonal terms are equal and all of the mutual terms are equal. Again, this is an indication of the transposition assumption.

From Example 6.1 the voltages and currents at *Node m* are

$$[VLG_{abc}]_m = \begin{bmatrix} 7199.56 / 0 \\ 7199.56 / -120 \\ 7199.56 / 120 \end{bmatrix} \text{ V}$$

$$[I_{abc}]_m = \begin{bmatrix} 277.79 / -25.84 \\ 277.79 / -145.84 \\ 277.79 / 94.16 \end{bmatrix} \text{ A}$$

Using Equation 6.47:

$$[VLG_{abc}]_n = [a] \cdot [VLG_{abc}]_m + [b] \cdot [I_{abc}]_m = \begin{bmatrix} 7491.65 / -1.73 \\ 7491.65 / -118.27 \\ 7491.65 / -121.73 \end{bmatrix} \text{ V}$$

Note that the computed voltages are balanced. In Example 6.1 it was shown that when the line is modeled accurately, there is a voltage unbalance of 0.6275%. It should also be noted that the average value of the voltages at *Node n* in Example 6.1 was 7491.69 volts.

The V_{ag} at *Node n* can also be computed using Equation 6.48:

$$Vag_n = Vag_m + Z_+ \cdot I_a + \frac{(Z_0 - Z_+)}{3} \cdot (I_a + I_b + I_c)$$

Since the currents are balanced, this equation reduces to:

$$Vag_n = Vag_m + Z_+ \cdot I_a$$

$$= 7199.56 / 0 + (0.5797 + j1.1875) \cdot 277.79 / -25.84 = 7491.65 / -1.73 \text{ V}$$

It can be noted that when the loads are balanced and transposition has been assumed, the three-phase line can be analyzed as a simple single-phase equivalent, as was done in the calculation above.

Example 6.4

Use the balanced voltages and unbalanced currents at *Node n* in Example 6.2 and the approximate line model to compute the voltages and currents at *Node m*.

SOLUTION

From Example 6.2 the voltages and currents at *Node n* are given as:

$$[VLG_{abc}]_n = \begin{bmatrix} 7199.56 / 0 \\ 7199.56 / -120 \\ 7199.56 / 120 \end{bmatrix} \text{ V}$$

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}_n = \begin{bmatrix} 249.97 / -24.5 \\ 277.56 / -145.8 \\ 305.54 / 95.2 \end{bmatrix} \text{ A}$$

The [A] and [B] matrices for the approximate line model are

where

[A] = Unity matrix

[B] = [Z_{approx}]

The voltages at *Node m* are determined by:

$$[VLG_{abc}]_m = [A] \cdot [VLG_{abc}]_n - [B] \cdot [I_{abc}]_n = \begin{bmatrix} 6993.12 / -1.93 \\ 6881.16 / -121.61 \\ 6880.24 / 117.50 \end{bmatrix} \text{ V}$$

The voltage unbalance for this case is computed by:

$$V_{\text{average}} = \frac{6993.12 + 6881.16 + 6880.24}{3} = 6918.17$$

$$V_{\text{deviation}}_{\text{max}} = |6993.12 - 6918.17| = 74.94$$

$$V_{\text{unbalance}} = \frac{74.94}{6918.17} \cdot 100 = 1.0833\%$$

Note that the approximate model has led to a higher voltage unbalance than the exact model.

6.4 Summary

This chapter has developed the exact, modified, and approximate line segment models. The exact model uses no approximations; that is, the phase impedance matrix, assuming no transposition is used, as well as the shunt admittance matrix. The modified model ignores the shunt admittance. The approximate line model ignores the shunt admittance and assumes that the positive and zero sequence impedances of the line are the known parameters. This is paramount to assuming the line is transposed. For the three line models, generalized matrix equations have been developed. The equations utilize the generalized matrices $[a]$, $[b]$, $[c]$, $[d]$, $[A]$, and $[B]$. The example problems demonstrate that, because the shunt admittance is very small, the generalized matrices can be computed neglecting the shunt admittance with very little if any error. In most cases the shunt admittance can be neglected; however, there are situations where the shunt admittances should not be neglected. This is particularly true for long, rural, lightly loaded lines, and for many underground lines.

References

1. Glover, J.D. and Sarma, M., *Power System Analysis and Design*, 2nd ed., PWS-Kent Publishing, Boston, 1995.
2. ANSI/NEMA Standard Publication No. MG1-1978, National Electrical Manufacturers Association, Washington, D.C.

Problems

6.1 A two-mile-long, three-phase line uses the configuration of Problem 4.1. The phase impedance matrix and shunt admittance matrix for the configuration are

$$[z_{abc}] = \begin{bmatrix} 0.3375 + j1.0478 & 0.1560 + j0.5017 & 0.1535 + j0.3849 \\ 0.1560 + j0.5017 & 0.3465 + j1.0179 & 0.1580 + j0.4236 \\ 0.1535 + j0.3849 & 0.1580 + j0.4236 & 0.3414 + j1.0348 \end{bmatrix} \Omega/\text{mile}$$

$$[y_{abc}] = \begin{bmatrix} j5.9540 & -j2.0030 & -j0.7471 \\ -j2.0030 & j6.3962 & -j1.2641 \\ -j0.7471 & -j1.2641 & j5.6322 \end{bmatrix} \mu\text{S}/\text{mile}$$

The line is serving a balanced three-phase load of 10,000 kVA, with balanced voltages of 13.2 kV line-to-line and a power factor of 0.85 lagging.

- (1) Determine the generalized matrices.
- (2) For the given load, compute the line-to-line and line-to-neutral voltages at the source end of the line.
- (3) Compute the voltage unbalance at the source end.
- (4) Compute the source end complex power per phase.
- (5) Compute the power loss by phase over the line. (*Hint: Power loss is defined as power in minus power out.*)

6.2 The positive and zero sequence impedances for the line of Problem 6.1 are

$$z_+ = 0.186 + j0.5968 \, \Omega/\text{mile} \quad z_0 = 0.6534 + j1.907 \, \Omega/\text{mile}$$

Repeat Problem 6.1 using the approximate line model.

6.3 The line of Problem 6.1 serves an unbalanced, grounded wye, connected constant impedance load of:

$$Z_{ag} = 15/\underline{30} \, \Omega, \quad Z_{bg} = 17/\underline{36.87} \, \Omega, \quad Z_{cg} = 20/\underline{25.84} \, \Omega$$

The line is connected to a balanced three-phase 13.2-kV source.

- (1) Determine the load currents.
- (2) Determine the load line-to-ground voltages.
- (3) Determine the complex power of the load by phase.
- (4) Determine the source complex power by phase.
- (5) Determine the power loss by phase and the total three-phase power loss.

6.4 Repeat Problem 6.3, but change the impedance on *Phase b* to $50/\underline{36.87} \, \Omega$

6.5 The two-phase line of Problem 4.2 has the following phase impedance matrix:

$$[z_{abc}] = \begin{bmatrix} 0.4576 + j1.0780 & 0 & 0.1535 + j0.3849 \\ 0 & 0 & 0 \\ 0.1535 + j0.3849 & 0 & 0.4615 + j1.0651 \end{bmatrix} \, \Omega/\text{mile}$$

The line is two miles long and serves a two-phase load such that:

$$S_{ag} = 2000 \, \text{kVA at } 0.9 \text{ lagging power factor and voltage of } 7620/\underline{0} \, \text{V}$$

$$S_{cg} = 1500 \, \text{kVA at } 0.95 \text{ lagging power factor and voltage of } 7620/\underline{120} \, \text{V}$$

Neglect the shunt admittance and determine the following:

- (1) The source line-to-ground voltages using the generalized matrices.
(Hint: Even though Phase *b* is physically not present, assume that it is with a value of $7620/\underline{-120}$ V and is serving a 0 kVA load.)
- (2) The complex power by phase at the source.
- (3) The power loss by phase on the line.

6.6 The single-phase line of Problem 4.3 has the following phase impedance matrix:

$$[z_{abc}] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1.3292 + j1.3475 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Omega/\text{mile}$$

The line is one mile long and is serving a single-phase load of 2000 kVA, 0.95 lagging power factor at a voltage of $7500/\underline{-120}$ V. Determine the source voltage and power loss on the line. (Hint: As in the previous problem, even though Phases *a* and *c* are not physically present, assume they are, and, along with Phase *b*, make up a balanced three-phase set of voltages.)

6.7 The three-phase concentric neutral cable configuration of Problem 4.10 is two miles long and serves a balanced three-phase load of 10,000 kVA, 13.2 kV, 0.85 lagging power factor. The phase impedance and shunt admittance matrices for the cable line are

$$[z_{abc}] = \begin{bmatrix} 0.8040 + j0.4381 & 0.3176 + j0.0276 & 0.2824 - j0.0184 \\ 0.3176 + j0.0276 & 0.7939 + j0.3966 & 0.3176 + j0.0276 \\ 0.2824 - j0.0184 & 0.2824 - j0.0184 & 0.8040 + j0.4381 \end{bmatrix} \Omega/\text{mile}$$

$$[y_{abc}] = \begin{bmatrix} j117.52 & 0 & 0 \\ 0 & j117.52 & 0 \\ 0 & 0 & j117.52 \end{bmatrix} \mu\text{S}/\text{mile}$$

- (1) Determine the generalized matrices.
- (2) For the given load, compute the line-to-line and line-to-neutral voltages at the source end of the line.
- (3) Compute the voltage unbalance at the source end.
- (4) Compute the source end complex power per phase.
- (5) Compute the power loss by phase over the line. (Hint: Power loss is defined as power in minus power out.)

6.8 The line of Problem 6.7 serves an unbalanced grounded wye connected constant impedance load of:

$$Z_{ag} = 15/\underline{30} \text{ } \Omega, \quad Z_{bg} = 50/\underline{36.87} \text{ } \Omega, \quad Z_{cg} = 20/\underline{25.84} \text{ } \Omega$$

The line is connected to a balanced three-phase 13.2 kV source.

- (1) Determine the load currents.
- (2) Determine the load line-to-ground voltages.
- (3) Determine the complex power of the load by phase.
- (4) Determine the source complex power by phase.
- (5) Determine the power loss by phase and the total three-phase power loss.

6.9 The tape-shielded cable single-phase line of Problem 4.12 is two miles long and serves a single-phase load of 3000 kVA, at 8.0 kV and 0.9 lagging power factor. The phase impedance and shunt admittances for the line are

$$[z_{abc}] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0.5287 + j0.5717 \end{bmatrix} \Omega/\text{mile}$$

$$[y_{abc}] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & j140.39 \end{bmatrix} \mu\text{S}/\text{mile}$$

Determine the source voltage and the power loss for the loading condition.

7

Regulation of Voltages

The regulation of voltages is an important function on a distribution feeder. As the loads on the feeders vary, there must be some means of regulating the voltage so that every customer's voltage remains within an acceptable level. Common methods of regulating the voltage are the application of step-type voltage regulators, load tap changing transformers (LTC), and shunt capacitors.

7.1 Standard Voltage Ratings

The American National Standards Institute (ANSI) standard ANSI C84.1-1995 for "Electric Power Systems and Equipment Voltage Ratings (60 Hertz)" provides the following definitions for system voltage terms:¹

- *System Voltage*: the root mean square (rms) phasor voltage of a portion of an alternating-current electric system. Each system voltage pertains to a portion of the system that is bounded by transformers or utilization equipment.
- *Nominal System Voltage*: the voltage by which a portion of the system is designated, and to which certain operating characteristics of the system are related. Each nominal system voltage pertains to a portion of the system bounded by transformers or utilization equipment.
- *Maximum System Voltage*: the highest system voltage that occurs under normal operating conditions, and the highest system voltage for which equipment and other components are designed for satisfactory continuous operation without derating of any kind.
- *Service Voltage*: the voltage at the point where the electrical system of the supplier and the electrical system of the user are connected.
- *Utilization Voltage*: the voltage at the line terminals of utilization equipment.

- *Nominal Utilization Voltage*: the voltage rating of certain utilization equipment used on the system.

The ANSI standard specifies two voltage ranges. An oversimplification of the voltage ranges is

- *Range A*: Electric supply systems shall be so designated and operated such that most service voltages will be within the limits specified for Range A. The occurrence of voltages outside of these limits should be infrequent.
- *Range B*: Voltages above and below Range A. When these voltages occur, corrective measures shall be undertaken within a reasonable time to improve voltages to meet Range A.

For a normal three-wire 120/240 volt service to a user, the Range A and Range B voltages are

- Range A
 - Nominal utilization voltage = 115 V
 - Maximum utilization and service voltage = 126 V
 - Minimum service voltage = 114 V
 - Minimum utilization voltage = 110 V
- Range B
 - Nominal utilization voltage = 115 V
 - Maximum utilization and service voltage = 127 V
 - Minimum service voltage = 110 V
 - Minimum utilization voltage = 107 V

These ANSI standards give the distribution engineer a range of normal steady-state voltages (Range A) and a range of emergency steady-state voltages (Range B) that must be supplied to all users.

In addition to the acceptable voltage magnitude ranges, the ANSI standard recommends that the “electric supply systems should be designed and operated to limit the maximum voltage unbalance to 3% when measured at the electric-utility revenue meter under a no-load condition.” Voltage unbalance is defined as:

$$\text{Voltage}_{\text{unbalance}} = \frac{\text{Max. deviation from average voltage}}{\text{Average voltage}} \cdot 100\% \quad (7.1)$$

The task for the distribution engineer is to design and operate the distribution system so that under normal steady-state conditions the voltages at the

meters of all users will lie within Range A, and that the voltage unbalance will not exceed 3%.

A common device used to maintain system voltages is the step-voltage regulator. Step-voltage regulators can be single-phase or three-phase. Single-phase regulators can be connected in wye, delta, or open delta, in addition to operating as single-phase devices. The regulators and their controls allow the voltage output to vary as the load varies.

A step-voltage regulator is basically an autotransformer with a load tap changing mechanism on the series winding. The voltage change is obtained by changing the number of turns (tap changes) of the series winding of the autotransformer.

An autotransformer can be visualized as a two-winding transformer with a solid connection between a terminal of the primary side of the transformer to a terminal on the secondary. Before proceeding to the autotransformer, a review of two-winding transformer theory and the development of generalized constants will be presented.

7.2 Two-Winding Transformer Theory

The exact equivalent circuit for a two-winding transformer is shown in Figure 7.1. In Figure 7.1 the high-voltage transformer terminals are denoted by H_1 and H_2 , and the low-voltage terminals are denoted by X_1 and X_2 . The standards for these markings are such that at no load, the voltage between H_1 and H_2 will be in phase with the voltage between X_1 and X_2 . Under a steady-state load condition, the currents I_1 and I_2 will be in phase.

Without introducing a significant error, the exact equivalent circuit of Figure 7.1 is modified by referring the primary impedance (Z_1) to the secondary side as shown in Figure 7.2. Referring to Figure 7.2, the total leakage

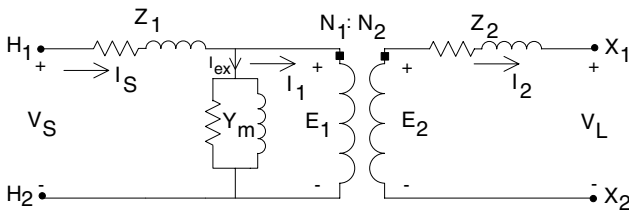
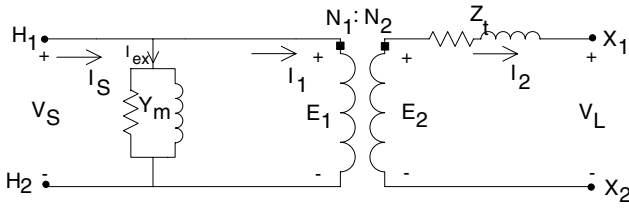


FIGURE 7.1

Two-winding transformer exact equivalent circuit.

**FIGURE 7.2**

Two-winding transformer approximate equivalent circuit.

impedance of the transformer is given by:

$$Z_t = n_t^2 \cdot Z_1 + Z_2 \quad (7.2)$$

where

$$n_t = \frac{N_2}{N_1} \quad (7.3)$$

In order to better understand the model for the step-regulator, a model for the two-winding transformer will first be developed. Referring to Figure 7.2, the equations for the ideal transformer become:

$$E_2 = \frac{N_2}{N_1} \cdot E_1 = n_t \cdot E_1 \quad (7.4)$$

$$I_1 = \frac{N_2}{N_1} \cdot I_2 = n_t \cdot I_2 \quad (7.5)$$

Applying KVL in the secondary circuit:

$$E_2 = V_L + Z_t \cdot I_2 \quad (7.6)$$

$$V_S = E_1 = \frac{1}{n_t} \cdot E_2 = \frac{1}{n_t} \cdot V_L + \frac{Z_t}{n_t} \cdot I_2$$

In general form, Equation 7.6 can be written as:

$$V_S = a \cdot V_L + b \cdot I_2 \quad (7.7)$$

where

$$a = \frac{1}{n_t} \quad (7.8)$$

$$b = \frac{Z_t}{n_t} \quad (7.9)$$

The input current to the two-winding transformer is given by:

$$I_S = Y_m \cdot V_S + I_1 \quad (7.10)$$

Substitute Equations 7.6 and 7.5 into Equation 7.10:

$$\begin{aligned} I_S &= Y_m \cdot \frac{1}{n_t} \cdot V_L + Y_m \cdot \frac{Z_t}{n_t} \cdot I_2 + n_t \cdot I_2 \\ I_S &= \frac{Y_m}{n_t} \cdot V_L + \left(\frac{Y_m \cdot Z_t}{n_t} + n_t \right) \cdot I_2 \end{aligned} \quad (7.11)$$

In general form, Equation 7.11 can be written as:

$$I_S = c \cdot V_L + d \cdot I_2 \quad (7.12)$$

where

$$c = \frac{Y_m}{n_t} \quad (7.13)$$

$$d = \frac{Y_m \cdot Z_t}{n_t} + n_t \quad (7.14)$$

Equations 7.7 and 7.12 are used to compute the input voltage and current to a two-winding transformer when the load voltage and current are known. These two equations are of the same form as Equations 6.8 and 6.16 that were derived in Chapter 6 for the three-phase line models. The only difference at this point is that only a single-phase two-winding transformer is being modeled. Later in this chapter the terms a , b , c , and d will be expanded to 3×3 matrices for all possible three-phase regulator connections.

Sometimes, particularly in an iterative process, the output voltage needs to be computed knowing the input voltage and the load current. Solving Equation 7.7 for the load voltage yields:

$$V_L = \frac{1}{a} \cdot V_S - \frac{b}{a} \cdot I_2 \quad (7.15)$$

Substituting Equations 7.8 and 7.9 into Equation 7.15 results in:

$$V_L = A \cdot V_S - B \cdot I_2 \quad (7.16)$$

where

$$A = n_t \quad (7.17)$$

$$B = Z_t \quad (7.18)$$

Again, Equation 7.16 is of the same form as Equation 6.26. Later in this chapter the expressions for A and B will be expanded to 3×3 matrices for all possible three-phase transformer connections.

Example 7.1

A single-phase transformer is rated 75 kVA, 2400-240 volts. The transformer has the following impedances and shunt admittance:

$$Z_1 = 0.612 + j1.2 \, \Omega \text{ (high-voltage winding impedance)}$$

$$Z_2 = 0.0061 + j0.0115 \, \Omega \text{ (low-voltage winding impedance)}$$

$$Y_m = 1.92 \times 10^{-4} - j8.52 \times 10^{-4} \text{ S (referred to the high-voltage winding)}$$

Determine the generalized a , b , c , d constants and the A and B constants. The transformer "turns ratio" is

$$n_t = \frac{N_2}{N_1} = \frac{V_{\text{rated } 2}}{V_{\text{rated } 1}} = \frac{240}{2400} = 0.1$$

The equivalent transformer impedance referred to the low-voltage side:

$$Z_t = Z_2 + n_t^2 \cdot Z_1 = 0.0122 + j0.0235$$

The generalized constants are

$$a = \frac{1}{n_t} = \frac{1}{0.1} = 10$$

$$b = \frac{Z_t}{0.1} = 0.1222 + j0.235$$

$$c = \frac{Y_m}{n_t} = 0.0019 - j0.0085$$

$$d = \frac{Y_m \cdot Z_t}{n_t} + n_t = 0.1002 - j0.0001$$

$$A = n_t = 0.1$$

$$B = Z_t = 0.0122 + j0.0235$$

Assume that the transformer is operated at rated load (75 kVA) and rated voltage (240 V) with a power factor of 0.9 lagging. Determine the source voltage and current using the generalized constants.

$$V_L = 240/\underline{0}$$

$$I_2 = \frac{75 \cdot 1000}{240} / \underline{-\cos^{-1}(0.9)} = 312.5/\underline{-25.84}$$

Applying the values of the a , b , c , and d parameters computed above:

$$V_S = a \cdot V_L + b \cdot I_2 = 2466.9/\underline{1.15} \text{ V}$$

$$I_S = c \cdot V_L + d \cdot I_2 = 32.67/\underline{-28.75} \text{ A}$$

Using the computed source voltage and the load current, determine the load voltage.

$$V_L = A \cdot V_S - B \cdot I_S = (0.1) \cdot (2466.9/\underline{1.15})$$

$$- (0.0122 + j0.0235) \cdot (312.5/\underline{-25.84})$$

$$V_L = 240.0/\underline{0} \text{ V}$$

For future reference, the per-unit impedance of the transformer is computed by:

$$Z_{\text{base}_2} = \frac{V_{\text{rated } 2}^2}{\text{kVA} \cdot 1000} = \frac{240^2}{75,000} = 0.768 \ \Omega$$

$$Z_{pu} = \frac{Z_t}{Z_{\text{base}_2}} = \frac{0.0122 + j0.0115}{0.768} = 0.0345/\underline{62.5} \text{ per-unit}$$

The per-unit shunt admittance is computed by:

$$Y_{\text{base}_1} = \frac{\text{kVA}}{\text{kV}_1^2 \cdot 1000} = 0.013 \text{ S}$$

$$Y_{pu} = \frac{Y_m}{Y_{\text{base}}} = \frac{1.92 \cdot 10^{-4} - j8.52 \cdot 10^{-4}}{0.013} = 0.0148 - j0.0655 \text{ per-unit}$$

Example 7.1 demonstrates that the generalized constants provide a quick method for analyzing the operating characteristics of a two-winding transformer.

7.3 The Two-Winding Autotransformer

A two-winding transformer can be connected as an autotransformer. Connecting the high-voltage terminal H_1 to the low-voltage terminal X_2 as shown in Figure 7.3 can create a “step-up” autotransformer. The source is connected to terminals H_1 and H_2 , while the load is connected between the X_1 terminal and the extension of H_2 . In Figure 7.3, V_S is the source voltage and V_L is the load voltage. The low-voltage winding of the two-winding transformer will be referred to as the “series” winding of the autotransformer, and the high-voltage winding will be referred to as the “shunt” winding.

Generalized constants similar to those of the two-winding transformer can be developed for the autotransformer. The total equivalent transformer impedance is referred to the “series” winding. The ideal transformer equations of 7.4 and 7.5 still apply.

Applying KVL in the secondary circuit:

$$E_1 + E_2 = V_L + Z_t \cdot I_2 \quad (7.19)$$

Using the ideal transformer relationship of Equation 7.5:

$$E_1 + n_t \cdot E_1 = (1 + n_t) \cdot E_1 = V_L + Z_t \cdot I_2 \quad (7.20)$$

Since the source voltage V_S is equal to E_1 , and I_2 is equal to I_L , Equation 7.20 can be modified to:

$$V_S = \frac{1}{1 + n_t} \cdot V_L + \frac{Z_t}{1 + n_t} \cdot I_L \quad (7.21)$$

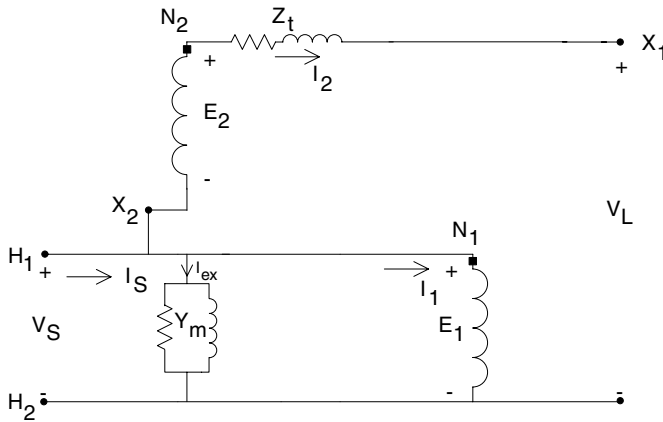


FIGURE 7.3
Step-up autotransformer.

$$V_S = a \cdot V_L + b \cdot I_L \quad (7.22)$$

where

$$a = \frac{1}{1 + n_t} \quad (7.23)$$

$$b = \frac{Z_t}{1 + n_t}. \quad (7.24)$$

Applying KCL at input node H_1 :

$$\begin{aligned} I_S &= I_1 + I_2 + I_{ex} \\ I_S &= (1 + n_t) \cdot I_2 + Y_m \cdot V_S \end{aligned} \quad (7.25)$$

Substitute Equation 7.21 into Equation 7.25:

$$\begin{aligned} I_S &= (1 + n_t) \cdot I_2 + Y_m \left(\frac{1}{1 + n_t} \cdot V_L + \frac{Z_t}{1 + n_t} \cdot I_2 \right) \\ I_S &= \frac{Y_m}{1 + n_t} \cdot V_L + \left(\frac{Y_m \cdot Z_t}{1 + n_t} + n_t + 1 \right) \cdot I_2 \\ I_S &= c \cdot V_L + d \cdot I_2 \end{aligned} \quad (7.26)$$

where

$$c = \frac{Y_m}{1 + n_t} \quad (7.27)$$

$$d = \frac{Y_m \cdot Z_t}{1 + n_t} + n_t + 1 \quad (7.28)$$

Equations 7.23, 7.24, 7.27, and 7.28 define the generalized constants relating the source voltage and current as functions of the output voltage and current for the step-up autotransformer.

The two-winding transformer can also be connected in the step-down connection by reversing the connection between the shunt and series winding as shown in Figure 7.4. Generalized constants can be developed for the step-down connection following the same procedure as that for the step-up connection.

Applying KVL in the secondary circuit:

$$E_1 - E_2 = V_L + Z_t \cdot I_2 \quad (7.29)$$

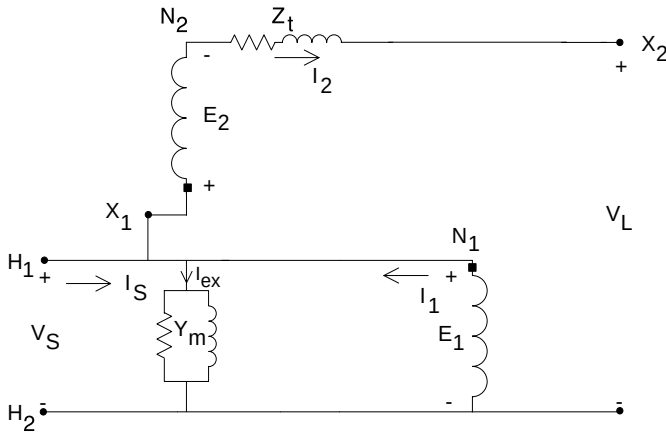


FIGURE 7.4
Step-down autotransformer.

Using the ideal transformer relationship of Equation 7.5:

$$E_1 - n_t \cdot E_2 = (1 - n_t) \cdot E_1 = V_L + Z_t \cdot I_2 \quad (7.30)$$

Since the source voltage V_S is equal to E_1 , and I_2 is equal to I_L , Equation 7.30 can be modified to:

$$V_S = \frac{1}{1 - n_t} \cdot V_L + \frac{Z_t}{1 - n_t} \cdot I_L \quad (7.31)$$

$$= a \cdot V_L + b \cdot I_L \quad (7.32)$$

where

$$a = \frac{1}{1 - n_t} \quad (7.33)$$

$$b = \frac{Z_t}{1 - n_t} \quad (7.34)$$

It is observed at this point that the only difference between the a and b constants of Equations 7.23 and 7.24 for the step-up connection, and Equations 7.33 and 7.44 for the step-down connection, is the sign in front of the turns ratio (n_t). This will also be the case for the c and d constants. Therefore, for the step-down connection, the c and d constants are defined by:

$$c = \frac{Y_m}{1 - n_t} \quad (7.35)$$

$$d = \frac{Y_m \cdot Z_t}{1 - n_t} + 1 - n_t \quad (7.36)$$

The only difference between the definitions of the generalized constants is the sign of the turns ratio n_t . In general, then, the generalized constants can be defined by:

$$a = \frac{1}{1 \pm n_t} \quad (7.37)$$

$$b = \frac{Z_t}{1 \pm n_t} \quad (7.38)$$

$$c = \frac{Y_m}{1 \pm n_t} \quad (7.39)$$

$$d = \frac{Y_m \cdot Z_t}{1 \pm n_t} 1 \pm n_t \quad (7.40)$$

In Equations 7.37 through 7.40, the sign in the equations will be positive for the step-up connection, and negative for the step-down connection.

As with the two-winding transformer, it is sometimes necessary to relate the output voltage as a function of the source voltage and the output current. Solving Equation 7.32 for the output voltage:

$$V_L = \frac{1}{a} \cdot V_S - \frac{b}{a} \cdot I_2 \quad (7.41)$$

$$V_L = A \cdot V_S - B \cdot I_2 \quad (7.42)$$

where

$$A = \frac{1}{a} = 1 \pm n_t \quad (7.43)$$

$$B = \frac{b}{a} = Z_t \quad (7.44)$$

The generalized equations for the step-up and step-down autotransformers have been developed. They are of exactly the same form as was derived for

The generalized equations for the step-up and step-down autotransformers have been developed. They are of exactly the same form as was derived for the two-winding transformer and for the line segment in Chapter 6. For the single-phase autotransformer the generalized constants are single values, but will be expanded later to 3×3 matrices for three-phase autotransformers.

7.3.1 Autotransformer Ratings

The kVA rating of the autotransformer is the product of the rated input voltage V_S times the rated input current I_S , or the rated load voltage V_L times the rated load current I_L . Define the rated kVA and rated voltages of the two-winding transformer and autotransformer as:

$$\begin{aligned} kVA_{xfm} &= \text{kVA rating of the two-winding transformer} \\ kVA_{\text{auto}} &= \text{kVA rating of the autotransformer} \\ V_{\text{rated } 1} &= E_1 = \text{rated source voltage of the two-winding transformer} \\ V_{\text{rated } 2} &= E_2 = \text{rated load voltage of the two-winding transformer} \\ V_{\text{auto } S} &= \text{rated source voltage of the autotransformer} \\ V_{\text{auto } L} &= \text{rated load voltage of the autotransformer} \end{aligned}$$

For the following derivation, neglect the voltage drop through the series winding impedance:

$$V_{\text{auto } L} = E_1 \pm E_2 = (1 \pm n_t) \cdot E_1 \quad (7.45)$$

The rated output kVA is then:

$$kVA_{\text{auto}} = V_{\text{auto } L} \cdot I_2 = (1 \pm n_t) \cdot E_1 \cdot I_2 \quad (7.46)$$

but
$$I_2 = \frac{I_1}{n_t}$$

therefore

$$kVA_{\text{auto}} = \frac{(1 \pm n_t)}{n_t} \cdot E_1 \cdot I_1 \quad (7.47)$$

but
$$E_1 \cdot I_1 = kVA_{xfm}$$

therefore

$$kVA_{\text{auto}} = \frac{(1 \pm n_t)}{n_t} \cdot kVA_{xfm} \quad (7.48)$$

Equation 7.48 gives the kVA rating of a two-winding transformer when connected as an autotransformer. For the step-up connection, the sign of n_t will be positive while the step-down will use the negative sign. In general, the turns ratio n_t will be a relatively small value, so the kVA rating of the autotransformer will be considerably greater than the kVA rating of the two-winding transformer.

Example 7.2

The two-winding transformer of Example 7.1 is connected as a step-up autotransformer. Determine the kVA and voltage ratings of the autotransformer.

From Example 7.1 the turns ratio was determined to be $n_t = 0.1$. The rated kVA of the autotransformer using Equation 7.48 is given by:

$$kVA_{\text{auto}} = \frac{1 + 0.1}{0.1} \cdot 75 = 825 \text{ kVA}$$

The voltage ratings are

$$V_{\text{auto } S} = V_{\text{rated } 1} = 2400 \text{ V}$$

$$V_{\text{auto } L} = V_{\text{rated } 1} + V_{\text{rated } 2} = 2400 + 240 = 2640 \text{ V}$$

Therefore, the autotransformer would be rated as 825 kVA, 2400-2640 V.

Suppose now that the autotransformer is supplying rated kVA at rated voltage with a power factor of 0.9 lagging. Determine the source voltage and current:

$$V_L = V_{\text{auto } L} = 2640 \text{ V}$$

$$I_2 = \frac{kVA_{\text{auto}} \cdot 1000}{V_{\text{auto } L}} = \frac{825,000}{2640} \angle -\cos^{-1}(0.9) = 312.5 \angle -25.84^\circ \text{ A}$$

Determine the generalized constants:

$$a = \frac{1}{1 + 0.1} = 0.9091$$

$$b = \frac{0.0122 + j0.0235}{1 + 0.1} = 0.0111 + j0.0214$$

$$c = \frac{(1.92 - j8.52) \cdot 10^{-4}}{1 + 0.1} = (1.7364 - j7.7455) \cdot 10^{-4}$$

$$d = \left[\frac{(1.92 - j8.52) \cdot 10^{-4} \cdot (0.0122 + j0.0235)}{1 + 0.1} \right] + 1 + 0.1 = 1.1002 - j0.000005$$

Applying the generalized constants:

$$V_s = a \cdot 2640 / 0 + b \cdot 312.5 / -25.84 = 2406.0 / 0.1 \text{ V}$$

$$I_s = c \cdot 2640 / 0 + d \cdot 312.5 / -25.84 = 345.06 / -26.11 \text{ A}$$

When the load side voltage is determined knowing the source voltage and load current, the A and B parameters are needed:

$$A = 1 + n_t = 1.1$$

$$B = Z_t = 0.0111 + j0.0235$$

The load voltage is then:

$$V_L = A \cdot 2406.04 / 0.107 - B \cdot 312.5 / -25.84 = 2640.00 / 0 \text{ V}$$

7.3.2 Per-Unit Impedance

The per-unit impedance of the autotransformer based upon the autotransformer kVA and kV ratings can be developed as a function of the per-unit impedance of the two-winding transformer based upon the two-winding transformer ratings.

Let $Z_{pu_{xfm}}$ = Per-unit impedance of the two-winding transformer based upon the two-winding kVA and kV ratings,

$V_{\text{rated } 2}$ = Rated load voltage of the two-winding transformer.

The base impedance of the two-winding transformer referred to the low-voltage winding (series winding of the autotransformer) is

$$Z_{base_{xfm}} = \frac{V_{\text{rated } 2}^2}{kVA_{xfm} \cdot 1000} \quad (7.49)$$

The actual impedance of the transformer referred to the low-voltage (series) winding is

$$Z_{t_{\text{actual}}} = Z_{t_{pu}} \cdot Z_{base_{xfm}} = Z_{t_{pu}} \cdot \frac{V_{\text{auto series}}^2}{kVA_{xfm} \cdot 1000} \quad (7.50)$$

Assume that the rated source voltage of the autotransformer is the nominal voltage of the system; that is

$$V_{\text{nominal}} = V_{\text{rated } 1} = \frac{V_{\text{rated } 2}}{n_t} \quad (7.51)$$

The base impedance for the autotransformer referenced to the nominal system voltage is

$$Zbase_{auto} = \frac{V_{nominal}^2}{kVA_{auto} \cdot 1000} \quad (7.52)$$

Substitute Equations 7.48 and 7.51 into Equation 7.52:

$$Zbase_{auto} = \frac{V_{nominal}^2}{kVA_{auto} \cdot 1000} = \frac{\left(\frac{V_{rated\ 2}}{n_t}\right)^2}{\frac{1 \pm n_t}{n_t} \cdot kVA_{xfm} \cdot 1000} \quad (7.53)$$

$$Zbase_{auto} = \frac{V_{rating\ 2}^2}{n_t \cdot (1 \pm n_t) \cdot kVA_{xfm} \cdot 1000}$$

The per-unit impedance of the autotransformer based upon the rating of the autotransformer is

$$Zauto_{pu} = \frac{Zt_{actual}}{Zbase_{auto}} \quad (7.54)$$

Substitute Equations 7.50 and 7.53 into Equation 7.54:

$$Zauto_{pu} = Zt_{pu} \cdot \frac{\frac{V_{rating\ 2}^2}{kVA_{xfm} \cdot 1000}}{\frac{V_{rating\ 2}^2}{n_t \cdot (1 \pm n_t) \cdot kVA_{xfm} \cdot 1000}} = n_t \cdot (1 \pm n_t) \cdot Zt_{pu} \quad (7.55)$$

Equation 7.55 gives the relationship between the per-unit impedance of the autotransformer and the per-unit impedance of the two-winding transformer. The point is that the per-unit impedance of the autotransformer is very small compared to that of the two-winding transformer. When the autotransformer is connected to boost the voltage 10%, the value of n_t is 0.1, and Equation 7.57 becomes:

$$Zauto_{pu} = 0.1 \cdot (1 + 0.1) \cdot Zt_{pu} = 0.11 \cdot Zt_{pu} \quad (7.56)$$

The per-unit shunt admittance of the autotransformer can be developed as a function of the per-unit shunt admittance of the two-winding transformer. Recall that the shunt admittance is represented on the source side of the two-winding transformer.

Let

$Yt_{pu} = Ym_{pu}$ = per-unit admittance of the two-winding transformer based upon the transformer ratings,

$Yauto_{pu}$ = per-unit admittance of the autotransformer based upon the autotransformer ratings.

The base admittance of the two-winding transformer referenced to the source side is given by:

$$Ybase_{source} = \frac{kVA_{xfm} \cdot 1000}{V_{rating\ 1}^2} \quad (7.57)$$

The actual shunt admittance referred to the source side of the two-winding transformer is

$$Yt_{source} = Yt_{pu} \cdot Ybase_{source} = Yt_{pu} \cdot \frac{kVA_{xfm} \cdot 1000}{V_{rating\ 1}^2} \quad (7.58)$$

The per-unit shunt admittance for the autotransformer is given by:

$$Yauto_{pu} = \frac{Yt_{source}}{Ybase_{auto}} = Yt_{source} \cdot \frac{V_{rated\ 1}^2}{kVA_{auto} \cdot 1000} \quad (7.59)$$

Substitute Equation 7.58 into 7.59:

$$Yauto_{pu} = Yt_{pu} \cdot \frac{kVA_{xfm} \cdot 1000}{V_{rated\ 1}^2} \cdot \frac{V_{rated\ 1}^2}{kVA_{auto} \cdot 1000} \quad (7.60)$$

$$Yauto_{pu} = Yt_{pu} \cdot \frac{kVA_{xfm}}{kVA_{auto}} = Yt_{pu} \cdot \frac{kVA_{xfm}}{\frac{(1 \pm n_t)}{n_t} \cdot kVA_{xfm}} = \frac{n_t}{(1 \pm n_t)} \cdot Yt_{pu}$$

Equation 7.60 shows that the per-unit admittance based upon the autotransformer ratings is much smaller than the per-unit admittance of the two-winding transformer. For an autotransformer in the raise connection with $n_t = 0.1$, Equation 7.60 becomes:

$$Ya_{pu} = \frac{0.1}{1 + 0.1} \cdot Yt_{pu} = 0.0909 \cdot Yt_{pu}$$

It has been shown that the per-unit impedance and admittance values based upon the autotransformer kVA rating and nominal voltage are approximately one-tenth that of the values for the two-winding transformer.

Example 7.3

The shunt admittance referred to the source side of the two-winding transformer of Example 7.2 is

$$Y_t = Y_m = 1.92 \cdot 10^{-4} - j8.52 \cdot 10^{-4} \text{ S}$$

1. Determine the per-unit shunt admittance based upon the two-winding transformer ratings:

$$Y_{base_{\text{source}}} = \frac{75 \cdot 1000}{2400^2} = 0.013$$

$$Y_{t_{pu}} = \frac{1.92 \cdot 10^{-4} - j8.52 \cdot 10^{-4}}{0.013} = 0.014746 - j0.065434$$

2. In Example 7.2 the kVA rating of the two-winding transformer connected as an autotransformer was computed to be 825 kVA, and the voltage ratings 2400–2640 V. Determine the per-unit admittance based upon the autotransformer kVA rating and a nominal voltage of 2400 V, and determine the ratio of the per-unit admittance of the autotransformer to the per-unit admittance of the two-winding transformer:

$$Y_{base_{\text{auto}}} = \frac{825 \cdot 1000}{2400^2} = 0.1432$$

$$Y_{auto_{pu}} = \frac{1.92 \cdot 10^{-4} - j8.52 \cdot 10^{-4}}{0.1432} = 0.001341 - j0.005949$$

$$\text{Ratio} = \frac{0.001341 - j0.005949}{0.014746 - j0.065434} = 0.0909$$

In this section the equivalent circuit of an autotransformer has been developed for the “raise” and “lower” connections. These equivalent circuits included the series impedance and shunt admittance. If a detailed analysis of the autotransformer is desired, the series impedance and shunt admittance should be included. However, it has been shown that these values are very small, and when the autotransformer is to be a component of a system, very little error will be made by neglecting both the series impedance and shunt admittance of the equivalent circuit.

7.4 Step-Voltage Regulators

A step-voltage regulator consists of an autotransformer and a load tap changing mechanism. The voltage change is obtained by changing the taps of the series winding of the autotransformer. The position of the tap is determined by a control circuit (line drop compensator). Standard step regulators contain a reversing switch enabling a $\pm 10\%$ regulator range, usually in 32 steps. This amounts to a $5/8\%$ change per step, or 0.75-V change per step, on a 120-V base. Step regulators can be connected in a Type A or Type B connection according to the ANSI/IEEE C57.15-1986 standard.² The more common Type B connection is shown in Figure 7.5. The block diagram circuit shown in Figure 7.6 controls tap changing on a step-voltage regulator. The step-voltage regulator control circuit requires the following settings:

1. *Voltage Level*: the desired voltage (on a 120-V base) to be held at the load center. The load center may be the output terminal of the regulator or a remote node on the feeder.
2. *Bandwidth*: the allowed variance of the load center voltage from the set voltage level. The voltage held at the load center will be $\pm$ one-half the bandwidth. For example, if the voltage level is set to 122 V

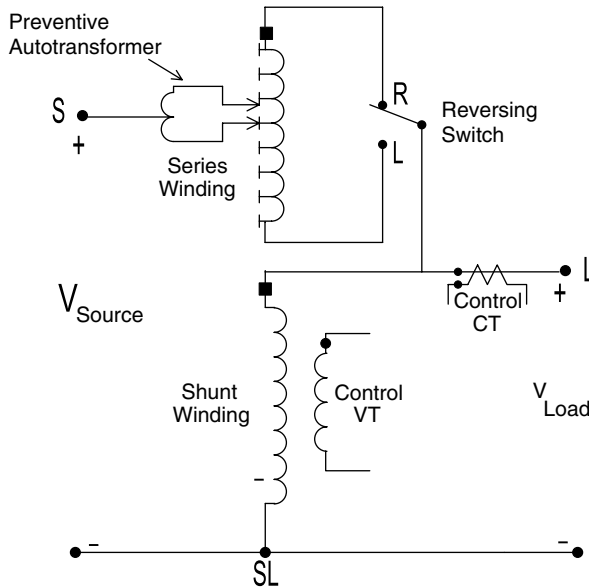


FIGURE 7.5

Type B step-voltage regulator.

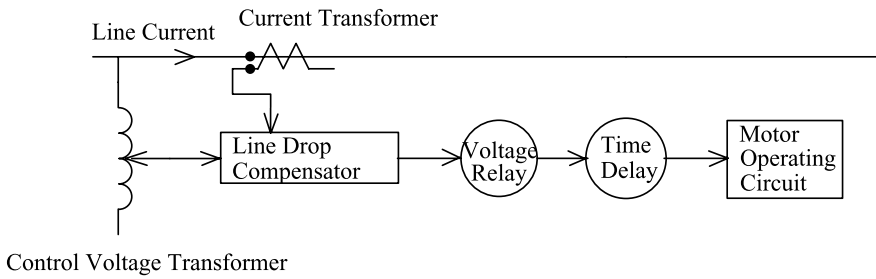


FIGURE 7.6
Step-voltage regulator control circuit.

and the bandwidth is set to 2 V, the regulator will change taps until the load center voltage lies between 121 and 123 V.

3. *Time Delay*: length of time that a raise or lower operation is called for before the actual execution of the command. This prevents taps changing during a transient or short time change in current.
4. *Line Drop Compensator*: set to compensate for the voltage drop (line drop) between the regulator and the load center. The settings consist of R and X settings in volts corresponding to the *equivalent* impedance between the regulator and the load center. This setting may be zero if the regulator output terminals are the load center.

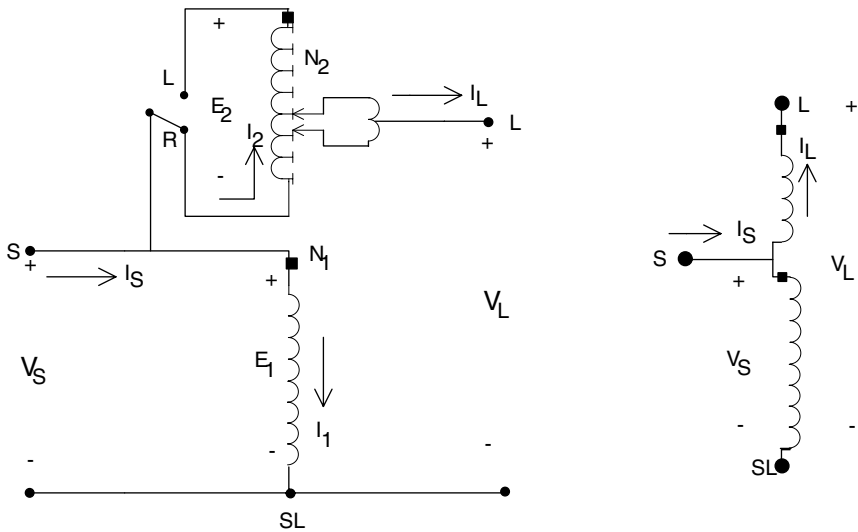
The required rating of a step-voltage regulator is based upon the kVA transformed, not the kVA rating of the line. In general, this will be 10% of the line rating since rated current flows through the series winding which represents the $\pm 10\%$ voltage change. The kVA rating of the step-voltage regulator is determined in the same manner as that of the previously discussed autotransformer.

7.4.1 Single-Phase Step-Voltage Regulators

Because the series impedance and shunt admittance values of step-voltage regulators are so small, they will be neglected in the following equivalent circuits. It should be pointed out, however, that if it is desired to include the impedance and admittance, they can be incorporated into the following equivalent circuits in the same way they were originally modeled in the autotransformer equivalent circuit.

7.4.1.1 Type A Step-Voltage Regulator

The detailed equivalent circuit and abbreviated equivalent circuit of a Type A step-voltage regulator in the raise position is shown in Figure 7.7.

**FIGURE 7.7**

Type A step-voltage regulator in the raise position.

As shown in Figure 7.7, the primary circuit of the system is connected directly to the shunt winding of the Type A regulator. The series winding is connected to the shunt winding and, in turn, via taps, to the regulated circuit. In this connection the core excitation varies because the shunt winding is connected directly across the primary circuit.

When the Type A connection is in the lower position, the reversing switch is connected to the L terminal. The effect of this reversal is to reverse the direction of the currents in the series and shunt windings. Figure 7.8 shows the equivalent circuit and abbreviated circuit of the Type A regulator in the lower position.

7.4.1.2 Type B Step-Voltage Regulator

The more common connection for step-voltage regulators is the Type B. Since this is the more common connection, the defining voltage and current equations for the voltage regulator will be developed only for the Type B connection.

The detailed and abbreviated equivalent circuits of a Type B step-voltage regulator in the raise position is shown in Figure 7.9. The primary circuit of the system is connected, via taps, to the series winding of the regulator in the Type B connection. The series winding is connected to the shunt winding, which is connected directly to the regulated circuit. In a Type B regulator the core excitation is constant because the shunt winding is connected across the regulated circuit.

The defining voltage and current equations for the regulator in the raise position are as follows:

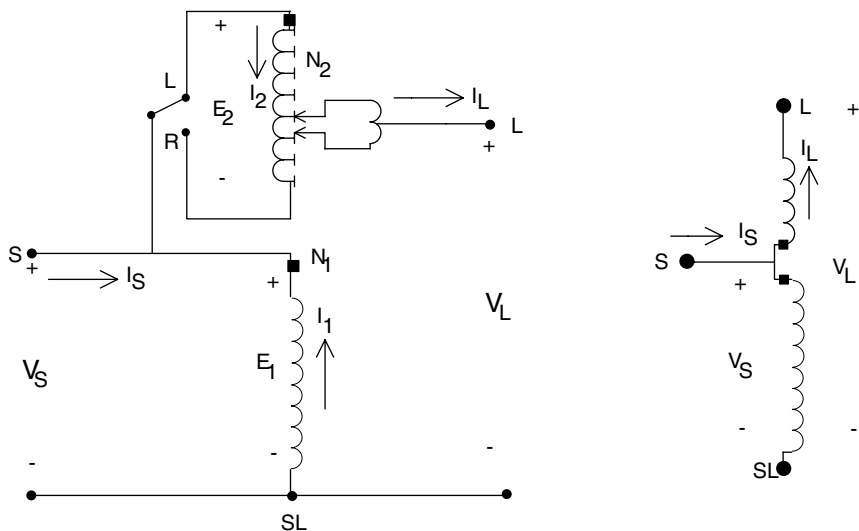


FIGURE 7.8
Type A step-voltage regulator in the lower position.

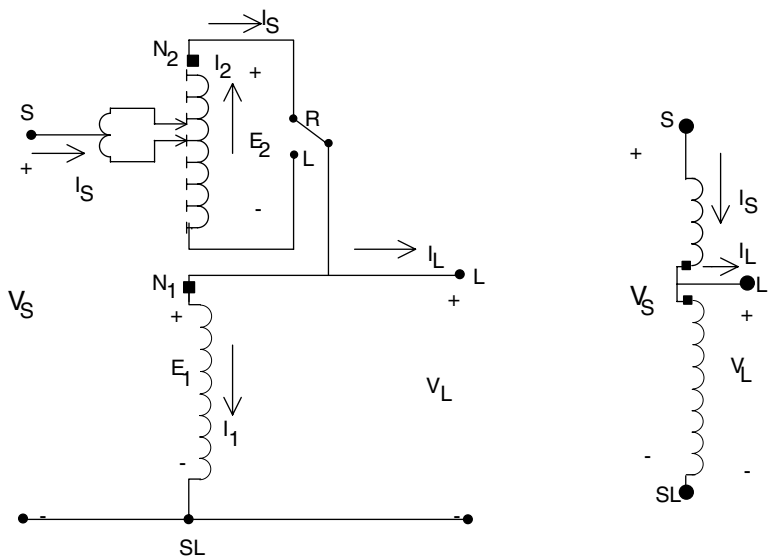


FIGURE 7.9
Type B step-voltage regulator in the raise position.

VOLTAGE EQUATIONS

$$\frac{E_1}{N_1} = \frac{E_2}{N_2}$$

$$V_S = E_1 - E_2$$

$$V_L = E_1$$

$$E_2 = \frac{N_2}{N_1} \cdot E_1 = \frac{N_2}{N_1} \cdot V_L$$

$$V_S = \left(1 - \frac{N_2}{N_1}\right) \cdot V_L$$

$$V_S = a_R \cdot V_L$$

CURRENT EQUATIONS

$$N_1 \cdot I_1 = N_2 \cdot I_2 \quad (7.61)$$

$$I_L = I_S - I_1 \quad (7.62)$$

$$I_2 = I_S \quad (7.63)$$

$$I_1 = \frac{N_2}{N_1} \cdot I_2 = \frac{N_2}{N_1} \cdot I_S \quad (7.64)$$

$$I_L = \left(1 - \frac{N_2}{N_1}\right) \cdot I_S \quad (7.65)$$

$$I_L = a_R \cdot I_S \quad (7.66)$$

$$a_R = 1 - \frac{N_2}{N_1} \quad (7.67)$$

Equations 7.66 and 7.67 are the necessary defining equations for modeling a regulator in the raise position.

The Type B step-voltage connection in the lower position is shown in Figure 7.10. As in the Type A, connection note that the direction of the currents through the series and shunt windings change, but the voltage polarity of the two windings remain the same.

The defining voltage and current equations for the Type B step-voltage regulator in the lower position are as follows:

VOLTAGE EQUATIONS

$$\frac{E_1}{N_1} = \frac{E_2}{N_2}$$

$$V_S = E_1 + E_2$$

$$V_L = E_1$$

$$E_2 = \frac{N_2}{N_1} \cdot E_1 = \frac{N_2}{N_1} \cdot V_L$$

$$V_S = \left(1 + \frac{N_2}{N_1}\right) \cdot V_L$$

$$V_S = a_R \cdot V_L$$

CURRENT EQUATIONS

$$N_1 \cdot I_1 = N_2 \cdot I_2 \quad (7.68)$$

$$I_L = I_S + I_1 \quad (7.69)$$

$$I_2 = I_S \quad (7.70)$$

$$I_1 = \frac{N_2}{N_1} \cdot I_2 = \frac{N_2}{N_1} \cdot I_S \quad (7.71)$$

$$I_L = \left(1 + \frac{N_2}{N_1}\right) \cdot I_S \quad (7.72)$$

$$I_L = a_R \cdot I_S \quad (7.73)$$

$$a_R = 1 + \frac{N_2}{N_1} \quad (7.74)$$

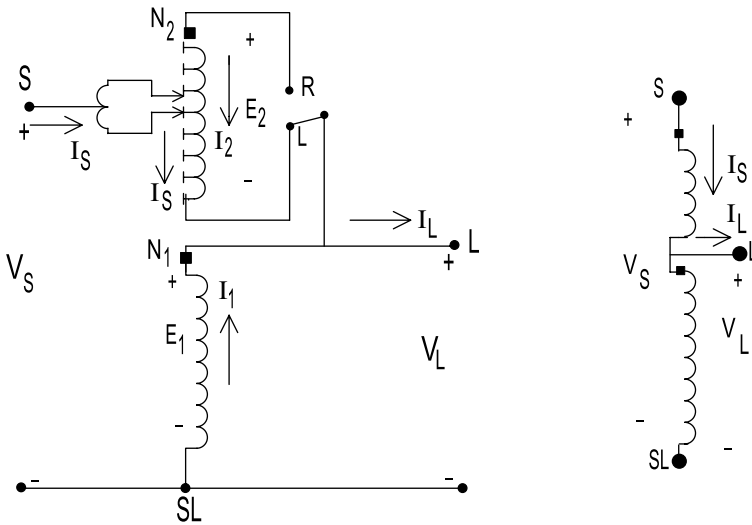


FIGURE 7.10
Type B step-voltage regulator in the lower position.

Equations 7.67 and 7.74 give the value of the effective regulator ratio as a function of the ratio of the number of turns on the series winding (N_2) to the number of turns on the shunt winding (N_1).

In the final analysis, the only difference between the voltage and current equations for the Type B regulator in the raise and lower positions is the sign of the turns ratio (N_2/N_1). The actual turns ratio of the windings is not known; however, the particular tap position will be known. Equations 7.67 and 7.74 can be modified to give the effective regulator ratio as a function of the tap position. Each tap changes the voltage by 5/8% or 0.00625 per-unit. Therefore, the effective regulator ratio can be given by:

$$a_R = 1 \mp 0.00625 \cdot \text{Tap} \quad (7.75)$$

In Equation 7.75, the minus sign applies for the raise position and the plus sign for the lower position.

7.4.1.3 Generalized Constants

In previous chapters and sections of this text generalized $abcd$ constants have been developed for various devices. It can now be shown that the generalized $abcd$ constants can also be applied to the step-voltage regulator. For both the Type A and Type B regulators, the relationship between the source voltage and current to the load voltage and current are

TABLE 7.1

Sign Convention Table for Equation 7.75

	Type A	Type B
Raise	+	-
Lower	-	+

of the form:

$$\text{Type A:} \quad V_S = \frac{1}{a_R} \cdot V_L \quad I_S = a_R \cdot I_L \quad (7.76)$$

$$\text{Type B:} \quad V_S = a_R \cdot V_L \quad I_S = \frac{1}{a_R} \cdot I_L \quad (7.77)$$

Therefore, the generalized constants for a single-phase step-voltage regulator become:

$$\text{Type A:} \quad a = \frac{1}{a_R} \quad b = 0 \quad c = 0 \quad d = a_R \quad (7.78a)$$

$$\text{Type B:} \quad a = a_R \quad b = 0 \quad c = 0 \quad d = \frac{1}{a_R} \quad (7.78b)$$

where a_R is given by Equation 7.75 and the sign convention is given in Table 7.1.

7.4.1.4 The Line Drop Compensator

The changing of taps on a regulator is controlled by the line drop compensator. Figure 7.11 shows a simplified sketch of the compensator circuit and how it is connected to the distribution line through a potential transformer and a current transformer. The purpose of the line drop compensator is to model the voltage drop of the distribution line from the regulator to the load center. The compensator is an analog circuit that is a scale model of the line circuit. The compensator input voltage is typically 120 volts, which requires the voltage transformer in Figure 7.11 to reduce the rated voltage to 120 volts. For a regulator connected line-to-ground the rated voltage is the nominal line-to-neutral voltage, while for a regulator connected line-to-line the rated voltage is the line-to-line voltage. The current transformer turns ratio is specified as $CT_p:CT_s$, where the primary rating (CT_p) will typically be the rated current of the feeder. The setting that is most critical is that of R' and X' calibrated in volts. These values must represent the equivalent impedance from the regulator to the load center. The basic requirement is to force the per-unit line impedance to be equal to the per-unit compensator impedance. In order to cause this to happen, it is essential that a consistent set of base values be developed wherein the per-unit voltage and currents in the line and in the compensator are equal. The consistent set of base values is determined by

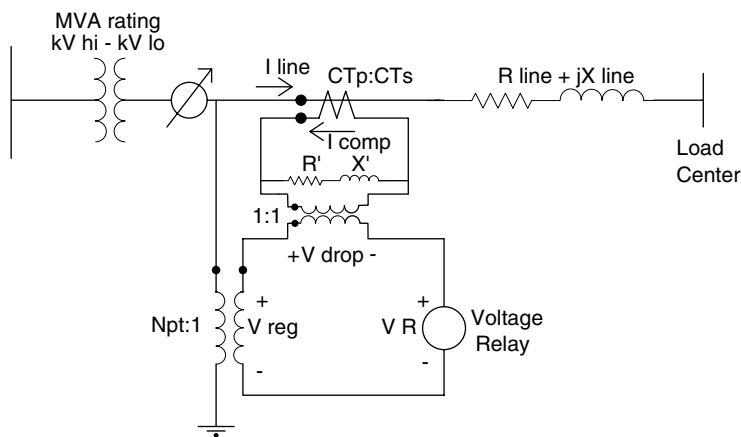


FIGURE 7.11
Line drop compensator circuit.

TABLE 7.2

Table of Base Values

Base	Line Circuit	Compensator Circuit
Voltage	V_{LN}	$\frac{V_{LN}}{N_{PT}}$
Current	CT_P	CT_S
Impedance	$Z_{base_{line}} = \frac{V_{LN}}{CT_P}$	$Z_{base_{comp}} = \frac{V_{LN}}{N_{PT} \cdot CT_S}$

selecting a base voltage and current for the line circuit, and then computing the base voltage and current in the compensator by dividing the system base values by the voltage transformer ratio and current transformer ratio, respectively. For regulators connected line-to-ground, the base system voltage is selected as the rated line-to-neutral voltage (V_{LN}) and the base system current is selected as the rating of the primary winding of the current transformer (CT_P). Table 7.2 gives a table of base values and employs these rules for a regulator connected line-to-ground. With the table of base values developed, the compensator R and X settings in ohms can be computed by first computing the per-unit line impedance:

$$R_{pu} + jX_{pu} = \frac{Rline_{\Omega} + jXline_{\Omega}}{Z_{base_{line}}} \quad (7.79)$$

$$R_{pu} + jX_{pu} = (Rline_{\Omega} + jXline_{\Omega}) \cdot \frac{CT_P}{V_{LN}}$$

The per-unit impedance of Equation 7.79 must be the same in the line and in the compensator. The compensator impedance in ohms is computed by multiplying the per-unit impedance by the base compensator impedance:

$$\begin{aligned}
 R_{comp\Omega} + jX_{comp\Omega} &= (R_{pu} + jX_{pu}) \cdot Z_{base_{comp}} \\
 R_{comp\Omega} + jX_{comp\Omega} &= (R_{line\Omega} + jX_{line\Omega}) \cdot \frac{CT_P}{V_{LN}} \cdot \frac{V_{LN}}{N_{PT} \cdot CT_S} \quad (7.80) \\
 R_{comp\Omega} + jX_{comp\Omega} &= (R_{line\Omega} + jX_{line\Omega}) \cdot \frac{CT_P}{N_{PT} \cdot CT_S} \quad \Omega
 \end{aligned}$$

Equation 7.80 gives the value of the compensator R and X settings in ohms. The compensator R and X settings in volts are determined by multiplying the compensator R and X in ohms times the rated secondary current in amps (CT_S) of the current transformer:

$$\begin{aligned}
 R' + jX' &= (R_{comp\Omega} + jX_{comp\Omega}) \cdot CT_S \\
 R' + jX' &= (R_{line\Omega} + jX_{line\Omega}) \cdot \frac{CT_P}{N_{PT} \cdot CT_S} \cdot CT_S \quad (7.81) \\
 R' + jX' &= (R_{line\Omega} + jX_{line\Omega}) \cdot \frac{CT_P}{N_{PT}} \quad V
 \end{aligned}$$

Knowing the equivalent impedance in ohms from the regulator to the load center, the required value for the compensator settings in volts is determined by using Equation 7.81. This is demonstrated in Example 7.4.

Example 7.4

Refer to Figure 7.11.

The substation transformer is rated 5000 kVA, 115 delta – 4.16 grounded wye, and the equivalent line impedance from the regulator to the load center is $0.3 + j0.9 \Omega$.

- (1) Determine the voltage transformer and current transformer ratings for the compensator circuit.

The rated line-to-ground voltage of the substation transformer is

$$V_S = \frac{4160}{\sqrt{3}} = 2401.8$$

In order to provide approximately 120 V to the compensator, the voltage transformer ratio is

$$N_{PT} = \frac{2400}{120} = 20$$

The rated current of the substation transformer is

$$I_{\text{rated}} = \frac{5000}{\sqrt{3} \cdot 4.16} = 693.9$$

The primary rating of the CT is selected as 700 A, and if the compensator current is reduced to 5 A, the CT ratio is

$$CT = \frac{CT_P}{CT_S} = \frac{700}{5} = 140$$

- (2) Determine the R and X settings of the compensator in ohms and volts.

Applying Equation 7.78 to determine the settings in volts:

$$R' + jX' = (0.3 + j0.9) \cdot \frac{700}{20} = 10.5 + j31.5 \text{ V}$$

The R and X settings in ohms are determined by dividing the settings in volts by the rated secondary current of the current transformer:

$$R_{\text{ohms}} + jX_{\text{ohms}} = \frac{10.5 + j31.5}{5} = 2.1 + j6.3 \text{ } \Omega$$

Understand that the R and X settings on the compensator control board are calibrated in volts.

Example 7.5

The substation transformer in Example 7.4 is supplying 2500 kVA at 4.16 kV and 0.9 power factor lag. The regulator has been set so that:

$$R' + jX' = 10.5 + j31.5 \text{ V}$$

Voltage Level = 120 V (desired voltage to be held at the load center)

Bandwidth = 2 V

Determine the tap position of the regulator that will hold the load center voltage at the desired voltage level and within the bandwidth. This means that the tap on the regulator needs to be set so that the voltage at the load center lies between 119 and 121 V.

The first step is to calculate the actual line current:

$$I_{\text{line}} = \frac{2500}{\sqrt{3} \cdot 4.16} / \cos^{-1}(0.9) = 346.97 / \underline{-25.84} \text{ A}$$

The current in the compensator is then:

$$I_{\text{comp}} = \frac{346.97 / \underline{-25.84}}{140} = 2.4783 / \underline{-25.84} \text{ A}$$

The input voltage to the compensator is

$$V_{\text{reg}} = \frac{2401.8 / 0}{20} = 120.09 / \underline{0} \text{ V}$$

The voltage drop in the compensator circuit is equal to the compensator current times the compensator R and X values in ohms:

$$V_{\text{drop}} = (2.1 + j6.3) \cdot 2.4783 / \underline{-25.84} = 16.458 / \underline{45.72} \text{ V}$$

The voltage across the voltage relay is

$$V_R = V_{\text{reg}} - V_{\text{drop}} = 120.09 / \underline{0} - 16.458 / \underline{45.72} = 109.24 / \underline{-6.19} \text{ V}$$

The voltage across the voltage relay represents the voltage at the load center. Since this is well below the minimum voltage level of 119, the voltage regulator will have to change taps in the raise position to bring the load center voltage up to the required level. Recall that on a 120-V base, one step change on the regulator changes the voltage 0.75 V. The number of required tap changes can then be approximated by

$$\text{Tap} = \frac{119 - 109.24}{0.75} = 13.02$$

This shows that the final tap position of the regulator will be “raise 13.” With the tap set at +13, the effective regulator ratio assuming a Type B regulator is

$$a_R = 1 - 0.00625 \cdot 13 = 0.9188$$

The generalized constants for modeling the regulator for this operating condition are

$$a = a_R = 0.9188$$

$$b = 0$$

$$c = 0$$

$$d = \frac{1}{0.9188} = 1.0884$$

Example 7.6

Using the results of Examples 7.5, calculate the actual voltage at the load center assuming the 2500 kVA at 4.16 kV is measured at the substation transformer low-voltage terminals.

The actual line-to-ground voltage and line current at the load side terminals of the regulator are

$$V_L = \frac{V_S}{a} = \frac{2401.8/\underline{0}}{0.9188} = 2614.2/\underline{0} \text{ V}$$

$$I_L = \frac{I_S}{d} = \frac{346.97/\underline{-25.84}}{1.0884} = 318.77/\underline{-25.84} \text{ A}$$

The actual line-to-ground voltage at the load center is

$$\begin{aligned} V_{LC} &= V_L - Z_{\text{line}} \cdot I_L = 2614.2/\underline{0} - (0.3 + j0.9) \cdot 318.77/\underline{-25.84} \\ &= 2412.8/\underline{-5.15} \text{ V} \end{aligned}$$

On a 120-V base, the load center voltage is

$$V_{LC_{120}} = \frac{V_{LC}}{N_{pt}} = \frac{2412.8/\underline{-5.15}}{20} = 120.6/\underline{-5.15} \text{ V}$$

Therefore, the +13 tap on the regulator has provided the desired voltage at the load center.

As an exercise, the student should go back and, using the output voltage and current of the regulator on the +13 tap, calculate the voltage across the voltage relay in the compensator circuit. That value will be computed to be exactly the same as the load center voltage on the 120-V base.

It is important to understand that the value of equivalent line impedance is not the actual impedance of the line between the regulator and the load center.

Typically, the load center is located down the primary main feeder after several laterals have been tapped. As a result, the current measured by the CT of the regulator is not the current that flows all the way from the regulator to the load center. The only way to determine the equivalent line impedance value is to run a power-flow program of the feeder without the regulator operating. From the output of the program the voltages at the regulator output and the load center are known. Now the equivalent line impedance can be computed as:

$$Rline_{\Omega} + jXline_{\Omega} = \frac{V_{\text{regulator output}} - V_{\text{load center}}}{I_{\text{line}}} \Omega \quad (7.82)$$

In Equation 7.82 the voltages must be specified in system volts and the current in system amperes.

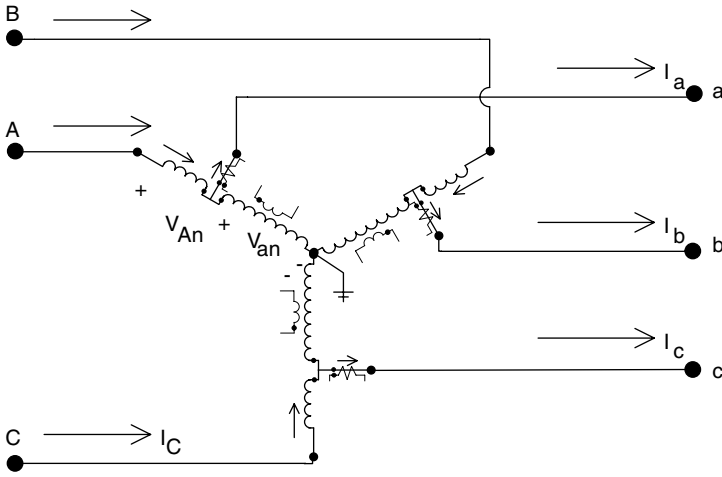
This section has developed the model and generalized constants for Type A and Type B single-phase step-voltage regulators. The compensator control circuit has been developed, and it has been demonstrated how this circuit controls the tap changing of the regulator. The next section will discuss the various three-phase step-type voltage regulators.

7.4.2 Three-Phase Step-Voltage Regulators

Three single-phase step-voltage regulators can be connected externally to form a three-phase regulator. When three single-phase regulators are connected together, each regulator has its own compensator circuit and, therefore, the taps on each regulator are changed separately. Typical connections for single-phase step-regulators are

1. Single-phase
2. Two regulators connected in “open-wye” (sometimes referred to as “V” phase)
3. Three regulators connected in grounded wye
4. Two regulators connected in open delta
5. Three regulators connected in closed delta

A three-phase regulator has the connections between the single-phase windings internal to the regulator housing. The three-phase regulator is gang operated so that the taps on all windings change the same and, as a result, only one compensator circuit is required. For this case it is up to the engineer to determine which phase current and voltage will be sampled by the compensator circuit. Three-phase regulators will only be connected in a three-phase wye or closed delta.

**FIGURE 7.12**

Wye-connected type B regulators.

In the regulator models to be developed in the next sections, the phasing on the source side of the regulator will use capital letters A, B, and C. The load side phasing will use lowercase letters a, b, and c.

7.4.2.1 Wye-Connected Regulators

Three Type B single-phase regulators connected in wye are shown in Figure 7.12. In Figure 7.12 the polarities of the windings are shown in the raise position. When the regulator is in the lower position a reversing switch will have reconnected the series winding so that the polarity on the series winding is now at the output terminal. Regardless of whether the regulator is raising or lowering the voltage, the following equations apply:

VOLTAGE EQUATIONS

$$\begin{bmatrix} V_{An} \\ V_{Bn} \\ V_{Cn} \end{bmatrix} = \begin{bmatrix} a_{Ra} & 0 & 0 \\ 0 & a_{Rb} & 0 \\ 0 & 0 & a_{Rc} \end{bmatrix} \cdot \begin{bmatrix} V_{an} \\ V_{bn} \\ V_{cn} \end{bmatrix} \quad (7.83)$$

where a_{Ra} , a_{Rb} , and a_{Rc} represent the effective turns ratios for the three single-phase regulators. Equation 7.83 is of the form:

$$[VLN_{ABC}] = [a] \cdot [VLN_{abc}] + [b] \cdot [I_{abc}] \quad (7.84)$$

CURRENT EQUATIONS

$$\begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix} = \begin{bmatrix} \frac{1}{a_{Ra}} & 0 & 0 \\ 0 & \frac{1}{a_{Rb}} & 0 \\ 0 & 0 & \frac{1}{a_{Rc}} \end{bmatrix} = \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \quad (7.85)$$

or
$$[I_{ABC}] = [c] \cdot [VLG_{abc}] + [d][I_{abc}] \quad (7.86)$$

Equations 7.84 and 7.86 are of the same form as the generalized equations that were developed for the three-phase line segment of Chapter 6. For a three-phase wye-connected step-voltage regulator, neglecting the series impedance and shunt admittance, the generalized matrices are defined as:

$$[a] = \begin{bmatrix} a_{Ra} & 0 & 0 \\ 0 & a_{Rb} & 0 \\ 0 & 0 & a_{Rc} \end{bmatrix} \quad (7.87)$$

$$[b] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (7.88)$$

$$[c] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (7.89)$$

$$[d] = \begin{bmatrix} \frac{1}{a_{Ra}} & 0 & 0 \\ 0 & \frac{1}{a_{Rb}} & 0 \\ 0 & 0 & \frac{1}{a_{Rc}} \end{bmatrix} \quad (7.90)$$

In Equations 7.87 and 7.90, the effective turns ratio for each regulator must satisfy:

$$0.9 \leq a_{Rabc} \leq 1.1 \text{ in 32 steps of } 0.625\%/\text{step} \text{ (0.75 V/step on 120-V base)}$$

The effective turn ratios (a_{Ra} , a_{Rb} , and a_{Rc}) can take on different values when three single-phase regulators are connected in wye. It is also possible to have a three-phase regulator connected in wye where the voltage and current are

sampled on only one phase, and then all three phases are changed by the same number of taps.

Example 7.7

An unbalanced three-phase load is served at the end of a 10,000-ft., 12.47-kV distribution line segment. The phase generalized matrices for the line segment were computed in Example 6.1 as

$$[a] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[b] = \begin{bmatrix} 0.8667 + j2.0417 & 0.2955 + j0.9502 & 0.2907 + j0.7290 \\ 0.2955 + j0.9502 & 0.8837 + j1.9852 & 0.2992 + j0.8023 \\ 0.2907 + j0.7290 & 0.2992 + j0.8023 & 0.8741 + j2.0172 \end{bmatrix}$$

For this line the A and B matrices are defined as:

$$[A] = [a]^{-1}$$

$$[B] = [a]^{-1} \cdot [b] = [Z_{abc}]$$

The line-to-neutral voltages at the substation are balanced three-phase:

$$[VLN_{ABC}] = \begin{bmatrix} 7200/\underline{0} \\ 7200/\underline{-120} \\ 7200/\underline{120} \end{bmatrix} \text{ V}$$

The line currents at the substation for the unbalanced loading are

$$[I_{abc}] = [I_{ABC}] = \begin{bmatrix} 258/\underline{-20} \\ 288/\underline{-147} \\ 324/\underline{86} \end{bmatrix} \text{ A}$$

For the measured substation voltages and currents, the line-to-neutral voltages at the load are computed as:

$$[Vload_{abc}] = [A] \cdot [VLN_{ABC}] - [B] \cdot [I_{abc}] = \begin{bmatrix} 6965.1/\underline{-2.1} \\ 6943.1/\underline{-121.2} \\ 6776.7/\underline{117.8} \end{bmatrix} \text{ V}$$

The load voltages on a 120-V base are determined by dividing by the potential transformer ratio that transforms rated line-to-neutral voltage

down to 120 V:

$$N_{pt} = \frac{7200}{120} = 60$$

The load voltages on a 120-V base are

$$[V_{120}] = \frac{1}{60} \cdot [Vload_{abc}] = \begin{bmatrix} 116.1/-2.1 \\ 115.7/-121.2 \\ 112.9/117.8 \end{bmatrix} \text{ V}$$

Three single-phase Type B step-voltage regulators are connected in wye and installed in the substation. The regulators are to be set so that each line-to-neutral load voltage on a 120-V base will lie between 119 and 121 volts.

The current transformers of the regulators are rated:

$$CT = \frac{600}{5} = \frac{CT_P}{CT_S} = 120$$

The equivalent line impedance for each phase can be determined by applying Equation 7.79:

$$Zline_a = \frac{7200/0 - 6965.7/-2.1}{258/-20} = 0.5346 + j1.2385 \Omega$$

$$Zline_b = \frac{7200/-120 - 6943.1/-121.2}{288/-147} = 0.5628 + j0.8723 \Omega$$

$$Zline_c = \frac{7200/120 - 6776.7/117.8}{324/86} = 0.6386 + j1.418 \Omega$$

Even though the three regulators will change taps independently, it is the usual practice to set the R and X settings of the three regulators the same. The average value of the three line impedances above can be used for this purpose:

$$Zline_{average} = 0.5787 + j1.1763$$

The compensator R and X settings are computed according to Equation 7.78:

$$R' + jX' = (Rline_{\Omega} + jXline_{\Omega}) \cdot \frac{CT_P}{N_{PT}} = (0.5787 + j1.1763) \cdot \frac{600}{60}$$

$$R' + jX' = 5.787 + j11.763 \text{ V}$$

The compensator controls are not calibrated to that many significant figures, so the values set are

$$R' + jX' = 6 + j12 \text{ V}$$

The compensator control will be set such that the voltage level = 120 V with a bandwidth of 2 V.

For the same unbalanced loading, and with the three-phase wye-connected regulators in service, the approximate tap settings are

$$\begin{aligned} Tap_a &= \frac{|119 - |Vload_a||}{0.75} = \frac{|119 - 116.1|}{0.75} = 3.8736 \\ Tap_b &= \frac{|119 - |Vload_b||}{0.75} = \frac{|119 - 115.7|}{0.75} = 4.3746 \\ Tap_c &= \frac{|119 - |Vload_c||}{0.75} = \frac{|119 - 112.9|}{0.75} = 8.0723 \end{aligned}$$

Since the taps must be integers, the actual tap settings will be

$$\begin{aligned} Tap_a &= +4 \\ Tap_b &= +5 \\ Tap_c &= +9 \end{aligned}$$

The effective turns ratio for the three regulators and the resulting generalized matrices are determined by applying Equation 7.75 for each phase:

$$\begin{aligned} [a] &= \begin{bmatrix} 1 - 0.00625 \cdot 4 & 0 & 0 \\ 0 & 1 - 0.00625 \cdot 5 & 0 \\ 0 & 0 & 1 - 0.00625 \cdot 8 \end{bmatrix} \\ &= \begin{bmatrix} 0.975 & 0 & 0 \\ 0 & 0.9688 & 0 \\ 0 & 0 & 0.9500 \end{bmatrix} \\ [d] = [a]^{-1} &= \begin{bmatrix} 1.0256 & 0 & 0 \\ 0 & 1.0323 & 0 \\ 0 & 0 & 1.0526 \end{bmatrix} \end{aligned}$$

The metered substation voltages and currents are the inputs to the three voltage regulators. The output voltages and currents of the regulators are

$$[Vreg_{abc}] = [a]^{-1} \cdot [VLN_{ABC}] = \begin{bmatrix} 7384.6/\underline{0} \\ 7432.3/\underline{-120} \\ 7578/\underline{120} \end{bmatrix} \text{ V}$$

The output currents of the regulators are

$$[Ireg_{abc}] = [d]^{-1} \cdot [I_{sub}] = \begin{bmatrix} 251.6 / -20 \\ 279.0 / -147 \\ 307.8 / 86 \end{bmatrix} \text{ A}$$

With the regulators adjusted, the load voltages can be computed to be

$$[Vload_{abc}] = [A] \cdot [Vreg_{abc}] - [B] \cdot [Ireg_{abc}] = \begin{bmatrix} 7150.7 / -2.0 \\ 7185.5 / 121.2 \\ 7179.1 / 118.1 \end{bmatrix} \text{ V}$$

On a 120-volt base the load voltages are

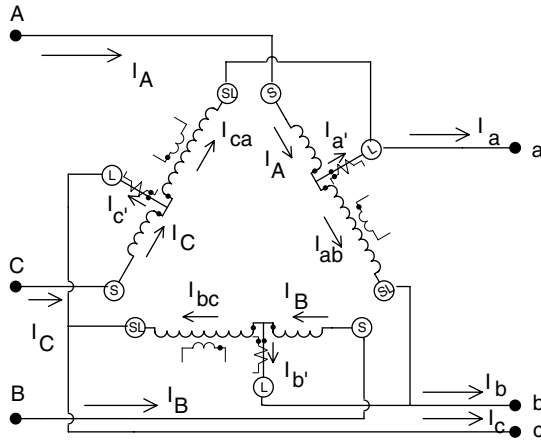
$$[V_{120}] = \frac{1}{60} \cdot \begin{bmatrix} 7150.7 / -2.0 \\ 7185.5 / -121.2 \\ 7179.1 / 118.2 \end{bmatrix} = \begin{bmatrix} 119.2 / -2.0 \\ 119.8 / -121.2 \\ 119.7 / 118.1 \end{bmatrix} \text{ V}$$

With the given regulator taps, the load voltages all now lie between the desired voltage limits of 119 and 121 V per the voltage level and bandwidth settings of the compensator circuit.

Example 7.7 is a long example intended to demonstrate how the engineer can determine the correct compensator R and X settings knowing the substation and load voltages and the currents leaving the substation. Generally, it will be necessary to run a power-flow study to determine these values. The example demonstrates that with the regulator tap settings, the load voltages lie within the desired limits. The regulator has automatically set the taps for this load condition and, as the load changes, the taps will continue to change in order to hold the load voltages within the desired limits.

7.4.2.2 Closed Delta-Connected Regulators

Three single-phase Type B regulators can be connected in a closed delta as shown in Figure 7.13. In the figure, the regulators are shown in the raise position. The closed delta connection is typically used in three-wire delta feeders. Note that the voltage transformers for this connection are monitoring the load side line-to-line voltages, and the current transformers are *not* monitoring the load side line currents.

**FIGURE 7.13**

Closed delta-connected regulators.

The relationships between the source side and currents and voltages are needed. Equations 7.64 through 7.67 define the relationships between the series and shunt winding voltages, and currents for a step-voltage regulator. These must be satisfied no matter how the regulators are connected.

Kirchhoff's voltage law is first applied around a closed loop, starting with the line-to-line voltage between phases A and B on the source side. Refer to Figure 7.13.

$$V_{AB} + V_{Bb} + V_{ba} + V_{aA} = 0 \quad (7.91)$$

but
$$V_{Bb} = -\frac{N_2}{N_1} \cdot V_{bc} \quad (7.92)$$

$$V_{aA} = \frac{N_2}{N_1} \cdot V_{ab} \quad (7.93)$$

$$V_{ba} = -V_{ab} \quad (7.94)$$

Substitute Equations 7.92, 7.93, and 7.94 into Equation 7.91 and simplify:

$$V_{AB} = \left(1 - \frac{N_2}{N_1}\right) \cdot V_{ab} + \frac{N_2}{N_1} \cdot V_{bc} = a_{Rab} \cdot V_{ab} + (1 - a_{Rbc}) \cdot V_{bc} \quad (7.95)$$

The same procedure can be followed to determine the relationships between the other line-to-line voltages. The final three-phase equation is

$$\begin{bmatrix} V_{AB} \\ V_{BC} \\ V_{CA} \end{bmatrix} = \begin{bmatrix} a_{Rab} & 1 - a_{Rbc} & 0 \\ 0 & a_{Rbc} & 1 - a_{Rca} \\ 1 - a_{Rab} & 0 & a_{Rca} \end{bmatrix} \cdot \begin{bmatrix} V_{ab} \\ V_{bc} \\ V_{ca} \end{bmatrix} \quad (7.96)$$

Equation 7.96 is of the generalized form:

$$[VLL_{ABC}] = [a] \cdot [VLL_{abc}] + [b] \cdot [I_{abc}] \quad (7.97)$$

The relationship between source and load line currents starts with applying KCL at the load side terminal a:

$$I_a = I'_a + I_{ca} = I_A - I_{ab} + I_{ca} \quad (7.98)$$

but

$$I_{ab} = \frac{N_2}{N_1} \cdot I_A \quad (7.99)$$

$$I_{ca} = \frac{N_2}{N_1} \cdot I_C \quad (7.100)$$

Substitute Equations 7.96 and 7.97 into Equation 7.95 and simplify:

$$I_a = \left(1 - \frac{N_2}{N_1}\right) \cdot I_A + \frac{N_2}{N_1} I_C = a_{Rab} \cdot I_A + (1 - a_{Rca}) \cdot I_C \quad (7.101)$$

The same procedure can be followed at the other two load side terminals. The resulting three-phase equation is

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} a_{Rab} & 0 & 1 - a_{Rca} \\ 1 - a_{Rab} & a_{Rbc} & 0 \\ 0 & 1 - a_{Rbc} & a_{Rca} \end{bmatrix} \cdot \begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix} \quad (7.102)$$

Equation 7.102 is of the general form:

$$[I_{abc}] = [d]^{-1} \cdot [I_{ABC}] \quad (7.103)$$

where

$$[d]^{-1} = \begin{bmatrix} a_{Rab} & 0 & 1 - a_{Rca} \\ 1 - a_{Rab} & a_{Rbc} & 0 \\ 0 & 1 - a_{Rbc} & a_{Rca} \end{bmatrix}$$

With the exception of the case when all three regulators are in the neutral position ($a_R = 1$), $[d]^{-1}$ does not give a simple expression for each of the elements. In general, all of the elements of $[d]$ will be nonzero. However, when the tap positions of each regulator are known, the elements of the inverse matrix can be determined. The inverse of the resulting matrix gives the matrix $[d]$ so that the generalized form of the current equations can still be applied.

$$[I_{ABC}] = [c] \cdot [VLL_{ABC}] + [d] \cdot [I_{abc}] \quad (7.104)$$

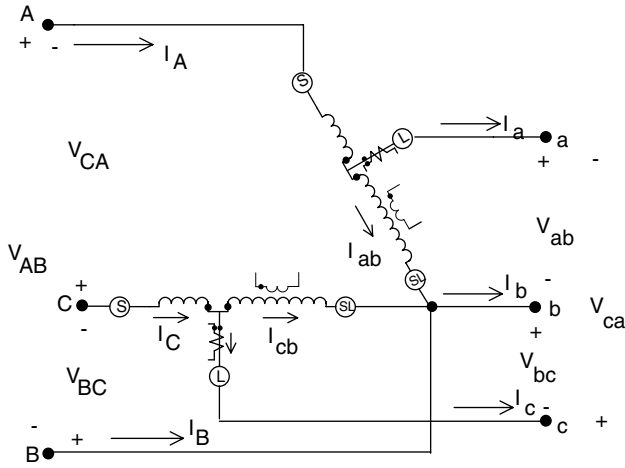
As with the wye-connected regulators, the matrices $[b]$ and $[c]$ are zero as long as the series impedance and shunt admittance of each regulator are neglected.

The closed delta connection can be difficult to apply. Note that in both the voltage and current equations, a change of the tap position in one regulator will affect voltages and currents in two phases. As a result, increasing the tap in one regulator will affect the tap position of the second regulator. Therefore, in most cases the bandwidth setting for the closed delta connection will have to be wider than that for wye-connected regulators.

7.4.2.3 Open Delta-Connected Regulators

Two Type B single-phase regulators can be connected in the open delta connection. Shown in Figure 7.14 is an open delta connection where two single-phase regulators have been connected between phases AB and CB . Two additional open connections can be made by connecting the single-phase regulators between phases BC and AC , and also between phases CA and BA .

The open delta connection is typically applied to three-wire delta feeders. Note that the potential transformers monitor the line-to-line voltages and the current transformers monitor the line currents. Once again, the basic voltage and current relations of the individual regulators are used to determine the relationships between the source side and load side voltages and currents. The connection shown in Figure 7.14 will be used to derive the relationships, and then the relationships of the other two possible connections will merely be stated.

**FIGURE 7.14**

Open delta connection.

The voltage drop V_{AB} across the first regulator consists of the drop across the series winding plus the drop across the shunt winding:

$$V_{AB} = V_{AL} + V_{ab} \quad (7.105)$$

where V_{AL} = drop across the series winding.

Paying attention to the polarity marks on the series and shunt windings, the drop across the series winding is

$$V_{AL} = -\frac{N_2}{N_1} \cdot V_{ab} \quad (7.106)$$

Substituting Equation 7.106 into Equation 7.105 yields:

$$V_{AB} = \left(1 - \frac{N_2}{N_1}\right) \cdot V_{ab} = a_{Rab} \cdot V_{ab} \quad (7.107)$$

Following the same procedure for the regulator connected across V_{BC} , the voltage equation is

$$V_{BC} = \left(1 - \frac{N_2}{N_1}\right) \cdot V_{bc} = a_{Rcb} \cdot V_{bc} \quad (7.108)$$

Kirchhoff's voltage law must be satisfied so that:

$$V_{CA} = -(V_{AB} + V_{BC}) = -a_{Rab} \cdot V_{ab} - a_{Rcb} \cdot V_{bc} \quad (7.109)$$

Equations 7.107, 7.108, and 7.109 can be put into matrix form:

$$\begin{bmatrix} V_{AB} \\ V_{BC} \\ V_{CA} \end{bmatrix} = \begin{bmatrix} a_{Rab} & 0 & 0 \\ 0 & a_{Rcb} & 0 \\ -a_{Rab} & -a_{Rcb} & 0 \end{bmatrix} \cdot \begin{bmatrix} V_{ab} \\ V_{bc} \\ V_{ca} \end{bmatrix} \quad (7.110)$$

Equation 7.110 in generalized form is

$$[VLL_{ABC}] = [a_{LL}] \cdot [VLL_{abc}] + [b_{LL}] \cdot [I_{abc}] \quad (7.111)$$

where

$$[a_{LL}] = \begin{bmatrix} a_{Rab} & 0 & 0 \\ 0 & a_{Rcb} & 0 \\ -a_{Rab} & -a_{Rcb} & 0 \end{bmatrix} \quad (7.112)$$

The effective turns ratio of each regulator is given by Equation 7.75. Again, as long as the series impedance and shunt admittance of the regulators are neglected, $[b_{LL}]$ is zero. Equation 7.111 gives the line-to-line voltages on the source side as functions of the line-to-line voltages on the load side of the open delta using the generalized matrices. Up to this point the relationships between the voltages have been in terms of line-to-neutral voltages. In Chapter 10 it will be shown how to convert this equation using equivalent line-to-neutral voltages.

When the load side line-to-line voltages are needed as functions of the source side line-to-line voltages, the necessary equation is

$$\begin{bmatrix} V_{ab} \\ V_{bc} \\ V_{ca} \end{bmatrix} = \begin{bmatrix} \frac{1}{a_{Rab}} & 0 & 0 \\ 0 & \frac{1}{a_{Rcb}} & 0 \\ -\frac{1}{a_{Rab}} & -\frac{1}{a_{Rcb}} & 0 \end{bmatrix} \cdot \begin{bmatrix} V_{AB} \\ V_{BC} \\ V_{CA} \end{bmatrix} \quad (7.113)$$

$$[VLL_{abc}] = [A_{LL}] \cdot [VLL_{ABC}] \quad (7.114)$$

where

$$[A_{LL}] = \begin{bmatrix} \frac{1}{a_{Rab}} & 0 & 0 \\ 0 & \frac{1}{a_{Rcb}} & 0 \\ -\frac{1}{a_{Rab}} & -\frac{1}{a_{Rcb}} & 0 \end{bmatrix} \quad (7.115)$$

The current equations are derived by applying KCL at the L node of each regulator:

$$I_A = I_a + I_{ab} \quad (7.116)$$

but

$$I_{ab} = \frac{N_2}{N_1} \cdot I_A$$

Therefore, Equation 7.116 becomes:

$$\left(1 - \frac{N_2}{N_1}\right) I_A = I_a \quad (7.117)$$

therefore

$$I_A = \frac{1}{a_{Rab}} \cdot I_a \quad (7.118)$$

In a similar manner, the current equation for the second regulator is given by:

$$I_C = \frac{1}{a_{Rcb}} \cdot I_c \quad (7.119)$$

Because this is a three-wire delta line, then:

$$I_B = -(I_A + I_C) = -\frac{1}{a_{Rab}} \cdot I_a - \frac{1}{a_{Rcb}} \cdot I_c \quad (7.120)$$

In matrix form, the current equations become:

$$\begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix} = \begin{bmatrix} \frac{1}{a_{Rab}} & 0 & 0 \\ -\frac{1}{a_{Rab}} & 0 & -\frac{1}{a_{Rcb}} \\ 0 & 0 & \frac{1}{a_{Rcb}} \end{bmatrix} \cdot \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \quad (7.121)$$

In generalized form Equation 7.121 becomes:

$$[I_{ABC}] = [c_{LL}] \cdot [VLL_{ABC}] + [d_{LL}] \cdot [I_{abc}] \quad (7.122)$$

where

$$[d_{LL}] = \begin{bmatrix} \frac{1}{a_{Rab}} & 0 & 0 \\ -\frac{1}{a_{Rab}} & 0 & -\frac{1}{a_{Rcb}} \\ 0 & 0 & \frac{1}{a_{Rcb}} \end{bmatrix} \quad (7.123)$$

When the series impedances and shunt admittances are neglected, the constant matrix $[c_{LL}]$ will be zero.

The load side line currents as functions of the source line currents are given by:

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} a_{Rab} & 0 & 0 \\ -a_{Rab} & 0 & -a_{Rcb} \\ 0 & 0 & a_{Rcb} \end{bmatrix} \cdot \begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix} \quad (7.124)$$

$$[I_{abc}] = [D_{LL}] \cdot [I_{ABC}] \quad (7.125)$$

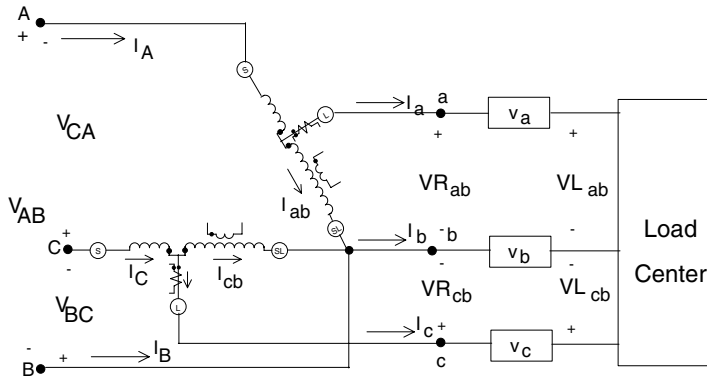
where

$$[D_{LL}] = \begin{bmatrix} a_{Rab} & 0 & 0 \\ -a_{Rab} & 0 & -a_{Rcb} \\ 0 & 0 & a_{Rcb} \end{bmatrix} \quad (7.126)$$

The determination of the R and X compensator settings for the open delta follows the same procedure as that of the wye-connected regulators. However, care must be taken to recognize that in the open delta connection, the voltages applied to the compensator are line-to-line and the currents are line currents. The open delta-connected regulators will maintain only two of the line-to-line voltages at the load center within defined limits. The third line-to-line voltage will be dictated by the other two (Kirchhoff's voltage law). Therefore, it is possible that the third voltage may not be within the defined limits. With reference to Figure 7.15, an equivalent impedance between the regulators and the load center must be computed. Since each regulator is sampling line-to-line voltages and a line current, the equivalent impedance is computed by taking the appropriate line-to-line voltage drop and dividing by the sampled line current. For the open delta connection shown in Figure 7.15, the equivalent impedances are computed as:

$$Zeq_a = \frac{VR_{ab} - VL_{ab}}{I_a} \quad (7.127)$$

$$Zeq_c = \frac{VR_{cb} - VL_{cb}}{I_c} \quad (7.128)$$

**FIGURE 7.15**

Open delta connected to a load center.

The units of these impedances will be in system ohms. They must be converted to compensator volts by applying Equation 7.78. For the open delta connection, the potential transformer will transform the system line-to-line rated voltage down to 120 V. Example 7.8 demonstrates how the compensator R and X settings are determined knowing the line-to-line voltages at the regulator and at the load center.

Example 7.8

A power-flow study has been run on a system prior to the installation of an open delta regulator bank at *Node R*. The load center is at *Node L*, as shown in Figure 7.16. The results of the power-flow study are

$$\text{Node R: } VR_{ab} = 12,470/\underline{0} \quad VR_{bc} = 12,470/\underline{-120} \quad VR_{ca} = 12,470/\underline{120}$$

$$I_a = 308.2/\underline{-58.0} \quad I_b = 264.2/\underline{-176.1} \quad I_c = 297.0/\underline{70.3}$$

$$\text{Node L: } VL_{ab} = 11,911/\underline{-1.4} \quad VL_{bc} = 12,117/\underline{-122.3} \quad VL_{ca} = 11,859/\underline{117.3}$$

**FIGURE 7.16**

Circuit for Example 7.8.

For this connection, the potential transformer ratio and current transformer ratios are selected to be

$$N_{pt} = \frac{12,470}{120} = 103.92$$

$$CT = \frac{500}{5} = 100$$

On a 120-V base, the load center voltages are

$$\begin{bmatrix} V120_{ab} \\ V120_{bc} \\ V120_{ca} \end{bmatrix} = \frac{1}{103.92} \cdot \begin{bmatrix} 11,911 / -1.4 \\ 12,117 / -122.3 \\ 11,859 / 117.3 \end{bmatrix} = \begin{bmatrix} 114.6 / -1.4 \\ 116.6 / -122.3 \\ 114.3 / 117.5 \end{bmatrix} \text{ V}$$

Two single-phase Type B regulators are to be installed in an open delta connection. The regulators are to be connected between phases *A-B* and *B-C* as shown in Figure 7.15. The voltage level will be set at 120 V with a bandwidth of 2 V. As computed above, the load center voltages are not within the desired limits of 120 ± 1 V.

The compensator *R* and *X* settings must first be determined using the results of the power-flow study. The first regulator monitors the voltage V_{ab} and the line current I_a . The equivalent line impedance for this regulator is

$$Z_{eq_a} = \frac{VR_{ab} - VL_{ab}}{I_a} = \frac{12,470 / 0 - 11,911 / -1.4}{308.2 / -58.0} = 0.1665 + j2.0483 \, \Omega$$

The second regulator monitors the voltage V_{cb} and the line current I_c . In the computation of the equivalent line impedance, it is necessary to use the *c-b* voltages, which are the negatives of the given *b-c* voltages:

$$Z_{eq_c} = \frac{VR_{cb} - VL_{cb}}{I_c} = \frac{12,470 / 60 - 12,117 / -57.7}{297.0 / 70.3} = 1.4945 + j1.3925 \, \Omega$$

Unlike the wye-connected regulators, the compensator settings for the two regulators will be different. The settings calibrated in volts are

$$R'_{ab} + jX'_{ab} = Z_a \cdot \frac{CT_P}{N_{PT}} = (0.1665 + j2.0483) \cdot \frac{500}{103.92} = 0.8012 + j9.8555 \text{ V}$$

$$R'_{cb} + jX'_{cb} = Z_c \cdot \frac{CT_P}{N_{PT}} = (0.1495 + j1.3925) \cdot \frac{500}{103.92} = 7.1908 + j6.7002 \text{ V}$$

The compensator settings will be

$$R'_{ab} + X'_{ab} = 0.8 + j9.9$$

$$R'_{cb} + jX'_{cb} = 7.2 + j6.7$$

With regulators installed and in the neutral position, and with the same loading, the currents and voltages in the compensator circuits are

$$V_{comp_{ab}} = \frac{VR_{ab}}{N_{pt}} = \frac{12,470 / 0}{103.92} = 120 / \underline{0} \text{ V}$$

$$V_{comp_{cb}} = \frac{-VR_{bc}}{N_{pt}} = \frac{12,470 / 60}{103.92} = 120 / \underline{60} \text{ V}$$

$$I_{comp_a} = \frac{I_a}{CT} = \frac{308.2 / -58.0}{100} = 3.082 / \underline{-58.0} \text{ A}$$

$$I_{comp_c} = \frac{I_c}{CT} = \frac{297.0 / 70.3}{100} = 2.97 / \underline{70.3} \text{ A}$$

The compensator impedances in ohms are determined by dividing the settings in volts by the secondary rating of the current transformer:

$$R_{ab} + jX_{ab} = \frac{R'_{ab} + jX'_{ab}}{CT_{secondary}} = \frac{0.8 + j9.9}{5} = 0.16 + j1.98$$

$$R_{cb} + jX_{cb} = \frac{R'_{cb} + jX'_{cb}}{CT_{secondary}} = \frac{7.2 + j6.7}{5} = 1.44 + j1.34$$

The voltages across the voltage relays in the two compensator circuits are

$$V_{relay_{ab}} = V_{comp_{ab}} - (R_{ab} + jX_{ab}) \cdot I_{comp_a} = 114.6 / \underline{-1.4} \text{ V}$$

$$V_{relay_{cb}} = V_{comp_{cb}} - (R_{cb} + jX_{cb}) \cdot I_{comp_c} = 116.6 / \underline{57.7} \text{ V}$$

Since the voltages are below the lower limit of 119, the control circuit will send “raise” commands to change the taps on both regulators. The number of tap changes necessary to bring the load center voltage into the lower limit of the bandwidth for each regulator will be

$$Tap_{ab} = \frac{|119 - 114.6|}{0.75} = 5.47 \approx 6$$

$$Tap_{cb} = \frac{|119 - 116.6|}{0.75} = 3.20 \approx 4$$

With the taps set at 6 and 4, a check can be made to determine if the voltages at the load center are now within the limits.

With the taps adjusted, the regulator ratios are

$$a_{Rab} = 1.0 - 0.00625 \cdot Tap_{ab} = 0.9625$$

$$a_{Rcb} = 1.0 - 0.00625 \cdot Tap_{cb} = 0.975$$

In order to determine the load side regulator voltages and currents, the matrices $[A_{LL}]$ (Equation 7.112) and $[D_{LL}]$ (Equation 7.123), must be defined:

$$[A_{LL}] = \begin{bmatrix} \frac{1}{0.9625} & 0 & 0 \\ 0 & \frac{1}{0.975} & 0 \\ -\frac{1}{0.9625} & -\frac{1}{0.975} & 0 \end{bmatrix} = \begin{bmatrix} 1.039 & 0 & 0 \\ 0 & 1.0256 & 0 \\ -1.039 & -1.0256 & 0 \end{bmatrix}$$

$$[D_{LL}] = \begin{bmatrix} 0.9625 & 0 & 0 \\ -0.9625 & 0 & -0.975 \\ 0 & 0 & 0.975 \end{bmatrix}$$

The output voltages from the regulators are

$$[Vreg_{abc}] = [A_{LL}] \cdot [VLL_{ABC}] = \begin{bmatrix} 12,956/\underline{0} \\ 12,790/\underline{-120} \\ 12,874/\underline{120} \end{bmatrix} \text{ V}$$

The output currents from the regulators are

$$[I_{abc}] = [D_{LL}] \cdot [I_{ABC}] = \begin{bmatrix} 296.6/\underline{-58} \\ 255.7/\underline{-175.3} \\ 289.6/\underline{70.3} \end{bmatrix} \text{ A}$$

There are two ways to test if the voltages at the load center are within the limits. The first method is to compute the relay voltages in the compensator circuits. The procedure is the same as was done initially to determine the load center voltages. First the voltages and currents in the compensator

circuits are computed:

$$\begin{aligned}
 V_{comp_{ab}} &= \frac{VR_{ab}}{N_{pt}} = \frac{12,956/0}{103.92} = 124.7/\underline{0} \text{ V} \\
 V_{comp_{cb}} &= \frac{-VR_{bc}}{N_{pt}} = \frac{12,790/60}{103.92} = 123.1/\underline{60} \text{ V} \\
 I_{comp_a} &= \frac{I_a}{CT} = \frac{296.6/-58.0}{100} = 2.966/\underline{-58.0} \text{ A} \\
 I_{comp_c} &= \frac{I_c}{CT} = \frac{289.6/70.3}{100} = 2.896/\underline{70.3} \text{ A}
 \end{aligned}$$

The voltages across the voltage relays are computed to be

$$\begin{aligned}
 V_{relay_{ab}} &= V_{comp_{ab}} - (R_{ab} + jX_{ab}) \cdot I_{comp_a} = 119.5/\underline{-1.3} \text{ V} \\
 V_{relay_{cb}} &= V_{comp_{cb}} - (R_{cb} + jX_{cb}) \cdot I_{comp_a} = 119.8/\underline{57.8} \text{ V}
 \end{aligned}$$

Since both voltages are within the bandwidth, no further tap changing will be necessary.

The actual voltages at the load center can be computed using the output voltages and currents from the regulator and then computing the voltage drop to the load center.

The phase impedance matrix for the line between the regulator and the load center is

$$[Z_{abc}] = \begin{bmatrix} 0.7604 + j2.6762 & 0.1804 + j1.6125 & 0.1804 + j1.2762 \\ 0.1804 + j1.6125 & 0.7604 + j2.6762 & 0.1804 + j1.4773 \\ 0.1804 + j1.2762 & 0.1804 + j1.4773 & 0.7604 + j2.6762 \end{bmatrix} \Omega$$

With reference to Figure 7.15, the voltage drops per phase are

$$[v_{abc}] = [Z_{abc}] \cdot [I_{abc}] = \begin{bmatrix} 450.6/\underline{-1.5} \\ 309.4/\underline{-106.6} \\ 402.8/\underline{142.8} \end{bmatrix} \text{ V}$$

The load center line-to-line voltages are

$$\begin{aligned}
 VL_{ab} &= Vreg_{ab} - v_a + v_b = 12,420/\underline{-1.3} \text{ V} \\
 VL_{bc} &= Vreg_{bc} - v_b + v_c = 12,447/\underline{-122.2} \text{ V} \\
 VL_{ca} &= Vreg_{ca} - v_c + v_a = 12,273/\underline{-118.1} \text{ V}
 \end{aligned}$$

Dividing the load center line-to-line voltages by the potential transformer ratio gives the voltages on the 120-V base as:

$$V120_{ab} = 119.5 / -1.3 \text{ V}$$

$$V120_{bc} = 119.8 / -122.2 \text{ V}$$

$$V120_{ca} = 118.1 / 118.1 \text{ V}$$

The desired voltages are being held on the two phases that have the voltage regulators. The third line-to-line voltage is below the limit. This cannot be helped since that voltage is being dictated by the other two line-to-line voltages. The only way to bring that voltage up is to set a higher voltage level on the two regulators.

This example is very long but has been included to demonstrate how the compensator circuit is set, and then how it will adjust taps so that the voltages at a remote load center node will be held within the set limits. In actual practice, the only responsibilities of the engineer will be to correctly determine the R and X settings of the compensator circuit, and to determine the desired voltage level and bandwidth.

The open delta regulator connection using phases A - B and C - B has been presented. There are two other possible open delta connections using phases B - C and A - C , and then C - A and B - A . Generalized matrices for these additional two connections can be developed using the procedures presented in this section.

7.5 Summary

It has been shown that all possible connections for Type B step-voltage regulators can be modeled using the generalized matrices. The derivations in this chapter were limited to three-phase connections. If a single-phase regulator is connected line-to-neutral or two regulators connected in open wye, then the $[a]$ and $[d]$ matrices will be of the same form as that of the wye-connected regulators, only the terms in the rows and columns associated with the missing phases will be zero. The same can be said for a single-phase regulator connected line-to-line. Again, the rows and columns associated with the missing phases would be set to zero in the matrices developed for the open delta connection.

The generalized matrices developed in this chapter are of exactly the same form as those developed for the three-phase line segments. In the next chapter, the generalized matrices for all three-phase transformers will be developed.

References

1. *American National Standard for Electric Power – Systems and Equipment Voltage Ratings (60 Hertz)*, ANSI C84.1-1995, National Electrical Manufacturers Association, Rosslyn, VA, 1996.
 2. *IEEE Standard Requirements, Terminology, and Test Code for Step-Voltage and Induction-Voltage Regulators*, ANSI/IEEE C57.15-1986, Institute of Electrical and Electronic Engineers, New York, 1988.
-

Problems

7.1 A single-phase transformer is rated 100 kVA, 2400-240 V. The impedances and admittance of the transformer are

$$Z_1 = 0.65 + j0.95 \, \Omega \text{ (high-voltage winding impedance)}$$

$$Z_2 = 0.0052 + j0.0078 \, \Omega \text{ (low-voltage winding impedance)}$$

$$Y_m = 2.56 \times 10^{-4} - j11.37 \times 10^{-4} \text{ S (referred to the high-voltage winding)}$$

- (1) Determine the a , b , c , d constants and the A and B constants.
- (2) The transformer is serving an 80-kW, 0.85 lagging power factor load at 230 V. Determine the primary voltage, current, complex power, and percent voltage drop.
- (3) Determine the per-unit transformer impedance and shunt admittance based upon the transformer ratings.

7.2 The single-phase transformer of Problem 7.1 is to be connected as a step-down autotransformer to transform the voltage from 2400 V to 2160 V.

- (1) Draw the connection diagram, including the series impedance and shunt admittance.
- (2) Determine the autotransformer kVA rating.
- (3) Determine the a , b , c , d , A , and B generalized constants.
- (4) The autotransformer is serving a load of 800 kVA, 0.95 lagging power factor at a voltage of 2000 V. Including the impedance and shunt admittance, determine the input voltage, current, complex power, and percent voltage drop.
- (5) Determine the per-unit impedance and shunt admittance based upon the autotransformer rating. How do these values compare to the per-unit values of Problem 7.1?

7.3 A Type B step-voltage regulator is installed to regulate the voltage on a 7200-V single-phase lateral. The potential transformer and current transformer ratios connected to the compensator circuit are

Potential transformer: 7200:120 V

Current transformer: 500:5 A

The R and X settings in the compensator circuit are: $R = 5$ V and $X = 10$ V. The regulator taps are set on the +10 position when the voltage and current on the source side of the regulator are

$V_{\text{source}} = 7200$ V and $I_{\text{source}} = 375$ at a power factor of 0.866 lagging power factor.

- (1) Determine the voltage at the load center.
- (2) Determine the equivalent line impedance between the regulator and the load center.
- (3) Assuming that the voltage level on the regulator has been set at 120 V with a bandwidth of 2 V, what tap will the regulator move to?

7.4 Refer to Figure 7.11. The substation transformer is rated 24 MVA, 230 kV delta-13.8 kV wye. Three single-phase Type B regulators are connected in wye. The equivalent line impedance between the regulators and the load center node is

$$Z_{\text{line}} = 0.264 + j0.58 \, \Omega/\text{mile}$$

The distance to the load center node is 10,000 ft.

- (1) Determine the appropriate PT and CT ratios.
- (2) Determine the R' and X' settings in ohms and volts for the compensator circuit.
- (3) The substation is serving a balanced three-phase load of 16 MVA, 0.9 lagging power factor when the output line-to-line voltages of the substation are balanced 13.8 kV and the regulators are set in the neutral position. Assume the voltage level is set at 121 V and a bandwidth of 2 V. Determine the final tap position for each regulator (they will be the same). The regulators have 32 – 5/8% taps (16 raise and 16 lower).
- (4) What would be the regulator tap settings for a load of 24 MVA, 0.9 lagging power factor with the output voltages of the substation transformer balanced three-phase 13.8 kV?
- (5) What would be the load center voltages for the load of Part 4 above?

7.5 Three Type B step-voltage regulators are connected in wye and located on the secondary bus of a 12.47-kV substation. The feeder is serving an

unbalanced load. A power-flow study has been run and the voltages at the substation and the load center node are

$$[V_{sub_{abc}}] = \begin{bmatrix} 7200 / 0 \\ 7200 / -120 \\ 7200 / 120 \end{bmatrix} \text{ V}$$

$$[VLC_{abc}] = \begin{bmatrix} 6890.6 / -1.49 \\ 6825.9 / -122.90 \\ 6990.5 / 117.05 \end{bmatrix} \text{ V}$$

The currents at the substation are

$$[I_{abc}] = \begin{bmatrix} 362.8 / -27.3 \\ 395.4 / -154.7 \\ 329.0 / 98.9 \end{bmatrix} \text{ A}$$

The regulator potential transformer ratio is 7200-120 and the current transformer ratio is 500:5. The voltage level of the regulators is set at 121 V and the bandwidth at 2 V.

- (1) Determine the equivalent line impedance per phase between the regulator and the load center.
- (2) The compensators on each regulator are to be set with the same R and X values. Specify these values in volts and in ohms.

7.6 The impedance compensator settings for the three step regulators of Problem 7.5 have been set as:

$$R' = 3.0 \text{ V} \quad X' = 9.3 \text{ V}$$

The voltages and currents at the substation bus are

$$[V_{sub_{abc}}] = \begin{bmatrix} 7200 / 0 \\ 7200 / -120 \\ 7200 / 120 \end{bmatrix} \text{ V}$$

$$[I_{abc}] = \begin{bmatrix} 320.6 / -27.4 \\ 409.0 / -155.1 \\ 331.5 / 98.2 \end{bmatrix} \text{ A}$$

Determine the final tap settings for each regulator.

7.7 For a different load condition for the system of Problem 7.5, the taps on the regulators have been automatically set by the compensator circuit to:

$$Tap_a = +8 \quad Tap_b = +11 \quad Tap_c = +6$$

The load reduces so that the voltages and currents at the substation bus are

$$[V_{sub_{abc}}] = \begin{bmatrix} 7200 / 0 \\ 7200 / -120 \\ 7200 / 120 \end{bmatrix} \text{ V}$$

$$[I_{abc}] = \begin{bmatrix} 177.1 / -28.5 \\ 213.4 / -156.4 \\ 146.8 / 98.3 \end{bmatrix} \text{ A}$$

Determine the new final tap settings for each regulator.

7.8 The load center node for the regulators described in Problem 7.5 is located 1.5 miles from the substation. There are no lateral taps between the substation and the load center. The phase impedance matrix of the line segment is

$$[z_{abc}] = \begin{bmatrix} 0.3465 + j1.0179 & 0.1560 + j0.5017 & 0.1580 + j0.4236 \\ 0.1560 + j0.5017 & 0.3375 + j1.0478 & 0.1535 + j0.3849 \\ 0.1580 + j0.4236 & 0.1535 + j0.3849 & 0.3414 + j1.0348 \end{bmatrix} \Omega/\text{mile}$$

A wye-connected, unbalanced constant impedance load is located at the load center node. The load impedances are

$$ZL_a = 19 + j11 \, \Omega, \quad ZL_b = 22 + j12 \, \Omega, \quad ZL_c = 18 + j10 \, \Omega$$

The voltages at the substation are balanced three-phase of 7200 V line-to-neutral. The regulators are set on neutral.

- (1) Determine the line-to-neutral voltages at the load center.
- (2) Determine the R and X settings in volts for the compensator.
- (3) Determine the required tap settings in order to hold the load center voltages within the desired limits.

7.9 The phase impedance matrix for a three-wire line segment is

$$[z_{abc}] = \begin{bmatrix} 0.4013 + j1.4133 & 0.0953 + j0.8515 & 0.0953 + j0.7802 \\ 0.0953 + j0.8515 & 0.4013 + j1.4133 & 0.0953 + j0.7266 \\ 0.0953 + j0.7802 & 0.0953 + j0.7266 & 0.4013 + j1.4133 \end{bmatrix} \Omega/\text{mile}$$

The line is two miles long and serving an unbalanced load, with the substation transformer line-to-line voltages and output currents:

$$[VLL_{abc}] = \begin{bmatrix} 12,470/\underline{0} \\ 12,470/\underline{-120} \\ 12,470/\underline{120} \end{bmatrix} \text{ V}$$

$$[I_{abc}] = \begin{bmatrix} 307.9/\underline{-54.6} \\ 290.6/\underline{178.6} \\ 268.2/\underline{65.3} \end{bmatrix} \text{ A}$$

Two Type B step-voltage regulators are connected in open delta at the substation using phases *A-B* and *C-B*. The potential transformer ratios are 12,470/120, and the current transformer ratios are 500:5. The voltage level is set at 121 V with a 2-V bandwidth.

- (1) Determine the line-to-line voltages at the load center.
- (2) Determine the *R* and *X* compensator settings in volts. For the open delta connection, the *R* and *X* settings will be different on each regulator.
- (3) Determine the final tap positions of the two voltage regulators.

7.10 The regulator in Problem 7.9 has gone to the +9 tap on both regulators for a particular load. The load is reduced so that the currents leaving the substation transformer with the regulators in the +9 position are

$$[I_{abc}] = \begin{bmatrix} 144.3/\underline{-53.5} \\ 136.3/\underline{179.6} \\ 125.7/\underline{66.3} \end{bmatrix} \text{ A}$$

Determine the final tap settings on each regulator for this new load condition.

8

Three-Phase Transformer Models

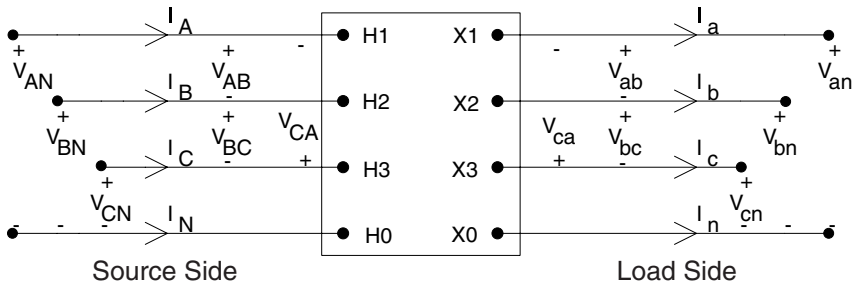
Three-phase transformer banks are found in the distribution substation where the voltage is transformed from the transmission or subtransmission level to the distribution feeder level. In most cases the substation transformer will be a three-phase unit, perhaps with high-voltage no-load taps and, perhaps, low-voltage load tap changing (LTC). For a four-wire wye feeder, the most common substation transformer connection is the delta-grounded wye. A three-wire delta feeder will typically have a delta-delta transformer connection in the substation. Three-phase transformer banks out on the feeder will provide the final voltage transformation to the customer's load. A variety of transformer connections can be applied. The load can be pure three-phase or a combination of single-phase lighting load and a three-phase load such as an induction motor. In the analysis of a distribution feeder, it is important that the various three-phase transformer connections be modeled correctly.

Unique models of three-phase transformer banks applicable to radial distribution feeders will be developed in this chapter. Models for the following three-phase connections are included:

- Delta-Grounded Wye
- Ungrounded Wye-Delta
- Grounded Wye-Grounded Wye
- Delta-Delta
- Open Wye-Open Delta

8.1 Introduction

Figure 8.1 defines the various voltages and currents for all transformer banks connected between the source side *Node n* and the load side *Node m*. In Figure 8.1 the models can represent a step-down (source side to load side) or a step-up (source side to load side) transformer bank. The notation is such

**FIGURE 8.1**

General three-phase transformer bank.

that the capital letters A, B, C, N will always refer to the source side (*Node n*) of the bank and the lower case letters a, b, c, n will always refer to the load side (*Node m*) of the bank. It is assumed that all variations of the wye-delta connections are connected in the “American Standard Thirty-Degree” connection. With the described phase notation, the standard phase shifts for positive sequence voltages and currents are

Step-Down Connection

$$V_{AB} \text{ leads } V_{ab} \text{ by 30 degrees} \quad (8.1)$$

$$I_A \text{ leads } I_a \text{ by 30 degrees} \quad (8.2)$$

Step-Up Connection

$$V_{ab} \text{ leads } V_{AB} \text{ by 30 degrees} \quad (8.3)$$

$$I_a \text{ leads } I_A \text{ by 30 degrees} \quad (8.4)$$

8.2 Generalized Matrices

The models to be used in power-flow and short-circuit studies are generalized for the connections in the same form as have been developed for line segments (Chapter 6) and voltage regulators Chapter 7). The matrix equations for computing the voltages and currents at *Node n* as a function of the voltages and currents at *Node m* are given by:

$$[VLN_{ABC}] = [a_t] \cdot [VLN_{abc}] + [b_t] \cdot [I_{abc}] \quad (8.5)$$

$$[I_{ABC}] = [c_t] \cdot [VLN_{abc}] + [d_t] \cdot [I_{abc}] \quad (8.6)$$

The ladder iterative technique described in Chapter 10 requires that the voltages at *Node m* to be a function of the voltages at *Node n* and the currents at *Node m*. The required equation is

$$[VLN_{abc}] = [A_t] \cdot [VLN_{ABC}] - [B_t] \cdot [I_{abc}] \quad (8.7)$$

In Equations 8.5, 8.6, and 8.7, the matrices $[VLN_{ABC}]$ and $[VLN_{abc}]$ represent the line-to-neutral voltages for an ungrounded wye connection, or the line-to-ground voltages for a grounded wye connection. For a delta connection the voltage matrices represent equivalent line-to-neutral voltages. The current matrices represent the line currents regardless of the transformer winding connection.

8.3 The Delta–Grounded Wye Step-Down Connection

The delta–grounded wye step-down connection is a popular connection that is typically used in a distribution substation serving a four-wire wye feeder system. Another application of the connection is to provide service to a load that is primarily single-phase. Because of the wye connection, three single-phase circuits are available, thereby making it possible to balance the single-phase loading on the transformer bank.

Three single-phase transformers can be connected to a delta–grounded wye in a standard thirty-degree step-down connection as shown in Figure 8.2.

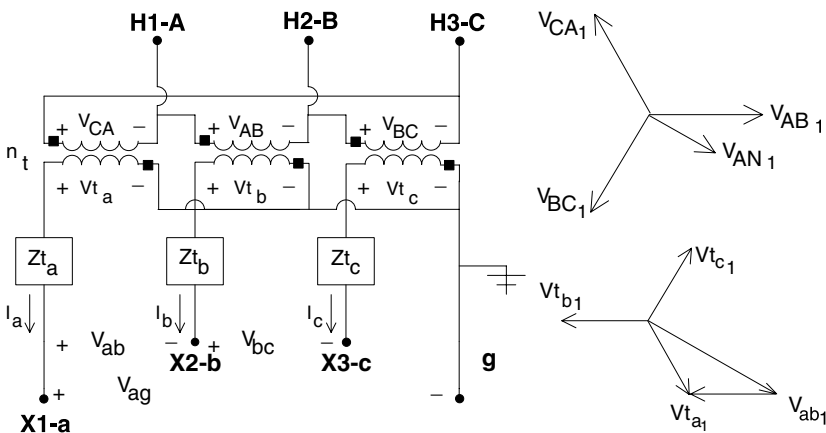


FIGURE 8.2
Standard delta–grounded wye connection with voltages.

8.3.1 Voltages

The positive sequence phasor diagrams of the voltages in Figure 8.2 show the relationships between the various positive sequence voltages. Care must be taken to observe the polarity marks on the individual transformer windings. In order to simplify the notation it is necessary to label the “ideal” voltages with voltage polarity markings as shown in Figure 8.2. Observing the polarity markings of the transformer windings, the voltage Vt_a will be 180 degrees out of phase with the voltage V_{CA} , and the voltage Vt_b will be 180 degrees out of phase with the voltage V_{AB} . Kirchhoff’s voltage law gives the line-to-line voltage between phases a and b as:

$$V_{ab} = Vt_a - Vt_b \quad (8.8)$$

The phasors of the positive sequence voltages of Equation 8.8 are shown in Figure 8.2.

The magnitude changes between the voltages can be defined in terms of the actual winding turns ratio (n_t) or the transformer ratio (a_t). With reference to Figure 8.2, these ratios are defined as follows:

$$n_t = \frac{VLL_{\text{Rated High Side}}}{VLN_{\text{Rated Low Side}}} \quad (8.9)$$

Applying Equation 8.9, the magnitude of the line-to-line voltage relative to the ideal transformer voltage is

$$|VLL| = n_t \cdot |Vt| \quad (8.10)$$

The magnitude of the positive sequence equivalent line-to-neutral voltage on the high-voltage side is given by:

$$|VLN| = \frac{|VLL|}{\sqrt{3}} = \frac{n_t}{\sqrt{3}} \cdot |Vt| = a_t |Vt| \quad (8.11)$$

where

$$a_t = \frac{n_t}{\sqrt{3}} = \frac{VLL_{\text{Rated High Side}}}{\sqrt{3} \cdot VLN_{\text{Rated Low Side}}} \quad (8.12)$$

With reference to Figure 8.2, the line-to-line voltages on the primary side of the transformer connection as a function of the ideal secondary side voltages

is given by:

$$\begin{bmatrix} V_{AB} \\ V_{BC} \\ V_{CA} \end{bmatrix} = \begin{bmatrix} 0 & -n_t & 0 \\ 0 & 0 & -n_t \\ -n_t & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} Vt_a \\ Vt_b \\ Vt_c \end{bmatrix} \quad (8.13)$$

In condensed form, Equation 8.13 is

$$[VLL_{ABC}] = [AV] \cdot [Vt_{abc}] \quad (8.14)$$

where

$$[AV] = \begin{bmatrix} 0 & -n_t & 0 \\ 0 & 0 & -n_t \\ -n_t & 0 & 0 \end{bmatrix} \quad (8.15)$$

Equation 8.14 gives the primary line-to-line voltages at *Node n* as functions of the ideal secondary voltages. However, what is needed is a relationship between equivalent line-to-neutral voltages at *Node n* and the ideal secondary voltages. The question is how is the equivalent line-to-neutral voltages determined knowing the line-to-line voltages? One approach is to apply the theory of symmetrical components.

The known line-to-line voltages are transformed to their sequence voltages by:

$$[VLL_{012}] = [A_s]^{-1} \cdot [VLL_{ABC}] \quad (8.16)$$

where

$$[A_s] = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a_s^2 & a_s \\ 1 & a_s & a_s^2 \end{bmatrix} \quad (8.17)$$

$$a_s = 1.0 / \underline{120}$$

By definition, the zero sequence line-to-line voltage is always zero. The relationship between the positive and negative sequence line-to-neutral and line-to-line voltages is known. These relationships in matrix form are given by:

$$\begin{bmatrix} VLN_0 \\ VLN_1 \\ VLN_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & t_s^* & 0 \\ 0 & 0 & t_s \end{bmatrix} \cdot \begin{bmatrix} VLL_0 \\ VLL_1 \\ VLL_2 \end{bmatrix} \quad (8.18)$$

$$[VLN_{012}] = [T] \cdot [VLL_{012}] \quad (8.19)$$

where

$$t_s = \frac{1}{\sqrt{3}} / \underline{30} \quad (8.20)$$

Since the zero sequence line-to-line voltage is zero, the (1,1) term of the matrix $[T]$ can be any value. For the purposes here, the (1,1) term is chosen to have a value of 1.0. Knowing the sequence line-to-neutral voltages, the equivalent line-to-neutral voltages can be determined.

The equivalent line-to-neutral voltages as functions of the sequence line-to-neutral voltages are

$$[VLN_{ABC}] = [A_s] \cdot [VLN_{012}] \quad (8.21)$$

Substitute Equation 8.19 into Equation 8.21:

$$[VLN_{ABC}] = [A_s] \cdot [T] \cdot [VLL_{012}] \quad (8.22)$$

Substitute Equation 8.16 into Equation 8.22:

$$[VLN_{ABC}] = [W] \cdot [VLL_{ABC}] \quad (8.23)$$

where

$$[W] = [A_s] \cdot [T] \cdot [A_s]^{-1} = \frac{1}{3} \cdot \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 2 \end{bmatrix} \quad (8.24)$$

Equation 8.23 provides a method of computing the equivalent line-to-neutral voltages from knowledge of the line-to-line voltages. This is an important relationship that will be used in a variety of ways as other three-phase transformer connections are studied.

To continue, Equation 8.14 can now be substituted into Equation 8.23:

$$[VLN_{ABC}] = [W] \cdot [AV] \cdot [Vt_{abc}] = [a_t] \cdot [Vt_{abc}] \quad (8.25)$$

where

$$[a_t] = [W] \cdot [AV] = \frac{-n_t}{3} \cdot \begin{bmatrix} 0 & 2 & 1 \\ 1 & 0 & 2 \\ 2 & 1 & 0 \end{bmatrix} \quad (8.26)$$

Equation 8.25 defines the generalized $[a]$ matrix for the delta-grounded wye step-down connection.

The ideal secondary voltages as functions of the secondary line-to-ground voltages and the secondary line currents are

$$[Vt_{abc}] = [VLG_{abc}] + [Zt_{abc}] \cdot [I_{abc}] \quad (8.27)$$

where

$$[Zt_{abc}] = \begin{bmatrix} Zt_a & 0 & 0 \\ 0 & Zt_b & 0 \\ 0 & 0 & Zt_c \end{bmatrix} \quad (8.28)$$

Notice in Equation 8.28 there is no restriction that the impedances of the three transformers be equal.

Substitute Equation 8.27 into Equation 8.25:

$$\begin{aligned} [VLN_{ABC}] &= [a_t] \cdot ([VLG_{abc}] + [Zt_{abc}] \cdot [I_{abc}]) \\ [VLN_{ABC}] &= [a_t] \cdot [VLG_{abc}] + [b_t] \cdot [I_{abc}] \end{aligned} \quad (8.29)$$

where $[b_t] = [a_t] \cdot [Zt_{abc}] = \frac{-n_t}{3} \cdot \begin{bmatrix} 0 & 2 \cdot Zt_b & Zt_c \\ Zt_a & 0 & 2 \cdot Zt_c \\ 2 \cdot Zt_a & Zt_b & 0 \end{bmatrix}$ (8.30)

The generalized matrices $[a]$ and $[b]$ have now been defined. The derivation of the generalized matrices $[A]$ and $[B]$ begins with solving Equation 8.14 for the ideal secondary voltages:

$$[Vt_{abc}] = [AV]^{-1} \cdot [VLL_{ABC}] \quad (8.31)$$

The line-to-line voltages as functions of the equivalent line-to-neutral voltages are

$$[VLL_{ABC}] = [D] \cdot [VLN_{ABC}] \quad (8.32)$$

where $[D] = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix}$ (8.33)

Substitute Equation 8.32 into Equation 8.31:

$$[Vt_{abc}] = [AV]^{-1} \cdot [D] \cdot [VLN_{ABC}] = [A_t] \cdot [VLN_{ABC}] \quad (8.34)$$

where $[A_t] = [AV]^{-1} \cdot [D] = \frac{1}{n_t} \cdot \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$ (8.35)

Substitute Equation 8.27 into Equation 8.34:

$$[VLG_{abc}] + [Zt_{abc}] \cdot [I_{abc}] = [A_t] \cdot [VLN_{ABC}] \quad (8.36)$$

Rearrange Equation 8.36:

$$[VLG_{abc}] = [A_t] \cdot [VLN_{ABC}] - [B_t] \cdot [I_{abc}] \quad (8.37)$$

where

$$[B_t] = [Zt_{abc}] = \begin{bmatrix} Zt_a & 0 & 0 \\ 0 & Zt_b & 0 \\ 0 & 0 & Zt_c \end{bmatrix} \quad (8.38)$$

Equations 8.29 and 8.37 are the generalized voltage equations for the step-down delta-grounded wye transformer. These equations are in exactly the same form as those derived in earlier chapters for line segments and step-voltage regulators.

8.3.2 Currents

The 30-degree connection specifies that the positive sequence current entering the H1 terminal will lead the positive sequence current leaving the X1 terminal by 30 degrees. Figure 8.3 shows the same connection as Figure 8.2, but with the currents instead of the voltages displayed. As with the voltages, the polarity marks on the transformer windings must be observed for the currents. For example, in Figure 8.3 the current I_a is entering the polarity mark on the low-voltage winding, so the current I_{AC} on the high-voltage winding

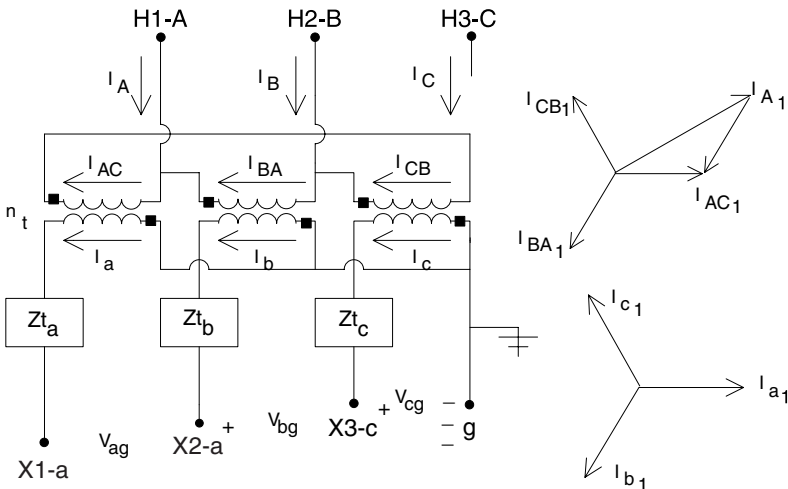


FIGURE 8.3
Delta-grounded wye connection with currents.

winding will be in phase with I_a . This relationship is shown in the phasor diagrams for positive sequence currents in Figure 8.3.

The line currents can be determined as functions of the delta currents by applying Kirchhoff's current law:

$$\begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} I_{AC} \\ I_{BA} \\ I_{CB} \end{bmatrix} \quad (8.39)$$

In condensed form, Equation 8.39 is

$$[I_{ABC}] = [D] \cdot [ID_{ABC}] \quad (8.40)$$

The matrix equation relating the delta primary currents to the secondary line currents is given by:

$$\begin{bmatrix} I_{AC} \\ I_{BA} \\ I_{CB} \end{bmatrix} = \frac{1}{n_t} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \quad (8.41)$$

$$[ID_{ABC}] = [AI] \cdot [I_{abc}] \quad (8.42)$$

Substitute Equation 8.42 into Equation 8.40:

$$[I_{ABC}] = [D] \cdot [AI] \cdot [I_{abc}] = [c_t] \cdot [VLG_{abc}] + [d_t] \cdot [I_{abc}] \quad (8.43)$$

where $[d_t] = [D] \cdot [AI] = \frac{1}{n_t} \cdot \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix}$ (8.44)

$$[c_t] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (8.45)$$

Equation 8.43 provides a direct method of computing the phase line currents at *Node n* knowing the phase line currents at *Node m*. Again, this equation

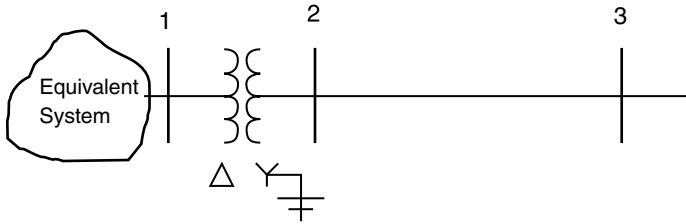


FIGURE 8.4
Example system.

is in the same form as that previously derived for three-phase line segments and three-phase step-voltage regulators.

The equations derived in this section are for the step-down connection. If the transformer bank is a step-up, the connection between transformer windings will be different, as will the definitions of the generalized matrices. The procedure for the derivation of the generalized matrices for the step-up connection will be the same as that developed in this section.

Example 8.1

In the Example System of Figure 8.4, an unbalanced constant impedance load is being served at the end of a one-mile section of a three-phase line. The line is being fed from a substation transformer rated 5000 kVA, 138 kV delta–12.47 kV grounded wye with a per-unit impedance of $0.085/\underline{85}$. The phase conductors of the line are 336,400 26/7 ACSR with a neutral conductor 4/0 ACSR. The configuration and computation of the phase impedance matrix is given in Example 4.1 From that example, the phase impedance matrix was computed to be

$$[Zline_{abc}] = \begin{bmatrix} 0.4576 + j1.0780 & 0.1560 + j0.5017 & 0.1535 + j0.3849 \\ 0.1560 + j0.5017 & 0.4666 + j1.0482 & 0.1580 + j0.4236 \\ 0.1535 + j0.3849 & 0.1580 + j0.4236 & 0.4615 + j1.0651 \end{bmatrix} \Omega/\text{mile}$$

The transformer impedance needs to be converted to per-unit referenced to the low-voltage side of the transformer. The base impedance is

$$Z_{\text{base}} = \frac{12.47^2 \cdot 1000}{5000} = 31.1$$

The transformer impedance referenced to the low-voltage side is

$$Zt = (0.085/\underline{85}) \cdot 31.3 = 0.2304 + j2.6335 \Omega$$

The transformer phase impedance matrix is

$$[Zt_{abc}] = \begin{bmatrix} 0.2304 + j2.6335 & 0 & 0 \\ 0 & 0.2304 + j2.6335 & 0 \\ 0 & 0 & 0.2304 + j2.6335 \end{bmatrix} \Omega$$

The unbalanced constant impedance load is connected in grounded wye. The load impedance matrix is specified to be

$$[Zload_{abc}] = \begin{bmatrix} 12 + j6 & 0 & 0 \\ 0 & 13 + j4 & 0 \\ 0 & 0 & 14 + j5 \end{bmatrix} \Omega$$

The unbalanced line-to-line voltages at Node 1 serving the substation transformer are given as:

$$[VLL_{ABC}] = \begin{bmatrix} 138,000/\underline{0} \\ 135,500/\underline{-115.5} \\ 145,959/\underline{123.1} \end{bmatrix} \text{ V}$$

(1) Determine the generalized matrices for the transformer.

$$\text{The transformer turns ratio is } n_t = \frac{kVLL_{\text{high}}}{kVLL_{\text{low}}} = \frac{138}{\frac{12.47}{\sqrt{3}}} = 19.1678$$

$$\text{The transformer ratio is } a_t = \frac{kVLL_{\text{high}}}{kVLL_{\text{low}}} = \frac{138}{12.47} = 11.0666$$

From Equation 8.26:

$$[a_t] = \frac{-19.1678}{3} \cdot \begin{bmatrix} 0 & 2 & 1 \\ 1 & 0 & 2 \\ 2 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -12.7786 & -6.3893 \\ -6.3893 & 0 & -12.7786 \\ -12.7786 & -6.3893 & 0 \end{bmatrix}$$

From Equation 8.30:

$$[b_t] = \frac{-19.1678}{3} \cdot \begin{bmatrix} 0 & 2 \cdot (0.2304 + j2.6335) & (0.2304 + j2.6335) \\ (0.2304 + j2.6335) & 0 & 2 \cdot (0.2304 + j2.6335) \\ 2 \cdot (0.2304 + j2.6335) & (0.2304 + j2.6335) & 0 \end{bmatrix}$$

$$[b_t] = \begin{bmatrix} 0 & -2.9441 - j33.6518 & -1.4721 - j16.8259 \\ -1.4721 - j16.8259 & 0 & -2.9441 - j33.6518 \\ -2.9441 - j33.6518 & -1.4721 - j16.8259 & 0 \end{bmatrix}$$

From Equation 8.44:

$$[d_t] = \frac{1}{19.1678} \cdot \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.0522 & -0.0522 & 0 \\ 0 & 0.0522 & -0.0522 \\ -0.0522 & 0 & 0.0522 \end{bmatrix}$$

From Equation 8.35:

$$[A_t] = \frac{1}{19.1678} \cdot \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 0.0522 & 0 & -0.0522 \\ -0.0522 & 0.0522 & 0 \\ 0 & -0.0522 & 0.0522 \end{bmatrix}$$

From Equation 8.38:

$$[B_t] = [Zt_{abc}] = \begin{bmatrix} 0.2304 + j2.6335 & 0 & 0 \\ 0 & 0.2304 + j2.6335 & 0 \\ 0 & 0 & 0.2304 + j2.6335 \end{bmatrix}$$

- (2) Given the line-to-line voltages at *Node 1*, determine the ideal transformer voltages. From Equation 8.15:

$$[AV] = \begin{bmatrix} 0 & -n_t & 0 \\ 0 & 0 & -n_t \\ -n_t & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -19.1678 & 0 \\ 0 & 0 & -19.1678 \\ -19.1678 & 0 & 0 \end{bmatrix}$$

$$[Vt_{abc}] = [AV]^{-1} \cdot [VLL_{ABC}] = \begin{bmatrix} 7614.8 / -56.9 \\ 7199.6 / 180 \\ 7069 / 64.5 \end{bmatrix} \text{ V}$$

(3) Determine the load currents.

Kirchhoff's voltage law gives:

$$[Vt_{abc}] = ([Zt_{abc}] + [Zline_{abc}] + [Zload_{abc}]) \cdot [I_{abc}] = [Ztotal_{abc}] \cdot [I_{abc}]$$

$$[Ztotal_{abc}] = \begin{bmatrix} 12.688 + j9.7115 & 0.156 + j0.5017 & 0.1535 + j0.3849 \\ 0.156 + j0.5017 & 13.697 + j7.6817 & 0.158 + j0.4236 \\ 0.1535 + j0.3849 & 0.158 + j0.4236 & 14.6919 + j8.6986 \end{bmatrix} \Omega$$

The line currents can now be computed:

$$[I_{abc}] = [Ztotal_{abc}]^{-1} \cdot [Vt_{abc}] = \begin{bmatrix} 484.1 / -93.0 \\ 470.7 / 151.5 \\ 425.4 / 34.8 \end{bmatrix} \text{ A}$$

(4) Determine the line-to-ground voltages at the load and at Node 2.

$$[Vload_{abc}] = [Zload_{abc}] \cdot [I_{abc}] = \begin{bmatrix} 6494.8 / -66.4 \\ 6401.6 / 168.6 \\ 6323.5 / 54.4 \end{bmatrix} \text{ V}$$

$$[VLG_{abc}] = [Vload_{abc}] + [Zline_{abc}] \cdot [I_{abc}] = \begin{bmatrix} 6842.2 / -65.0 \\ 6594.5 / 171.0 \\ 6594.9 / 56.3 \end{bmatrix} \text{ V}$$

(5) Using the generalized matrices, determine the equivalent line-to-neutral voltages and the line-to-line voltages at Node 1.

$$[VLN_{ABC}] = [a_t] \cdot [VLG_{abc}] + [b_t] \cdot [I_{abc}] = \begin{bmatrix} 83,224 / -29.3 \\ 77,103 / -148.1 \\ 81,843 / 95.0 \end{bmatrix} \text{ V}$$

$$[VLL_{ABC}] = [D] \cdot [VLN_{ABC}] = \begin{bmatrix} 138,000 / 0 \\ 135,500 / -115.5 \\ 145,959 / 123.1 \end{bmatrix} \text{ V}$$

It is always comforting to be able to work back and compute what was initially given. In this case, the line-to-line voltages at Node 1 have been computed and the same values result that were given at the start of the problem.

- (6) Use the reverse equation to verify that the line-to-ground voltages at Node 2 can be computed knowing the equivalent line-to-neutral voltages at Node 1 and the currents leaving Node 2.

$$[VLG_{abc}] = [A_t] \cdot [VLN_{ABC}] - [B_t] \cdot [I_{abc}] = \begin{bmatrix} 6842.2 / -65.0 \\ 6594.5 / 171.0 \\ 6494.9 / 56.3 \end{bmatrix} \text{ V}$$

These are the same values for the line-to-ground voltages at Node 2 that were determined working from the load towards the source.

Example 8.1 has demonstrated the application of the generalized constants. The example also provides verification that the same voltages and currents result when working from the load toward the source or from the source toward the load. This will become very important for the iterative technique to be developed in the Chapter 10.

8.4 The Ungrounded Wye-Delta Step-Down Connection

Three single-phase transformers can be connected in a wye-delta connection. The neutral of the wye can be grounded or ungrounded. The grounded wye connection is rarely used because:

- The grounded wye provides a path for zero sequence currents for line-to-ground faults upstream from the transformer bank. This causes the transformers to be susceptible to burnouts on the upstream faults.
- If one phase of the primary circuit is opened, the transformer bank will continue to provide three-phase service by operating as an open wye-open delta bank. However, the two remaining transformers may be subject to an overload condition leading to burnout.

The most common connection is the ungrounded wye-delta. This connection is typically used to provide service to a combination single-phase “lighting” load and a three-phase “power” load such as an induction motor. The generalized constants for only the ungrounded wye-delta transformer connection

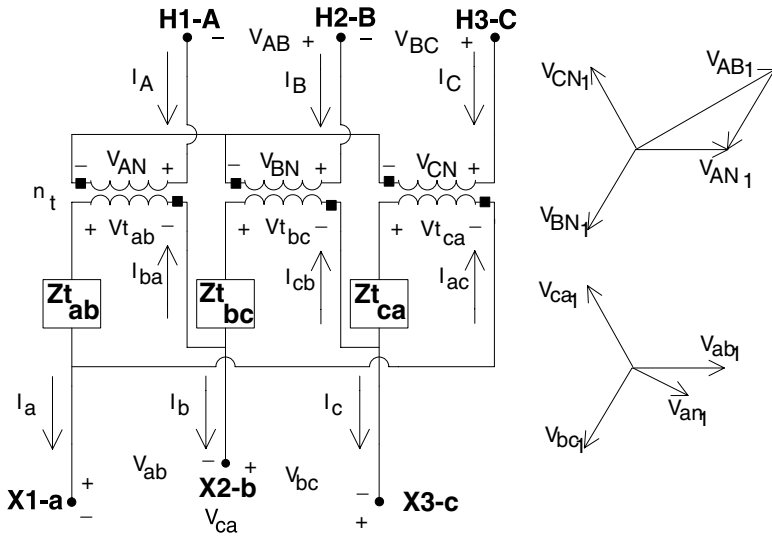


FIGURE 8.5
Standard ungrounded wye-delta connection.

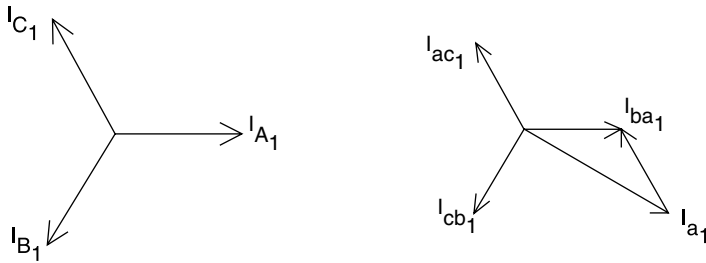


FIGURE 8.6
Positive sequence current phasors.

will be developed following the same procedure as was used for the delta-grounded wye.

Three single-phase transformers can be connected in an ungrounded-wye standard 30-degree connection as shown in Figure 8.5. The voltage phasor diagrams in Figure 8.5 illustrate that the high side positive sequence line-to-line voltage leads the low side positive sequence line-to-line voltage by 30 degrees. Also, the same phase shift occurs between the high side line-to-neutral voltage and the low side equivalent line-to-neutral voltage. The negative sequence phase shift is such that the high side negative sequence voltage will lag the low side negative sequence voltage by 30 degrees.

The positive sequence current phasor diagrams for the connection in Figure 8.5 are shown in Figure 8.6. Figure 8.6 illustrates that the positive

sequence line current on the high side of the transformer (*Node n*) leads the low side line current (*Node m*) by 30 degrees. It can also be shown that the negative sequence high side line current will lag the negative sequence low side line current by 30 degrees.

The definition for the “turns ratio n_t ” will be the same as Equation 8.9, with the exception that the numerator will be the line-to-neutral voltage and the denominator will be the line-to-line voltage. The “transformer ratio a_t ,” as given in Equation 8.12, will apply for this connection. It should be noted in Figure 8.5 that the ideal low side transformer voltages for this connection will be line-to-line voltages. Also, the ideal low side currents are the currents flowing inside the delta.

The basic ideal transformer voltage and current equations as functions of the turns ratio are

$$\begin{bmatrix} V_{AN} \\ V_{BN} \\ V_{CN} \end{bmatrix} = \begin{bmatrix} n_t & 0 & 0 \\ 0 & n_t & 0 \\ 0 & 0 & n_t \end{bmatrix} \cdot \begin{bmatrix} Vt_{ab} \\ Vt_{bc} \\ Vt_{ca} \end{bmatrix} \quad (8.46)$$

where

$$n_t = \frac{VLN_{\text{Rated High Side}}}{VLL_{\text{Rated Low Side}}}$$

$$[VLN_{ABC}] = [AV] \cdot [Vt_{abc}] \quad (8.47)$$

$$\begin{bmatrix} I_{ba} \\ I_{cb} \\ I_{ac} \end{bmatrix} = \begin{bmatrix} n_t & 0 & 0 \\ 0 & n_t & 0 \\ 0 & 0 & n_t \end{bmatrix} \cdot \begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix} \quad (8.48)$$

$$[ID_{abc}] = [AI] \cdot [I_{ABC}] \quad (8.49)$$

Solving Equation 8.47 for the ideal delta transformer voltages:

$$[Vt_{abc}] = [AV]^{-1} \cdot [VLN_{ABC}] \quad (8.50)$$

The line-to-line voltages at *Node m* as functions of the ideal transformer voltages and the delta currents are given by:

$$\begin{bmatrix} V_{ab} \\ V_{bc} \\ V_{ca} \end{bmatrix} = \begin{bmatrix} Vt_{ab} \\ Vt_{bc} \\ Vt_{ca} \end{bmatrix} - \begin{bmatrix} Zt_{ab} & 0 & 0 \\ 0 & Zt_{ac} & 0 \\ 0 & 0 & Zt_{ca} \end{bmatrix} \cdot \begin{bmatrix} ID_{ba} \\ ID_{cb} \\ ID_{ac} \end{bmatrix} \quad (8.51)$$

$$[VLL_{abc}] = [Vt_{abc}] - [Zt_{abc}] \cdot [ID_{abc}] \quad (8.52)$$

Substitute Equations 8.49 and 8.50 into Equation 8.52:

$$[VLL_{abc}] = [AV]^{-1} \cdot [VLN_{ABC}] - [ZNt_{abc}] \cdot [I_{ABC}] \quad (8.53)$$

$$\text{where} \quad [ZNt_{abc}] = [Zt_{abc}] \cdot [AI] = \begin{bmatrix} n_t \cdot Zt_{ab} & 0 & 0 \\ 0 & n_t \cdot Zt_{bc} & 0 \\ 0 & 0 & n_t \cdot Zt_{ca} \end{bmatrix} \quad (8.54)$$

The line currents on the delta side of the transformer bank as functions of the wye transformer currents are given by:

$$[I_{abc}] = [DI] \cdot [ID_{abc}] \quad (8.55)$$

$$\text{where} \quad [DI] = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \quad (8.56)$$

Substitute Equation 8.49 into Equation 8.55:

$$[I_{abc}] = [DI] \cdot [AI] \cdot [I_{ABC}] = [DY] \cdot [I_{ABC}] \quad (8.57)$$

$$\text{where} \quad [DY] = [DI] \cdot [AI] = \begin{bmatrix} n_t & 0 & -n_t \\ -n_t & n_t & 0 \\ 0 & -n_t & n_t \end{bmatrix} \quad (8.58)$$

Because the matrix $[DY]$ is singular, it is not possible to use Equation 8.57 to develop an equation relating the wye side line currents at *Node n* to the delta side line currents at *Node m*. In order to develop the necessary matrix equation, three independent equations must be written. Two independent KCL equations at the vertices of the delta can be used. Because there is no path for the high side currents to flow to ground, they must sum to zero and, therefore, so must the delta currents sum to zero. This provides the third independent equation. The resulting three independent equations in matrix form are given by:

$$\begin{bmatrix} I_a \\ I_b \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} I_{ba} \\ I_{cb} \\ I_{ac} \end{bmatrix} \quad (8.59)$$

Solving Equation 8.59 for the delta currents:

$$\begin{bmatrix} I_{ba} \\ I_{cb} \\ I_{ac} \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}^{-1} \cdot \begin{bmatrix} I_a \\ I_b \\ 0 \end{bmatrix} = \frac{1}{3} \cdot \begin{bmatrix} 1 & -1 & 1 \\ 1 & 2 & 1 \\ -2 & -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} I_a \\ I_b \\ 0 \end{bmatrix} \quad (8.60)$$

$$[ID_{abc}] = [L0] \cdot [I_{ab0}] \quad (8.61)$$

Equation 8.61 can be modified to include the Phase c current by setting the third column of the $[L0]$ matrix to zero:

$$\begin{bmatrix} I_{ba} \\ I_{cb} \\ I_{ac} \end{bmatrix} = \frac{1}{3} \cdot \begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 0 \\ -2 & -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \quad (8.62)$$

$$[ID_{abc}] = [L] \cdot [I_{abc}] \quad (8.63)$$

Solve Equation 8.49 for $[I_{ABC}]$ and substitute into Equation 7.63:

$$[I_{ABC}] = [AI]^{-1} \cdot [L] \cdot [I_{abc}] = [d_t] \cdot [I_{abc}] \quad (8.64)$$

where

$$[d_t] = [AI]^{-1} \cdot [L] = \frac{1}{3 \cdot n_T} \cdot \begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 0 \\ -2 & -1 & 0 \end{bmatrix} \quad (8.65)$$

Equation 8.65 defines the generalized constant matrix $[d_t]$ for the ungrounded wye–delta transformer connection. In the process of the derivation, a very convenient equation (8.62) evolved that can be used anytime the currents in a delta need to be determined when the line currents are known. However, it must be understood that this equation will only work when the delta currents sum to zero, which means an ungrounded neutral on the primary. The generalized matrices $[a_i]$ and $[b_i]$ can now be developed. Solve Equation 8.53 for $[VLN_{ABC}]$:

$$[VLN_{ABC}] = [AV] \cdot [VLL_{abc}] + [AV] \cdot [ZNt_{abc}] \cdot [I_{ABC}] \quad (8.66)$$

Substitute Equation 8.64 into Equation 8.66:

$$\begin{aligned}
 [VLN_{ABC}] &= [AV] \cdot [VLL_{abc}] + [AV] \cdot [ZNT_{abc}] \cdot [d_t] \cdot [I_{abc}] \\
 \text{but } [VLL_{abc}] &= [D] \cdot [VLN_{abc}] \\
 [VLN_{ABC}] &= [AV] \cdot [D] \cdot [VLN_{abc}] + [AV] \cdot [ZNT_{abc}] \cdot [d_t] \cdot [I_{abc}] \\
 [VLN_{ABC}] &= [a_t] \cdot [VLN_{abc}] + [b_t] \cdot [I_{abc}]
 \end{aligned} \tag{8.67}$$

$$\text{where } [a_t] = [AV] \cdot [D] = n_t \cdot \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix} \tag{8.68}$$

$$[b_t] = [AV] \cdot [ZNT_{abc}] \cdot [d_t] = \frac{n_t}{3} \cdot \begin{bmatrix} Zt_{ab} & -Zt_{ab} & 0 \\ Zt_{bc} & 2 \cdot Zt_{bc} & 0 \\ -2 \cdot Zt_{ca} & -Zt_{ca} & 0 \end{bmatrix} \tag{8.69}$$

The generalized constant matrices have been developed for computing voltages and currents from the load toward the source. The reverse generalized matrices can be developed by referring back to Equation 8.53, which is repeated as Equation 8.70 for convenience:

$$[VLL_{abc}] = [AV]^{-1} \cdot [VLN_{ABC}] - [ZNT_{abc}] \cdot [I_{ABC}] \tag{8.70}$$

Equation 8.23 is used to compute the equivalent line-to-neutral voltages as functions of the line-to-line voltages:

$$[VLN_{abc}] = [W] \cdot [VLL_{abc}] \tag{8.71}$$

Substitute Equations 8.70 and 8.64 into Equation 8.71:

$$\begin{aligned}
 [VLN_{abc}] &= [W] \cdot [AV]^{-1} \cdot [VLN_{ABC}] - [W] \cdot [ZNT_{abc}] \cdot [d_t] \cdot [I_{abc}] \\
 [VLN_{abc}] &= [A_t] \cdot [VLN_{ABC}] - [B_t] \cdot [I_{abc}]
 \end{aligned} \tag{8.72}$$

$$\text{where } [A_t] = [W] \cdot [AV]^{-1} = \frac{1}{3 \cdot n_t} \cdot \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 2 \end{bmatrix} \tag{8.73}$$

$$[B_t] = [W] \cdot [ZNT_{abc}] \cdot [d_t] = \frac{1}{9} \cdot \begin{bmatrix} 2 \cdot Zt_{ab} + Zt_{bc} & 2 \cdot Zt_{bc} - 2 \cdot Zt_{ab} & 0 \\ 2 \cdot Zt_{bc} - 2 \cdot Zt_{ca} & 4 \cdot Zt_{bc} - Zt_{ca} & 0 \\ Zt_{ab} - 4 \cdot Zt_{ca} & -Zt_{ab} - 2 \cdot Zt_{ca} & 0 \end{bmatrix} \tag{8.74}$$

The generalized matrices have been developed for the ungrounded wye–delta transformer connection. The derivation has applied basic circuit theory and the basic theories of transformers. The end result of the derivations is to provide an easy method of analyzing the operating characteristics of the transformer connection. Example 8.2 will demonstrate the application of the generalized matrices for this transformer connection.

Example 8.2

Figure 8.7 shows three single-phase transformers in an ungrounded wye–delta connection serving a combination single-phase and three-phase load in a delta connection. The voltages at the load are balanced three-phase of 240 V line-to-line. The net loading by phase is

$$S_{ab} = 100 \text{ kVA at } 0.9 \text{ lagging power factor}$$

$$S_{bc} = S_{ca} = 50 \text{ kVA at } 0.8 \text{ lagging power factor}$$

In sizing the three transformers it will be assumed that the lighting transformer serves all of the single-phase load and a third of the three-phase load, while the two power transformers will each serve a third of the three-phase load. With those assumptions, the selected ratings of the transformers are

Lighting: 100 kVA, 7200–240 V, $Z = 0.01 + j0.04$ per-unit

Power: 50 kVA, 7200–240 V, $Z = 0.015 + j0.035$ per-unit

Determine the following:

1. The currents in the load
2. The secondary line currents
3. The equivalent line-to-neutral secondary voltages
4. The primary line-to-neutral and line-to-line voltages
5. The primary line currents

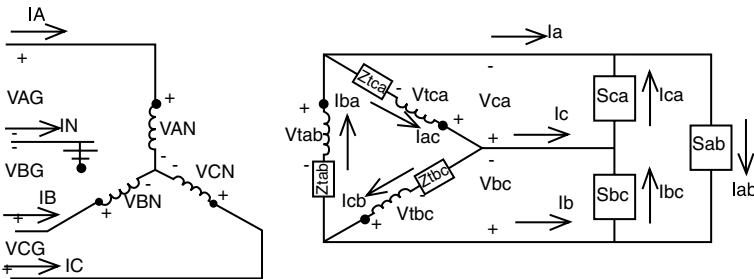


FIGURE 8.7

Ungrounded wye–delta serving an unbalanced load.

Before the analysis can start, the transformer impedances must be converted to actual values in ohms and located inside the delta-connected secondary windings.

Lighting transformer:

$$Z_{\text{base}} = \frac{0.24^2 \cdot 1000}{100} = 0.576$$

$$Zt_{ab} = (0.01 + j0.4) \cdot 0.576 = 0.0058 + j0.23 \Omega$$

Power transformers:

$$Z_{\text{base}} = \frac{0.24^2 \cdot 1000}{50} = 1.152$$

$$Zt_{bc} = Zt_{ca} = (0.015 + j0.35) \cdot 1.152 = 0.0173 + j0.0403 \Omega$$

The transformer impedance matrix can now be defined:

$$[Zt_{abc}] = \begin{bmatrix} 0.0058 + j0.23 & 0 & 0 \\ 0 & 0.017 + j0.0403 & 0 \\ 0 & 0 & 0.0173 + j0.0403 \end{bmatrix} \Omega$$

The turns ratio of the transformers is: $n_t = \frac{7200}{240} = 30$

Define all of the matrices:

$$[W] = \frac{1}{3} \cdot \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 2 \end{bmatrix} \quad [D] = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix} \quad [DI] = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$[a_t] = n_t \cdot \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 30 & -30 & 0 \\ 0 & 30 & -30 \\ -30 & 0 & 30 \end{bmatrix}$$

$$[b_t] = \frac{n_t}{3} \cdot \begin{bmatrix} Zt_{ab} & -Zt_{ab} & 0 \\ Zt_{bc} & 2 \cdot Zt_{bc} & 0 \\ -2 \cdot Zt_{ca} & -Zt_{ca} & 0 \end{bmatrix} = \begin{bmatrix} 0.576 + j0.2304 & -0.576 - j0.2304 & 0 \\ 0.1728 + j0.4032 & 0.3456 + j0.8064 & 0 \\ -0.3456 + j0.8064 & -0.1728 - j0.4032 & 0 \end{bmatrix}$$

$$[c_t] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[d_t] = \frac{1}{3 \cdot n_T} \cdot \begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 0 \\ -2 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 0.0111 & -0.0111 & 0 \\ 0.0111 & 0.0222 & 0 \\ -0.0222 & -0.0111 & 0 \end{bmatrix}$$

$$[A_t] = \frac{1}{3 \cdot n_T} \cdot \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 0.0222 & 0.0111 & 0 \\ 0 & 0.0222 & 0.0111 \\ 0.0111 & 0 & 0.0222 \end{bmatrix}$$

$$[B_t] = \frac{1}{9} \cdot \begin{bmatrix} 2 \cdot Zt_{ab} + Zt_{bc} & 2 \cdot Zt_{bc} - 2 \cdot Zt_{ab} & 0 \\ 2 \cdot Zt_{bc} - 2 \cdot Zt_{ca} & 4 \cdot Zt_{bc} - Zt_{ca} & 0 \\ Zt_{ab} - 4 \cdot Zt_{ca} & -Zt_{ab} - 2 \cdot Zt_{ca} & 0 \end{bmatrix}$$

$$[B_t] = \begin{bmatrix} 0.0032 + j0.0096 & 0.0026 + j0.0038 & 0 \\ 0 & 0.0058 + j0.0134 & 0 \\ -0.007 - j0.0154 & -0.0045 - j0.0115 & 0 \end{bmatrix}$$

Define the line-to-line load voltages:

$$[VLL_{abc}] = \begin{bmatrix} 240/\underline{0} \\ 240/\underline{-120} \\ 240/\underline{120} \end{bmatrix} \text{ V}$$

Define the loads:

$$[SD_{abc}] = \begin{bmatrix} 100/\cos^{-1}(0.9) \\ 50/\cos^{-1}(0.8) \\ 50/\cos^{-1}(0.8) \end{bmatrix} = \begin{bmatrix} 90 + j43.589 \\ 40 + j30 \\ 40 + j30 \end{bmatrix} \text{ kVA}$$

Calculate the delta load currents:

$$ID_i = \left(\frac{SD_i \cdot 1000}{VLL_{abc_i}} \right)^* \text{ A}$$

$$[ID_{abc}] = \begin{bmatrix} I_{ab} \\ I_{bc} \\ I_{ca} \end{bmatrix} = \begin{bmatrix} 416.7/\underline{-25.84} \\ 208.3/\underline{-156.87} \\ 208.3/\underline{83.13} \end{bmatrix} \text{ A}$$

Compute the secondary line currents:

$$[I_{abc}] = [DI] \cdot [ID_{abc}] = \begin{bmatrix} 522.9 / -47.97 \\ 575.3 / -119.06 \\ 360.8 / 53.13 \end{bmatrix} \text{ A}$$

Compute the equivalent secondary line-to-neutral voltages:

$$[VLN_{abc}] = [W] \cdot [VLL_{abc}] = \begin{bmatrix} 138.56 / -30 \\ 138.56 / -150 \\ 138.56 / 90 \end{bmatrix} \text{ A}$$

Use the generalized constant matrices to compute the primary line-to-neutral voltages and line-to-line voltages:

$$[VLN_{ABC}] = [a_t] \cdot [VLN_{abc}] + [b_t] \cdot [I_{abc}] = \begin{bmatrix} 7367.6 / 1.4 \\ 7532.3 / -119.1 \\ 7406.2 / 121.7 \end{bmatrix} \text{ V}$$

$$[VLL_{ABC}] = \frac{[D] \cdot [VLN_{ABC}]}{1000} = \begin{bmatrix} 12.94 / 31.54 \\ 12.88 / -88.95 \\ 12.81 / 151.50 \end{bmatrix} \text{ kV}$$

The high primary line currents are

$$[I_{ABC}] = [d_t] \cdot [I_{abc}] = \begin{bmatrix} 11.54 / -28.04 \\ 8.95 / -166.43 \\ 7.68 / 101.16 \end{bmatrix} \text{ A}$$

It is interesting to compute the operating kVA of the three transformers. Taking the product of the transformer voltage times the conjugate of the current gives the operating kVA of each transformer:

$$ST_i = \frac{VLN_{ABC_i} \cdot (I_{ABC_i})^*}{1000} = \begin{bmatrix} 85.02 / 29.46 \\ 67.42 / 47.37 \\ 56.80 / 20.58 \end{bmatrix} \text{ kVA}$$

The operating power factors of the three transformers are

$$[PF] = \begin{bmatrix} \cos(29.46) \\ \cos(47.37) \\ \cos(20.58) \end{bmatrix} = \begin{bmatrix} 87.1 \\ 67.7 \\ 93.6 \end{bmatrix} \% \text{ lagging}$$

Note that the operating kVAs do not match very closely the rated kVAs of the three transformers. In particular, the lighting transformer did not serve the total load of 100 kVA that is directly connected its terminals. The lighting transformer is operating below rated kVA while the two power transformers are overloaded. In fact, the transformer connected to Phase B is operating 35% above rated kVA. Because of this overload, the ratings of the three transformers should be changed so that the two power transformers are rated 75 kVA. Finally, the operating power factors of the three transformers bear little resemblance to the load power factors.

Example 8.2 demonstrates how the generalized constant matrices can be used to determine the operating characteristics of the transformers. In addition, the example shows that the obvious selection of transformer ratings will lead to an overload condition on the two power transformers. The beauty in this is that if the generalized constant matrices have been applied in a computer program, it is a simple task to change the transformer kVA ratings and be assured that none of the transformers will be operating in an overload condition.

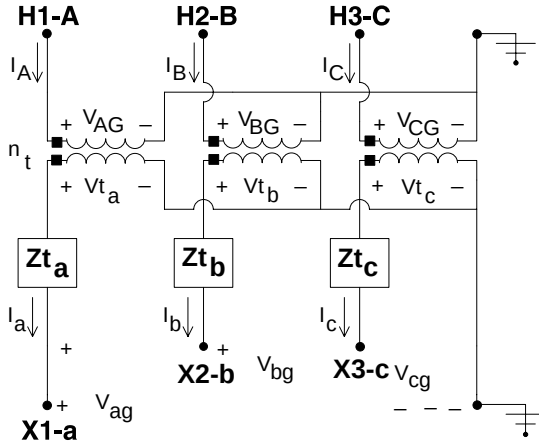
8.5 The Grounded Wye–Grounded Wye Connection

The grounded wye–grounded wye connection is primarily used to supply single-phase and three-phase loads on four-wire multigrounded systems. The grounded wye–grounded wye connection is shown in Figure 8.8. Unlike the delta–wye and wye–delta connections, there is no phase shift between the voltages and the currents on the two sides of the bank. This makes the derivation of the generalized constant matrices much easier.

With reference to Figure 8.8, the ideal transformer voltages on the secondary windings can be computed by:

$$\begin{bmatrix} Vt_a \\ Vt_b \\ Vt_c \end{bmatrix} = \begin{bmatrix} V_{ag} \\ V_{bg} \\ V_{cg} \end{bmatrix} + \begin{bmatrix} Zt_a & 0 & 0 \\ 0 & Zt_b & 0 \\ 0 & 0 & Zt_c \end{bmatrix} \cdot \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \quad (8.75)$$

$$[Vt_{abc}] = [VLG_{abc}] + [Zt_{abc}] \cdot [I_{abc}] \quad (8.76)$$

**FIGURE 8.8**

Grounded wye-grounded wye connection.

The line-to-ground voltages on the primary side are related to the ideal transformer voltages by the turns ratio:

$$\begin{bmatrix} V_{AG} \\ V_{BG} \\ V_{CG} \end{bmatrix} = \begin{bmatrix} n_t & 0 & 0 \\ 0 & n_t & 0 \\ 0 & 0 & n_t \end{bmatrix} \cdot \begin{bmatrix} V_{t_a} \\ V_{t_b} \\ V_{t_c} \end{bmatrix} \quad (8.77)$$

$$[VLG_{ABC}] = [AV] \cdot [V_{t_{abc}}] \quad (8.78)$$

Substitute Equation 8.76 into Equation 8.78:

$$[VLG_{ABC}] = [AV] \cdot [VLG_{abc}] + [AV] \cdot [Z_{t_{abc}}] \cdot [I_{abc}] \quad (8.79)$$

Equation 8.79 is in the generalized form with the $[a]$ and $[b]$ matrices defined by:

$$[a_t] = [AV] = \begin{bmatrix} n_t & 0 & 0 \\ 0 & n_t & 0 \\ 0 & 0 & n_t \end{bmatrix} \quad (8.80)$$

$$[b_t] = [AV] \cdot [Z_{t_{abc}}] = \begin{bmatrix} n_t \cdot Z_{t_a} & 0 & 0 \\ 0 & n_t \cdot Z_{t_b} & 0 \\ 0 & 0 & n_t \cdot Z_{t_c} \end{bmatrix} \quad (8.81)$$

The primary line currents as a function of the secondary line currents are given by:

$$[I_{ABC}] = [d_t] \cdot [I_{abc}] \quad (8.82)$$

where

$$[d_t] = \begin{bmatrix} \frac{1}{n_t} & 0 & 0 \\ 0 & \frac{1}{n_t} & 0 \\ 0 & 0 & \frac{1}{n_t} \end{bmatrix} \quad (8.83)$$

The reverse equation is determined solving Equation 8.79 for the secondary line-to-ground voltages:

$$[VLG_{abc}] = [AV]^{-1} \cdot [VLG_{ABC}] - [Zt_{abc}] \cdot [I_{abc}] \quad (8.84)$$

Constant matrices for the reverse equation are given by:

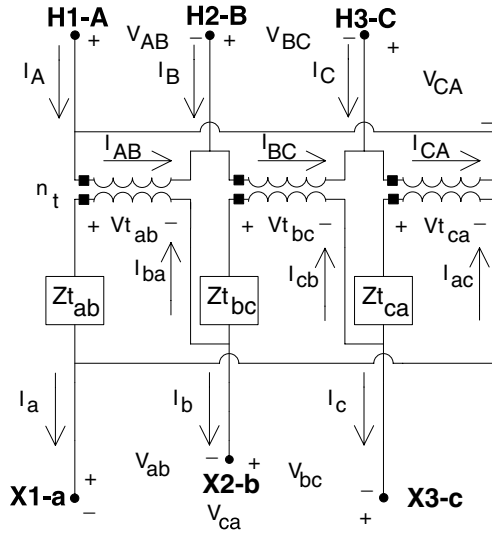
$$[A_t] = [AV]^{-1} = \begin{bmatrix} \frac{1}{n_t} & 0 & 0 \\ 0 & \frac{1}{n_t} & 0 \\ 0 & 0 & \frac{1}{n_t} \end{bmatrix} \quad (8.85)$$

$$[B_t] = \begin{bmatrix} Zt_a & 0 & 0 \\ 0 & Zt_b & 0 \\ 0 & 0 & Zt_c \end{bmatrix} \quad (8.86)$$

The modeling and analysis of the grounded wye-grounded wye connection does not present any problems. Without the phase shift and a closed delta connection, there is a direct relationship between the primary and secondary voltages and currents, as has been demonstrated in the derivation of the generalized constant matrices.

8.6 The Delta-Delta Connection

The delta-delta connection is primarily used on three-wire delta systems to provide service to a three-phase load or a combination of three-phase and single-phase loads. Three single-phase transformers connected in a delta-delta

**FIGURE 8.9**

Delta-delta connection.

are shown in Figure 8.9. The basic ideal transformer voltage and current equations as functions of the turns ratio are

$$\begin{bmatrix} VLL_{AB} \\ VLL_{BC} \\ VLL_{CA} \end{bmatrix} = \begin{bmatrix} n_t & 0 & 0 \\ 0 & n_t & 0 \\ 0 & 0 & n_t \end{bmatrix} \cdot \begin{bmatrix} Vt_{ab} \\ Vt_{bc} \\ Vt_{ca} \end{bmatrix} \quad (8.87)$$

$$[VLL_{ABC}] = [AV] \cdot [Vt_{abc}] \quad (8.88)$$

$$\begin{bmatrix} I_{ba} \\ I_{cb} \\ I_{ca} \end{bmatrix} = \begin{bmatrix} n_t & 0 & 0 \\ 0 & n_t & 0 \\ 0 & 0 & n_t \end{bmatrix} \cdot \begin{bmatrix} I_{AB} \\ I_{BC} \\ I_{CA} \end{bmatrix} \quad (8.89)$$

$$[ID_{abc}] = [AI] \cdot [ID_{ABC}] \quad (8.90)$$

where

$$n_t = \frac{VLL_{\text{Rated High Side}}}{VLL_{\text{Rated Low Side}}} \quad (8.91)$$

Solve Equation 8.90 for the source side delta currents:

$$[ID_{ABC}] = [AI]^{-1} \cdot [ID_{abc}] \quad (8.92)$$

The line currents as functions of the delta currents on the source side are given by:

$$\begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} I_{AB} \\ I_{BC} \\ I_{CA} \end{bmatrix} \quad (8.93)$$

$$[I_{ABC}] = [DI] \cdot [ID_{ABC}] \quad (8.94)$$

Substitute Equation 8.92 into Equation 8.94:

$$[I_{ABC}] = [DI] \cdot [AI]^{-1} \cdot [ID_{abc}] \quad (8.95)$$

Since $[AI]$ is a diagonal matrix, Equation 8.95 can be rewritten as:

$$[I_{ABC}] = [AI]^{-1} \cdot [DI] \cdot [ID_{abc}] \quad (8.96)$$

Determine the load side line currents as functions of the load side delta currents:

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} I_{ba} \\ I_{cb} \\ I_{ac} \end{bmatrix} \quad (8.97)$$

$$[I_{abc}] = [DI] \cdot [ID_{abc}] \quad (8.98)$$

Applying Equation 8.98, Equation 8.96 becomes:

$$[I_{ABC}] = [AI]^{-1} \cdot [I_{abc}] \quad (8.99)$$

Turn Equation 8.99 around to solve for the load side line currents as functions of the source side line currents:

$$[I_{abc}] = [AI] \cdot [I_{ABC}] \quad (8.100)$$

Equations 8.99 and 8.100 merely demonstrate that the line currents on the two sides of the transformer are in phase and differ only by the turns ratio of the transformer windings. In the per-unit system, the per-unit line currents on the two sides of the transformer are exactly equal.

The ideal delta voltages on the load side as functions of the line-to-line voltages, the delta currents, and the transformer impedances are given by:

$$[Vt_{abc}] = [VLL_{abc}] + [Zt_{abc}] \cdot [ID_{abc}] \quad (8.101)$$

where

$$[Zt_{abc}] = \begin{bmatrix} Zt_{ab} & 0 & 0 \\ 0 & Zt_{bc} & 0 \\ 0 & 0 & Zt_{ca} \end{bmatrix} \quad (8.102)$$

Substitute Equation 8.101 into Equation 8.88:

$$[VLL_{ABC}] = [AV] \cdot [VLL_{abc}] + [AV] \cdot [Zt_{abc}] \cdot [ID_{abc}] \quad (8.103)$$

Solve Equation 8.103 for the load side line-to-line voltages:

$$[VLL_{abc}] = [AV]^{-1} \cdot [VLL_{ABC}] - [Zt_{abc}] \cdot [ID_{abc}] \quad (8.104)$$

The delta currents $[ID_{abc}]$ in Equations 8.103 and 8.104 need to be replaced by the load side line currents $[I_{abc}]$. In order to develop the needed relationship, three independent equations are needed. The first two come from applying KCL at two vertices of the delta-connected secondary.

$$I_a = I_{ba} - I_{ac} \quad (8.105)$$

$$I_b = I_{cb} - I_{ba} \quad (8.106)$$

The third equation comes from recognizing that the sum of the primary line-to-line voltages and, therefore, the secondary ideal transformer voltages must sum to zero. KVL around the delta windings gives:

$$Vt_{ab} - Zt_{ab} \cdot I_{ba} + Vt_{bc} - Zt_{bc} \cdot I_{cb} + Vt_{ca} - Zt_{ca} \cdot I_{ac} = 0 \quad (8.107)$$

Replacing the ideal delta voltages with the source side line-to-line voltages:

$$\frac{V_{AB}}{n_t} + \frac{V_{BC}}{n_t} + \frac{V_{CA}}{n_t} = Zt_{ab} \cdot I_{ba} + Zt_{bc} \cdot I_{cb} + Zt_{ca} \cdot I_{ac} \quad (8.108)$$

Since the sum of the line-to-line voltages must equal zero (KVL) and the turns ratio of the three transformers are equal, Equation 8.108 is simplified to:

$$0 = Zt_a \cdot I_{ba} + Zt_b \cdot I_{cb} + Zt_c \cdot I_{ac} \quad (8.109)$$

Note in Equation 8.109 that if the three transformer impedances are equal, then the sum of the delta currents will add to zero, meaning that the zero sequence delta currents will be zero.

Equations 8.105, 8.106, and 8.109 can be put into matrix form:

$$\begin{bmatrix} I_a \\ I_b \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ Zt_{ab} & Zt_{bc} & Zt_{ca} \end{bmatrix} \cdot \begin{bmatrix} I_{ba} \\ I_{cb} \\ I_{ac} \end{bmatrix} \quad (8.110)$$

$$[IO_{abc}] = [F] \cdot [ID_{abc}] \quad (8.111)$$

where

$$[IO_{abc}] = \begin{bmatrix} I_a \\ I_b \\ 0 \end{bmatrix} \quad (8.112)$$

$$[F] = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ Zt_{ab} & Zt_{bc} & Zt_{ca} \end{bmatrix} \quad (8.113)$$

Solve Equation 8.111 for the load side delta currents:

$$[ID_{abc}] = [F]^{-1} \cdot [IO_{abc}] = [G] \cdot [IO_{abc}] \quad (8.114)$$

where

$$[G] = [F]^{-1} = \frac{1}{Zt_{ab} + Zt_{bc} + Zt_{ca}} \cdot \begin{bmatrix} Zt_{ca} & -Zt_{bc} & 1 \\ Zt_{ca} & Zt_{ab} + Zt_{ca} & 1 \\ -Zt_{ab} - Zt_{bc} & -Zt_{bc} & 1 \end{bmatrix} \quad (8.115)$$

Writing Equation 8.114 in matrix form gives:

$$\begin{bmatrix} I_{ba} \\ I_{cb} \\ I_{ac} \end{bmatrix} = \begin{bmatrix} G_{11} & G_{12} & G_{13} \\ G_{21} & G_{22} & G_{23} \\ G_{31} & G_{32} & G_{33} \end{bmatrix} \cdot \begin{bmatrix} I_a \\ I_b \\ 0 \end{bmatrix} \quad (8.116)$$

From Equations 8.115 and 8.116 it is seen that the delta currents are functions of the transformer impedances and just the line currents in Phases *a* and *b*. Equation 8.116 can be modified to include the line current in Phase *c* by

setting the last column of the $[G]$ matrix to zeros:

$$\begin{bmatrix} I_{ba} \\ I_{cb} \\ I_{ac} \end{bmatrix} = \begin{bmatrix} G_{11} & G_{12} & 0 \\ G_{21} & G_{22} & 0 \\ G_{31} & G_{32} & 0 \end{bmatrix} \cdot \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \quad (8.117)$$

The final form of Equation 8.117 can be written as:

$$[ID_{abc}] = [G1] \cdot [I_{abc}] \quad (8.118)$$

$$\text{where} \quad [G1] = \frac{1}{Zt_{ab} + Zt_{bc} + Zt_{ca}} \cdot \begin{bmatrix} Zt_{ca} & -Zt_{bc} & 0 \\ Zt_{ca} & Zt_{ab} + Zt_{ca} & 0 \\ -Zt_{ab} - Zt_{bc} & -Zt_{bc} & 0 \end{bmatrix} \quad (8.119)$$

If the three transformers have equal impedances, Equation 8.117 becomes:

$$\begin{bmatrix} I_{ba} \\ I_{cb} \\ I_{ac} \end{bmatrix} = \frac{1}{3} \cdot \begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 0 \\ -2 & -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \frac{1}{3} \cdot \begin{bmatrix} I_a - I_b \\ I_a + 2 \cdot I_b \\ -2 \cdot I_a - I_b \end{bmatrix} \quad (8.120)$$

Note in Equation 8.120 that the sum of the delta currents will be zero, meaning that there is no circulating zero sequence current in the delta windings.

It is a common practice to serve a combination of single-phase and three-phase loads from the delta–delta connection. The single-phase loads are served by center tapping one of the three transformers (lighting transformer), thereby giving the standard 120/240 V service. The kVA rating, and thus the impedance of the lighting transformer, will be different from the two power transformers. Therefore, there will be a circulating zero sequence in the delta windings.

The general form of Equation 8.118 will be used to define the generalized constant matrices for the delta–delta connection. Substitute Equation 8.118 into Equation 8.103:

$$[VLL_{ABC}] = [AV] \cdot [VLL_{abc}] + [AV][Zt_{abc}] \cdot [G1] \cdot [I_{abc}] \quad (8.121)$$

The generalized matrices are defined in terms of the line-to-neutral voltages on the two sides of the transformer bank. Equation 8.121 is modified to be in terms of equivalent line-to-neutral voltages:

$$\begin{aligned} [VLN_{ABC}] &= [W] \cdot [VLL_{ABC}] \\ [VLN_{ABC}] &= [W][AV] \cdot [D] \cdot [VLN_{abc}] + [W] \cdot [AV] \cdot [Zt_{abc}] \cdot [G1] \cdot [I_{abc}] \end{aligned} \quad (8.122)$$

Equation 8.122 is in the general form:

$$[VLN_{ABC}] = [a_t] \cdot [VLN_{abc}] + [b_t] \cdot [I_{abc}] \quad (8.123)$$

where

$$[a_t] = [W] \cdot [AV] \cdot [D] = \frac{n_t}{3} \cdot \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \quad (8.124)$$

$$[b_t] = [W] \cdot [AV] \cdot [Zt_{abc}] \cdot [G1] \quad (8.125)$$

Equation 8.125 does not reduce to a simple form, so the quadruple matrix multiplication must be used to compute the matrix $[b_t]$.

Equation 8.99 gives the generalized equation for currents:

$$[I_{ABC}] = [AI]^{-1} \cdot [I_{abc}] = [d_t] \cdot [I_{abc}] \quad (8.126)$$

where

$$[d_t] = \begin{bmatrix} \frac{1}{n_t} & 0 & 0 \\ 0 & \frac{1}{n_t} & 0 \\ 0 & 0 & \frac{1}{n_t} \end{bmatrix} \quad (8.127)$$

The reverse generalized equation can be derived by modifying Equation 8.104 in terms of equivalent line-to-neutral voltages:

$$\begin{aligned} [VLN_{abc}] &= [W] \cdot [VLL_{abc}] \\ [VLN_{abc}] &= [W] \cdot [AV]^{-1} \cdot [D] \cdot [VLN_{ABC}] - [W] \cdot [Zt_{abc}] \cdot [G1] \cdot [I_{abc}] \end{aligned} \quad (8.128)$$

The generalized equation is

$$[VLN_{abc}] = [A_t] \cdot [VLN_{ABC}] - [B_t] \cdot [I_{abc}] \quad (8.129)$$

where

$$[A_t] = [W] \cdot [AV]^{-1} \cdot [D] = \frac{1}{3 \cdot n_t} \cdot \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & 2 & -1 \end{bmatrix} \quad (8.130)$$

$$[B_t] = [W] \cdot [Zt_{abc}] \cdot [G1] \quad (8.131)$$

The generalized matrices for the delta–delta connection have been derived. Once again it has been a long process to get to the final six equations that define the matrices. The derivation provides an excellent exercise in the application of basic transformer theory and circuit theory. Once the matrices have been defined for a particular transformer connection the analysis of the connection is a relatively simple task. Example 8.3 will demonstrate the analysis of this connection using the generalized matrices.

Example 8.3

Figure 8.10 shows three single-phase transformers in a delta–delta connection serving a combination single-phase and three-phase load connected in delta. The voltages at the load are balanced three-phase of 240 V line-to-line:

$$[VLL_{abc}] = \begin{bmatrix} 240/0 \\ 240/-120 \\ 240/120 \end{bmatrix} \text{ V}$$

The net loading by phase is

$$S_{ab} = 100 \text{ kVA at } 0.9 \text{ lagging power factor}$$

$$S_{bc} = S_{ca} = 50 \text{ kVA at } 0.8 \text{ lagging power factor.}$$

As in Example 8.2, it will be assumed that the lighting transformer serves all of the single-phase load and a third of the three-phase load, while the two power transformers will each serve a third of the three-phase load. With those assumptions, the selected ratings of the transformers are

Lighting: 100 kVA, 12,470–240 V, $Z = 0.01 + j0.04$ per-unit

Power: 50 kVA, 12,470–240 V, $Z = 0.015 + j0.035$ per-unit

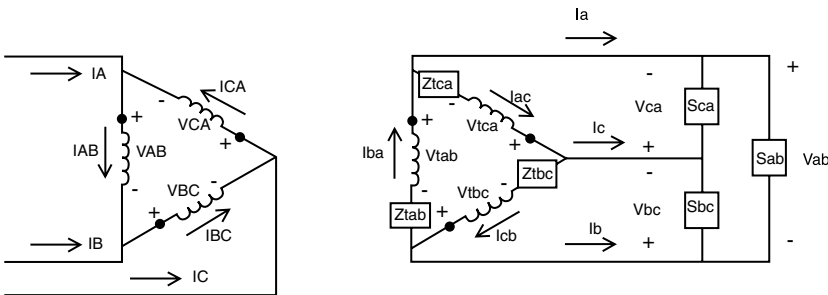


FIGURE 8.10

Delta–delta bank serving and unbalanced delta-connected load.

Determine the following:

1. The currents in the load
2. The secondary line currents
3. The equivalent line-to-neutral secondary voltages
4. The primary line-to-neutral and line-to-line voltages
5. The primary line currents
6. The delta currents in the primary and secondary windings

Before the analysis can start, the transformer impedances must be converted to actual values in ohms and located inside the delta-connected secondary windings:

Lighting transformer:

$$Z_{\text{base}} = \frac{0.24^2 \cdot 1000}{100} = 0.576 \, \Omega$$

$$Zt_{ab} = (0.01 + j0.4) \cdot 0.576 = 0.0058 + j0.023 \, \Omega$$

Power transformers:

$$Z_{\text{base}} = \frac{0.24^2 \cdot 1000}{50} = 1.152 \, \Omega$$

$$Zt_{bc} = Zt_{ca} = (0.015 + j0.35) \cdot 1.152 = 0.0173 + j0.0403 \, \Omega$$

The transformer impedance matrix can now be defined:

$$[Zt_{abc}] = \begin{bmatrix} 0.0058 + j0.023 & 0 & 0 \\ 0 & 0.0173 + j0.0403 & 0 \\ 0 & 0 & 0.0173 + j0.0403 \end{bmatrix} \, \Omega$$

The turns ratio of the transformers is $n_t = \frac{12,470}{240} = 51.9583$.

Define all of the matrices:

$$[W] = \frac{1}{3} \cdot \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 2 \end{bmatrix} \quad [D] = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix} \quad [DI] = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$[AV] = \begin{bmatrix} n_t & 0 & 0 \\ 0 & n_t & 0 \\ 0 & 0 & n_t \end{bmatrix} = \begin{bmatrix} 51.9583 & 0 & 0 \\ 0 & 51.9583 & 0 \\ 0 & 0 & 51.9583 \end{bmatrix}$$

$$[AI] = \begin{bmatrix} \frac{1}{n_t} & 0 & 0 \\ 0 & \frac{1}{n_t} & 0 \\ 0 & 0 & \frac{1}{n_t} \end{bmatrix} = \begin{bmatrix} 0.0192 & 0 & 0 \\ 0 & 0.0192 & 0 \\ 0 & 0 & 0.0192 \end{bmatrix}$$

$$[G1] = \frac{1}{Zt_{ab} + Zt_{bc} + Zt_{ca}} \cdot \begin{bmatrix} Zt_{ca} & -Zt_{bc} & 0 \\ Zt_{ca} & Zt_{ab} + Zt_{ca} & 0 \\ -Zt_{ab} - Zt_{bc} & -Zt_{bc} & 0 \end{bmatrix}$$

$$[G1] = \begin{bmatrix} 0.3941 - j0.0134 & -0.3941 + j0.0134 & 0 \\ 0.3941 - j0.0134 & 0.6059 + j0.0134 & 0 \\ -0.6059 - j0.0134 & -0.3941 + j0.0134 & 0 \end{bmatrix}$$

$$[a_t] = \frac{51.9583}{3} \cdot \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 34.6389 & -17.3194 & -17.3194 \\ -17.3194 & 34.6389 & -17.3194 \\ -17.3194 & -17.3194 & 34.6389 \end{bmatrix}$$

$$[b_1] = [AV] \cdot [W] \cdot [Zt_{abc}] \cdot [G1] = \begin{bmatrix} 0.2166 + j0.583 & 0.0826 + j0.1153 & 0 \\ 0.0826 + j0.1153 & 0.2166 + j0.583 & 0 \\ -0.2993 - j0.6983 & -0.2993 - j0.6983 & 0 \end{bmatrix}$$

$$[A_t] = \frac{1}{3 \cdot 51.983} \cdot \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 0.0128 & -0.0064 & -0.0064 \\ -0.0064 & 0.0128 & -0.0064 \\ -0.0064 & -0.0064 & 0.0128 \end{bmatrix}$$

$$[B_t] = [W] \cdot [Zt_{abc}] \cdot [G1] = \begin{bmatrix} 0.0042 + j0.0112 & 0.0016 + j0.0022 & 0 \\ 0.0016 + j0.0022 & 0.0042 + j0.0112 & 0 \\ -0.0058 - j0.0134 & -0.0058 - j0.0134 & 0 \end{bmatrix}$$

With the generalized constant matrices defined, the delta load currents can first be computed:

$$(1) \quad IL_i = \left(\frac{SL_i}{VLL_i} \right)^* = \begin{bmatrix} 416.67 / -25.84 \\ 208.33 / -156.87 \\ 208.33 / 83.13 \end{bmatrix} \text{ A}$$

(2) Compute the line currents:

$$[I_{abc}] = [DI] \cdot [IL] = \begin{bmatrix} 522.93 / -47.97 \\ 575.31 / 170.01 \\ 360.84 / 53.13 \end{bmatrix} \text{ A}$$

(3) Compute the equivalent line-to-neutral secondary voltages:

$$[VLN_{abc}] = [W] \cdot [VLL_{abc}]$$

$$[VLN_{abc}] = \begin{bmatrix} 138.56 / -30 \\ 138.56 / -150 \\ 138.56 / 90 \end{bmatrix} \text{ V}$$

(4) Use the generalized constant matrices to compute the primary equivalent line-to-neutral voltages:

$$[VLN_{ABC}] = [a_t] \cdot [VLN_{abc}] + [b_t] \cdot [I_{abc}] = \begin{bmatrix} 7381.53 / -28.63 \\ 7445.51 / -148.41 \\ 7438.40 / 91.05 \end{bmatrix} \text{ V}$$

The primary line-to-line voltages are

$$[VLL_{ABC}] = [D] \cdot [VLN_{ABC}] = \begin{bmatrix} 12,826 / 1.62 \\ 12,925 / -118.69 \\ 12,814 / 121.08 \end{bmatrix} \text{ V}$$

(5) The primary line currents are

$$[I_{ABC}] = [d_t] \cdot [I_{abc}] = \begin{bmatrix} 10.06 / -47.97 \\ 11.07 / 170.01 \\ 6.95 / 53.13 \end{bmatrix} \text{ A}$$

(6) Equation 8.118 can be used to compute the secondary delta currents:

$$[ID_{abc}] = [G1] \cdot [I_{abc}] = \begin{bmatrix} 409.54 / -29.99 \\ 236.43 / -153.65 \\ 183.81 / 88.56 \end{bmatrix} \text{ A}$$

The primary delta currents are computed from Equation 8.92:

$$[ID_{ABC}] = [AI]^{-1} \cdot [ID_{abc}] = \begin{bmatrix} 7.88 / -29.99 \\ 4.55 / -153.65 \\ 3.54 / 88.56 \end{bmatrix} \text{ A}$$

A check on the primary line currents can be made by applying Equation 8.94:

$$[I_{ABC}] = [DI] \cdot [ID_{ABC}] = \begin{bmatrix} 10.06 / -47.98 \\ 11.07 / 170.01 \\ 6.95 / 53.13 \end{bmatrix} \text{ A}$$

These are the same primary line currents previously computed. This provides a method of checking on the many calculations that have taken place to get to this point. Another check can be made on the accuracy of the calculations and the models by applying the reverse Equation 8.129:

$$[VLN_{abc}] = [A_t] \cdot [VLN_{ABC}] - [B_t] \cdot [I_{abc}] = \begin{bmatrix} 138.56 / -30 \\ 138.56 / -150 \\ 138.56 / 90 \end{bmatrix} \text{ V}$$

Again, these are the same values of the equivalent secondary line-to-neutral voltages that were used at the start of the analysis.

Last but not least, a check on the operating kVA and power factor for each transformer can be determined:

$$ST_i = \left| \frac{VLL_{ABC_i} \cdot (ID_{ABC_i})^*}{1000} \right| = \begin{bmatrix} 101.1 \\ 58.8 \\ 45.3 \end{bmatrix} \text{ kVA}$$

$$PF_i = \begin{bmatrix} 0.85 \\ 0.82 \\ 0.84 \end{bmatrix} \text{ lagging}$$

From this example it is seen that the operating kVA is very close to the rating of each transformer.

8.7 Open Wye–Open Delta Connection

A common load to be served on a distribution feeder is a combination of a single-phase lighting load and a three-phase power load. Many times the three-phase power load will be an induction motor. This combination load can be served by an ungrounded wye–delta connection as previously described, or by an open wye–open delta connection. When the three-phase load is small compared to the single-phase load, the open wye–open delta connection is commonly used. The open wye–open delta connection requires only two transformers, but the connection will provide three-phase line-to-line voltages to the combination load. Figure 8.11 shows the open wye–open delta connection and the primary and secondary positive sequence voltage phasors. With reference to Figure 8.11, the basic ideal transformer voltages as functions of the turns ratio are

$$\begin{bmatrix} V_{AG} \\ V_{BG} \\ V_{CG} \end{bmatrix} = \begin{bmatrix} n_t & 0 & 0 \\ 0 & n_t & 0 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} V_{t_{ab}} \\ V_{t_{bc}} \\ V_{t_{ca}} \end{bmatrix} \quad (8.132)$$

$$[VLG_{ABC}] = [AV] \cdot [V_{t_{abc}}] \quad (8.133)$$

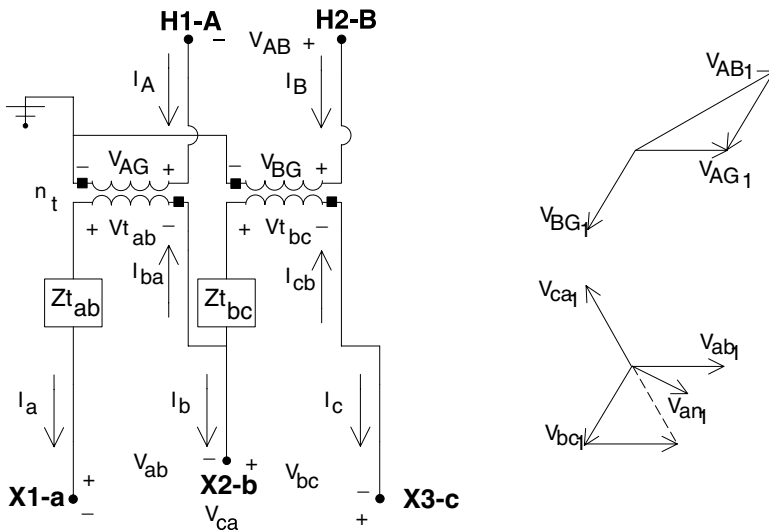


FIGURE 8.11
Open wye–open delta connection.

The currents as functions of the turns ratio are given by:

$$\begin{aligned} I_{ba} &= n_T \cdot I_A = I_a \\ I_{cb} &= n_T \cdot I_B = -I_c \\ I_b &= -(I_a + I_c) \end{aligned} \quad (8.134)$$

Equation 8.134 can be expressed in matrix form by:

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} n_t & 0 & 0 \\ -n_t & n_t & 0 \\ 0 & -n_t & 0 \end{bmatrix} \cdot \begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix} \quad (8.135)$$

$$[I_{abc}] = [AI] \cdot [I_{ABC}] \quad (8.136)$$

The ideal voltages on the delta windings can be determined by:

$$\begin{aligned} Vt_{ab} &= V_{ab} + Zt_{ab} \cdot I_a \\ Vt_{bc} &= V_{bc} - Zt_{bc} \cdot I_c \end{aligned} \quad (8.137)$$

Applying Equation 8.132 to Equations 8.137:

$$\begin{aligned} V_{AG} &= n_t \cdot Vt_{ab} = n_t \cdot V_{ab} + n_t \cdot Zt_{ab} \cdot I_a \\ V_{BG} &= n_t \cdot Vt_{bc} = n_t \cdot V_{bc} - n_t \cdot Zt_{bc} \cdot I_c \end{aligned} \quad (8.138)$$

Equation 8.138 can be put into three-phase matrix form as:

$$\begin{bmatrix} V_{AG} \\ V_{BG} \\ V_{CG} \end{bmatrix} = \begin{bmatrix} n_t & 0 & 0 \\ 0 & n_t & 0 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} V_{ab} \\ V_{bc} \\ V_{ca} \end{bmatrix} + \begin{bmatrix} n_t \cdot Zt_{ab} & 0 & 0 \\ 0 & 0 & -n_t \cdot Zt_{bc} \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \quad (8.139)$$

$$[VLG_{ABC}] = [AV] \cdot [VLL_{abc}] + [b_t] \cdot [I_{abc}] \quad (8.140)$$

The secondary line-to-line voltages in Equation 8.140 can be replaced by the equivalent line-to-neutral secondary voltages:

$$[VLG_{ABC}] = [AV] \cdot [D] \cdot [VLN_{abc}] + [b_t] \cdot [I_{abc}] \quad (8.141)$$

The generalized matrix equation becomes:

$$[VLG_{ABC}] = [a_t] \cdot [VLN_{abc}] + [b_t] \cdot [I_{abc}] \quad (8.142)$$

where

$$[a_t] = \begin{bmatrix} n_t & -n_t & 0 \\ 0 & n_t & -n_t \\ 0 & 0 & 0 \end{bmatrix} \quad (8.143)$$

$$[b_t] = \begin{bmatrix} n_t \cdot Zt_{ab} & 0 & 0 \\ 0 & 0 & -n_t \cdot Zt_{bc} \\ 0 & 0 & 0 \end{bmatrix} \quad (8.144)$$

The source side line currents as functions of the load side line currents are given by:

$$\begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix} = \begin{bmatrix} \frac{1}{n_t} & 0 & 0 \\ 0 & 0 & -\frac{1}{n_t} \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \quad (8.145)$$

$$[I_{ABC}] = [d_t] \cdot [I_{abc}] \quad (8.146)$$

where

$$[d_t] = \begin{bmatrix} \frac{1}{n_t} & 0 & 0 \\ 0 & 0 & -\frac{1}{n_t} \\ 0 & 0 & 0 \end{bmatrix} \quad (8.147)$$

The reverse equations relating the secondary equivalent line-to-neutral voltages as functions of the source side line-to-ground voltages and line currents is needed. Solving Equation 8.138 for the two line-to-line secondary voltages:

$$\begin{aligned} V_{ab} &= \frac{1}{n_t} \cdot V_{AG} - Zt_{ab} \cdot I_a \\ V_{bc} &= \frac{1}{n_t} \cdot V_{BG} - Zt_{bc} \cdot I_c \end{aligned} \quad (8.148)$$

The third line-to-line voltage (V_{ca}) must be equal to the negative sum of the other two line-to-line voltages (KVL). In matrix form the desired equation is

$$\begin{bmatrix} V_{ab} \\ V_{bc} \\ V_{ca} \end{bmatrix} = \begin{bmatrix} \frac{1}{n_t} & 0 & 0 \\ 0 & \frac{1}{n_t} & 0 \\ -\frac{1}{n_t} & -\frac{1}{n_t} & 0 \end{bmatrix} \cdot \begin{bmatrix} V_{AG} \\ V_{BG} \\ V_{CG} \end{bmatrix} - \begin{bmatrix} Zt_{ab} & 0 & 0 \\ 0 & 0 & -Zt_{bc} \\ -Zt_{ab} & 0 & Zt_{bc} \end{bmatrix} \cdot \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \quad (8.149)$$

$$[VLL_{abc}] = [BV] \cdot [VLG_{ABC}] - [Zt_{abc}] \cdot [I_{abc}] \quad (8.150)$$

The equivalent secondary line-to-neutral voltages are then given by:

$$[VLN_{abc}] = [W] \cdot [VLL_{abc}] = [W][BV] \cdot [VLG_{ABC}] - [W] \cdot [Zt_{abc}] \cdot [I_{abc}] \quad (8.151)$$

The generalized reverse equation is given by:

$$[VLN_{abc}] = [A_t] \cdot [VLG_{ABC}] - [B_t] \cdot [I_{abc}] \quad (8.152)$$

where

$$[A_t] = [W] \cdot [BV] = \frac{1}{3 \cdot n_t} \cdot \begin{bmatrix} 2 & 1 & 0 \\ -1 & 1 & 0 \\ -1 & -2 & 0 \end{bmatrix} \quad (8.153)$$

$$[B_t] = [W] \cdot [Zt_{abc}] = \frac{1}{3} \cdot \begin{bmatrix} 2 \cdot Zt_{ab} & 0 & -Zt_{bc} \\ -Zt_{ab} & 0 & -Zt_{bc} \\ -Zt_{ab} & 0 & 2 \cdot Zt_{bc} \end{bmatrix} \quad (8.154)$$

The open wye–open delta connection derived in this section utilized Phases *A* and *B* on the primary. This is just one of three possible connections. The other two possible connections would use Phases *B* and *C*, and then Phases *C* and *A*. The terms “leading” and “lagging” connection are also associated with the open wye–open delta connection. The generalized matrices will be different from those just derived. The same procedure can be used to derive the matrices for the other two connections.

When the lighting transformer is connected across the leading of the two phases, the connection is referred to as the “leading” connection. Similarly, when the lighting transformer is connected across the lagging of the two phases, the connection is referred to as the “lagging” connection. For example, if the bank is connected to Phases *A* and *B* and the lighting transformer is connected from Phase *A* to ground, this would be referred to as the leading connection because the voltage *A-G* leads the voltage *B-G* by 120 degrees. Reverse the connection and it would now be called the lagging connection. Obviously, there is a leading and lagging connection for each of the three possible open wye–open delta connections.

Example 8.4

The unbalanced load of Example 8.2 is to be served by the leading open wye–open delta connection using Phases *A* and *B*. Assume that the voltages again are balanced three-phase at the load so that the voltages and line currents of Example 8.2 will apply for this example. It will also be assumed that the lighting transformer is rated 100 kVA and the power transformer is rated 50 kVA. Calculate the primary voltages, currents, and the operating kVA of each transformer.

Compute the generalized constant matrices:

$$[a_t] = \begin{bmatrix} n_t & -n_t & 0 \\ 0 & n_t & -n_t \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 30 & -30 & 0 \\ 0 & 30 & -30 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[b_t] = \begin{bmatrix} n_t \cdot Zt_{ab} & 0 & 0 \\ 0 & 0 & -n_t \cdot Zt_{bc} \\ 0 & 0 & 0 \end{bmatrix}$$

$$[b_t] = \begin{bmatrix} 0.1728 + j0.6912 & 0 & 0 \\ 0 & 0 & -0.5184 - j1.2096 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[d_t] = \begin{bmatrix} \frac{1}{n_t} & 0 & 0 \\ 0 & 0 & -\frac{1}{n_t} \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0.0333 & 0 & 0 \\ 0 & 0 & -0.0333 \\ 0 & 0 & 0 \end{bmatrix}$$

The equivalent line-to-neutral secondary voltages and line currents from Example 8.2 are

$$[VLN_{abc}] = \begin{bmatrix} 138.56 / -30 \\ 138.56 / -150 \\ 138.56 / 90 \end{bmatrix} \text{ V}$$

$$[I_{abc}] = \begin{bmatrix} 522.93 / -47.97 \\ 575.31 / 170.01 \\ 360.84 / 53.13 \end{bmatrix} \text{ A}$$

The primary line-to-ground voltages are computed to be

$$[VLG_{ABC}] = [a_t] \cdot [VLN_{abc}] + [b_t] \cdot [I_{abc}] = \begin{bmatrix} 7531 / 1.33 \\ 7449 / -116.84 \\ 0 \end{bmatrix} \text{ V}$$

The voltage V_{CG} has been computed to be zero above. That should not be interpreted to mean that the actual primary voltage from C to ground is zero. Rather, it must be interpreted to mean that there is no transformer connected between Phase C and ground. If Phase C is present on the primary, it will certainly have a value other than zero.

The primary transformer currents are

$$[I_{ABC}] = [d_t] \cdot [I_{abc}] = \begin{bmatrix} 17.43 / -47.97 \\ 12.03 / -126.87 \\ 0 \end{bmatrix} \text{ A}$$

The operating kVA and power factor of each transformer is

$$ST_i = \left| \frac{VLG_i \cdot (I_{ABC_i})^*}{1000} \right| = \begin{bmatrix} 131.27 \\ 89.60 \\ 0 \end{bmatrix} \text{ kVA}$$

$$PF_i = \begin{bmatrix} 0.652 \\ 0.985 \\ 0 \end{bmatrix} \text{ lagging}$$

Note in this case that the two transformers are operating far above their ratings. If this load is to be served by an open wye–open delta connection, the lighting transformer would need to be rated 167 kVA and the power transformer rated 100 kVA. Note also that the operating power factors of the two transformers are quite diverse.

8.8 The Thevenin Equivalent Circuit

The six generalized constant matrices for each of five common three-phase transformer connections have been developed. In Chapter 10, the section for short-circuit analysis will require the Thevenin equivalent circuit referenced to the secondary terminals of the transformer. This equivalent circuit must take into account the equivalent impedance between the primary terminals of the transformer and the feeder source. Figure 8.12 is a general circuit showing the feeder source down to the secondary bus. The Thevenin equivalent circuit needs to be determined at the secondary node of the transformer bank. This basically is the same as referring the source voltage and the source impedance to the secondary side of the transformer. The desired Thevenin equivalent circuit at the transformer secondary node is shown in Figure 8.13. A general Thevenin equivalent circuit that can be used for all connections employs the generalized matrices.

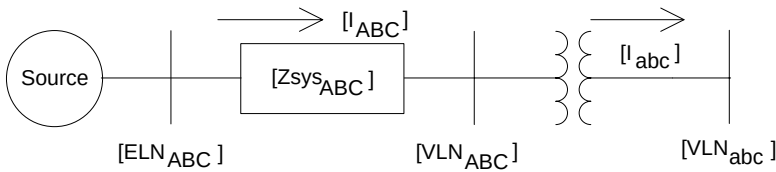


FIGURE 8.12
Equivalent system.

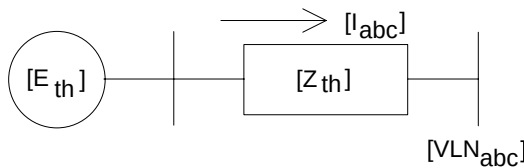


FIGURE 8.13
The Thevenin equivalent circuit.

In Figure 8.12 the primary transformer equivalent line-to-neutral voltages as functions of the source voltages and the equivalent high side impedance are given by:

$$[VLN_{ABC}] = [ELN_{ABC}] - [Zsys_{ABC}] \cdot [I_{ABC}] \quad (8.155)$$

but $[I_{ABC}] = [d_t] \cdot [I_{abc}]$

therefore $[VLN_{ABC}] = [ELN_{ABC}] + [Zsys_{ABC}] \cdot [d_t] \cdot [I_{abc}] \quad (8.156)$

The general reverse equation gives the secondary line-to-neutral voltages as functions of the primary line-to-neutral voltages and secondary currents:

$$[VLN_{abc}] = [A_t] \cdot [VLN_{ABC}] - [B_t] \cdot [I_{abc}] \quad (8.157)$$

Substitute Equation 8.156 into Equation 8.157:

$$[VLN_{abc}] = [A_t] \cdot \{[ELN_{ABC}] - [Zsys_{ABC}] \cdot [d_t] \cdot [I_{abc}]\} - [B_t] \cdot [I_{abc}] \quad (8.158)$$

With reference to Equation 8.158, the Thevenin equivalent voltages and impedances can be defined as:

$$[Eth_{abc}] = [A_t] \cdot [ELN_{ABC}] \quad (8.159)$$

$$[Zth_{abc}] = [A_t] \cdot [Zsys_{ABC}] \cdot [d_t] + [B_t] \quad (8.160)$$

The definitions of the Thevenin equivalent voltages and impedances as given in Equations 8.159 and 8.160 are general and can be used for all transformer connections. Example 8.5 is used to demonstrate the computation and application of the Thevenin equivalent circuit.

Example 8.5

The ungrounded wye–delta transformer bank of Example 8.2 is connected to a source through a one-mile section of a four-wire three-phase line. The phase impedance matrix for the one mile of line is given by:

$$[Zsys_{ABC}] = \begin{bmatrix} 0.4576 + j1.0780 & 0.1559 + j0.5017 & 0.1535 + j0.3849 \\ 0.1559 + j0.5017 & 0.4666 + j1.0482 & 0.1580 + j0.4236 \\ 0.1535 + j0.3849 & 0.1580 + j0.4236 & 0.4615 + j1.0651 \end{bmatrix} \Omega$$

For the unbalanced load and balanced voltage in Example 8.2, the primary line-to-neutral voltages and primary currents were computed as:

$$[VLN_{ABC}] = \begin{bmatrix} 7367.6/\underline{1.4} \\ 7532.3/\underline{-119.1} \\ 7406.2/\underline{121.7} \end{bmatrix} \text{ V}$$

$$[I_{ABC}] = [d_t] \cdot [I_{abc}] = \begin{bmatrix} 11.54/\underline{-28.04} \\ 8.95/\underline{-166.43} \\ 7.68/\underline{101.16} \end{bmatrix} \text{ A}$$

The source line-to-neutral voltage for this load condition is computed to be

$$[ELN_{ABC}] = [VLN_{ABC}] + [Zsys_{ABC}] \cdot [I_{ABC}] = \begin{bmatrix} 7374.8/\underline{1.45} \\ 7537.4/\underline{-119.1} \\ 7409.9/\underline{121.8} \end{bmatrix} \text{ V}$$

Now the Thevenin equivalent circuit at the transformer bank secondary terminals is

$$[Eth_{abc}] = [A_t] \cdot [ELN_{ABC}] = \begin{bmatrix} 141.2/\underline{-29.3} \\ 146.2/\underline{-148.5} \\ 142.2/\underline{91.9} \end{bmatrix} \text{ V}$$

$$[Zth_{abc}] = [A_t] \cdot [Zsys_{ABC}] \cdot [d_t] + [B_t]$$

$$[Zth_{abc}] = \begin{bmatrix} 0.0033 + j0.0099 & 0.0026 + j0.0039 & 0 \\ 0 & 0.0059 + j0.0137 & 0 \\ -0.0072 - j0.0156 & -0.0046 - j0.0117 & 0 \end{bmatrix} 0 \text{ } \Omega$$

The load currents computed in Example 8.2 were

$$[I_{abc}] = \begin{bmatrix} 522.9/\underline{-47.97} \\ 575.3/\underline{170.0} \\ 360.8/\underline{53.13} \end{bmatrix} \text{ A}$$

Using the Thevenin equivalent circuit and the previously computed line currents, the equivalent line-to-neutral voltages can be computed:

$$[VLN_{abc}] = [Eth_{abc}] - [Zth_{abc}] \cdot [I_{abc}] = \begin{bmatrix} 138.56 / -30 \\ 138.56 / -150 \\ 138.56 / 90 \end{bmatrix} \text{ V}$$

This example is only intended to demonstrate that it is possible to compute the Thevenin equivalent circuit at the secondary terminals of the transformer bank. The example shows that using the Thevenin equivalent circuit and the original secondary line currents, the original equivalent line-to-neutral load voltages are computed. The major application of the Thevenin equivalent circuit will be in the short-circuit analysis of a distribution that will be developed in Chapter 10.

8.9 Summary

In this chapter the generalized matrices have been developed for five common three-phase transformer bank connections. The generalized matrices can be derived for all of the other common three-phase transformer bank connections using the same approach taken in this chapter.

The generalized matrices can be used for two purposes:

1. A quick computation of the operating conditions of the individual transformers for a specified load condition.
2. With the generalized matrices in a power-flow iterative routine, it is not necessary to build into the program the ability to determine the type of transformer bank connection.

For short-circuit calculations it is necessary to develop the three-phase Thevenin equivalent circuit at the point of the fault. Since all transformer connections are defined by the constant matrices, the computation of the equivalent circuit is straightforward.

Problems

8.1 A three-phase substation transformer is connected delta-grounded wye and rated:

5000 kVA, 115 kV delta–12.47 kV grounded wye, $Z = 1.0 + j7.5\%$

The transformer serves an unbalanced wye-connected load of:

Phase a: 1384.5 kVA, 89.2% lagging power factor, at $6922.5 / -33.1$ V

Phase b: 1691.2 kVA, 80.2% lagging power factor, at $6776.8 / -153.4$ V

Phase c: 1563.0 kVA, unity power factor, at $7104.7 / 85.9$ V

- (1) Determine the generalized matrices for the transformer.
- (2) Compute the primary equivalent line-to-neutral voltages.
- (3) Compute the primary line-to-line voltages.
- (4) Compute the primary line currents.
- (5) Compute the currents flowing in the high side delta windings.
- (6) Compute the real power loss in the transformer for this load condition.

8.2 Three single-phase transformers are connected in delta-grounded wye and are serving an unbalanced load. The ratings of three transformers are

Phase A-B: 100 kVA, 12,470–120 V, $Z = 1.3 + j1.7\%$

Phase B-C: 50 kVA, 12,470–120 V, $Z = 1.1 + j1.4\%$

Phase C-A: same as Phase B-C transformer

The unbalanced wye-connected loads are

Phase a: 40 kVA, 0.8 lagging power factor, $V = 117.5 / -32.5$ V

Phase b: 85 kVA, 0.95 lagging power factor, $V = 115.7 / -147.3$ V

Phase c: 50 kVA, 0.8 lagging power factor, $V = 117.0 / -95.3$ V

- (1) Determine the generalized matrices for this connection.
- (2) Compute the load currents.
- (3) Compute the primary line-to-neutral voltages.
- (4) Compute the primary line-to-line voltages.
- (5) Compute the primary currents.
- (6) Compute the currents in the delta primary windings.
- (7) Compute the transformer bank real power loss.

8.3 The three single-phase transformers of Problem 8.2 are serving an unbalanced constant impedance wye-connected load of:

Phase a: $0.32 + j0.14 \Omega$

Phase b: $0.21 + j0.08 \Omega$

Phase c: $0.28 + j0.12 \Omega$

The transformers are connected to a balanced 12.47-kV source.

- (1) Determine the load currents.
- (2) Determine the load voltages.
- (3) Compute the complex power of each load.
- (4) Compute the primary currents.
- (5) Compute the operating kVA of each transformer.

8.4 A three-phase transformer connected wye–delta is rated:

$$500 \text{ kVA}, 4160\text{--}240 \text{ V}, Z = 1.1 + j3.0\%$$

The primary neutral is ungrounded. The transformer is serving a balanced load of 480 kW with balanced voltages of 235 V line-to-line and a lagging power factor of 0.9.

- (1) Compute the secondary line currents.
- (2) Compute the primary line currents.
- (3) Compute the currents flowing in the secondary delta windings.
- (4) Compute the real power loss in the transformer for this load.

8.5 The transformer of Problem 8.4 is serving an unbalanced delta load of:

$$S_{ab} = 150 \text{ kVA}, \quad 0.95 \text{ lagging power factor}$$

$$S_{bc} = 125 \text{ kVA}, \quad 0.90 \text{ lagging power factor}$$

$$S_{ca} = 160 \text{ kVA}, \quad 0.8 \text{ lagging power factor}$$

The magnitudes of the line-to-line voltages are

$$V_{ab} = 240 \text{ V}, \quad V_{bc} = 237 \text{ V}, \quad V_{ca} = 235 \text{ V}$$

- (1) Use the Law of Cosines to compute the angles on the voltages. Set the voltage a-b as reference.
- (2) Determine the delta load currents.
- (3) Compute the generalized matrices for the transformer bank.
- (4) Compute the primary line-to-neutral and line-to-line voltages.
- (5) Compute the primary line currents.
- (6) Compute the operating kVA of each transformer winding.

8.6 Three single-phase transformers are connected in an ungrounded wye–delta connection and serving an unbalanced delta connected load. The transformers are rated:

Phase A: 15 kVA, 2400–240 V, $Z = 1.3 + j1.0\%$

Phase B: 25 kVA, 2400–240 V, $Z = 1.1 + j1.1\%$

Phase C: Same as phase A transformer

The transformers are connected to a balanced source of 4.16 kV line-to-line. The primary currents entering the transformer are

$I_A = 4.62$ A, 0.95 lagging power factor

$I_B = 6.92$ A, 0.88 lagging power factor

$I_C = 5.28$ A, 0.69 lagging power factor

- (1) Determine the primary line-to-neutral voltages.
- (2) Determine the line currents entering the delta-connected load.
- (3) Determine the line-to-line voltages at the load.
- (4) Determine the operating kVA of each transformer.
- (5) Is it possible to determine the load currents in the delta-connected load? If so, do it. If not, why not?

8.7 Three single-phase transformers are connected in grounded wye-grounded wye and are serving an unbalanced constant impedance load. The transformer bank is connected to a balanced three-phase 12.47 kV line-to-line voltage source. The transformers are rated:

Phase A: 100 kVA, 7200–120 V, $Z = 0.9 + j1.8\%$

Phase B and Phase C: 37.5 kVA, 7200–120 V, $Z = 1.1 + j1.4\%$

The constant impedance wye-connected loads are

Phase a: $0.14 + j0.08 \Omega$

Phase b: $0.40 + j0.14 \Omega$

Phase c: $0.50 + j0.20 \Omega$

- (1) Compute the generalized matrices for this transformer bank.
- (2) Determine the load currents.
- (3) Determine the load voltages.
- (4) Determine the kVA and power factor of each load.
- (5) Determine the primary line currents.
- (6) Determine the operating kVA of each transformer.

8.8 Three single-phase transformers are connected in delta–delta and are serving a balanced three-phase motor rated 150 kVA, 0.8 lagging power factor, and a single-phase lighting load of 25 kVA, 0.95 lagging power factor connected across Phases A-B. The transformers are rated:

Phase A-B: 75 kVA, 4800–240 V, $Z = 1.0 + j1.5\%$

Phase B-C: 50 kVA, 4800–240 V, $Z = 1.1 + j1.4\%$

Phase C-A: same as Phase B-C

The load voltages are balanced three-phase of 240 V line-to-line.

- (1) Determine the generalized matrices.
- (2) Compute the motor input currents.
- (3) Compute the single-phase lighting load current.
- (4) Compute the primary line currents.
- (5) Compute the primary line-to-line voltages.
- (6) Compute the currents flowing in the primary and secondary delta windings.

8.9 The three-phase motor and single-phase lighting loads of Problem 8.8 are being served by an open wye–open delta transformer bank. The load voltages are balanced three-phase of 240 V line-to-line. The transformers are rated:

Lighting transformer connected to Phase A to ground:

$$100 \text{ kVA, } 7200\text{--}240 \text{ V, } Z = 0.8 + j1.5\%$$

Power transformer connected to Phase B to ground

$$75 \text{ kVA, } 7200\text{--}240 \text{ V, } Z = 0.8 + j1.2\%$$

- (1) Determine the generalized matrices for this connection.
- (2) Compute the primary line-to-ground and line-to-line voltages.
- (3) Compute the primary line currents.
- (4) Compute the operating kVA of each transformer. Are they overloaded?

8.10 The unbalanced loads of Problem 8.5 are served from an open wye–open delta connection. The line-to-line load voltages are the same as in Problem 8.5. Each transformer is rated:

$$167 \text{ kVA, } 2400\text{--}240 \text{ V, } Z = 1.1 + j1.3\%$$

- (1) Determine the operating kVA and power factor for each transformer.
- (2) What is the percentage overload on each transformer? Is this overload okay?

9

Load Models

The loads on a distribution system are typically specified by the complex power consumed. With reference to Chapter 2, the specified load will be the maximum diversified demand. This demand can be specified as kVA and power factor, kW and power factor, or kW and kvar. The voltage specified will always be the voltage at the low-voltage terminals of the distribution substation. This creates a problem since the current requirement of the loads cannot be determined without knowing the voltage. For this reason, some form of an iterative technique must be employed. An iterative technique will be presented in Chapter 10.

Loads on a distribution feeder can modeled as wye-connected or delta-connected. The loads can be three-phase, two-phase, or single-phase with any degree of unbalance, and can be modeled as:

- Constant real and reactive power (constant PQ)
- Constant current
- Constant impedance
- Any combination of the above

The load models developed are to be used in the iterative process of a power-flow program where the load voltages are initially assumed. One of the results of the power-flow analysis is to replace the assumed voltages with the actual operating load voltages. All models are initially defined by a complex power per phase and an assumed line-to-neutral voltage (wye load) or an assumed line-to-line voltage (delta load). The units of the complex power can be in volt-amperes and volts, or per-unit volt-amperes and per-unit voltages. For all loads the line currents entering the load are required in order to perform the power-flow analysis.

9.1 Wye-Connected Loads

Figure 9.1 is the model of a wye-connected load. The notation for the specified complex powers and voltages are as follows:

$$\text{Phase a: } |S_a|/\underline{\theta_a} = P_a + jQ_a \quad \text{and} \quad |V_{an}|/\underline{\delta_a} \quad (9.1)$$

$$\text{Phase b: } |S_b|/\underline{\theta_b} = P_b + jQ_b \quad \text{and} \quad |V_{bn}|/\underline{\delta_b} \quad (9.2)$$

$$\text{Phase c: } |S_c|/\underline{\theta_c} = P_c + jQ_c \quad \text{and} \quad |V_{cn}|/\underline{\delta_c} \quad (9.3)$$

9.1.1 Constant Real and Reactive Power Loads

The line currents for constant real and reactive power loads (PQ loads) are given by:

$$\begin{aligned} IL_a &= \left(\frac{S_a}{V_{an}} \right)^* = \frac{|S_a|}{|V_{an}|} / \underline{\delta_a - \theta_a} = |IL_a| / \underline{\alpha_a} \\ IL_b &= \left(\frac{S_b}{V_{bn}} \right)^* = \frac{|S_b|}{|V_{bn}|} / \underline{\delta_b - \theta_b} = |IL_b| / \underline{\alpha_b} \\ IL_c &= \left(\frac{S_c}{V_{cn}} \right)^* = \frac{|S_c|}{|V_{cn}|} / \underline{\delta_c - \theta_c} = |IL_c| / \underline{\alpha_c} \end{aligned} \quad (9.4)$$

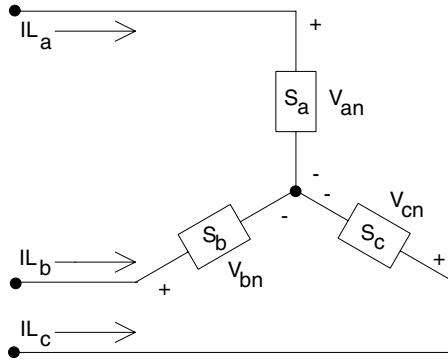


FIGURE 9.1
Wye-connected load.

In this model the line-to-neutral voltages will change during each iteration until convergence is achieved.

9.1.2 Constant Impedance Loads

The constant load impedance is first determined from the specified complex power and assumed line-to-neutral voltages:

$$\begin{aligned} Z_a &= \frac{|V_{an}|^2}{S_a^*} = \frac{|V_{an}|^2}{|S_a|} / \underline{\theta_a} = |Z_a| / \underline{\theta_a} \\ Z_b &= \frac{|V_{bn}|^2}{S_b^*} = \frac{|V_{bn}|^2}{|S_b|} / \underline{\theta_b} = |Z_b| / \underline{\theta_b} \\ Z_c &= \frac{|V_{cn}|^2}{S_c^*} = \frac{|V_{cn}|^2}{|S_c|} / \underline{\theta_c} = |Z_c| / \underline{\theta_c} \end{aligned} \quad (9.5)$$

The load currents as a function of the constant load impedances are given by:

$$\begin{aligned} IL_a &= \frac{V_{an}}{Z_a} = \frac{|V_{an}|}{|Z_a|} / \underline{\delta_a - \theta_a} = |IL_a| / \underline{\alpha_a} \\ IL_b &= \frac{V_{bn}}{Z_b} = \frac{|V_{bn}|}{|Z_b|} / \underline{\delta_b - \theta_b} = |IL_b| / \underline{\alpha_b} \\ IL_c &= \frac{V_{cn}}{Z_c} = \frac{|V_{cn}|}{|Z_c|} / \underline{\delta_c - \theta_c} = |IL_c| / \underline{\alpha_c} \end{aligned} \quad (9.6)$$

In this model the line-to-neutral voltages will change during each iteration, but the impedance computed in Equation 9.5 will remain constant.

9.1.3 Constant Current Loads

In this model the magnitudes of the currents are computed according to Equations 9.4 and are then held constant while the angle of the voltage (δ) changes, resulting in a changing angle on the current so that the power factor of the load remains constant:

$$\begin{aligned} IL_a &= |IL_a| / \underline{\delta_a - \theta_a} \\ IL_b &= |IL_b| / \underline{\delta_b - \theta_b} \\ IL_c &= |IL_c| / \underline{\delta_c - \theta_c} \end{aligned} \quad (9.7)$$

where δ_{abc} = Line-to-neutral voltage angles
 θ_{abc} = Power factor angles.

9.1.4 Combination Loads

Combination loads can be modeled by assigning a percentage of the total load to each of the three above load models. The total line current entering the load is the sum of the three components.

Example 9.1

The complex powers of a wye-connected load are

$$[S_{abc}] = \begin{bmatrix} 2236.1 / 26.6 \\ 2506.0 / 28.6 \\ 2101.4 / 25.3 \end{bmatrix} \text{ kVA}$$

The load is specified to be 50% constant complex power, 20% constant impedance, and 30% constant current. The nominal line-to-line voltage of the feeder is 12.47 kV.

- (1) Assume the nominal voltage and compute the component of load current attributed to each component of the load and the total load current.

The assumed line-to-neutral voltages at the start of the iterative routine are

$$[VLN_{abc}] = \begin{bmatrix} 7200 / 0 \\ 7200 / -120 \\ 7200 / 120 \end{bmatrix} \text{ V}$$

The component of currents due to the constant complex power are

$$Ipq_i = \left(\frac{S_i \cdot 1000}{VLN_i} \right)^* \cdot 0.5 = \begin{bmatrix} 155.3 / -26.6 \\ 174.0 / -148.6 \\ 146.0 / 94.7 \end{bmatrix} \text{ A}$$

The constant impedances for that part of the load are computed as

$$Z_i = \frac{[VLN_i]^2}{S_i^* \cdot 1000} = \begin{bmatrix} 20.7 + j10.4 \\ 18.2 + j9.9 \\ 22.3 + j10.6 \end{bmatrix} \Omega$$

For the first iteration, the currents due to the constant impedance portion of the load are

$$I_{z_i} = \left(\frac{VLN_i}{Z_i} \right) \cdot 0.2 = \begin{bmatrix} 62.1 / -26.6 \\ 69.6 / -148.6 \\ 58.4 / 94.7 \end{bmatrix} \text{ A}$$

The magnitudes of the constant current portion of the load are

$$IM_i = \left| \left(\frac{S_i \cdot 1000}{VLN_i} \right)^* \right| \cdot 0.3 = \begin{bmatrix} 93.2 \\ 104.4 \\ 87.6 \end{bmatrix} \text{ A}$$

The contribution of the load currents due to the constant current portion of the load are

$$II_i = IM_i / \underline{\delta_i - \theta_i} = \begin{bmatrix} 93.2 / -26.6 \\ 104.4 / -148.6 \\ 87.6 / 94.7 \end{bmatrix} \text{ A}$$

The total load current is the sum of the three components

$$[I_{abc}] = [I_{pq}] + [I_z] + [I_I] = \begin{bmatrix} 310.6 / -26.6 \\ 348.1 / -148.6 \\ 292.0 / 94.7 \end{bmatrix} \text{ A}$$

- (2) Determine the currents at the start of the second iteration. The voltages at the load after the first iteration are

$$[VLN] = \begin{bmatrix} 6850.0 / -1.9 \\ 6972.7 / -122.1 \\ 6886.1 / 117.5 \end{bmatrix} \text{ V}$$

The steps are repeated with the exceptions that the impedances of the constant impedance portion of the load will not be changed and the magnitude of the currents for the constant current portion of the load change will not change.

The constant complex power portion of the load currents are

$$I_{pq_i} = \left(\frac{S_i \cdot 1000}{VLN_i} \right)^* \cdot 0.5 = \begin{bmatrix} 163.2 / -28.5 \\ 179.7 / -150.7 \\ 152.7 / 92.1 \end{bmatrix} \text{ A}$$

The currents due to the constant impedance portion of the load are

$$I_{z_i} = \left(\frac{VLN_i}{Z_i} \right) \cdot 0.2 = \begin{bmatrix} 59.1 / -28.5 \\ 67.4 / -150.7 \\ 55.9 / 92.1 \end{bmatrix} \text{ A}$$

The currents due to the constant current portion of the load are

$$I_{I_i} = IM_i / \underline{\delta_i - \theta_i} = \begin{bmatrix} 93.2 / -28.5 \\ 104.4 / -150.7 \\ 87.6 / 92.1 \end{bmatrix} \text{ A}$$

The total load currents at the start of the second iteration will be

$$[I_{abc}] = [I_{pq}] + [I_z] + [I_I] = \begin{bmatrix} 315.5 / -28.5 \\ 351.5 / -150.7 \\ 296.2 / 92.1 \end{bmatrix} \text{ A}$$

Observe how these currents have changed from the original currents. The currents for the constant complex power loads have increased because the voltages are reduced from the original assumption. The currents for the constant impedance portion of the load have decreased because the impedance stayed constant but the voltages are reduced. Finally, the magnitude of constant current portion of the load has indeed remained constant. Again, all three components of the load have the same phase angles since the power factor of the load has not changed.

9.2 Delta-Connected Loads

The model for a delta-connected load is shown in Figure 9.2. The notation for the specified complex powers and voltages in Figure 9.2 are as follows:

$$\text{Phase ab: } |S_{ab}| / \underline{\theta_{ab}} = P_{ab} + jQ_{ab} \quad \text{and} \quad |V_{ab}| / \underline{\delta_{ab}} \quad (9.8)$$

$$\text{Phase bc: } |S_{bc}| / \underline{\theta_{bc}} = P_{bc} + jQ_{bc} \quad \text{and} \quad |V_{bc}| / \underline{\delta_{bc}} \quad (9.9)$$

$$\text{Phase ca: } |S_{ca}| / \underline{\theta_{ca}} = P_{ca} + jQ_{ca} \quad \text{and} \quad |V_{ca}| / \underline{\delta_{ca}} \quad (9.10)$$

9.2.1 Constant Real and Reactive Power Loads

The currents in the delta connected loads are

$$\begin{aligned} IL_{ab} &= \left(\frac{S_{ab}}{V_{ab}} \right)^* = \frac{|S_{ab}|}{|V_{ab}|} / \underline{\delta_{ab} - \theta_{ab}} = |IL_{ab}| / \underline{\alpha_{ab}} \\ IL_{bc} &= \left(\frac{S_{bc}}{V_{bc}} \right)^* = \frac{|S_{bc}|}{|V_{bc}|} / \underline{\delta_{bc} - \theta_{bc}} = |IL_{bc}| / \underline{\alpha_{bc}} \\ IL_{ca} &= \left(\frac{S_{ca}}{V_{ca}} \right)^* = \frac{|S_{ca}|}{|V_{ca}|} / \underline{\delta_{ca} - \theta_{ca}} = |IL_{ca}| / \underline{\alpha_{ac}} \end{aligned} \quad (9.11)$$

In this model the line-to-line voltages will change during each iteration resulting in new current magnitudes and angles at the start of each iteration.

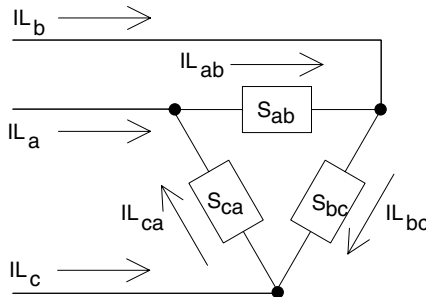


FIGURE 9.2
Delta-connected load.

9.2.2 Constant Impedance Loads

The constant load impedance is first determined from the specified complex power and line-to-line voltages:

$$\begin{aligned} Z_{ab} &= \frac{|V_{ab}|^2}{S_{ab}^*} = \frac{|V_{ab}|^2}{|S_{ab}|} / \underline{\theta_{ab}} = |Z_{ab}| / \underline{\theta_{ab}} \\ Z_{bc} &= \frac{|V_{bc}|^2}{S_{bc}^*} = \frac{|V_{bc}|^2}{|S_{bc}|} / \underline{\theta_{bc}} = |Z_{bc}| / \underline{\theta_{bc}} \\ Z_{ca} &= \frac{|V_{ca}|^2}{S_{ca}^*} = \frac{|V_{ca}|^2}{|S_{ca}|} / \underline{\theta_{ca}} = |Z_{ca}| / \underline{\theta_{ca}} \end{aligned} \quad (9.12)$$

The delta load currents as functions of the constant load impedances are

$$\begin{aligned} IL_{ab} &= \frac{V_{ab}}{Z_{ab}} = \frac{|V_{ab}|}{|Z_{ab}|} / \underline{\delta_{ab} - \theta_{ab}} = |IL_{ab}| / \underline{\alpha_{ab}} \\ IL_{bc} &= \frac{V_{bc}}{Z_{bc}} = \frac{|V_{bc}|}{|Z_{bc}|} / \underline{\delta_{bc} - \theta_{bc}} = |IL_{bc}| / \underline{\alpha_{bc}} \\ IL_{ca} &= \frac{V_{ca}}{Z_{ca}} = \frac{|V_{ca}|}{|Z_{ca}|} / \underline{\delta_{ca} - \theta_{ca}} = |IL_{ca}| / \underline{\alpha_{ca}} \end{aligned} \quad (9.13)$$

In this model the line-to-line voltages will change during each iteration until convergence is achieved.

9.2.3 Constant Current Loads

In this model the magnitudes of the currents are computed according to Equations 9.11 and then held constant while the angle of the voltage (δ) changes during each iteration. This keeps the power factor of the load constant:

$$\begin{aligned} IL_{ab} &= |IL_{ab}| / \underline{\delta_{ab} - \theta_{ab}} \\ IL_{bc} &= |IL_{bc}| / \underline{\delta_{bc} - \theta_{bc}} \\ IL_{ca} &= |IL_{ca}| / \underline{\delta_{ca} - \theta_{ca}} \end{aligned} \quad (9.14)$$

9.2.4 Combination Loads

Combination loads can be modeled by assigning a percentage of the total load to each of the three above load models. The total delta current for each load is the sum of the three components.

9.2.5 Line Currents Serving a Delta-Connected Load

The line currents entering the delta-connected load are determined by applying Kirchhoff's current law at each of the nodes of the delta. In matrix form the equations are

$$\begin{bmatrix} IL_a \\ IL_b \\ IL_c \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} IL_{ab} \\ IL_{bc} \\ IL_{ca} \end{bmatrix} \quad (9.15)$$

9.3 Two-Phase and Single-Phase Loads

In both the wye- and delta-connected loads, single-phase and two-phase loads are modeled by setting the currents of the missing phases to zero. The currents in the phases present are computed using the same appropriate equations for constant complex power, constant impedance, and constant current.

9.4 Shunt Capacitors

Shunt capacitor banks are commonly used in distribution systems to help in voltage regulation and to provide reactive power support. The capacitor banks are modeled as constant susceptances connected in either wye or delta. Similar to the load model, all capacitor banks are modeled as three-phase banks with the currents of the missing phases set to zero for single-phase and two-phase banks.

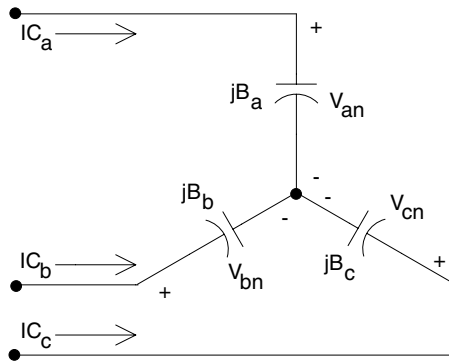
9.4.1 Wye-Connected Capacitor Bank

The model of a three-phase wye connected shunt capacitor bank is shown in Figure 9.3. The individual phase capacitor units are specified in kvar and kV. The constant susceptance for each unit can be computed in Siemens. The susceptance of a capacitor unit is computed by:

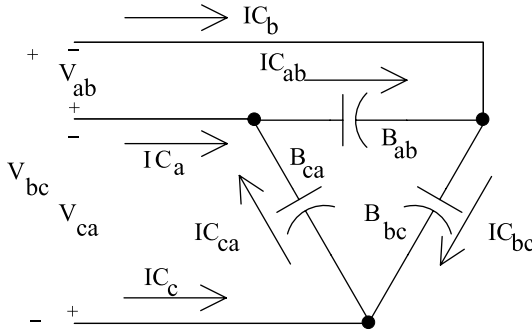
$$B_c = \frac{kvar}{kV_{LN}^2 \cdot 1000} \text{ S} \quad (9.16)$$

With the susceptance computed, the line currents serving the capacitor bank are given by:

$$\begin{aligned} IC_a &= jB_a \cdot V_{an} \\ IC_b &= jB_b \cdot V_{bn} \\ IC_c &= jB_c \cdot V_{cn} \end{aligned} \quad (9.17)$$

**FIGURE 9.3**

Wye-connected capacitor bank.

**FIGURE 9.4**

Delta-connected capacitor bank.

9.4.2 Delta-Connected Capacitor Bank

The model for a delta-connected shunt capacitor bank is shown in Figure 9.4. The individual phase capacitor units are specified in kvar and kV. For the delta-connected capacitors the kV must be the line-to-line voltage. The constant susceptance for each unit can be computed in Siemens. The susceptance of a capacitor unit is computed by:

$$B_c = \frac{kvar}{kV_{LL}^2 \cdot 1000} \text{ S} \quad (9.18)$$

With the susceptance computed, the delta currents serving the capacitor bank are given by:

$$\begin{aligned} I_{C_{ab}} &= jB_a \cdot V_{ab} \\ I_{C_{bc}} &= jB_b \cdot V_{bc} \\ I_{C_{ca}} &= jB_c \cdot V_{ca} \end{aligned} \quad (9.19)$$

The line currents flowing into the delta-connected capacitors are computed by applying Kirchhoff's current law at each node. In matrix form the equations are

$$\begin{bmatrix} IC_a \\ IC_b \\ IC_c \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} IC_{ab} \\ IC_{bc} \\ IC_{ca} \end{bmatrix} \quad (9.20)$$

9.5 The Three-Phase Induction Motor

The analysis of an induction motor when operating under unbalanced voltage conditions has traditionally been performed using the method of symmetrical components. Using this approach, the positive and negative sequence equivalent circuits of the machine are developed and then, given the sequence line-to-neutral voltages, the sequence currents are computed. The zero sequence network is not required since the machines are typically connected delta or ungrounded wye, which means there will not be any zero sequence currents or voltages. The phase currents are determined by transforming back to the phase line currents. The internal operating conditions are determined by the complete analysis of the sequence networks.

The sequence line-to-neutral equivalent circuit of a wye-connected three-phase induction machine is shown in Figure 9.5. The circuit in Figure 9.5 applies to both the positive and negative sequence networks. The only difference between the two is the value of the load resistance RL defined by:

$$RL_i = \frac{1 - s_i}{s_i} \cdot Rr_i \quad (9.21)$$

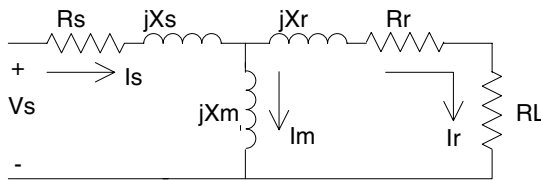


FIGURE 9.5
Sequence equivalent circuit.

where

$i = 1$ for positive sequence
 $i = 2$ for negative sequence.

$$\text{Positive sequence slip: } s_1 = \frac{n_s - n}{n_s} \quad (9.22)$$

n_s = Synchronous speed
 n_r = Rotor speed

$$\text{Negative sequence slip: } s_2 = 2 - s_1 \quad (9.23)$$

Note that the negative sequence load resistance RL_2 will be a negative value. This will lead to a negative shaft power in the negative sequence. The negative sequence currents are attempting to make the motor rotate in the reverse direction. This negative power results in additional power losses and heating in the motor. As a result, according to the ANSI C84.1-1995 standard, the motor must be de-rated when the voltage unbalance is greater than 1%.¹

If the value of positive sequence slip (s_1) is known, then the input sequence impedances for the positive and negative sequence networks can be determined as:

$$ZM_i = Rs_i + jXs_i + \frac{(jXm_i)(Rr_i + RL_i + jXr_i)}{Rr_i + RL_i + j(Xm_i + Xr_i)} \quad (9.24)$$

The input sequence impedances are converted to input sequence admittances by taking the reciprocal of the impedances:

$$YM_i = \frac{1}{ZM_i} \quad (9.25)$$

The sequence motor currents are

$$IM_i = YM_i \cdot VLN_i \quad (9.26)$$

A wye-connected induction motor will not have the neutral grounded. With no ground, the zero sequence current must be zero. The line-to-neutral zero sequence voltage will also be zero. Therefore, it is possible to write an equation equating the zero sequence current to the zero sequence line-to-neutral voltage:

$$IM_0 = VLN_0 = 0 \quad (9.27)$$

Equations 9.26 and 9.27 can be put into matrix form:

$$\begin{bmatrix} IM_0 \\ IM_1 \\ IM_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & YM_1 & 0 \\ 0 & 0 & YM_2 \end{bmatrix} \cdot \begin{bmatrix} VLN_0 \\ VLN_1 \\ VLN_2 \end{bmatrix} \quad (9.28)$$

$$[IM_{012}] = [YM_{012}] \cdot [VLN_{012}] \quad (9.29)$$

With the neutral ungrounded, the only voltages that can be measured are the line-to-line voltages at the motor terminals. In symmetrical component theory, the relationship between sequence line-to-neutral and sequence line-to-line voltages is given by:

$$\begin{bmatrix} VLN_0 \\ VLN_1 \\ VLN_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & t_s^* & 0 \\ 0 & 0 & t_s \end{bmatrix} \cdot \begin{bmatrix} VLL_0 \\ VLL_1 \\ VLL_2 \end{bmatrix} \quad (9.30)$$

where
$$t_s = \frac{1}{\sqrt{3}} \cdot \underline{/30} \quad (9.31)$$

In condensed form, Equation 9.30 becomes:

$$[VLN_{012}] = [T] \cdot [VLL_{012}] \quad (9.32)$$

Substitute Equation 9.32 into Equation 9.29:

$$[IM_{012}] = [YM_{012}] \cdot [T] \cdot [VLL_{012}] \quad (9.33)$$

Equation 9.33 is used to compute the sequence motor currents when the sequence line-to-line voltages are known. What is really needed is a relationship between the phase motor currents and the terminal phase line-to-line voltages. Recognize that in symmetrical component theory:

$$[IM_{abc}] = [A_s] \cdot [IM_{012}] \quad (9.34)$$

$$[VLL_{012}] = [A_s]^{-1} \cdot [VLL_{abc}] \quad (9.35)$$

Applying Equations 9.34 and 9.35 to Equation 9.33 gives the final desired result:

$$\begin{aligned} [IM_{abc}] &= [A_s] \cdot [IM_{012}] = [A_s] \cdot [YM_{012}] \cdot [T] \cdot [A_s]^{-1} \cdot [VLL_{abc}] \\ &= [YM_{abc}] \cdot [VLL_{abc}] \end{aligned} \quad (9.36)$$

where

$$[YM_{abc}] = [A_s] \cdot [YM_{012}] \cdot [T] \cdot [A_s]^{-1} \quad (9.37)$$

$$[YM_{abc}] = \begin{bmatrix} YM_{aa} & YM_{ab} & YM_{ac} \\ YM_{ba} & YM_{bb} & YM_{bc} \\ YM_{ca} & YM_{cb} & YM_{cc} \end{bmatrix} \quad (9.38)$$

The phase admittance matrix $[YM_{abc}]$ will be a packed matrix and can therefore be modeled as a wye-connected constant admittance load with mutual admittances between phases. Even when the motor is being modeled as connected in wye, the applied voltages must be line-to-line. The constant admittance will only be constant for a given value of slip. However, this model allows the motor to be modeled under all load conditions from start (slip = 1) to full load (slip ≈ 0.03).

Example 9.2

To demonstrate the analysis of an induction motor in the phase frame, the following induction motor will be used:

25 Hp, 240 V operating with slip = 0.035

$$Z_s = 0.0774 + j0.1843 \, \Omega$$

$$Z_m = 0 + j4.8384 \, \Omega$$

$$Z_r = 0.0908 + j0.1843 \, \Omega$$

The load resistances are

$$RL_1 = \left(\frac{1 - 0.035}{0.035} \right) \cdot 0.0908 = 2.5029 \, \Omega$$

$$RL_2 = \left(\frac{1 - (1.965)}{1.965} \right) \cdot 0.0908 = -0.0446 \, \Omega$$

The input sequence impedances are

$$ZM_1 = Z_s + \frac{Z_m \cdot (Z_r + RL_1)}{Z_m + Z_r + RL_1} = 1.9775 + j1.3431 \, \Omega$$

$$ZM_2 = Z_s + \frac{Z_m \cdot (Z_r + RL_2)}{Z_m + Z_r + RL_2} = 0.1203 + j0.3623 \, \Omega$$

The positive and negative sequence input admittances are

$$YM_1 = \frac{1}{ZM_1} = 0.3461 - j0.2350 \text{ S}$$

$$YM_2 = \frac{1}{ZM_2} = 0.8255 - j2.4863 \text{ S}$$

The sequence admittance matrix is

$$[YM_{012}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.3461 - j0.2350 & 0 \\ 0 & 0 & 0.8255 - j2.4863 \end{bmatrix} \text{ S}$$

Applying Equation 9.37, the phase admittance matrix is

$$[YM_{abc}] = \begin{bmatrix} 0.7452 - j0.4074 & -0.0999 - j0.0923 & 0.3547 + j0.4997 \\ -0.0999 + j0.0923 & 0.7452 - j0.4074 & -0.0999 - j0.0923 \\ -0.0999 - j0.0923 & 0.3547 + j0.4497 & 0.7452 - j0.4074 \end{bmatrix} \text{ S}$$

The line-to-line terminal voltages are measured to be

$$V_{ab} = 235 \text{ V}, \quad V_{bc} = 240 \text{ V}, \quad V_{ca} = 245 \text{ V}$$

Since the sum of the line-to-line voltages must sum to zero, the law of cosines can be used to determine the angles on the voltages. Applying the law of cosines results in:

$$[VLL_{abc}] = \begin{bmatrix} 235/\underline{0} \\ 240/\underline{-117.9} \\ 245/\underline{120.0} \end{bmatrix} \text{ V}$$

The phase motor currents can now be computed:

$$[IM_{abc}] = [YM_{abc}] \cdot [VLL_{abc}] = \begin{bmatrix} 53.15/\underline{-71.0} \\ 55.15/\underline{-175.1} \\ 66.6/\underline{55.6} \end{bmatrix} \text{ A}$$

It is obvious that the currents are very unbalanced. The measure of unbalance for the voltages and currents can be computed as:

$$V_{\text{unbalance}} = \left(\frac{\text{Max. deviation}}{|V_{\text{avg}}|} \right) \cdot 100 = \left(\frac{5}{240} \right) \cdot 100 = 2.08\%$$

$$I_{\text{unbalance}} = \left(\frac{\text{Max. deviation}}{|I_{\text{avg}}|} \right) \cdot 100 = \left(\frac{8.3232}{58.31} \right) \cdot 100 = 14.27\%$$

This example demonstrates that the current unbalance is seven times greater than the voltage unbalance. This ratio of current unbalance to voltage unbalance is typical. The actual operating characteristics including stator and rotor losses of the motor can be determined using the method developed in Reference 2.

References

1. *American National Standard for Electric Power Systems and Equipment—Voltage Ratings (60 Hertz)*, ANSI C84.1-1995, National Electrical Manufacturers Association, Rosslyn, VA, 1996.
 2. Kersting, W.H. and Phillips, W.H., Phase frame analysis of the effects of voltage unbalance on induction machines, *IEEE Transactions on Industry Applications*, March/April 1997.
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Problems

9.1 A 12.47-kV feeder provides service to an unbalanced wye-connected load specified to be

Phase a: 1000 kVA, 0.9 lagging power factor

Phase b: 800 kVA, 0.95 lagging power factor

Phase c: 1100 kVA, 0.85 lagging power factor

- (1) Compute the initial load currents assuming the loads are modeled as constant complex power.
- (2) Compute the magnitude of the load currents that will be held constant assuming the loads are modeled as constant current.
- (3) Compute the impedance of the load to be held constant assuming the loads are modeled as constant impedance.
- (4) Compute the initial load currents assuming that 60% of the load is complex power, 25% constant current, and 15% constant impedance.

9.2 Using the results of Problem 9.1, rework the problem at the start of the second iteration if the load voltages after the first iteration have been computed to be

$$[VLN_{abc}] = \begin{bmatrix} 6851 / -1.9 \\ 6973 / -122.1 \\ 6886 / 117.5 \end{bmatrix} \text{ V}$$

9.3 A 12.47-kV feeder provides service to an unbalanced delta-connected load specified to be

Phase ab: 1500 kVA, 0.95 lagging power factor

Phase bc: 1000 kVA, 0.85 lagging power factor

Phase ca: 950 kVA, 0.9 lagging power factor

- (1) Compute the load and line currents if the load is modeled as constant complex power.
- (2) Compute the magnitude of the load current to be held constant if the load is modeled as constant current.
- (3) Compute the impedance to be held constant if the load is modeled as constant impedance.
- (4) Compute the line currents if the load is modeled as 25% constant complex power, 20% constant current, and 55% constant impedance.

9.4 After the first iteration of the system of Problem 9.5, the load voltages are

$$[VLL_{abc}] = \begin{bmatrix} 11,981 / 28.3 \\ 12,032 / -92.5 \\ 11,857 / 147.7 \end{bmatrix} \text{ V}$$

- (1) Compute the load and line currents if the load is modeled as constant complex power.
- (2) Compute the load and line currents if the load is modeled as constant current.
- (3) Compute the load and line currents if the load is modeled as constant impedance.
- (4) Compute the line currents if the load mix is 25% constant complex power, 20% constant current, and 55% constant impedance.

9.5 The motor in Example 9.2 is operating with a slip of 0.03 with balanced voltages of 240 V line-to-line. Determine the following:

- (1) The input line currents and complex three-phase power.
- (2) The currents in the rotor circuit.
- (3) The developed shaft power in Hp.

9.6 The motor in Example 9.2 is operating with a slip of 0.03, and the line-to-line voltage magnitudes are

$$V_{ab} = 240 \text{ V}, \quad V_{bc} = 230 \text{ V}, \quad V_{ca} = 250 \text{ V}.$$

- (1) Compute the angles on the line-to-line voltages assuming the voltage a - b is referenced.
- (2) For the given voltages and slip, determine the input line currents and complex three-phase power.
- (3) Compute the rotor currents.
- (4) Compute the developed shaft power in Hp.

9.7 For the motor in Example 9.2, determine the starting line current if the motor terminal voltages are those given in Problem 9.6.

10

Distribution Feeder Analysis

The analysis of a distribution feeder will typically consist of a study of the feeder under normal steady-state operating conditions (power-flow analysis), and a study under short-circuit conditions (short-circuit analysis). Models of all of the components of a distribution feeder have been developed in previous chapters. These models will be applied for the analysis under steady-state and short-circuit conditions.

10.1 Power-Flow Analysis

The power-flow analysis of a distribution feeder is similar to that of an interconnected transmission system. Typically, what will be known prior to the analysis will be the three-phase voltages at the substation and the complex power of all of the loads and the load model (constant complex power, constant impedance, constant current, or a combination). Sometimes the input complex power supplied to the feeder from the substation is also known.

In Chapters 6, 7, and 8 phase frame models were developed for the series components of a distribution feeder. In Chapter 9 models were developed for the shunt components (loads and capacitor banks). These models are used in the power-flow analysis of a distribution feeder.

A power-flow analysis of a feeder can determine the following by phase and total three-phase:

- Voltage magnitudes and angles at all nodes of the feeder
- Line flow in each line section specified in kW and kvar, amps and degrees, or amps and power factor
- Power loss in each line section
- Total feeder input kW and kvar
- Total feeder power losses
- Load kW and kvar based upon the specified model for the load

10.1.1 The Ladder Iterative Technique

Because a distribution feeder is radial, iterative techniques commonly used in transmission network power-flow studies are not used because of poor convergence characteristics.¹ Instead, an iterative technique specifically designed for a radial system is used.

10.1.1.1 Linear Network

A modification of the ladder network theory of linear systems provides a robust iterative technique for power-flow analysis.² A distribution feeder is nonlinear because most loads are assumed to be constant kW and kvar. However, the approach taken for the linear system can be modified to take into account the nonlinear characteristics of the distribution feeder. Figure 10.1 shows a linear ladder network. For the ladder network it is assumed that all of the line impedances and load impedances are known along with the voltage at the source (V_s). The solution for this network is to assume a voltage at the most remote load (V_5). The load current I_5 is then determined as:

$$I_5 = \frac{V_5}{Z_{L5}} \quad (10.1)$$

For this end node case, the line current I_{45} is equal to the load current I_5 . Applying Kirchhoff's voltage law (KVL), the voltage at Node 4 (V_4) can be determined:

$$V_4 = V_5 + Z_{45} \cdot I_{45} \quad (10.2)$$

The load current I_4 can be determined, and then Kirchhoff's current law (KCL) applied to determine the line current I_{34} :

$$I_{34} = I_{45} + I_4 \quad (10.3)$$

Kirchhoff's voltage law is applied to determine the node voltage V_3 . This procedure is continued until a voltage (V_1) has been computed at the source.

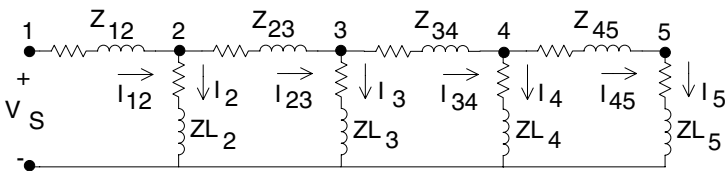


FIGURE 10.1
Linear ladder network.

The computed voltage V_1 is compared to the specified voltage V_s . There will be a difference between these two voltages. The ratio of the specified voltage to the compute voltage can be determined as:

$$\text{Ratio} = \frac{V_s}{V_1} \quad (10.4)$$

Since the network is linear, all of the line and load currents and node voltages in the network can be multiplied by the *Ratio* for the final solution to the network.

10.1.1.2 Nonlinear Network

The linear network of Figure 10.1 is modified to a nonlinear network by replacing all of the constant load impedances by constant complex power loads as shown in Figure 10.2. The procedure outlined for the linear network is applied initially to the nonlinear network. The only difference is that the load current at each node is computed by:

$$I_n = \left(\frac{S_n}{V_n} \right)^* \quad (10.5)$$

The forward sweep will determine a computed source voltage V_1 . As in the linear case, this first iteration will produce a voltage that is not equal to the specified source voltage V_s . Because the network is nonlinear, multiplying currents and voltages by the ratio of the specified voltage to the computed voltage will not give the solution. The most direct modification to the ladder network theory is to perform a backward sweep. The backward sweep commences by using the specified source voltage and the line currents from the forward sweep. Kirchhoff's voltage law is used to compute the voltage at Node 2 by:

$$V_2 = V_s - Z_{12} \cdot I_{12} \quad (10.6)$$

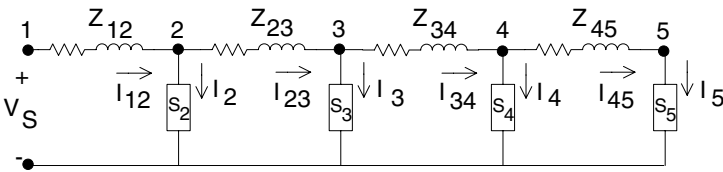


FIGURE 10.2
Nonlinear ladder network.

This procedure is repeated for each line segment until a new voltage is determined at Node 5. Using the new voltage at Node 5, a second forward sweep is started that will lead to a new computed voltage at the source.

The forward and backward sweep process is continued until the difference between the computed and specified voltage at the source is within a given tolerance.

Example 10.1

A single-phase lateral is shown in Figure 10.3. The line impedance is

$$z = 0.3 + j0.6 \, \Omega/\text{mile}$$

The impedance of the line segment 1-2 is

$$Z_{12} = (0.3 + j0.6) \cdot \frac{3000}{5280} = 0.1705 + j0.3409 \, \Omega$$

The impedance of the line segment 2-3 is

$$Z_{23} = (0.3 + j0.6) \cdot \frac{4000}{5280} = 0.2273 + j0.4545 \, \Omega$$

The loads are

$$\begin{aligned} S_2 &= 1500 + j750 \\ S_3 &= 900 + j500 \end{aligned} \quad (\text{kW} + j\text{kvar})$$

The source voltage at Node 1 is 7200 V.

Compute the node voltages after one full iteration.

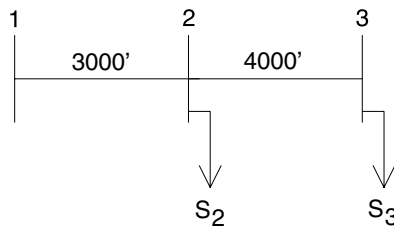


FIGURE 10.3
Single-phase lateral.

The forward sweep begins by assuming the voltage at Node 3 to be $7200/\underline{0}$. The load current at Node 3 is computed to be

$$I_3 = \left(\frac{(900 + j500) \cdot 1000}{7200/\underline{0}} \right)^* = 143.0/\underline{-29.0} \text{ A}$$

The current flowing in the line section 2-3 is

$$I_{23} = I_3 = 143.0/\underline{-29.0} \text{ A}$$

The voltage at Node 2 is computed to be

$$\begin{aligned} V_2 &= V_3 + Z_{23} \cdot I_{23} = 7200/\underline{0} + (0.2273 + j0.4545) \cdot 143.0/\underline{-29.1} \\ &= 7260.1/\underline{0.32} \text{ V} \end{aligned}$$

The load current at Node 2 is

$$I_2 = \left(\frac{(1500 + j750) \cdot 1000}{7260.1/\underline{0.32}} \right)^* = 231.0/\underline{-26.3} \text{ A}$$

The current in line segment 1-2 is

$$I_{12} = I_{23} + I_2 = 373.9/\underline{-27.3} \text{ A}$$

The computed voltage at the source Node 1 is

$$V_1 = V_2 + Z_{12} \cdot I_{12} = 7376.2/\underline{0.97} \text{ V}$$

At this point the magnitude of the computed voltage at Node 1 is compared to the magnitude of the specified source voltage:

$$\text{Error} = ||V_s| - |V_1|| = 176.2 \text{ V}$$

If the error is less than a specified tolerance, the solution has been achieved. If the error is greater than the tolerance, the backward sweep begins. A typical tolerance is 0.001 per unit, which on a 7200-V base is 7.2 V. Since the error in this case is greater than the tolerance, the backward sweep begins by setting the voltage at Node 1 to the specified source voltage:

$$V_1 = V_s = 7200/\underline{0} \text{ V}$$

Now the voltage at Node 2 is computed using this value of the Node 1 voltage and the computed line current in the forward sweep current:

$$\begin{aligned} V_2 &= V_1 - Z \cdot I_{12} = 7200 \angle 0 - (0.1705 + j0.3409) \cdot 373.9 \angle -27.2 \\ &= 7085.4 \angle -0.68 \text{ V} \end{aligned}$$

The backward sweep continues by computing the next downstream voltage. All of the currents computed in the forward sweep are used in the backward sweep:

$$V_3 = V_2 - Z \cdot I_{23} = 7026.0 \angle -1.02 \text{ V}$$

This completes the first iteration. At this point the forward sweep will be repeated, only this time starting with the new voltage at Node 3 rather than the initially assumed voltage.

10.1.2 The General Feeder

A typical distribution feeder will consist of the primary main, with laterals tapped off the primary main and sublaterals tapped off the laterals, etc. Figure 10.4 shows a typical feeder. The ladder iterative technique for the feeder of Figure 10.4 would proceed as follows:

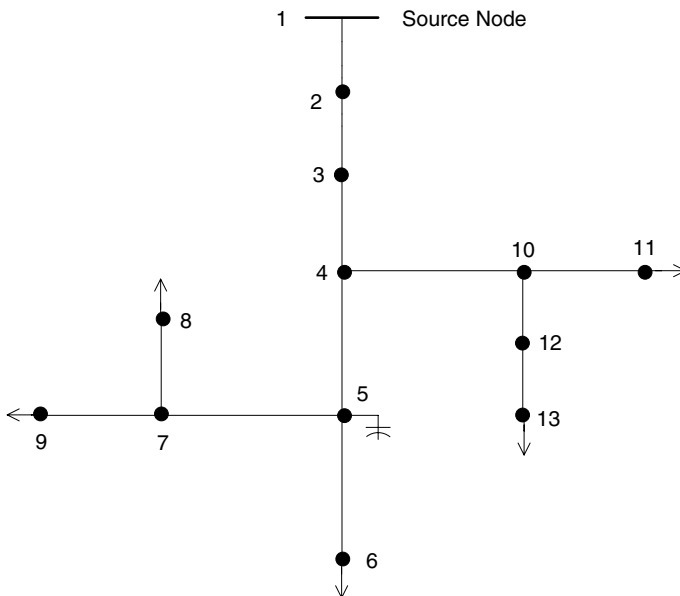


FIGURE 10.4
Typical distribution feeder.

1. Assume three-phase voltages at the end nodes (6,8,9,11, and 13). The usual assumption is to use the nominal voltages.
2. Starting at Node 13, compute the node current (load current plus capacitor current if present).
3. With this current, apply Kirchhoff's voltage law (KVL) to calculate the node voltages at 12 and 10.
4. Node 10 is referred to as a "junction" node since laterals branch in two directions from the node. For this feeder, go to Node 11 and compute the node current. Use that current to compute the voltage at Node 10. This will be referred to as "the most recent voltage at Node 10."
5. Using the most recent value of the voltage at Node 10, the node current at Node 10 (if any) is computed.
6. Apply Kirchhoff's current law (KCL) to determine the current flowing from Node 4 toward Node 10.
7. Compute the voltage at Node 4.
8. Node 4 is a junction node. An end node downstream from Node 4 is selected to start the forward sweep toward Node 4.
9. Select Node 6, compute the node current, and then compute the voltage at Junction Node 5.
10. Go to downstream end Node 8. Compute the node current and then the voltage at Junction Node 7.
11. Go to downstream end Node 9. Compute the node current and then the voltage at Junction Node 7.
12. Compute the node current at Node 7 using the most recent value of the Node 7 voltage.
13. Apply KCL at Node 7 to compute the current flowing on the line segment from Node 5 to Node 7.
14. Compute the voltage at Node 5.
15. Compute the node current at Node 5.
16. Apply KCL at Node 5 to determine the current flowing from Node 4 toward Node 5.
17. Compute the voltage at Node 4.
18. Compute the node current at Node 4.
19. Apply KCL at Node 4 to compute the current flowing from Node 3 to Node 4.
20. Calculate the voltage at Node 3.
21. Compute the node current at Node 3.
22. Apply KCL at Node 3 to compute the current flowing from Node 2 to Node 3.

23. Calculate the voltage at Node 2.
24. Compute the node current at Node 2.
25. Apply KCL at Node 2.
26. Calculate the voltage at Node 1.
27. Compare the calculated voltage at Node 1 to the specified source voltage.
28. If not within tolerance, use the specified source voltage and the forward sweep current flowing from Node 1 to Node 2, and compute the new voltage at Node 2.
29. The backward sweep continues, using the new upstream voltage and line segment current from the forward sweep to compute the new downstream voltage.
30. The backward sweep is completed when new voltages at all end nodes have been completed.
31. This completes the first iteration.
32. Repeat the forward sweep, only now using the new end voltages rather than the assumed voltages as was done in the first iteration.
33. Continue the forward and backward sweeps until the calculated voltage at the source is within a specified tolerance of the source voltage.
34. At this point the voltages are known at all nodes, and the currents flowing in all line segments are known. An output report can be produced giving all desired results.

10.1.3 The Unbalanced Three-Phase Distribution Feeder

The previous section outlined the general procedure for performing the ladder iterative technique. This section will address how that procedure can be used for an unbalanced three-phase feeder.

Figure 10.5 is the one-line diagram of an unbalanced three-phase feeder. The topology of the feeder in Figure 10.5 is the same as the feeder in Figure 10.4. Figure 10.5 shows more detail of the feeder, however. The feeder of Figure 10.5 can be broken into “series” components and “shunt” components.

10.1.3.1 Series Components

The series components of a distribution feeder are

- Line segments
- Transformers
- Voltage regulators

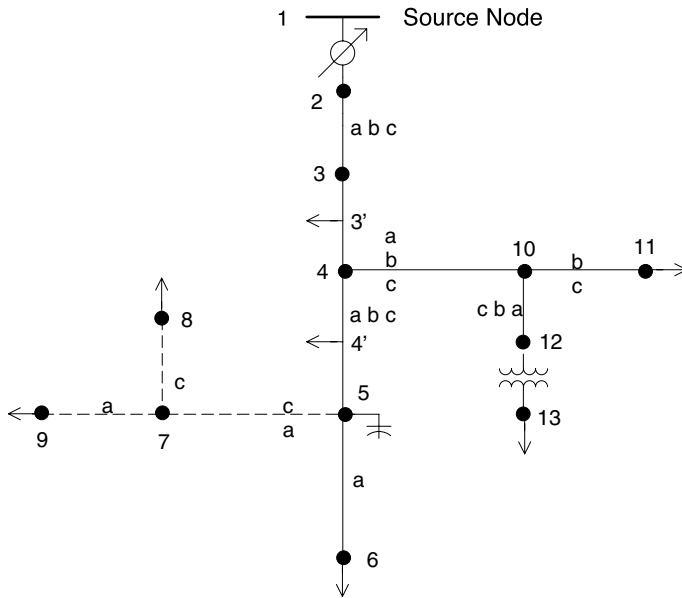


FIGURE 10.5
Unbalanced three-phase distribution feeder.



FIGURE 10.6
Series feeder component.

Models for each of the series components have been developed in earlier chapters. In all cases, models (three-phase, two-phase, and single-phase) were in terms of generalized matrices. Figure 10.6 shows the general model for each of the series components. The general equations defining the input (Node n) and output (Node m) voltages and currents are given by:

$$[V_{abc}]_n = [a] \cdot [V_{abc}]_m + [b] \cdot [I_{abc}]_m \quad (10.7)$$

$$[I_{abc}]_n = [c] \cdot [V_{abc}]_m + [d] \cdot [I_{abc}]_m \quad (10.8)$$

The general equation relating the output (Node m) and input (Node n) voltages are given by:

$$[V_{abc}]_m = [A] \cdot [V_{abc}]_n - [B] \cdot [I_{abc}]_m \quad (10.9)$$

In Equations 10.7, 10.8, and 10.9 the voltages are line-to-neutral for a four-wire wye feeder and equivalent line-to-neutral for a three-wire delta system. For voltage regulators, the voltages are line-to-neutral for terminals that are connected to a four-wire wye, and line-to-line when connected to a three-wire delta.

10.1.3.2 Shunt Components

The shunt components of a distribution feeder are

- Spot loads
- Distributed loads
- Capacitor banks

Spot loads are located at a node and can be three-phase, two-phase, or single phase, and connected in either a wye or a delta connection. The loads can be modeled as constant complex power, constant current, constant impedance, or a combination of the three.

Distributed loads are modeled in accordance with Figure 3.11. A distributed load is modeled when the loads on a line segment are uniformly distributed along the length of the segment. As in the spot load, the distributed load can be three-phase, two-phase, or single-phase, and connected in either a wye or delta. The loads can be modeled as constant complex power, constant current, constant impedance or a combination of the three. Two thirds of the load is connected at a dummy node located at the one-quarter point of the line and the remaining one third of the load is connected at the load end of the line segment.

Capacitor banks are located at a node and can be three-phase, two-phase, or single-phase, and can be connected in a wye or delta. Capacitor banks are modeled as constant admittances.

In Figure 10.5 the solid line segments represent overhead lines, while the dashed lines represent underground lines. Note that the phasing is shown for all of the line segments. In Chapter 4 the application of Carson's equations for computing the line impedances for overhead and underground lines was presented. In that chapter it was pointed out that two-phase and single-phase lines are represented by a three-by-three matrix with zeros set in the rows and columns of the missing phases.

In Chapter 5 the method for the computation of the shunt capacitive susceptance for overhead and underground lines was presented. Most of the time the shunt capacitance of the line segment can be ignored, however, for long underground line segments the shunt capacitance should be included.

The node currents may be three-phase, two-phase, or single-phase and consist of the sum of the load current at the node plus the capacitor current (if any) at the node.

10.1.4 Applying the Ladder Iterative Technique

Section 10.1.2 outlined the steps required for the application of the ladder iterative technique. Generalized matrices have been developed in Chapters 6, 7, and 8 for the series devices. By applying the generalized matrices, the computation of the voltage drops along a segment will always be the same regardless of whether the segment represents a line, voltage regulator, or transformer.

In the preparation of data for a power-flow study, it is extremely important that the impedances and admittances of the line segments be computed using the exact spacings and phasing. Because of the unbalanced loading and resulting unbalanced line currents, the voltage drops due to the mutual coupling of the lines become very important. It is not unusual to observe a voltage rise on a lightly loaded phase of a line segment that has an extreme current unbalance.

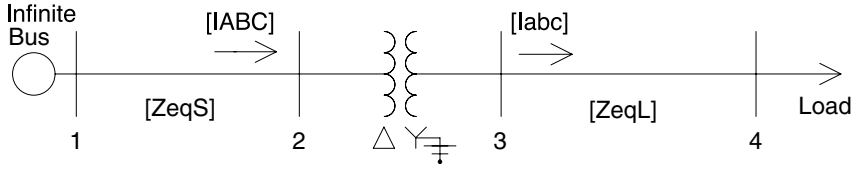
The real power loss in a device should not be computed by using the phase current squared times the phase resistance. In a balanced system that works, however, in an unbalanced system, the real power losses of a line segment must be computed as the difference (by phase) of the input power in a line segment minus the output power of the line segment. It is possible to have a negative power loss on a phase that is lightly loaded compared to the other two phases. Computing power loss as the phase current squared times the phase resistance does not give the actual real power loss in the phases.

10.1.5 Putting It All Together

At this point of the text, the models for all components of a distribution feeder have been developed. The modified ladder iterative technique has also been developed. It is time to put them all together and demonstrate the power-flow analysis of a very simple system. Example 10.2 will be long but will demonstrate how the models of the components work together in applying the modified ladder technique to achieve a final solution of the operating characteristics of an unbalanced feeder.

Example 10.2

A very simple distribution feeder is shown in Figure 10.7. For the system in Figure 10.7, the infinite bus voltages are balanced three-phase of 12.47 kV line-to-line. The source line segment from Node 1 to Node 2 is a three-wire delta 2000 ft. long and is constructed on the pole configuration of Figure 4.7. The load line segment from Node 3 to Node 4 is 2500 ft. long and is also constructed on the pole configuration of Figure 4.7, but is a four-wire wye.

**FIGURE 10.7**

Example 10.2 feeder.

Both line segments use 336,400 26/7 ACSR phase conductors, and the neutral conductor on the four-wire wye line is 4/0 6/1 ACSR. Since the lines are short, the shunt admittance will be neglected. The phase impedance matrices for the two line segments are

$$[ZeqS_{ABC}] = \begin{bmatrix} 0.1414 + j0.5353 & 0.0361 + j0.3225 & 0.0361 + j0.2752 \\ 0.0361 + j0.3225 & 0.1414 + j0.5353 & 0.0361 + j0.2955 \\ 0.0361 + j0.2752 & 0.0361 + j0.2955 & 0.1414 + j0.5353 \end{bmatrix} \Omega$$

$$[ZeqL_{abc}] = \begin{bmatrix} 0.1907 + j0.5035 & 0.0607 + j0.2302 & 0.0598 + j0.1751 \\ 0.0607 + j0.2302 & 0.1939 + j0.4885 & 0.0614 + j0.1931 \\ 0.0598 + j0.1751 & 0.0614 + j0.1931 & 0.1921 + j0.4970 \end{bmatrix} \Omega$$

The transformer bank is connected delta-grounded wye and consists of three single-phase transformers each rated:

$$2000 \text{ kVA}, \quad 12.47 - 2.4 \text{ kV}, \quad Z = 1.0 + j6.0\%$$

The feeder serves an unbalanced three-phase wye-connected load of:

$$S_a = 750 \text{ kVA at } 0.85 \text{ lagging power factor}$$

$$S_b = 1000 \text{ kVA at } 0.90 \text{ lagging power factor}$$

$$S_c = 1250 \text{ kVA at } 0.95 \text{ lagging power factor}$$

Before starting the iterative solution, the generalized matrices for the three series components must be defined.

SOURCE LINE SEGMENT

Equations 6.9 and 6.18:

$$[a_1] = [d_1] = [U] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Equation 6.10:

$$[b_1] = [ZeqS_{ABC}] = \begin{bmatrix} 0.1414 + j0.5353 & 0.0361 + j0.3225 & 0.0361 + j0.2752 \\ 0.0361 + j0.3225 & 0.1414 + j0.5353 & 0.0361 + j0.2955 \\ 0.0361 + j0.2752 & 0.0361 + j0.2955 & 0.1414 + j0.5353 \end{bmatrix}$$

Equation 6.17:

$$[c_1] = [0]$$

Equation 6.27:

$$[A_1] = [a_1]^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Equation 6.28:

$$\begin{aligned} [B_1] &= [a_1]^{-1} \cdot [b_1] \\ &= \begin{bmatrix} 0.1414 + j0.5353 & 0.0361 + j0.3225 & 0.0361 + j0.2752 \\ 0.0361 + j0.3225 & 0.1414 + j0.5353 & 0.0361 + j0.2955 \\ 0.0361 + j0.2752 & 0.0361 + j0.2955 & 0.1414 + j0.5353 \end{bmatrix} \end{aligned}$$

LOAD LINE SEGMENT (using the same equations as for the source segment)

$$[a_2] = [d_2] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[b_2] = \begin{bmatrix} 0.1907 + j0.5035 & 0.0607 + j0.2302 & 0.0598 + j0.1751 \\ 0.0607 + j0.2302 & 0.1939 + j0.4885 & 0.0614 + j0.1931 \\ 0.0598 + j0.1751 & 0.0614 + j0.1931 & 0.1921 + j0.4970 \end{bmatrix}$$

$$[c_2] = [0]$$

$$[A_2] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[B_2] = \begin{bmatrix} 0.1907 + j0.5035 & 0.0607 + j0.2302 & 0.0598 + j0.1751 \\ 0.0607 + j0.2302 & 0.1939 + j0.4885 & 0.0614 + j0.1931 \\ 0.0598 + j0.1751 & 0.0614 + j0.1931 & 0.1921 + j0.4970 \end{bmatrix}$$

TRANSFORMER

The transformer impedance must be converted to actual values in ohms referenced to the low-voltage windings.

$$Z_{\text{base}} = \frac{2.4^2 \cdot 1000}{2000} = 2.88 \, \Omega$$

$$Zt_{\text{low}} = (0.01 + j0.06) \cdot 2.88 = 0.0288 + j0.1728 \, \Omega$$

The transformer phase impedance matrix is

$$[Zt_{abc}] = \begin{bmatrix} 0.0288 + j0.1728 & 0 & 0 \\ 0 & 0.0288 + j0.1728 & 0 \\ 0 & 0 & 0.0288 + j0.1728 \end{bmatrix} \, \Omega$$

$$\text{The turns ratio: } n_t = \frac{12.47}{2.4} = 5.1958$$

$$\text{The transformer ratio: } a_t = \frac{12.47}{\sqrt{3} \cdot 2.4} = 2.9998$$

The generalized matrices are

Equation 8.26:

$$[a_t] = \frac{-n_t}{3} \cdot \begin{bmatrix} 0 & 2 & 1 \\ 1 & 0 & 2 \\ 2 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -3.4639 & -1.7319 \\ -1.7319 & 0 & -3.4639 \\ -3.4639 & -1.7319 & 0 \end{bmatrix}$$

Equation 8.30:

$$[b_t] = \frac{-n_t}{3} \cdot \begin{bmatrix} 0 & 2 \cdot Zt & Zt \\ Zt & 0 & 2 \cdot Zt \\ 2 \cdot Zt & Zt & 0 \end{bmatrix}$$

$$[b_t] = \begin{bmatrix} 0 & -0.0998 - j0.5986 & -0.0499 - j0.2993 \\ -0.0499 - j0.2993 & 0 & -0.0998 - j0.5986 \\ -0.0998 - j0.5986 & -0.0499 - j0.2993 & 0 \end{bmatrix}$$

Equation 8.45:

$$[c_t] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Equation 8.44:

$$[d_t] = \frac{1}{n_t} \cdot \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.1925 & -0.1925 & 0 \\ 0 & 0.1925 & -0.1925 \\ -0.1925 & 0 & 0.1925 \end{bmatrix}$$

Equation 8.35:

$$[A_t] = \frac{1}{n_t} \cdot \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 0.1925 & 0 & -0.1925 \\ -0.1925 & 0.1925 & 0 \\ 0 & -0.1925 & 0.1925 \end{bmatrix}$$

Equation 8.38:

$$[B_t] = [Zt_{abc}] = \begin{bmatrix} 0.0288 + j0.1728 & 0 & 0 \\ 0 & 0.0288 + j0.1728 & 0 \\ 0 & 0 & 0.0288 + j0.1728 \end{bmatrix}$$

Define infinite bus line-to-line and line-to-neutral voltages:

$$[ELL_s] = \begin{bmatrix} 12,470/\underline{30} \\ 12,470/\underline{-90} \\ 12,470/\underline{150} \end{bmatrix} \text{ V}$$

$$[ELN_s] = \begin{bmatrix} 7199.6/\underline{0} \\ 7199.6/\underline{-120} \\ 7199.6/\underline{120} \end{bmatrix} \text{ V}$$

Set the line-to-neutral voltages at Node 4 equal to the nominal voltage shifted by 30 degrees:

$$[V4] = \begin{bmatrix} 2400/\underline{-30} \\ 2400/\underline{-150} \\ 2400/\underline{90} \end{bmatrix} \text{ V}$$

Define the Node 4 loads:

$$[S4] = \begin{bmatrix} 750/\underline{31.79} \\ 1000/\underline{25.84} \\ 1250/\underline{18.19} \end{bmatrix} \text{ kVA}$$

Start the forward sweep by computing the load currents at Node 4:

$$I4_i = \left(\frac{S_i \cdot 1000}{V4_i} \right)^* = \begin{bmatrix} 312.5 / -61.8 \\ 416.7 / -175.8 \\ 520.8 / 71.8 \end{bmatrix} \text{ A}$$

Compute the voltages and currents at Node 3:

$$[V3] = [a_2] \cdot [V4] + [b_2] \cdot [I4] = \begin{bmatrix} 2470.9 / -29.5 \\ 2534.4 / -148.4 \\ 2509.5 / 94.1 \end{bmatrix} \text{ V}$$

$$[I3] = [c_2] \cdot [V4] + [d_2] \cdot [I4] = \begin{bmatrix} 312.5 / 61.8 \\ 416.7 / -175.8 \\ 520.8 / 71.8 \end{bmatrix} \text{ A}$$

Compute the voltages and currents at Node 2:

$$[V2] = [a_t] \cdot [V3] + [b_t] \cdot [I3] = \begin{bmatrix} 7956.4 / 3.3 \\ 7344.5 / -113.4 \\ 7643.0 / 120.5 \end{bmatrix} \text{ V}$$

$$[I2] = [c_t] \cdot [V3] + [d_t] \cdot [I3] = \begin{bmatrix} 118.2 / -23.5 \\ 150.3 / -137.8 \\ 148.3 / 88.9 \end{bmatrix} \text{ A}$$

Compute the equivalent LN voltages and line currents at Node 1:

$$[V1] = [a_1] \cdot [V2] + [b_1] \cdot [I2] = \begin{bmatrix} 7985.9 / 3.4 \\ 7370.6 / -113.2 \\ 7673.6 / 120.7 \end{bmatrix} \text{ V}$$

$$[I1] = [c_1] \cdot [V2] + [d_1] \cdot [I2] = \begin{bmatrix} 118.2 / -23.5 \\ 150.3 / -137.8 \\ 148.3 / 88.9 \end{bmatrix} \text{ A}$$

The computed line-to-line voltage at Node 1 is

$$[VLL1] = [D] \cdot [V1] = \begin{bmatrix} 13,067.5 / 33.7 \\ 13,411.4 / -85.7 \\ 13,375.9 / 152.7 \end{bmatrix} \text{ V}$$

Compute the magnitude of the line-to-line voltage errors:

$$[\text{Error}]_{pu} = \frac{[ELL_s - VLL1]}{12,470} = \begin{bmatrix} 0.0809 \\ 0.1086 \\ 0.0876 \end{bmatrix} \text{ per-unit}$$

Since these errors are greater than the usual tolerance of 0.001 per unit, the backward sweep begins. The backward sweep uses the equivalent line-to-neutral voltage at the source as the Node 1 voltage, and proceeds to Node 4 using the line currents from the forward sweep.

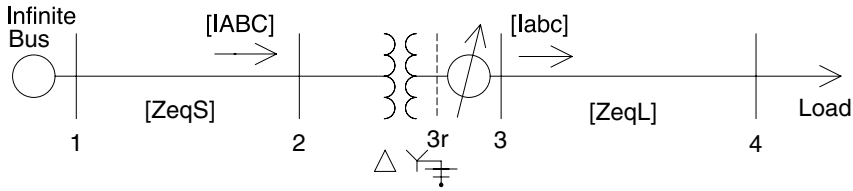
$$[V2] = [A_t] \cdot [ELN_s] - [B_t] \cdot [I2] = \begin{bmatrix} 7171.1 / -0.1 \\ 7176.7 / -120.2 \\ 7169.3 / 119.8 \end{bmatrix} \text{ V}$$

$$[V3] = [A_t] \cdot [V2] - [B_t] \cdot [I3] = \begin{bmatrix} 2354.0 / -31.2 \\ 2351.0 / -151.6 \\ 2349.9 / 87.8 \end{bmatrix} \text{ V}$$

$$[V4] = [A_2] \cdot [V3] - [B_2] \cdot [I4] = \begin{bmatrix} 2283.7 / -31.7 \\ 2221.4 / -153.6 \\ 2261.0 / 83.2 \end{bmatrix} \text{ V}$$

This completes the first iteration. The second iteration begins by computing the currents at the Node 4 load using the new values of the Node 4 voltages. The forward sweep uses these new currents. The forward and backward sweeps continue until the error at the source is less than the specified tolerance of 0.001 per-unit. After four iterations the solution has converged to a tolerance of 0.0003 per-unit. The resulting load voltages at Node 4 are

$$[V4_{\text{final}}] = \begin{bmatrix} 2278.7 / -31.8 \\ 2199.8 / -153.5 \\ 2211.2 / 83.1 \end{bmatrix} \text{ V}$$

**FIGURE 10.8**

Voltage regulator added to the system.

On a 120-V base the final voltages are

$$[V_{4_{120}}] = \begin{bmatrix} 113.9 / -31.8 \\ 110.0 / -153.5 \\ 110.6 / 83.1 \end{bmatrix} \text{ V}$$

The voltages at Node 4 are below the required 120 (121 ± 1) V. These low voltages can be corrected with the installation of three step-voltage regulators connected in wye on the secondary bus (Node 3) of the transformer. The new configuration of the feeder is shown in Figure 10.8. For the regulator, the potential transformer ratio will be 2400-120 V ($N_{pt} = 20$), and the CT ratio is selected to carry the rated current of the transformer bank. The rated current is

$$I_{\text{rated}} = \frac{6000}{\sqrt{3} \cdot 2.4} = 832.7$$

The CT ratio is selected to be 1000:5 = CT = 200.

The equivalent phase impedance between Node 3 and Node 4 is computed using the converged voltages at the two nodes. This is done so that the R and X settings of the compensator can be determined:

$$Zeq_i = \frac{V3_i - V4_i}{I3_i} = \begin{bmatrix} 0.1414 + j0.1829 \\ 0.2078 + j0.2826 \\ 0.0889 + j0.3833 \end{bmatrix} \Omega$$

The three regulators are to have the same R and X compensator settings. The average value of the computed impedances will be used:

$$Z_{\text{avg}} = \frac{1}{3} \cdot \sum_{k=1}^3 Zeq_k = 0.1461 + j0.2830 \Omega$$

The value of the compensator impedance in volts is given by Equation 7.81:

$$R' + jX' = (0.1461 + j0.2830) \cdot \frac{1000}{20} = 7.3 + j14.2 \text{ V}$$

The value of the compensator settings in ohms is

$$R_{\Omega} + jX_{\Omega} = \frac{7.3 + j14.2}{5} = 1.46 + j2.84 \text{ } \Omega$$

With the regulator in the neutral position, the voltages being input to the compensator circuit for the given conditions are

$$V_{reg_i} = \frac{V3_i}{PT} = \begin{bmatrix} 117.5 / -31.2 \\ 117.1 / -151.7 \\ 116.7 / 87.8 \end{bmatrix} \text{ V}$$

The compensator currents are

$$I_{comp_i} = \frac{I3_i}{CT} = \begin{bmatrix} 1.6456 / -63.6 \\ 2.2730 / -179.4 \\ 2.8267 / 64.9 \end{bmatrix} \text{ A}$$

With the input voltages and compensator currents, the voltages across the voltage relays in the compensator circuit are computed to be

$$[V_{\text{relay}}] = [V_{\text{reg}}] - [Z_{\text{comp}}] \cdot [I_{\text{comp}}] = \begin{bmatrix} 113.0 / -32.5 \\ 111.2 / -153.8 \\ 110.0 / 84.7 \end{bmatrix} \text{ V}$$

Notice how close these compare to the actual voltages on a 120-V base at Node 4. Assume that the voltage level has been set at 121 V with a bandwidth of 2 V. The regulators will change taps until the phase voltage is at least 120 V. With the computed voltages across the relays, the approximate number of steps that each regulator will move to can be computed by:

$$Tap_i = \frac{120 - |V4_i|}{0.75} = \begin{bmatrix} 9.3 \\ 11.7 \\ 13.4 \end{bmatrix} \approx \begin{bmatrix} 9 \\ 12 \\ 13 \end{bmatrix}$$

With the taps set at 9, 12, and 13, the $[a]$ matrix for the regulators is

$$[a_r] = \begin{bmatrix} 1.0 - 0.00625 \cdot Tap_1 & 0 & 0 \\ 0 & 1.0 - 0.00625 \cdot Tap_2 & 0 \\ 0 & 0 & 1.0 - 0.00625 \cdot Tap_3 \end{bmatrix}$$

$$[a_r] = \begin{bmatrix} 0.9438 & 0 & 0 \\ 0 & 0.9250 & 0 \\ 0 & 0 & 0.9188 \end{bmatrix}$$

The $[d]$ matrix for the regulators is

$$[d_r] = [a_r]^{-1} = \begin{bmatrix} 1.0596 & 0 & 0 \\ 0 & 1.0811 & 0 \\ 0 & 0 & 1.0884 \end{bmatrix}$$

The $[b]$, $[c]$, and $[B]$ matrices are zero and the $[A]$ matrix is given by:

$$[A_r] = [a_r]^{-1} = \begin{bmatrix} 1.0596 & 0 & 0 \\ 0 & 1.0811 & 0 \\ 0 & 0 & 1.0884 \end{bmatrix}$$

The procedure now is to go back and perform the same modified ladder iterative technique as before, only this time the regulators are an added element between Nodes $3r$ and 3 as shown in Figure 10.8.

With the taps on their specified settings, the system converges after four iterations with the voltages on a 120-V base at Node 4:

$$[V_{4_{120}}] = \begin{bmatrix} 121.0 / -31.8 \\ 120.1 / -153.3 \\ 121.5 / 83.9 \end{bmatrix} \text{ V}$$

The voltages on all phases are within the specified limits.

Example 10.2 has demonstrated how all the series components of a feeder can be modeled and then used in the modified ladder technique to achieve a final solution. The example has also demonstrated how the compensator circuit settings are determined, and then how the compensator circuit causes the taps to change on the individual regulators so that the final voltages at the load center (Node 4) will be within the specified limits.

10.1.6 Load Allocation

Many times the input complex power (kW and kVAR) to a feeder is known because of the metering at the substation. This information can be either total three-phase or for each individual phase. In some cases the metered data may be the current and power factor in each phase.

It is desirable to force the computed input complex power to the feeder to match the metered input. This can be accomplished (following a converged iterative solution) by computing the ratio of the metered input to the computed input. The phase loads can now be modified by multiplying the loads by this ratio. Because the losses of the feeder will change when the loads are changed, it is necessary to go through the ladder iterative process to determine a new computed input to the feeder. This new computed input will be closer to the metered input, but most likely not within a specified tolerance. Again, a ratio can be determined and the loads modified. This process is repeated until the computed input is within a specified tolerance of the metered input.

Load allocation does not have to be limited to matching metered readings at the substation. The same process can be performed at any point on the feeder where metered data is available. The only difference is that only the load at nodes downstream from the metered point will be modified.

10.1.7 Summary of Power-Flow Studies

This section has developed a method for performing power-flow studies on a distribution feeder. Models for the various components of the feeder have been developed in previous chapters. The purpose of this section has been to develop and demonstrate the modified ladder iterative technique using the generalized matrices for the series elements. It should be obvious that a study of a large feeder with many laterals and sublaterals cannot be performed without the aid of a computer program.

The development of the models and examples in this text has used actual values of voltage, current, impedance, and complex power. When per-unit values are used, it is imperative that all values be converted using a common set of base values. In the usual application of per-unit there will be a base line-to-line voltage and a base line-to-neutral voltage; also, there will be a base line current and a base delta current. For both the voltage and current there is a square-root-of-three relationship between the two base values. In all of the derivations of the models, and in particular those for the three-phase transformers, the square root of three has been used to relate the difference in magnitudes between line-to-line and line-to-neutral voltages, and between the line and delta currents. Because of this, when using the per-unit system, there should be only one base voltage, and that should be the base line-to-neutral voltage. When this is done, for example, the per-unit positive and negative sequence voltages will be the square root of three times the per-unit positive and negative sequence line-to-neutral voltages.

Similarly, the positive and negative sequence per-unit line currents will be the square of three times the positive and negative sequence per-unit delta currents. By using just one base voltage and one base current, the per-unit generalized matrices for all system models can be determined.

10.2 Short-Circuit Studies

The computation of short-circuit currents for unbalanced faults in a normally balanced three-phase system has traditionally been accomplished by the application of symmetrical components. However, this method is not well suited to a distribution feeder that is inherently unbalanced. The unequal mutual coupling between phases leads to mutual coupling between sequence networks. When this happens there is no advantage to using symmetrical components. Another reason for not using symmetrical components is that the phases between which faults occur are limited. For example, using symmetrical components, line-to-ground faults are limited to Phase a to ground. What happens if a single-phase lateral is connected to Phase b or c and the short-circuit current is needed? This section will develop a method for short-circuit analysis of an unbalanced three-phase distribution feeder using the phase frame.

10.2.1 General Theory

Figure 10.9 shows the unbalanced feeder as modeled for short-circuit calculations. Short circuits can occur at any one of the five points shown in Figure 10.9. Point 1 is the high-voltage bus of the distribution substation transformer. The values of the short-circuit currents at Point 1 are normally determined from a transmission system short-circuit study. The results of these studies are supplied in terms of the three-phase and single-phase short-circuit MVAs. Using the short-circuit MVAs, the positive and zero sequence impedances of the equivalent system can be determined. These values are needed for the short-circuit studies at the other four points in Figure 10.9.

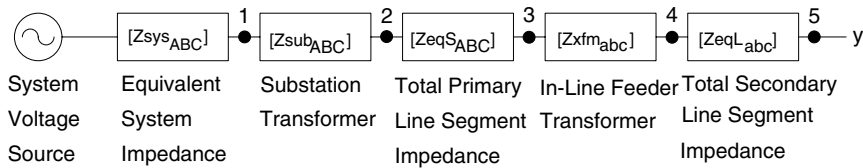
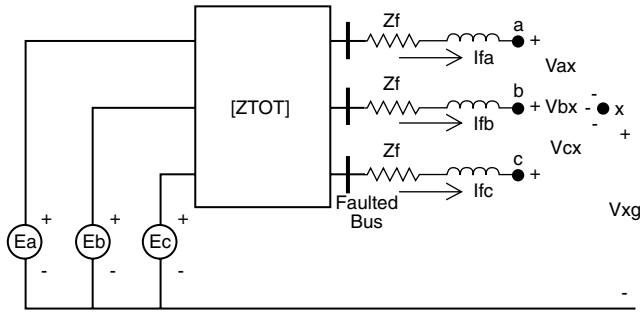


FIGURE 10.9 Unbalanced feeder short-circuit analysis model.

**FIGURE 10.10**

Thevenin equivalent circuit.

Given the three-phase short-circuit MVA magnitude and angle, the positive sequence equivalent system impedance in ohms is determined by:

$$Z_+ = \frac{kVLL^2}{(MVA_{3-phase})^*} \Omega \quad (10.10)$$

Given the single-phase short-circuit MVA magnitude and angle, the zero sequence equivalent system impedance in ohms is determined by:

$$Z_0 = \frac{3 \cdot kVLL^2}{(MVA_{1-phase})^*} - 2 \cdot Z_+ \Omega \quad (10.11)$$

In Equations 10.10 and 10.11, $kVLL$ is the nominal line-to-line voltage in kV of the transmission system.

The computed positive and zero sequence impedances need to be converted into the phase impedance matrix as defined in Equations 6.42 and 6.43 in Chapter 6.

For short circuits at points 2, 3, 4, and 5, it is going to be necessary to compute the Thevenin equivalent three-phase circuit at the short-circuit point. The Thevenin equivalent voltages will be the nominal line-to-ground voltages with the appropriate angles. For example, assume the equivalent system line-to-ground voltages are balanced three-phase of nominal voltage with the Phase a voltage at zero degrees. The Thevenin equivalent voltages at Points 2 and 3 will be computed by multiplying the system voltages by the generalized transformer matrix $[A_t]$ of the substation transformer. Carrying this further, the Thevenin equivalent voltages at Points 4 and 5 will be the voltages at Node 3 multiplied by the generalized matrix $[A_t]$ for the in-line transformer.

The Thevenin equivalent phase impedance matrices will be the sum of the phase impedance matrices of each device between the system voltage source

and the point of fault. Step-voltage regulators are assumed to be set in the neutral position so they do not enter into the short-circuit calculations. Anytime a delta-grounded wye transformer is encountered, the total phase impedance matrix on the primary side of the transformer must be referred to the secondary side using Equation 8.160.

Figure 10.10 illustrates the Thevenin equivalent circuit at the faulted node.³ In Figure 10.10, the voltage sources E_a , E_b , and E_c represent the Thevenin equivalent line-to-ground voltages at the faulted node. The matrix $[ZTOT]$ represents the Thevenin equivalent phase impedance matrix at the faulted node. The fault impedance is represented by Z_f in Figure 10.10.

Kirchhoff's voltage law in matrix form can be applied to the circuit of Figure 10.10:

$$\begin{bmatrix} E_a \\ E_b \\ E_c \end{bmatrix} = \begin{bmatrix} Z_{aa} & Z_{ab} & Z_{ac} \\ Z_{ba} & Z_{bb} & Z_{bc} \\ Z_{ca} & Z_{cb} & Z_{cc} \end{bmatrix} \cdot \begin{bmatrix} I_f^a \\ I_f^b \\ I_f^c \end{bmatrix} + \begin{bmatrix} Z_f & 0 & 0 \\ 0 & Z_f & 0 \\ 0 & 0 & Z_f \end{bmatrix} \cdot \begin{bmatrix} I_f^a \\ I_f^b \\ I_f^c \end{bmatrix} + \begin{bmatrix} V_{ax} \\ V_{bx} \\ V_{cx} \end{bmatrix} + \begin{bmatrix} V_{xg} \\ V_{xg} \\ V_{xg} \end{bmatrix} \quad (10.12)$$

Equation 10.12 can be written in compressed form as

$$[E_{abc}] = [ZTOT] \cdot [I_{fabc}] + [ZF] \cdot [I_{fabc}] + [V_{abcx}] + [V_{xg}] \quad (10.13)$$

Combine terms in Equation 10.13:

$$[E_{abc}] = [ZEQ] \cdot [I_{fabc}] + [V_{abcx}] + [V_{xg}] \quad (10.14)$$

where

$$[ZEQ] = [ZTOT] + [ZF] \quad (10.15)$$

Solve Equation 10.14 for the fault currents

$$[I_{fabc}] = [Y] \cdot [E_{abc}] - [Y] \cdot [V_{abcx}] - [Y] \cdot [V_{xg}] \quad (10.16)$$

where

$$[Y] = [ZEQ]^{-1} \quad (10.17)$$

Since the matrices $[Y]$ and $[E_{abc}]$ are known, define:

$$[IP_{abc}] = [Y] \cdot [E_{abc}] \quad (10.18)$$

Substituting Equation 10.18 into Equation 10.16 and rearranging results in

$$[IP_{abc}] = [I_{fabc}] + [Y] \cdot [V_{abcx}] + [Y] \cdot [V_{xg}] \quad (10.19)$$

Expanding Equation 10.19:

$$\begin{bmatrix} IP_a \\ IP_b \\ IP_c \end{bmatrix} = \begin{bmatrix} If_a \\ If_b \\ If_c \end{bmatrix} + \begin{bmatrix} Y_{aa} & Y_{ab} & Y_{ac} \\ Y_{ba} & Y_{bb} & Y_{bc} \\ Y_{ca} & Y_{cb} & Y_{cc} \end{bmatrix} \cdot \begin{bmatrix} V_{ax} \\ V_{bx} \\ V_{cx} \end{bmatrix} + \begin{bmatrix} Y_{aa} & Y_{ab} & Y_{ac} \\ Y_{ba} & Y_{bb} & Y_{bc} \\ Y_{ca} & Y_{cb} & Y_{cc} \end{bmatrix} + \begin{bmatrix} V_{xg} \\ V_{xg} \\ V_{xg} \end{bmatrix} \quad (10.20)$$

Performing the matrix operations in Equation 10.19:

$$\begin{aligned} IP_a &= If_a + (Y_{aa} \cdot V_{ax} + Y_{ab} \cdot V_{bx} + Y_{ac} \cdot V_{cx}) + Ys_a \cdot V_{xg} \\ IP_b &= If_b + (Y_{ba} \cdot V_{ax} + Y_{bb} \cdot V_{bx} + Y_{bc} \cdot V_{cx}) + Ys_b \cdot V_{xg} \\ IP_c &= If_c + (Y_{ca} \cdot V_{ax} + Y_{cb} \cdot V_{bx} + Y_{cc} \cdot V_{cx}) + Ys_c \cdot V_{xg} \end{aligned} \quad (10.21)$$

where

$$\begin{aligned} Ys_a &= Y_{aa} + Y_{ab} + Y_{ac} \\ Ys_b &= Y_{ba} + Y_{bb} + Y_{bc} \\ Ys_c &= Y_{ca} + Y_{cb} + Y_{cc} \end{aligned} \quad (10.22)$$

Equations 10.21 become the general equations that are used to simulate all types of short circuits. Basically, there are three equations and seven unknowns ($If_a, If_b, If_c, V_{ax}, V_{bx}, V_{cx}$ and V_{xg}). The other three variables in the equations (IP_a, IP_b and IP_c) are functions of the total impedance and the Thevenin voltages and are therefore known. In order to solve Equations 10.21, it will be necessary to specify four additional independent equations. These equations are functions of the type of fault being simulated. The additional required four equations for various types of faults are given below. These values are determined by placing short circuits in Figure 10.11 to simulate the particular type of fault. For example, a three-phase fault is simulated by placing a short circuit from Node a to x, Node b to x, and Node c to x. That gives three voltage equations. The fourth equation comes from applying Kirchhoff's current law at Node x, which gives the sum of the fault currents to be zero.

10.2.2 Specific Short Circuits

THREE-PHASE FAULTS:

$$\begin{aligned} V_{ax} &= V_{bx} = V_{cx} = 0 \\ I_a + I_b + I_c &= 0 \end{aligned} \quad (10.23)$$

THREE-PHASE-TO-GROUND FAULTS:

$$V_{ax} = V_{bx} = V_{cx} = V_{xg} = 0 \quad (10.24)$$

LINE-TO-LINE FAULTS (assume i - j fault with phase k unfaulted):

$$\begin{aligned} V_{ix} &= V_{jx} = 0 \\ I_{fk} &= 0 \\ I_{fi} + I_{fj} &= 0 \end{aligned} \quad (10.25)$$

LINE-TO-GROUND FAULTS (assume phase k fault with phases i and j unfaulted):

$$\begin{aligned} V_{kx} &= V_{xg} = 0 \\ I_{fi} &= I_{fj} = 0 \end{aligned} \quad (10.26)$$

Notice that Equations 10.25 and 10.26 will allow the simulation of line-to-line faults and line-to-ground faults for all phases. There is no limitation to b - c faults for line-to-line and a - g for line-to-ground, as is the case when the method of symmetrical components is employed.

A good way to solve the seven equations is to set them up in matrix form:

$$\begin{bmatrix} IP_a \\ IP_b \\ IP_c \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & Y_{1,1} & Y_{1,2} & Y_{1,3} & Y_{s1} \\ 0 & 1 & 0 & Y_{2,1} & Y_{2,2} & Y_{2,3} & Y_{s2} \\ 0 & 0 & 1 & Y_{3,1} & Y_{3,2} & Y_{3,3} & Y_{s3} \\ - & - & - & - & - & - & - \\ - & - & - & - & - & - & - \\ - & - & - & - & - & - & - \\ - & - & - & - & - & - & - \end{bmatrix} \cdot \begin{bmatrix} I_{fa} \\ I_{fb} \\ I_{fc} \\ V_{ax} \\ V_{bx} \\ V_{cx} \\ V_{xg} \end{bmatrix} \quad (10.27)$$

Equation 10.27 in condensed form:

$$[IP_s] = [C] \cdot [X] \quad (10.28)$$

Equation 10.28 can be solved for the unknowns in matrix $[X]$:

$$[X] = [C]^{-1} \cdot [IP_s] \quad (10.29)$$

The blanks in the last four rows of the coefficient matrix in Equation 10.27 are filled in with the known variables, depending upon what type of fault is to be simulated. For example, the elements in the $[C]$ matrix simulating a three-phase fault would be

$$\begin{aligned} C_{4,4} &= 1, \quad C_{5,5} = 1, \quad C_{6,6} = 1 \\ C_{7,1} &= C_{7,2} = C_{7,3} = 1 \end{aligned}$$

All of the other elements in the last four rows will be set to zero.

Example 10.3

Use the system of Example 10.2 and compute the short-circuit currents for a bolted ($Z_f = 0$) line-to-line fault between Phases a and b at Node 4.

The infinite bus balanced line-to-line voltages are 12.47 kV, which leads to balanced line-to-neutral voltages at 7.2 kV.

$$[ELL_s] = \begin{bmatrix} 12,470/\underline{30} \\ 12,470/\underline{-90} \\ 12,470/\underline{150} \end{bmatrix} \text{ V}$$

$$[ELN_s] = \begin{bmatrix} 7199.6/\underline{0} \\ 7199.6/\underline{-120} \\ 7199.6/\underline{120} \end{bmatrix} \text{ V}$$

The line-to-neutral Thevenin circuit voltages at Node 4 are determined using Equation 8.159:

$$[Eth_4] = [A_t] \cdot [ELN_s] = \begin{bmatrix} 2400/\underline{-30} \\ 2400/\underline{-150} \\ 2400/\underline{90} \end{bmatrix} \text{ V}$$

The Thevenin equivalent impedance at the secondary terminals (Node 3) of the transformer consists of the primary line impedances referred across the transformer plus the transformer impedances. Using Equation 8.160:

$$[Zth_3] = [A_t] \cdot [ZeqS_{ABC}] \cdot [d_t] + [Zt_{abc}]$$

$$[Zth_3] = \begin{bmatrix} 0.0366 + j0.1921 & -0.0039 - j0.0086 & -0.0039 - j0.0106 \\ -0.0039 - j0.0086 & 0.0366 + j0.1886 & -0.0039 - j0.0071 \\ -0.0039 - j0.0106 & -0.0039 - j0.0071 & 0.0366 + j0.1906 \end{bmatrix} \Omega$$

Note that the Thevenin impedance matrix is not symmetrical. This is a result, once again, of the unequal mutual coupling between the phases of the primary line segment.

The total Thevenin impedance at Node 4 is

$$[Zth_4] = [ZTOT] = [Zth_3] + [ZeqL_{abc}]$$

$$[ZTOT] = \begin{bmatrix} 0.2273 + j0.6955 & 0.0568 + j0.2216 & 0.0559 + j0.1645 \\ 0.0568 + j0.2216 & 0.2305 + j0.6771 & 0.0575 + j0.1860 \\ 0.0559 + j0.1645 & 0.0575 + j0.1860 & 0.2287 + j0.6876 \end{bmatrix} \Omega$$

The equivalent admittance matrix at Node 4 is

$$[Yeq_4] = [ZTOT]^{-1}$$

$$[Yeq_4] = \begin{bmatrix} 0.5031 - j1.4771 & -0.1763 + j0.3907 & -0.0688 + j0.2510 \\ -0.1763 + j0.3907 & 0.5501 - j1.5280 & -0.1148 + j0.3133 \\ -0.0688 + j0.2510 & -0.1145 + j0.3133 & 0.4843 - j1.4532 \end{bmatrix} \text{ S}$$

Using Equation 10.18, the equivalent injected currents at the point of fault are

$$[IP] = [Yeq_4] \cdot [Eth_4] = \begin{bmatrix} 4466.8 / -96.4 \\ 4878.9 / 138.0 \\ 4440.9 / 16.4 \end{bmatrix} \text{ A}$$

The sums of each row of the equivalent admittance matrix are computed according to Equation 10.22:

$$Y_i = \sum_{k=1}^3 Yeq_{i,k} = \begin{bmatrix} 0.2580 - j0.8353 \\ 0.2590 - j0.8240 \\ 0.3007 - j0.8889 \end{bmatrix} \text{ S}$$

For the a-b fault at Node 4, according to Equation 10.25:

$$If_a + If_b = 0$$

$$I_c = 0$$

$$V_{ax} = 0$$

$$V_{bx} = 0$$

The coefficient matrix [C] using Equation 10.27:

$$[C] = \begin{bmatrix} 1 & 0 & 0 & 0.501 - j1.477 & -0.176 + j0.390 & -0.069 + j0.252 & 0.258 - j0.835 \\ 0 & 1 & 0 & -0.176 + j0.390 & 0.550 - j1.528 & -0.115 + j0.314 & 0.259 - j0.824 \\ 0 & 0 & 1 & -0.069 + j0.251 & -0.115 + j0.313 & 0.484 - j1.452 & 0.301 - j0.889 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

The injected current matrix is

$$[IP_s] = \begin{bmatrix} 4466.8 / \underline{-96.4} \\ 4878.9 / \underline{138.0} \\ 4440.9 / \underline{16.4} \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

The unknowns are computed by:

$$[X] = [C]^{-1} \cdot [IP_s] = \begin{bmatrix} 8901.7 / \underline{-8.4} \\ 8901.7 / \underline{171.6} \\ 0 \\ 7740.4 / \underline{-90.6} \\ 0 \\ 0 \\ 2587.9 / \underline{89.1} \end{bmatrix}$$

The interpretation of the results are

$$If_a = X_1 = 8901.7 / \underline{-8.4}$$

$$If_b = X_2 = 8901.7 / \underline{171.6}$$

$$If_c = X_3 = 0$$

$$V_{ax} = X_4 = 7740.4 / \underline{-90.6}$$

$$V_{bx} = X_5 = 0$$

$$V_{cx} = X_6 = 0$$

$$V_{xg} = X_7 = 2587.9 / \underline{89.1}$$

Using the line-to-ground voltages at Node 4 and the short-circuit currents and working back to the source, using the generalized matrices will check the validity of these results. The line-to-ground voltages at Node 4 are

$$[VLG_4] = \begin{bmatrix} V_{ax} + V_{xg} \\ V_{bx} + V_{xg} \\ V_{cx} + V_{xg} \end{bmatrix} = \begin{bmatrix} 5153.4 / \underline{-90.4} \\ 2587.2 / \underline{89.1} \\ 2587.2 / \underline{89.1} \end{bmatrix} \text{ V}$$

The short-circuit currents in matrix form:

$$[I_4] = [I_3] = \begin{bmatrix} 8901.7 / -8.4 \\ 8901.7 / 171.6 \\ 0 \end{bmatrix} \text{ A}$$

The line-to-ground voltages at Node 3 are

$$[VLG_3] = [a_2] \cdot [VLG_4] + [b_1] \cdot [I_4] = \begin{bmatrix} 3261.1 / -63.4 \\ 1544.3 / 161.7 \\ 2430.9 / 89.9 \end{bmatrix} \text{ V}$$

The equivalent line-to-neutral voltages and line currents at the primary terminals (Node 2) of the transformer are

$$[VLN_2] = [a_t] \cdot [VLG_3] + [b_t] \cdot [I_3] = \begin{bmatrix} 6766.3 / -6.4 \\ 6833.7 / -119.6 \\ 7480.3 / 116.6 \end{bmatrix} \text{ V}$$

$$[I_2] = [d_t] \cdot [I_3] = \begin{bmatrix} 3426.4 / -8.4 \\ 1713.2 / 171.6 \\ 1713.2 / 171.6 \end{bmatrix} \text{ A}$$

Finally, the equivalent line-to-neutral voltages at the infinite bus can be computed:

$$[VLN_1] = [a_1] \cdot [VLN_2] + [b_1] \cdot [I_2] = \begin{bmatrix} 7199.6 / 0 \\ 7199.6 / -120 \\ 7199.6 / 120 \end{bmatrix} \text{ V}$$

These are the same equivalent line-to-neutral voltages that were used to start the short-circuit analysis.

10.3 Summary

This chapter has taken the component models of a distribution feeder and developed techniques for power-flow and short-circuit analyses. The techniques do not lend themselves to hand calculations, but have been developed with a computer program in mind. In developing the models and analyses

techniques, the importance of modeling the system components exactly has been emphasized. Because of the unbalanced nature of a distribution feeder, without exact models the results of the analyses are suspect.

The examples in this chapter have been very long and should be used as learning tools. Many of the interesting operating characteristics of a feeder can only be demonstrated through numerical examples. The examples were designed to illustrate some of these characteristics.

Armed with a computer program, using the models and techniques of this text provides the engineer with powerful tools for solving present-day problems and performing long-range planning studies.

References

1. Trevino, C., Cases of difficult convergence in load-flow problems, *IEEE Paper n. 71-62-PWR*, Presented at the IEEE Summer Power Meeting, Los Angeles, 1970.
2. Kersting, W.H. and Mendive, D.L., An application of ladder network theory to the solution of three-phase radial load-flow problems, *IEEE Conference Paper*, Presented at the IEEE Winter Power Meeting, New York, January 1976.
3. Kersting, W.H. and Phillips, W.H., Distribution system short-circuit analysis, *25th Intersociety Energy Conversion Engineering Conference*, Reno, NV, August 12–17, 1990.

Problems

The power-flow problems in this set require an iterative solution. Students should be encouraged to write their own computer program to solve the problems.

The first six problems of this set will be based upon the system in Figure 10.11.

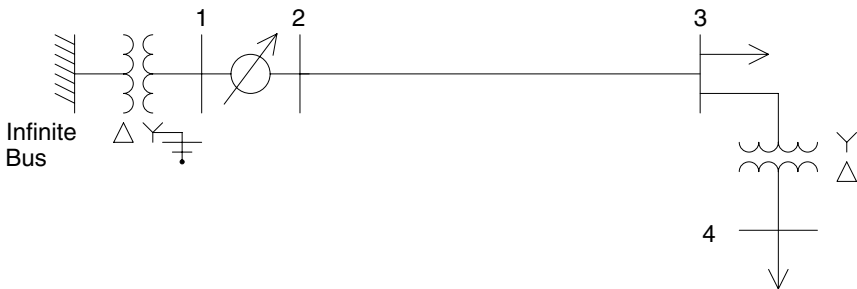


FIGURE 10.11

Wye homework system.

The substation transformer is connected to an infinite bus with balanced three-phase voltages of 69 kV. The substation transformer is rated:

$$5000 \text{ kVA, } 69 \text{ kV delta} - 4.16 \text{ grounded wye, } Z = 1.5 + j8.0\%$$

The phase impedance matrix for a four-wire wye line is

$$[z_{4\text{-wire}}] = \begin{bmatrix} 0.4576 + j1.0780 & 0.1560 + j0.5017 & 0.1535 + j0.3849 \\ 0.1560 + j0.5017 & 0.4666 + j1.0482 & 0.1580 + j0.4236 \\ 0.1535 + j0.3849 & 0.1580 + j0.4236 & 0.4615 + j1.0651 \end{bmatrix} \Omega/\text{mile}$$

The secondary voltages of the substation transformer are balanced and being held at 4.16 kV for all power-flow problems.

The four-wire wye feeder is 0.75 miles long. An unbalanced wye-connected load is located at Node 3 and has the following values:

Phase a: 750 kVA at 0.85 lagging power factor

Phase b: 500 kVA at 0.90 lagging power factor

Phase c: 850 kVA at 0.95 lagging power factor

The load at Node 4 is zero initially.

10.1 For the system as described above and assuming that the regulators are in the neutral position.

- (1) Determine the generalized matrices for the line segment.
- (2) Use the modified ladder technique to determine the line-to-ground voltages at Node 3. Use a tolerance of 0.001 per-unit. Give the voltages in actual values in volts and on a 120-volt base.

10.2 Three Type B step-voltage regulators are installed in a wye connection at the substation in order to hold the load voltages at a voltage level of 121 V and a bandwidth of 2 V.

- (1) Compute the actual equivalent line impedance between the substation and the load node.
- (2) Use a potential transformer ratio of 2400-120 V and a current transformer ratio of 500:5 A. Determine the R and X compensator settings calibrated in volts and ohms. The settings must be the same for all three regulators.
- (3) For the load conditions of Problem 10.1 and with the regulators in the neutral position, compute the voltages across the voltage relays in the compensator circuits.

- (4) Determine the appropriate tap settings for the three regulators to hold the Node 3 voltages at 121 V in a bandwidth of 2 V.
- (5) With the regulator taps set, compute the load voltages on a 120-V base.

10.3 A wye-connected three-phase shunt capacitor bank of 300 kvar per phase is installed at Node 3. With the regulator compensator settings from Problem 10.2, determine:

- (1) The new tap settings for the three regulators.
- (2) The voltages at the load on a 120-V base.
- (3) The voltages across the relays in the compensator circuits.

10.4 The load at Node 4 is served through an ungrounded wye–delta transformer bank. The load is connected in delta with the following values:

Phase a-b: 400 kVA at 0.9 factor power factor

Phase b-c: 150 kVA at 0.8 lagging power factor

Phase c-a: 150 kVA at 0.8 lagging power factor

The three single-phase transformers are rated as:

Lighting transformer: 500 kVA, 2400–240 V, $Z = 0.9 + j3.0\%$

Power transformers: 167 kVA, 2400–240 V, $Z = 1.0 + j1.6\%$

Use the original loads and the shunt capacitor bank at Node 3 and this new load at Node 4. Determine:

- (1) The voltages on a 120-V base at Node 3, assuming the regulators are in the neutral position.
- (2) The voltages on a 120-V base at Node 4, assuming the regulators are in the neutral position.
- (3) The new tap settings for the three regulators.
- (4) The Node 3 and 4 voltages on a 120-V base after the regulators have changed tap positions.

10.5 Under short-circuit conditions the infinite bus voltage is the only voltage that is constant. The voltage regulators in the substation are in the neutral position. Determine the short-circuit currents and voltages at Nodes 1, 2, 3 for the following short-circuits at Node 3:

- (1) Three-phase to ground
- (2) Phase b to ground
- (3) Line-to-line fault on Phases a-c

10.6 A line-to-line fault occurs at Node 4. Determine the currents in the fault and on the line segment between Nodes 2 and 3. Determine the voltages at Nodes 1, 2, 3, and 4.

10.7 A three-wire delta line of length 0.75 miles is serving an unbalanced delta load of:

Phase a-b: 600 kVA, 0.9 lagging power factor

Phase b-c: 800 kVA, 0.8 lagging power factor

Phase c-a: 500 kVA, 0.95 lagging power factor

The phase impedance matrix for the line is

$$[z_{3\text{-wire}}] = \begin{bmatrix} 0.4013 + j1.4133 & 0.0953 + j0.8515 & 0.0953 + j0.7802 \\ 0.0953 + j0.8515 & 0.4013 + j1.4133 & 0.0953 + j0.7266 \\ 0.0953 + j0.7802 & 0.0953 + j0.7266 & 0.4013 + j1.4133 \end{bmatrix} \Omega/\text{mile}$$

The line is connected to a constant balanced voltage source of 4.8 kV line-to-line. Determine the load voltages on a 120-V base.

10.8 Add two Type B step-voltage regulators in an open delta connection, using phases A-B and B-C, to the system in Problem 10.7. The regulator should be set to hold 121 ± 1 V. Determine the R and X settings and the final tap settings. For the open delta connection, the R and X settings will be different on the two regulators.

10.9 The three-wire line of Problem 10.6 is connected to a substation transformer connected delta-delta. The substation transformer is connected to a 69-kV infinite bus and is rated:

$$10,000 \text{ kVA}, \quad 69 \text{ kV delta} - 4.8 \text{ kV delta}, \quad Z = 1.6 + j7.8\%$$

Determine the short-circuit currents and substation transformer secondary voltages for the following short circuits at the end of the line:

- (1) Three-phase
- (2) Line-to-line between phases a-b

Appendix A

Conductor Data						
Size	Stranding	Material	DIAM Inches	GMR Feet	RES Ω/mile	Capacity Amps
1		ACSR	0.355	0.00418	1.38	200
1	7 STRD	Copper	0.328	0.00992	0.765	270
1	CLASS A	AA	0.328	0.00991	1.224	177
2	6/1	ACSR	0.316	0.00418	1.69	180
2	7 STRD	Copper	0.292	0.00883	0.964	230
2	7/1	ACSR	0.325	0.00504	1.65	180
2	AWG SLD	Copper	0.258	0.00836	0.945	220
2	CLASS A	AA	0.292	0.00883	1.541	156
3	6/1	ACSR	0.281	0.0043	2.07	160
3	AWG SLD	Copper	0.229	0.00745	1.192	190
4	6/1	ACSR	0.25	0.00437	2.57	140
4	7/1	ACSR	0.257	0.00452	2.55	140
4	AWG SLD	Copper	0.204	0.00663	1.503	170
4	CLASS A	AA	0.232	0.007	2.453	90
5	6/1	ACSR	0.223	0.00416	3.18	120
5	AWG SLD	Copper	0.1819	0.0059	1.895	140
6	6/1	ACSR	0.198	0.00394	3.98	100
6	AWG SLD	Copper	0.162	0.00526	2.39	120
6	CLASS A	AA	0.184	0.00555	3.903	65
7	AWG SLD	Copper	0.1443	0.00468	3.01	110
8	AWG SLD	Copper	0.1285	0.00416	3.8	90
9	AWG SLD	Copper	0.1144	0.00371	4.6758	80
10	AWG SLD	Copper	0.1019	0.00330	5.9026	75
12	AWG SLD	Copper	0.0808	0.00262	9.3747	40
14	AWG SLD	Copper	0.0641	0.00208	14.8722	20
16	AWG SLD	Copper	0.0508	0.00164	23.7262	10
18	AWG SLD	Copper	0.0403	0.00130	37.6726	5
19	AWG SLD	Copper	0.0359	0.00116	47.5103	4
20	AWG SLD	Copper	0.032	0.00103	59.684	3
22	AWG SLD	Copper	0.0253	0.00082	95.4835	2
24	AWG SLD	Copper	0.0201	0.00065	151.616	1
1/0		ACSR	0.398	0.00446	1.12	230
1/0	7 STRD	Copper	0.368	0.01113	0.607	310
1/0	CLASS A	AA	0.368	0.0111	0.97	202
2/0		ACSR	0.447	0.0051	0.895	270
2/0	7 STRD	Copper	0.414	0.01252	0.481	360
2/0	CLASS A	AA	0.414	0.0125	0.769	230
3/0	12 STRD	Copper	0.492	0.01559	0.382	420

(Continued)

Conductor Data (continued)

Size	Stranding	Material	DIAM Inches	GMR Feet	RES Ω/mile	Capacity Amps
3/0	6/1	ACSR	0.502	0.006	0.723	300
3/0	7 STRD	Copper	0.464	0.01404	0.382	420
3/0	CLASS A	AA	0.464	0.014	0.611	263
3/8	INCH STE	Steel	0.375	0.00001	4.3	150
4/0	12 STRD	Copper	0.552	0.0175	0.303	490
4/0	19 STRD	Copper	0.528	0.01668	0.303	480
4/0	6/1	ACSR	0.563	0.00814	0.592	340
4/0	7 STRD	Copper	0.522	0.01579	0.303	480
4/0	CLASS A	AA	0.522	0.0158	0.484	299
250,000	12 STRD	Copper	0.6	0.01902	0.257	540
250,000	19 STRD	Copper	0.574	0.01813	0.257	540
250,000	CON LAY	AA	0.567	0.0171	0.41	329
266,800	26/7	ACSR	0.642	0.0217	0.385	460
266,800	CLASS A	AA	0.586	0.0177	0.384	320
300,000	12 STRD	Copper	0.657	0.0208	0.215	610
300,000	19 STRD	Copper	0.629	0.01987	0.215	610
300,000	26/7	ACSR	0.68	0.023	0.342	490
300,000	30/7	ACSR	0.7	0.0241	0.342	500
300,000	CON LAY	AA	0.629	0.0198	0.342	350
336,400	26/7	ACSR	0.721	0.0244	0.306	530
336,400	30/7	ACSR	0.741	0.0255	0.306	530
336,400	CLASS A	AA	0.666	0.021	0.305	410
350,000	12 STRD	Copper	0.71	0.0225	0.1845	670
350,000	19 STRD	Copper	0.679	0.0214	0.1845	670
350,000	CON LAY	AA	0.679	0.0214	0.294	399
397,500	26/7	ACSR	0.783	0.0265	0.259	590
397,500	30/7	ACSR	0.806	0.0278	0.259	600
397,500	CLASS A	AA	0.724	0.0228	0.258	440
400,000	19 STRD	Copper	0.726	0.0229	0.1619	730
450,000	19 STRD	Copper	0.77	0.0243	0.1443	780
450,000	CON LAG	AA	0.77	0.0243	0.229	450
477,000	26/7	ACSR	0.858	0.029	0.216	670
477,000	30/7	ACSR	0.883	0.0304	0.216	670
477,000	CLASS A	AA	0.795	0.0254	0.216	510
500,000	19 STRD	Copper	0.811	0.0256	0.1303	840
500,000	37 STRD	Copper	0.814	0.026	0.1303	840
500,000	CON LAY	AA	0.813	0.026	0.206	483
556,500	26/7	ACSR	0.927	0.0313	0.1859	730
556,500	30/7	ACSR	0.953	0.0328	0.1859	730
556,500	CLASS A	AA	0.858	0.0275	0.186	560
600,000	37 STRD	Copper	0.891	0.0285	0.1095	940
600,000	CON LAY	AA	0.891	0.0285	0.172	520
605,000	26/7	ACSR	0.966	0.0327	0.172	760
605,000	54/7	ACSR	0.953	0.0321	0.1775	750
636,000	27/7	ACSR	0.99	0.0335	0.1618	780
636,000	30/19	ACSR	1.019	0.0351	0.1618	780
636,000	54/7	ACSR	0.977	0.0329	0.1688	770
636,000	CLASS A	AA	0.918	0.0294	0.163	620

Conductor Data (continued)

Size	Stranding	Material	DIAM Inches	GMR Feet	RES Ω /mile	Capacity Amps
666,600	54/7	ACSR	1	0.0337	0.1601	800
700,000	37 STRD	Copper	0.963	0.0308	0.0947	1040
700,000	CON LAY	AA	0.963	0.0308	0.148	580
715,500	26/7	ACSR	1.051	0.0355	0.1442	840
715,500	30/19	ACSR	1.081	0.0372	0.1442	840
715,500	54/7	ACSR	1.036	0.0349	0.1482	830
715,500	CLASS A	AA	0.974	0.0312	0.145	680
750,000	37 STRD	AA	0.997	0.0319	0.0888	1090
750,000	CON LAY	AA	0.997	0.0319	0.139	602
795,000	26/7	ACSR	1.108	0.0375	0.1288	900
795,000	30/19	ACSR	1.14	0.0393	0.1288	910
795,000	54/7	ACSR	1.093	0.0368	0.1378	900
795,000	CLASS A	AA	1.026	0.0328	0.131	720

Appendix B

Concentric Neutral 15 kV Cable

Conductor Size AWG or kcmil	Diameter over Insulation Inches	Diameter over Screen Inches	Outside Diameter Inches	Copper Neutral No. x AWG	Ampacity UG Duct Amps
<i>Full Neutral</i>					
2(7×)	0.78	0.85	0.98	10 × 14	120
1(19×)	0.81	0.89	1.02	13 × 14	135
1/0(19×)	0.85	0.93	1.06	16 × 14	155
2/0(19×)	0.90	0.97	1.13	13 × 12	175
3/0(19×)	0.95	1.02	1.18	16 × 12	200
4/0(19×)	1.01	1.08	1.28	13 × 10	230
250(37×)	1.06	1.16	1.37	16 × 10	255
350(37×)	1.17	1.27	1.47	20 × 10	300
<i>1/3 Neutral</i>					
2(7×)	0.78	0.85	0.98	6 × 14	135
1(19×)	0.81	0.89	1.02	6 × 14	155
1/0(19×)	0.85	0.93	1.06	6 × 14	175
2/0(19×)	0.90	0.97	1.10	7 × 14	200
3/0(19×)	0.95	1.02	1.15	9 × 14	230
4/0(19×)	1.01	1.08	1.21	11 × 14	240
250(37×)	1.06	1.16	1.29	13 × 14	260
350(37×)	1.17	1.27	1.39	18 × 14	320
500(37×)	1.29	1.39	1.56	16 × 12	385
750(61×)	1.49	1.59	1.79	15 × 10	470
1000(61×)	1.64	1.77	1.98	20 × 10	550

Tape-Shielded 15 kV Cable
Tape Thickness = 5 mils

Conductor Size AWG or kcmil	Diameter over Insulation Inches	Diameter over Screen Inches	Jacket Thickness mils	Outside Diameter Inches	Ampacity in UG Duct Amps
1/0	0.82	0.88	80	1.06	165
2/0	0.87	0.93	80	1.10	190
3/0	0.91	0.97	80	1.16	215
4/0	0.96	1.02	80	1.21	245
250	1.01	1.08	80	1.27	270
350	1.11	1.18	80	1.37	330
500	1.22	1.30	80	1.49	400
750	1.40	1.48	110	1.73	490
1000	1.56	1.66	110	1.91	565

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